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NATIONAL STANDARD
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GB 50010-2010

Code for Design of Concrete Structures
混凝土结构设计规范

Issued on August 18, 2010

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Jointly issued by **the Ministry of Housing and Urban-rural Development (MOHUD) and the General Administration of Quality Supervision, Inspection and Quarantine (AQSIQ) of the People's Republic of China**

National Standard of the People's Republic of China

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Announcement on Publishing the National Standard of "Code for Design of Concrete Structures"

"Code for Design of Concrete Structures" has been approved as a national standard with a serial number of GB 50010-2010, and it shall be implemented from July 1, 2011. Therein, Articles 3.1.7, 3.3.2, 4.1.3, 4.1.4, 4.2.2, 4.2.3, 8.5.1, 10.1.1, 11.1.3, 11.2.3, 11.3.1, 11.3.6, 11.4.12 and 11.7.14 are compulsory provisions and must be enforced strictly. The former "Code for Design of Concrete Structures" (GB 50010-2002) shall be abolished simultaneously.

Authorized by the Standard Rating Research Institute of the Ministry of Housing and Urban-Rural Development of the People's Republic of China, this code is published and distributed by China Architecture and Building Press.

Ministry of Housing and Urban-Rural Development of the People's Republic of China
August 18, 2010

Foreword

According to the requirements of Document Jian Biao [2006]NO.77 issued by the former Ministry of Construction (MOC) -“Notice on Printing the Development and Revision Plan (First One) of National Engineering Construction Standards in 2006”, China Academy of Building Research, in conjunction with the organizations concerned, revised this code through extensive investigations and studies by earnestly summarizing the experiences in actual practices and by referring to the relevant international standards and foreign advanced standards as well as the relevant opinions.

The main contents of this code are: General Provisions, Terms and Symbols, General Requirements, Materials, Structural Analysis, Calculation of Ultimate Limit States, Checking of Serviceability Limit States, Detailing Requirements, Fundamental Requirements for Structural Members, Prestressed Concrete Structural Members, Seismic Design of Reinforced Concrete Structural Members and the relevant appendixes.

There have been some significant changes in this code in the following technical aspects: 1. The provisions on the principles of structural scheme, progressive collapse protection of structure, design of existing structures and design of unbounded prestressing were supplemented; 2. The relevant provisions on the checking of serviceability limit states were amended; 3. The ribbed steel reinforcement of Grade 500MPa was added, and the steel reinforcement of Grade 235MPa was replaced by plain round steel reinforcement of Grade 300MPa; 4. The relevant provisions on the design of compound loaded members were supplemented, and the formulae for the calculation of shear and punching shear bearing capacity were amended; 5. The relevant provisions on the cover thickness and anchorage length of steel reinforcement as well as on the minimum ratio of reinforcement of longitudinal stressed steel reinforcement were adjusted; 6. The relevant provisions on the seismic design of two-way shear members of column, connecting beam, shear wall and other boundary members were supplemented and amended; 7. The relevant requirements of the seismic design of prestressed concrete members and slab-column joints were supplemented and amended.

The provisions printed in bold type in this code are compulsory ones and must be enforced strictly.

The Ministry of Housing and Urban-Rural Development is in charge of the administration of this code and the explanation of the compulsory provisions; the China Academy of Building Research is responsible for the explanation of specific technical contents. The relevant opinions and advice, whenever necessary, can be posted or passed on to the National Standard "Code for Design of Concrete Structures" Administrative Group of China Academy of Building Research (address: No. 30, Beisanhuan East Road, Beijing City, 100013, China).

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Contents

1	General Provisions	1
2	Terms and Symbols	2
2.1	Terms	2
2.2	Symbols	3
3	General Requirements	6
3.1	General	6
3.2	Structural Scheme	7
3.3	Calculation of Ultimate Limit States	7
3.4	Checking of Serviceability Limit States	9
3.5	Durability Design	11
3.6	Principles for Design of Preventing Progressive Collapse	14
3.7	Principles for Design of Existing Structures	15
4	Materials	17
4.1	Concrete	17
4.2	Steel Reinforcement	19
5	Structural Analysis	24
5.1	General	24
5.2	Analysis Model	24
5.3	Elastic Analysis	26
5.4	Analysis on Plastic Redistribution of Internal Forces	27
5.5	Elastic-Plastic Analysis	27
5.6	Plastic Limit Analysis	28
5.7	Indirect Action Effect Analysis	29
6	Calculation of Ultimate Limit States	30
6.1	General	30
6.2	Calculation of Normal Section Load-bearing Capacity	30
6.3	Calculation of Inclined Section Load-bearing Capacity	47
6.4	Calculation of Load-bearing Capacity of Distortion Section	55
6.5	Calculation of Punching Shear Bearing Capacity	64
6.6	Calculation of Partial Compression Load-bearing Capacity	67
6.7	Checking of Fatigue	69
7	Checking of Serviceability Limit States	77
7.1	Checking of Cracks	77
7.2	Checking of Deflection of Flexural Members	85
8	Detailing Requirements	89
8.1	Expansion Joint	89
8.2	Concrete Cover	90
8.3	Anchorage of Steel Reinforcement	91
8.4	Splices of Steel Reinforcement	93
8.5	Minimum Ratio of Reinforcement for Longitudinal Stressed Steel Reinforcement	96
9	Fundamental Requirements for Structural Members	98

9.1	Slabs	98
9.2	Beams	102
9.3	Columns, Beam-column Joints and Brackets	109
9.4	Walls.....	118
9.5	Composite Members	120
9.6	Precast Concrete Structures	121
9.7	Embedded Parts and Connecting Pieces.....	123
10	Prestressed Concrete Structural Members	127
10.1	General	127
10.2	Calculation of Value for Loss of Prestress.....	136
10.3	Detailing of Prestressed Concrete Members.....	141
11	Seismic Design of Reinforced Concrete Structural Members.....	148
11.1	General.....	148
11.2	Materials	151
11.3	Frame Beams	152
11.4	Frame Columns and Frame-supported Columns	156
11.5	Column of Hinged Bent.....	164
11.6	Nodes of Frame Beam Column.....	166
11.7	Shear Walls and Connecting Beams	171
11.8	Prestressed Concrete Structural Members	181
11.9	Slab-column Joints	183
Appendix A	Nominal Diameter, Nominal Sectional Area and Theoretical Weight of Steel Reinforcement.....	186
Appendix B	Amplified Coefficient Method for Approximate Calculation of Sway Second-order Effect of Eccentric Compression Members.....	187
Appendix C	Constitutive Relations for Steel Reinforcement and Concrete and the Multi-axial Strength Criterion for Concrete	189
Appendix D	Design of Plain Concrete Structural Members.....	201
Appendix E	Calculation for Load-bearing Capacity of Arbitrary Sections, Circular and Annular Normal Sections of Members.....	205
Appendix F	Design Value of Equivalent Concentrated Reaction Used for Calculation of Slab-column Joints	210
Appendix G	Deep Flexural Members.....	215
Appendix H	Composite Beam and Slab Without Support.....	221
Appendix J	Prestress Loss of Curved Post-tensioned Prestressing Tendons and/or Bars Due to Anchorage Deformation and Tendon Shrinkage	227
Appendix K	Time-dependent Loss of Prestress	232
	Explanation of Wording in This Code	235
	List of Quoted Standards	236

1 General Provisions

1.0.1 This code was formulated with a view to implementing the national technical and economic policies in the design of concrete structures, achieving safety, applicability and economy and guaranteeing quality.

1.0.2 This code is applicable to the design of reinforced concrete, prestressed concrete and plain concrete structures of buildings and general structures. However, it not applicable the design of lightweight aggregate concrete structures or structures using special concrete.

1.0.3 This code was formulated based on the principle of the current national standards "Unified Standard for Reliability Design of Engineering Structures" (GB 50153) and "Unified Standard Reliability Design of Building Structures" (GB 50068). This code gives the basic requirements for the design of concrete structures.

1.0.4 In addition to this code, the design of concrete structures also shall comply with those specified in the relevant current standards of China.

2 Terms and Symbols

2.1 Terms

2.1.1 Concrete structure

The structure is mainly made by concrete, including plain concrete structure, reinforced concrete structure and prestressed concrete structure, etc.

2.1.2 Plain concrete structure

The concrete structure without reinforcement or not provided with the stressed reinforcement.

2.1.3 Steel bar

It is the generic term of various nonprestressing tendons and/or bars used in the concrete structural members.

2.1.4 Prestressing tendon and/or bar

It is the generic term of those prestressed steel wires, steel strands and prestressed twisted bars used in the concrete structural members.

2.1.5 Reinforced concrete structure

The structure that is made of concrete and is provided with the stressed reinforcements.

2.1.6 Prestressed concrete structure

The structure that is made of concrete and is provided with stressed prestressing tendons and/or bars, which will be through stretching or by other methods to establish prestress.

2.1.7 Cast-in-situ concrete structure

The concrete structure that is built by erecting form at the original position on the spot as well as integral pouring.

2.1.8 Precast concrete structure

The concrete structure that is formed by assembling and connecting the precast concrete members or parts.

2.1.9 Assembled monolithic concrete structure

The integrally stressed concrete structure that is formed by connecting precast concrete members or parts with reinforcements, connection pieces or applied prestress as well as casting concrete at the connecting parts.

2.1.10 Composite member

The integrally-stressed structural member that is composed of precast concrete members (or existing concrete structural members) and post-cast concrete and is formed through two phases.

2.1.11 Deep flexural member

The flexural member of which the span-height ratio is less than 5.

2.1.12 Deep beam

The simply-supported single-span beam of which the span-height ratio is less than 2 or the multi-span continuous beam of which the span-height ratio is less than 2.5.

2.1.13 Pretensioned prestressed concrete structure

The concrete structure that is built by tensioning prestressing tendon and/or bar on pedestal following concreting and is established with prestress by loosening the prestressing

tendon and/or bar and transferring by bonding.

2.1.14 Post-tensioned prestressed concrete structure

The concrete structure of which the strength of concrete should be first reached the required strength, then stretching prestressing tendon and/or bar and anchoring them on the structure to establish prestress.

2.1.15 Unbonded prestressed concrete structure

The post-tensioned prestressed concrete structure that is built with unbonded prestressing tendon and/or bar that can maintain relative sliding with concrete.

2.1.16 Bonded prestressed concrete structure

The concrete structure of which the prestress is established by grouting or by directly contacting with concrete to form the mutual bonding between prestressing tendon and/or bar and concrete.

2.1.17 Structural joint

The generic term of the spaces taken to divide the concrete structure according to the requirements of structure design.

2.1.18 Concrete cover

The concrete applied from the outer edge of reinforcements in the structural members to the surface of members with a view to protecting the reinforcements, which is named as "protective layer" for short.

2.1.19 Anchorage length

The length required for the stressed reinforcement to reach the stress it shall be bear as specified in design depending on the bonding action of its surface with concrete or the squeezing action with the construction at end.

2.1.20 Splice of reinforcement

A structural form realizing the internal force transmission between reinforcements by such methods as banding and lapping, mechanical connection and welding.

2.1.21 Ratio of reinforcement

The ratio between area (or volume) of reinforcements allocated in concrete member and the specified sectional area (or volume) of concrete.

2.1.22 Ratio of shear span to effective depth

The ratio between the sectional bending moment and the product of shear force with effective height.

2.1.23 Transverse reinforcement

The stirrup or indirect reinforcement perpendicular to the longitudinal stressed reinforcement.

2.2 Symbols

2.2.1 Material properties

E_c ——the elastic modulus of concrete;

E_s ——the elastic modulus of steel reinforcement;

C30——the concrete strength grade of which the standard value for its cube compressive strength is 30N/mm^2 ;

HRB500——the ordinary hot-rolled ribbed steel bar of which the strength level is

500MPa;

HRBF400—the fine grain hot-rolled ribbed steel bar of which the strength level is 400MPa;

RRB400—the remained heat treatment ribbed steel bar of which the strength level is 400MPa;

HPB300—the hot rolled plain round steel bars of which the strength level is 300MPa;

HRB400E—the ordinary hot-rolled ribbed steel bar of which the strength level is 400MPa and having relatively high seismic performance;

f_{ck}, f_c —the standard value and design value of the axial compressive strength of concrete;

f_{tk}, f_t —the standard value and design value of the axial tensile strength of concrete;

f_{yk}, f_{pyk} —the standard values of the yield strength of steel bar and that of prestressing tendon and/or bar;

f_{stk}, f_{ptk} —the standard values of the ultimate strength of steel bar and that of prestressing tendon and/or bar;

f_y, f'_y —the design values of the tensile and compressive strength of steel bar;

f_{py}, f'_{py} —the design values of tensile and compressive strength of prestressing tendon and/or bar;

f_{yv} —the design value of the tensile strength of transverse reinforcement;

δ_{gt} —the overall elongation of reinforcement under the action of maximum force, which is also name as uniform elongation.

2.2.2 Actions and action effects

N —the design value of axial force;

N_k, N_q —the values of axial force calculated according to the characteristic combination and quasi-permanent combination of loads;

N_{u0} —the design value of the sectional axial compression or axial tension bearing capacity of the member;

N_{p0} —the pre-applied force when the concrete prestress in the normal direction of the is equal to zero;

M —the design value of bending moment;

M_k, M_q —the values of bending moment calculated according to the characteristic combination and quasi-permanent combination of loads;

M_u —the design value of the bend bearing capacity for normal section of member;

M_{cr} —the cracking bending moment value for the normal section of flexural member;

T —the design value of torsional moment;

V —the design value of shear force;

F_l —the design value of local load or the design value of concentrated counterforce;

σ_s, σ_p —the stress in longitudinal reinforcements and the stress in prestressing tendon and/or bar in the calculation of bearing capacity of normal section respectively;

σ_{pe} —the effective prestress of prestressing tendon and/or bar;

σ_l, σ'_l —the values of the prestress loss of prestressing tendon and/or bar in the tension

zone and compression zone in the corresponding phase;

τ ——the shear stress of concrete;

$w_{\max}$ ——the maximum crack width calculated according to the quasi-permanent combination or characteristic combination of loads, in consideration of the affect of long term action;

2.2.3 Geometric parameters

b ——the width of rectangular section, the web width of T-shaped and I-shaped section;

c ——the thickness of concrete cover;

d ——the nominal diameter of steel reinforcement (hereinafter referred to as "diameter") or the diameter of circular section;

h ——the depth of section;

h_0 ——the effective depth of section;

l_{ab}, l_a ——the basic anchorage length and anchorage length of the longitudinal tensile reinforcement;

l_0 ——the calculated span or the calculated length;

s ——the spacing of transverse reinforcements, spacing of spiral reinforcements or spacing of stirrups in the direction along axis of member;

x ——the depth of concrete compression zone;

A ——the sectional area of member;

A_s, A'_s ——the sectional area of longitudinal steel bar in the tension zone and in the compression zone respectively;

A_p, A'_p ——the sectional area of longitudinal prestressing tendon and/or bar in the tension zone and in the compression zone respectively;

A_l ——the local compression area of concrete;

A_{cor} ——the core sectional area of concrete surrounded by stirrups, spiral reinforcements or fabric reinforcement;

B ——the section rigidity of flexural member;

I ——the moment of inertia of section;

W ——the moment of elastic resistance of the tension edge of section;

W_t ——the moment of plastic resistance of torsional section.

2.2.4 Calculation coefficients and miscellaneous

a_E ——the ratio of the elastic modulus of steel reinforcement to the elastic modulus of concrete;

γ ——the plastic influence coefficient of section resistance moment of concrete members;

η ——the enhancement coefficient for eccentricity of axial force considering the influence of second-order effect for eccentric compression members;

λ ——the ratio of shear span to effective depth for calculated section, namely $M/(Vh_0)$;

ρ ——the ratio of reinforcement for longitudinal stressed reinforcement;

ρ_v ——the ratio of volumetric reinforcement for indirect steel reinforcement or stirrup;

ϕ ——the symbol representing the reinforcement diameter, $\phi 20$ represents the steel reinforcement in diameter of 20mm.

3 General Requirements

3.1 General

3.1.1 The design of concrete structures shall include the following contents:

- 1 Design of structural scheme, including the structure selection, member layout and force transmission route;
- 2 Action and action effect analysis;
- 3 Limit states design of the structure;
- 4 Construction and connection measures of structures and members;
- 5 Durability and construction requirements;
- 6 Special performance design of such structure meeting special requirements.

3.1.2 This code adopts the probability-based limit states design method, the degree of reliability of structural members is measured by the reliability index, and the design is carried out by adopting the design expressions of partial coefficients.

3.1.3 The limit states design of the concrete structures shall include:

- 1 Ultimate limit states: A structure or a structural member reaches the maximum load-bearing capacity and appears the fatigue failure or undue deformation unsuitable for loading continuously or has progressive collapse due to the partial failure of structure;

- 2 Serviceability limit states: A structure or a structural member reaches a certain specified limit value of serviceability or a certain specified state of durability.

3.1.4 The direct action (load) on a structure shall be determined in accordance with the current national standard "Load Code for the Design of Building Structures" (GB 50009) and the relevant standards; the earthquake action shall be determined in accordance with the current national standard "Code for Seismic Design of Buildings" (GB 50011).

The indirect action and accidental action shall be determined in accordance with the relevant standards or the specific conditions.

The structural member directly bearing the crane load shall take the dynamic coefficient of crane load into account. For the fabrication, transportation and installation of precast members, the corresponding dynamic coefficients shall be taken into account. For cast-in-situ structures, the loads during the construction stage shall be taken into account if necessary.

3.1.5 The safety class and design life of concrete structures shall meet those specified in the current national standard "Unified Standard for Reliability Design of Engineering Structures" (GB 50153).

The safety class of different structural members in the concrete structure should be same as the safety class of the whole structure. And the safety class of parts of the structural members may be adjusted properly according to their importance. For the important members and key force transmission positions in the structure, their safety classes should be improved appropriately.

3.1.6 The design of concrete structures shall take the technical level of construction and the feasibility of practical engineering condition into account. As for the concrete structure for special purpose, the corresponding construction requirements shall be proposed.

3.1.7 The design shall make clear the functions of structures and the functions and

service environment of the structure shall not be changed without technical evaluation or design approval within the design life.

3.2 Structural Scheme

3.2.1 The design scheme of concrete structures shall meet the following requirements:

- 1 Reasonable structural system, member form and layout shall be selected;
- 2 The flat and vertical planes of the structure should be arranged regularly, the mass and rigidity of all parts should be uniform and continuous;
- 3 The force transmission route of the structure shall be simple and definite, and the vertical member should penetrate continuously and be aligned;
- 4 The hyperstatic structure should be adopted, the important members and key force transmission positions shall be added with redundant constraint or have several force transmission routes;
- 5 Measures should be taken to reduce the effects of accidental actions.

3.2.2 The design of the structural joints in concrete structures shall meet the following requirements:

- 1 The position and structural form of structural joint shall be determined reasonably in accordance with the loading-bearing characteristics and the requirements on architectural scale, shape and use functions of the structure;
- 2 The quantity of structural joints should be controlled, and effective measures shall be taken to reduce the adverse impacts of joints on the functions of use;
- 3 The temporary structural joints in the construction stage may be arranged as required.

3.2.3 The connection of structural members shall meet the following requirements:

- 1 The load-bearing capacity of the connecting part shall guarantee the force transmission property between the connected members;
- 2 Where the concrete members are connected with those made of other materials, reliable measures shall be taken;
- 3 The impact caused by the deformation of a member on the connecting joint and the adjacent structure or member shall be considered.

3.2.4 The design of concrete structures shall meet the requirements on saving materials, being convenient for construction, reducing energy consumption and protecting environment.

3.3 Calculation of Ultimate Limit States

3.3.1 The calculation of ultimate limit states of concrete structures shall include the following contents:

- 1 The structural members shall be carried out with the calculation of load-bearing capacity (including instability);
- 2 The member directly bearing repeated loads shall be carried out checking of fatigue;
- 3 Where seismic protection is required, the calculation of seismic bearing capacity shall be carried out;
- 4 The checking of structural overturning, sliding and floating shall be carried out if necessary;

5 As for the important structures that may suffer from accidental actions and may cause serious consequences when collapsing, the design for preventing progressive collapse should be carried out.

3.3.2 As for the durable design conditions, temporary design conditions and seismic design conditions, if the internal force form is used to express, the structural members shall be adopted with the following design expressions of ultimate limit states:

$$\gamma_0 S \leq R \quad (3.3.2-1)$$

$$R = R(f_c, f_s, a_k, \dots) / \gamma_{Rd} \quad (3.3.2-2)$$

Where:

γ_0 —the significance coefficient of structure: under the durable design conditions and temporary design conditions, this coefficient of the structural members of Safety Class I shall not be less than 1.1, that of the structural members Safety Class II shall not be less than 1.0 and that of the structural members of Safety Class III shall not be less than 0.9; under the seismic design conditions, this coefficient shall be 1.0;

S —the design value of the effect for combination of actions at ultimate limit states, which shall be calculated according to the fundamental combination of actions under the durable design conditions and temporary design conditions and shall be calculated according to the seismic combination of actions under seismic design conditions;

R —the design value of resistance of structural member;

$R(\cdot)$ —the function of resistance of structural member;

γ_{Rd} —the uncertainty coefficient of the resistance model of structural member, which is taken as 1.0 for static design, taken as the values larger than 1.0 for the structural members with large uncertainty according to specific conditions; in the seismic design, γ_{Rd} shall be replaced by the seismic adjustment coefficient of bearing capacity γ_{RE} ;

f_c, f_s —the design values of the strength of concrete and steel reinforcement respectively, which shall be taken with values in accordance with Article 4.1.4 and Article 4.2.3 of this code;

a_k —the standard value of geometric parameter. When the variation of the geometric parameter has obvious adverse impact on the structural behavior, a_k may be increased or decreased by an additional value.

Note: $\gamma_0 S$ in Formula (3.3.2-1) is the design value of internal force and is expressed by N, M, V, T and other in the chapters of this code.

3.3.3 As for the two-dimensional and three-dimensional concrete structural members, if the analysis is carried out according to the elastic or elastic-plastic method and the stress form is applied in the expression, the concrete stress may be substituted into the design value of internal force in zones and be calculated according to Article 3.3.2 of this code; or the checking of design may be carried out by directly adopting the multi-axial strength criterion.

3.3.4 Where the ultimate limit states design of the structure under accidental actions is carried out, the design value S of the action effect in Formula (3.3.2-1) shall be calculated according to the accidental combination and the significance coefficient of structure (γ_0) shall be taken as a value no less than 1.0; the design values of strength of concrete and steel reinforcement (f_c and f_s) in Formula (3.3.2-2) shall be replaced by the standard values of

strength (f_{ck} and f_{yk}) (or f_{pyk}).

Where the checking of preventing progressive collapse of structure is carried out, the function of load-bearing capacity of structural member shall be determined according to the principles stated in Section 3.6 of this code.

3.3.5 The ultimate limit states design of existing structures shall be carried out according to the following requirements:

1 Where the checking of ultimate limit states is required with a view to conducting safety rechecking, changing the functions or extending the service life of existing structures, those specified Article 3.3.2 of this code should be met;

2 Where the existing structures are redesigned for purpose of renovation, extension or strengthening, the calculation of ultimate limit states shall meet those specified in Section 3.7 of this code.

3.4 Checking of Serviceability Limit States

3.4.1 The concrete structural members shall be carried out the checking of serviceability limit states based on the functions of use the appearance requirements according to the following requirements:

1 As for the members requiring deformation control, the checking of deformation shall be carried out;

2 As for the members that must not appear cracks, the checking of the concrete tension stress shall be carried out;

3 As for the members that may appear cracks, the width checking of the stressed crack shall be carried out;

4 as for the floor structures having the requirements on comfort degree, the checking of vertical natural vibration frequency shall be carried out.

3.4.2 For serviceability limit states, the reinforced concrete members and prestressed concrete members shall be checked respectively according to the quasi-permanent combination or characteristic combination of loads with consideration of the influence of long-term actions by adopting the following design expressions of limit states:

$$S \leq C \quad (3.4.2)$$

Where:

S ——the design value of the effect of load combination for serviceability limit states;

C ——the limit values of the specified deformation, stress, crack width and natural vibration frequency when the structural member meets the requirements for serviceability.

3.4.3 The maximum deflection of reinforced concrete flexural member shall be calculated according to the quasi-permanent combination of loads and that of the prestressed concrete flexural member shall be calculated according to the characteristic combination of loads and considering the influence of long-term action of loads, its calculated value shall not exceed the limit value of deflection specified in Table 3.4.3.

Table 3.4.3 Deflection Limit Values of Flexural Members

Member type		Limit value of deflection
Crane beam	Manual-operated crane	$l_0/500$
	Electric-operated crane	$l_0/600$

Roof, floors and stair members	When $l_0 < 7\text{m}$	$l_0/200(l_0/250)$
	When $7\text{m} \leq l_0 \leq 9\text{m}$	$l_0/250(l_0/300)$
	When $l_0 > 9\text{m}$	$l_0/300(l_0/400)$

Notes:

- 1 l_0 in this table is the calculated span of members; to calculate the limit value of deflection of cantilever members, its calculated span l_0 shall be adopted according to two times the actual cantilever length;
- 2 Values in parentheses in this table are applicable to the members that have higher requirement on deflection in application;
- 3 If the member is cambered before its fabrication and it is allowed in application, the camber value shall be deducted from the calculated deflection value during the checking of deflection; as for the prestressed concrete members, the inverted camber value caused by pre-applied force may be also be deducted;
- 4 The camber value during the fabrication of member and the inverted camber value caused by pre-applied force should not exceed the calculated deflection value of the member under the action of corresponding load combination.

3.4.4 The stressed crack control level for normal section of structural members shall be divided into three levels, and the classification and requirements of the control level shall meet the following requirements:

Level 1—members on which cracks are strictly prohibited, when the calculation is in accordance with the characteristic combination of loads, that the concrete at the tension edge of the member shall not produce tension stress.

Level 2—members on which cracks are generally prohibited, when the calculation is in accordance with the characteristic combination of loads, that the concrete tension stress at the tension edge of the member shall not be larger than the standard value of concrete tensile strength.

Level 3—members on which cracks are allowed: as for reinforced concrete members, when the calculation is in accordance with the quasi-permanent combination of loads and considering the influence of long-term actions of loads, that the maximum crack width of the member shall not exceed the limit values of maximum crack width as specified in Table 3.4.5 of this code. As for prestressed concrete members, when the calculation is in accordance with the characteristic combination of loads and considering the influence of long-term actions of loads, the maximum crack width of the member shall not exceed the limit values of maximum crack width as specified in Article 3.4.5 of this code; as for the prestressed concrete members of Environmental Category II-a, the calculation still shall be in accordance with the quasi-permanent combination of loads and the concrete tension stress at the tension edge of member shall not be larger than the standard value of the concrete tensile strength.

3.4.5 The different crack control levels and the limit values of maximum crack width W_{lim} of the structural members shall be selected from Table 3.4.5 according to the structure type and the environmental categories specified in Article 3.5.2 of this code.

Table 3.4.5 Crack Control Levels and Limit Values of Maximum Crack Width (mm) of Structural Members

Environmental category	Reinforced concrete structure		Prestressed concrete structure	
	Crack control level	W_{lim}	Crack control level	W_{lim}
I	Level 3	0.30(0.40)	Level 3	0.20
II-a		0.20		0.10
II-b			Level 2	—

III-a and III-b			Level 1	—
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Notes:

- 1 As for the flexural members of Environmental Category I in such areas where the annual average relative humidity is less than 60%, the limit value for their maximum crack width may be taken as the values in parentheses;
- 2 Under Category I environment, the limit values for the maximum crack width of reinforced concrete roof truss, bracket and crane beam requiring checking of fatigue shall be taken as 0.20mm; and that for reinforced concrete roof beam and joist shall be taken as 0.30mm;
- 3 Under Category I environment, the prestressed concrete roof truss, bracket and two-way slab system shall be carried out with checking according to the crack control level 2; under the Category I environment, the prestressed concrete roof beam, joist and one-way slab shall be carried out with checking according to the requirements for Category II-a environment as given in this table; under the Category I and II-a environments, the prestressed concrete crane beam requiring the checking of fatigue shall be carried out with checking according to the members with crack control level no less than Level 2;
- 4 The crack control levels and the limit values of maximum crack width for prestressed concrete members are only applicable to the checking of normal section; the checking of the crack control of inclined section of prestressed concrete members shall meet the relevant requirements stated in Chapter 7 of this code;
- 5 As for chimneys, silos and the structures under the liquid pressure, the crack control requirements shall meet the relevant provisions of special standards;
- 6 As for the structural members under Category IV and V environments, their crack control requirements shall meet the relevant provisions of special standards;
- 7 The limit values of maximum crack width in this Table are used for the checking of maximum crack width caused by the action of loads.

3.4.6 As for concrete floor structures, their vertical natural vibration frequency shall be checked according to the requirements of use function and should meet the following requirements:

- 1 The vertical natural vibration frequency of dwelling houses and apartments should not be less than 5Hz;
- 2 The vertical natural vibration frequency of office buildings and hotels should not be less than 4Hz;
- 3 The vertical natural vibration frequency of large-span public buildings should not be less than 3Hz.

3.5 Durability Design

3.5.1 The durability of concrete structures shall be designed in accordance with the design life and the environmental categories, and the durability design shall include the following contents:

- 1 The environmental category in which the structure is located shall be determined;
- 2 The basic requirements on the durability of concrete materials shall be proposed;
- 3 The thickness of concrete cover for the steel reinforcements in member shall be determined;
- 4 The technical measures for durability taken under different ambient conditions;
- 5 The inspection and maintenance requirements for the service stage of structure shall be proposed.

Note: As for the temporary concrete structures, the requirements for durability of concrete may not be considered.

3.5.2 The category of the environment to which the concrete structure is exposed shall be divided according to the requirements of Table 3.5.2.

Table 3.5.2 Environmental Categories of the Concrete Structures

Environmental category	Condition
I	Dry indoor environment; Environment without the submersion of aggressive dead water
II-a	Indoor humid environment; Open-air environments of the non-severe cold and non-cold areas; Environments in non-severe cold and non-cold areas, directly contacting with non-aggressive water or soil; Environments below the frost lines in severe cold and cold areas, directly contacting with non-aggressive water or soil
II-b	Alternating wetting and drying environment; Environment with frequently changing water levels; Open-air environments of the severe cold and cold areas; Environments above the frost lines in severe cold and cold areas, directly contacting with non-aggressive water or soil
III-a	Environments in such regions with varying water levels in winter in the severe cold and cold areas; Environment affected by deicing salt; Sea wind environment
III-b	Environment of salty soil; Environment under the action of deicing salt; Seacoast environment
IV	Sea water environment
V	Environment affected by human action or natural corrosive substance

Notes:

- 1 The indoor humid environment refers to the environment in which the member surface is at the dew or wet state frequently;
- 2 The division of severe cold and cold areas shall meet the relevant provisions of the current national standard "Thermal Design Code for Civil Building" (GB 50176).
- 3 The seacoast environment and sea wind environment should be determined by investigations, studies and engineering experiences based on the local circumstances, in consideration of the influence of such factors as the prevailing wind direction and the windward and leeward positions of the structure;
- 4 The environment affected by deicing salt refers to the environment that is affected by the mist of deicing salt; the environment under the action of deicing salt refers to the environment with spouting of deicing salt solution as well as the buildings, like car wash and parking structure in the areas where the deicing salt is used.
- 5 The environment to which the concrete structure is exposed refers to the environment where the surfaces of concrete structures exist.

3.5.3 As for the concrete structures of which the design life is 50 years, the concrete materials should be in accordance with those specified in Table 3.5.3.

Table 3.5.3 Basic Requirements on Durability of Concrete Materials of Structures

Environmental category	Maximum water-cement ratio	Minimum strength grade	Maximum chloride ion content (%)	Maximum alkali content (kg/m ³)
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I	0.60	C20	0.30	Unlimited
II-a	0.55	C25	0.20	3.0
II-b	0.50(0.55)	C30(C25)	0.15	
III-a	0.45(0.50)	C35(C30)	0.15	
III-b	0.40	0.40	0.10	

Notes:

- 1 Chloride ion content refers to the percentage of chloride ions in the total amount of cement;
- 2 The maximum chloride ion content in the concrete of prestressed member is 0.06%; the minimum concrete strength grade should be increased by two grades according to those specified in the table;
- 3 Requirements on the water-cement ratio and minimum strength grade of plain concrete members may be reduced appropriately;
- 4 When reliable engineering experience is available, the minimum concrete strength grade in the environmental category II may be reduced by one grade;
- 5 The concrete in the Category II-b and III-a environments of severe cold and cold areas shall be used with air entraining agent and may be adopted with the relevant parameters in parentheses;
- 6 Where the non-alkali activated aggregate is applied, the alkali content in the concrete may not be limited.

3.5.4 The concrete structures and members still shall be taken with the following technical measures for their durability:

1 The prestressing tendon and/or bar in the prestressed concrete structure shall be taken with such measures as surface protection, duct grouting and increasing the thickness of concrete cover according to specific conditions, the exposed anchored end shall be taken with effective measures, such as anchor seal and concrete surface treatment, etc;

2 For the concrete structures with requirement on impermeability, the impermeability grade of concrete shall meet the requirements of relevant standards;

3 In the humid environment in severe cold and cold areas, the structural concrete shall meet the requirements on freezing resistance, and the resistance class to freezing-thawing of concrete shall meet the requirements of relevant standards;

4 The cantilever members in the Category II and III environments should adopt the structure form of cantilever beam-plate or may be added with protective coating on their upper surfaces;

5 For the structural members in Category II and III environments, the metal parts, like embedded part, hook and connecting piece, of their surface shall be taken with reliable antirust measures; as for the exposed metal anchorage devices of post-tensioning prestressed concrete, their protection requirements are detailed in Article 10.3.13 of this code;

6 The concrete structural members in Category III environment may be adopted with corrosion inhibitor, epoxy coated steel reinforcement or other steel reinforcements having corrosion resistance, or may be taken with cathodic protection measure or other measures, like using replaceable members, etc.

3.5.5 In Category I environment, the concrete structures with design life up to 100 years shall meet the following requirements:

1 The minimum strength grade of reinforced concrete structures and prestressed concrete structures are C30 and C40 respectively;

2 The maximum chloride ion content in concrete is 0.06%;

3 The non-alkali activated aggregate should be used; if alkaline reaction aggregate is

used, the maximum alkali content in concrete shall be 3.0kg/m^3 ;

4 The thickness of concrete cover shall meet those specified in Article 8.2.1 of this code; where effective surface protection measures are taken, the thickness of concrete cover may be reduced appropriately.

3.5.6 In the Category II and III environments, the concrete structures with design life up to 100 years shall be taken with special effective measures.

3.5.7 As for the concrete structures of which the environmental categories of their durability are Category IV and V, their durability requirements shall meet those specified in the relevant standards.

3.5.8 The concrete structures still shall meet the following requirements within their design life:

- 1** Periodical detection and maintenance system shall be established;
- 2** The replaceable concrete members in the design shall be replaced as required;
- 3** The protective coating on the surface of member shall be maintained or replaced as required;
- 4** The visible durability defects of the structure, if any, shall be treated timely.

3.6 Principles for Design of Preventing Progressive Collapse

3.6.1 The design of preventing the progressive collapse of concrete structures should meet the following requirements:

- 1** Measures should be taken to reduce the effects of accidental actions;
- 2** Measures should be taken to protect the important members and key force transmission position from directly undergoing accidental actions;
- 3** Redundant constraint should be added and standby force transmission routes shall be arranged in the zones where the structure is easy to suffer from the effects of accidental actions;
- 4** The load-bearing capacity and deformability of the important structural members and key force transmission positions in the evacuation exit, waiting space and others should be reinforced;
- 5** The steel reinforcements should be allocated by penetrating the horizontal and vertical members to reliably anchor with the peripheral members;
- 6** The structural joints should be arranged and the scope having the possibility of progressive collapse should be controlled.

3.6.2 The design of preventing progressive collapse of important structures may be adopted with the following methods:

- 1** Local reinforcement: the emergency capacity of such vertical important members and key force transmission positions that may suffer from accidental actions and have partial failure shall be improved, or the design may be carried out in consideration of the accidental actions directly.
- 2** Member tying method: if the local vertical member of the structure is out of service, the checking of load-bearing capacity may be done respectively according to the beam-tie model, suspension cable-tie model and cantilever-tie model based on the specific conditions, in order to maintain the monolithic stability of the structure.

3 Member dismantling method: the main loaded members of the structure shall be dismantled according to certain regulations and the ultimate bearing capacity of the residual structural system shall be checked; or the design may also be carried out by adopting with the overall process analysis of collapse.

3.6.3 To do the checking of preventing progressive collapse of the structure under accidental actions, the actions should be considered with the dynamic coefficient caused by the collapse impact of the corresponding position of this structure. In the calculation of the function of resistance, the concrete strength shall be taken as the standard value of strength f_{ck} ; the strength of steel bars shall be taken as the standard value of ultimate strength f_{stk} , the strength of prestressing tendon and/or bar shall be taken as the standard value of ultimate strength f_{ptk} with the consideration of the influence of anchorage device. The influence of the collapse of structure due to the accidental actions on the geometric parameters of this structure should be taken into account. When necessary, the strengthening and brittleness of the material properties under dynamic actions also shall be considered, and the corresponding characteristic values of strength shall be taken.

3.7 Principles for Design of Existing Structures

3.7.1 The existing structures shall be evaluated, checked or redesigned with a view to extending their service life, changing their functions, renovation, extension or strengthening and repairing, etc.

3.7.2 To evaluate the safety, applicability, durability and disaster resistance of the existing structures, the requirements of the principles stated in the current national standard "Unified Standard for Reliability Design of Engineering Structures" (GB 50153) as well as the following requirements shall be met:

1 The design scheme of existing structure shall be determined according to the evaluation result, service requirements and continuous service life;

2 As for the existing structures, if their functions are changed or their service life is extended, the checking of ultimate limit states should meet the relevant provisions of this code;

3 Where the existing structures are redesigned for purpose of renovation, extension or reconstruction by strengthening, the calculation of ultimate limit states shall meet those specified in this code and the relevant standards;

4 For existing structures, the checking of serviceability limit states and the construction requirements should meet those specified in this code;

5 When necessary, the functions of use may be adjusted correspondingly, and the requirements for restriction of service shall be proposed.

3.7.3 The design of existing structures shall meet the following requirements:

1 The structural scheme shall be optimized to guarantee the monolithic stability of the structure;

2 The loads may be determined according to those specified in the current codes and may also be adjusted appropriately according to the functions of use;

3 The design values of strength of concrete and steel reinforcements in the existing portion of the structure shall be determined according to the measured strength value; where

the properties of materials meet the requirements of the original design, the strength may be also be taken values according to the requirements of the original design;

4 The design shall take the actual geometric dimensions, sectional reinforcements, connection construction of the members in existing structures as well as the influence of the existing defects into account; if they meet the requirements of the original design, values may be taken according the requirements of the original design;

5 The load bearing history of the existing structure and the influence of the construction state shall be taken into consideration; and the composite members formed in phase II may be designed according to those specified in Section 9.5 of this code.

4 Materials

4.1 Concrete

4.1.1 The concrete strength grade shall be determined according to the standard value of cube compressive strength. The standard value of cube compressive strength refers to the compressive strength value with a 95% guarantee rate measured from the standard tests on 150mm-side-long cube specimen that is fabricated and cured for 28d or a period specified in the design according to the standard methods.

4.1.2 The concrete strength grade of plain concrete structures shall not be less than C15 and that of reinforced concrete structures shall not be less than C20; when the steel reinforcements of or above strength grade 400MPa are used, the concrete strength grade shall not be less than C25.

The concrete strength grade of prestressed concrete structures should not be less than C40 and shall not be less than C30.

The concrete strength grade of such reinforced concrete members bearing repeated loads shall not be less than C30.

4.1.3 The standard value of axial compressive strength (f_{ck}) of concrete shall be selected according to Table 4.1.3-1; the standard value of axial tensile strength (f_{tk}) shall be selected according to Table 4.1.3-2.

Table 4.1.3-1 Standard Values of the Axial Compressive Strength of Concrete (N/mm²)

Strength	Concrete strength grade													
	C15	C20	C25	C30	C35	C40	C45	C50	C55	C60	C65	C70	C75	C80
f_{ck}	10.0	13.4	16.7	20.1	23.4	26.8	29.6	32.4	35.5	38.5	41.5	44.5	47.4	50.2

Table 4.1.3-2 Standard Values of the Axial Tensile Strength of Concrete (N/mm²)

Strength	Concrete strength grade													
	C15	C20	C25	C30	C35	C40	C45	C50	C55	C60	C65	C70	C75	C80
f_{tk}	1.27	1.54	1.78	2.01	2.20	2.39	2.51	2.64	2.74	2.85	2.93	2.99	3.05	3.11

4.1.4 The design value of axial compressive strength (f_c) of concrete shall be selected according to Table 4.1.4-1; the design value of axial tensile strength (f_t) shall be selected according to Table 4.1.4-2.

Table 4.1.4-1 Design value of the Axial Compressive Strength of Concrete (N/mm²)

Strength	Concrete strength grade													
	C15	C20	C25	C30	C35	C40	C45	C50	C55	C60	C65	C70	C75	C80
f_c	7.2	9.6	11.9	14.3	16.7	19.1	21.1	23.1	25.3	27.5	29.7	31.8	33.8	35.9

Table 4.1.4-2 Design value of the Axial Tensile Strength of Concrete (N/mm²)

Strength	Concrete strength grade													
	C15	C20	C25	C30	C35	C40	C45	C50	C55	C60	C65	C70	C75	C80
f_t	0.91	1.10	1.27	1.43	1.57	1.71	1.80	1.89	1.96	2.04	2.09	2.14	2.18	2.22

4.1.5 The elastic modulus of concrete E_c under compression or tension shall be selected according to Table 4.1.5.

The modulus of concrete shear deformation G_c may be adopted according to 40% of the values of corresponding elastic modulus.

The Poisson's ratio of concrete μ_c may be adopted according to 0.2.

Table 4.1.5 Elastic Modulus of Concrete ($\times 10^4 \text{N/mm}^2$)

Concrete strength grade	C15	C20	C25	C30	C35	C40	C45	C50	C55	C60	C65	C70	C75	C80
E_c	2.20	2.55	2.80	3.00	3.15	3.25	3.35	3.45	3.55	3.60	3.65	3.70	3.75	3.80

Notes:

- 1 Where reliable test basis is available, the elastic modulus may be determined according to the actual measured data;
- 2 Where the concrete is mixed with a large amount of admixtures, the elastic modulus of concrete may be determined according to the actual measured data based on the specified age.

4.1.6 The design values of the axial compressive fatigue strength and axial tensile fatigue strength of concrete, namely f_c^f and f_t^f , shall be determined respectively according to the design values of strength listed in Table 4.1.4-1 and Table 4.1.4-2 by multiplying the correction coefficient γ_p of fatigue strength. The correction coefficient γ_p for the compressive or tensile fatigue strength of concrete shall be selected respectively according to Table 4.1.6-1 and Table 4.1.6-2 based on the fatigue ratio ρ_c^f ; when the concrete is under the action of tensile-compressive fatigue stress, the correction coefficient γ_p of fatigue strength shall be taken as 0.60.

the fatigue ratio ρ_c^f shall be calculated according to the following formula:

$$\rho_c^f = \frac{\sigma_{c,\min}^f}{\sigma_{c,\max}^f} \quad (4.1.6)$$

Where:

$\sigma_{c,\min}^f$, $\sigma_{c,\max}^f$ —the minimum stress and maximum stress of concrete at the same fiber of the section during the checking of fatigue of members respectively.

Table 4.1.6-1 Correction Coefficient of Concrete Compressive Fatigue Strength γ_p

ρ_c^f	$0 \leq \rho_c^f < 0.1$	$0.1 \leq \rho_c^f < 0.2$	$0.2 \leq \rho_c^f < 0.3$	$0.3 \leq \rho_c^f < 0.4$	$0.4 \leq \rho_c^f < 0.5$	$\rho_c^f \geq 0.5$
γ_p	0.78	0.74	0.80	0.86	0.93	1.00

Table 4.1.6-1 Correction Coefficient of Concrete Tensile Fatigue Strength γ_p

ρ_c^f	$0 \leq \rho_c^f < 0.1$	$0.1 \leq \rho_c^f < 0.2$	$0.2 \leq \rho_c^f < 0.3$	$0.3 \leq \rho_c^f < 0.4$	$0.4 \leq \rho_c^f < 0.5$
γ_p	0.63	0.66	0.69	0.72	0.74
ρ_c^f	$0.5 \leq \rho_c^f < 0.6$	$0.6 \leq \rho_c^f < 0.7$	$0.7 \leq \rho_c^f < 0.8$	$\rho_c^f \geq 0.8$	—
γ_p	0.76	0.80	0.90	1.00	—

Note: In case that the concrete members directly bearing fatigue loads is adopted with steam curing, the curing temperature should not be larger than 60°C.

4.1.7 The fatigue deformation modulus of concrete E_c^f shall be selected according to Table 4.1.7.

Table 4.1.7 Fatigue Deformation Modulus of Concrete ($\times 10^4 \text{N/mm}^2$)

Strength grade	C30	C35	C40	C45	C50	C55	C60	C65	C70	C75	C80
E_c^f	1.30	1.40	1.50	1.55	1.60	1.65	1.70	1.75	1.80	1.85	1.90

4.1.8 When the temperature is within the range from 0°C to 100°C , the thermal parameters of concrete may be taken with value according to the following requirements:

Linear expansion coefficient α_c : $1 \times 10^{-5}/^\circ\text{C}$;

Coefficient of thermal conductivity λ : $10.6 \text{kJ}/(\text{m} \cdot \text{h} \cdot ^\circ\text{C})$;

Specific heat capacity c : $0.96 \text{kJ}/(\text{kg} \cdot ^\circ\text{C})$.

4.2 Steel Reinforcement

4.2.1 The steel reinforcements in concrete structures shall be selected according to the following provisions:

1 The longitudinal stressed steel bars should adopt the HRB400, HRB500, HRBF400 and HRBF500 steel reinforcements and may also adopt the HPB300, HRB335, HRBF335 and RRB400 steel reinforcements;

2 The longitudinal stressed steel bars for beam and column shall adopt HRB400, HRB500, HRBF400 and HRBF500 steel reinforcements;

3 Stirrups should adopt HRB400, HRBF400, HPB300, HRB500 and HRBF500 steel reinforcements and may also adopt HRB335 and HRBF335 steel reinforcements;

4 The prestressing tendon and/or bar should adopt prestressed steel wires, steel strands and prestressed twisted steel reinforcements.

4.2.2 The guarantee rate for the standard value of strength of steel reinforcements shall not be less than 95%.

The standard value of yield strength f_{yk} and the standard value of ultimate strength f_{stk} of steel bars shall be selected according to Table 4.2.2-1; the standard values of yield strength f_{pyk} and standard values of ultimate strength f_{ptk} of prestressed steel wires, steel strands and prestressed twisted steel reinforcements shall be selected according to Table 4.2.2-2.

Table 4.2.2-1 Standard Values of Strength of Steel Bars (N/mm^2)

Grade	Symbol	Nominal diameter $d(\text{mm})$	Standard value of yield strength f_{yk}	Standard value of ultimate strength f_{stk}
HPB300	Φ	6~22	300	420
HRB335 HRBF335	Φ ΦF	6~50	335	455
HRB400 HRBF400 RRB400	Φ ΦF ΦR	6~50	400	540
HRB500 HRBF500	Φ ΦF	6~50	500	630

Table 4.2.2-2 Standard Values of Strength of Prestressing Tendons and/or Bars (N/mm^2)

Type	Symbol	Nominal diameter	Standard value of yield strength f_{pyk}	Standard value of ultimate strength f_{ptk}
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			$d(\text{mm})$		
Medium-strength prestressed steel wire	PlainSpiral rib	Φ^{PM} Φ^{HM}	5, 7, 9	620	800
				780	970
				980	1270
Prestressed twisted steel reinforcement	Twisted	Φ^{T}	18, 25, 32, 40, 50	785	980
				930	1080
				1080	1230
Stress-relieved steel wire	PlainSpiral rib	Φ^{P} Φ^{H}	5	—	1570
				—	1860
			7	—	1570
			9	—	1470
—	1570				
Steel strand	1×3 (three-strand)	Φ^{S}	8.6, 10.8, 12.9	—	1570
				—	1860
				—	1960
	1×7 (seven-strand)		9.5, 12.7, 15.2, 17.8	—	1720
				—	1860
				—	1960
			21.6	—	1860

Note: Where the steel strands with standard value of ultimate strength of 1960N/mm² are used as the post-tensioned prestressed reinforcements, reliable engineering experiences shall be available.

4.2.3 The design values of tensile strength f_y and compressive strength f'_y of steel bars shall be selected according to Table 4.2.3-1; and the design values of tensile strength f_{py} and compressive strength f'_{py} of prestressing tendon and/or bar shall be selected according to Table 4.2.3-2.

If different kinds of steel reinforcements are used in the member, each kind of steel reinforcement shall be adopted with its respective design value of strength. The design value of tensile strength f_{yv} of transverse steel reinforcement shall be selected according to the f_y values given in the table; if f_{yv} is used in calculation of shear, torsion or punching bearing capacity, it shall be taken as 360N/mm² when it is larger than 360N/mm².

Table 4.2.3-1 Design Values of Strength of Steel Bars (N/mm²)

Grade	Design value of tensile strength f_y	Design value of compressive strength f'_y
HPB300	270	270
HRB335, HRBF335	300	300
HRB400, HRBF400, RRB400	360	360
HRB500, HRBF500	435	410

Table 4.2.3-2 Design Values of Strength of Prestressing Tendons and/or Bars (N/mm²)

Type	Standard value of ultimate strength f_{ptk}	Design value of tensile strength f_{py}	Design value of compressive
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			strength f'_{py}
Medium-strength prestressed steel wire	800	510	410
	970	650	
	1270	810	
Stress-relieved steel wire	1470	1040	410
	1570	1110	
	1860	1320	
Steel strand	1570	1110	390
	1720	1220	
	1860	1320	
	1960	1390	
Prestressed twisted steel reinforcement	980	650	410
	1080	770	
	1230	900	

Note: Where the standard value of strength of prestressing tendon and/or bar does not conform to those specified in Table 4.2.3-2, the design value of strength shall be converted by corresponding ratio.

4.2.4 The overall elongation δ_{gt} of steel bars and prestressing tendons and/or bars under the action of maximum force shall not be less than the values specified in Table 4.2.4.

Table 4.2.4 Limit Values for Overall Elongation of Steel Bars and Prestressing Tendons and/or Bars under Maximum Force

Type	Steel bar			Prestressing tendon and/or bar
	HPB300	HRB335, HRBF335, HRB400, HRBF400, HRB500, HRBF500	RRB400	
$\delta_{gt}(\%)$	10.0	7.5	5.0	3.5

4.2.5 The elastic modulus E_s of steel bar and prestressing tendon and/or bar shall be selected according to Table 4.2.5.

Table 4.2.5 Elastic Modulus of Steel Reinforcement ($\times 10^5 \text{N/mm}^2$)

Grade or type	Elastic modulus E_s
HPB300 steel reinforcement	2.10
HRB335, HRB400 and HRB500 steel reinforcements HRBF335, HRBF400 and HRBF500 steel reinforcements RRB400 steel reinforcement Prestressed twisted steel reinforcement	2.00
Stress-relieved steel wire and medium-strength prestressed steel wire	2.05
Steel strand	1.95

Note: When necessary, the measured elastic modulus may be adopted.

4.2.6 The limit values $\triangle f_y^f$ and $\triangle f_{py}^f$ for the fatigue stress amplitude of steel bars and prestressing tendons and/or bars shall be taken respectively from Table 4.2.6-1 and Table 4.2.6-2 by linear interpolation based on the fatigue ratio ρ_s^f and ρ_p^f of steel reinforcement.

Table 4.2.6-1 Limit Values for Fatigue Stress Amplitude of Steel Bars (N/mm^2)

Fatigue ratio ρ_s^f	Limit value for fatigue stress amplitude Δf_y^f	
	HRB335	HRB400
0	175	175
0.1	162	162
0.2	154	156
0.3	144	149
0.4	131	137
0.5	115	123
0.6	97	106
0.7	77	85
0.8	54	60
0.9	28	31

Note: When the longitudinal tension steel reinforcements are connected by flash butt welding, the limit value for the fatigue stress amplitude of steel reinforcement at the joint shall be the values listed in the table and be multiplied by 0.8.

Table 4.2.6-2 Limit Values for Fatigue Stress Amplitude of Prestressing Tendon and/or Bar (N/mm²)

Fatigue ratio ρ_p^f	Steel strand $f_{ptk}=1570$	Stress-relieved steel wire $f_{ptk}=1570$
0.7	144	240
0.8	118	168
0.9	70	88

Notes:

- 1 When ρ_p^f is no less than 0.9, there is no need for the checking of fatigue of prestressing tendon and/or bar;
- 2 When sufficient basis is available, the limit value for fatigue stress amplitude specified in the table may be adjusted appropriately.

The fatigue ratio ρ_s^f of steel bars shall be calculated according to the following formula:

$$\rho_s^f = \frac{\sigma_{s, \min}^f}{\sigma_{s, \max}^f} \quad (4.2.6-1)$$

Where:

$\sigma_{s, \min}^f$, $\sigma_{s, \max}^f$ —the minimum stress and maximum stress of the same layer of steel reinforcements during the checking of fatigue of members respectively.

The fatigue ratio ρ_p^f of prestressing tendon and/or bar shall be calculated according to the following formula:

$$\rho_p^f = \frac{\sigma_{p, \min}^f}{\sigma_{p, \max}^f} \quad (4.2.6-2)$$

Where:

$\sigma_p^f, \min$, $\sigma_p^f, \max$ ——the minimum stress and maximum stress of the same layer of

prestressing tendons and/or bars during the checking of fatigue of members respectively.

4.2.7 Steel reinforcements in the members may be arranged in form of twin bars. The quantity of the bundled steel reinforcements in diameter of or less than 28mm shall not exceed 3; that of steel reinforcements in diameter of 32mm should be 2; and the steel reinforcements in diameter of or above 36mm shall not be bundled. The twin bars shall be calculated according to the single equivalent steel reinforcement, and the equivalent diameter of this equivalent steel reinforcement shall be determined by conversion in accordance with the principle of equal sectional area.

4.2.8 When the steel reinforcements are replaced, in addition to the load-bearing capacity of members, the overall elongation under maximum force, the checking of crack width and the seismic provisions as stated in the design, the requirements on minimum ratio of reinforcement, spacing of bars, cover thickness, anchorage length of steel reinforcement, joint area percentage and overlapping length still shall be met.

4.2.9 When the precast reinforced welded mesh or the reinforcements for framework of steel reinforcement are used in the members, they shall comply with the relevant current standards of China.

4.2.10 As for the steel bars and prestressing tendons and/or bars of various nominal diameters, their nominal sectional area and theoretical weight shall be adopted in accordance with Appendix A of this code.

5 Structural Analysis

5.1 General

5.1.1 The integrity action effect analysis shall be carried out for the concrete structure, the special parts under stress conditions in the structure shall be analyzed in more details if necessary.

5.1.2 If there are diversified stress conditions during different stages of construction and usage period, the structural analysis shall be carried out respectively and the most unfavorable action combination shall be determined.

The corresponding structural analysis shall be carried out in accordance with the requirements of the current relevant national standards if the structure meets such accidental actions as fire, hurricane, explosion and impacting.

5.1.3 The model of the structural analysis shall meet the following requirements:

- 1** The calculation diagrams, geometric dimensions, calculation parameters, boundary conditions, performance index of structural materials and constructional measures adopted for the structural analysis shall meet the actual work states;

- 2** The possible action and its combination, primary stress and deformation conditions of the structure shall meet the actual state of the structure;

- 3** All kinds of approximate assumption and simplification adopted in the structural analysis shall be provided with theoretical and test criteria or be verified through engineering practice; the precision of calculated results shall meet the requirements of the engineering design.

5.1.4 The structural analysis shall meet the following requirements:

- 1** Meet mechanical equilibrium conditions;

- 2** Meet deformation compatibility conditions in varying degrees, including constraint conditions of nodes and the boundary;

- 3** Adopt reasonable material constitutive relation or the stressed-deformation relation of member unit.

5.1.5 In the structural analysis, the following analysis methods shall be selected according to the structure type, material property and stressed characteristics:

- 1** Elastic analysis method;

- 2** Analysis method on plastic redistribution of internal forces;

- 3** Elastic-plastic analysis method;

- 4** Plastic limit analysis method;

- 5** Test analysis method.

5.1.6 The calculation software adopted for the structural analysis shall pass the assessment and verification, the technical conditions shall meet the requirements of this code and current relevant national standards.

The analysis results shall be judged and checked, and they shall be applied to the engineering design only after the confirmation of their reasonableness and effectiveness.

5.2 Analysis Model

5.2.1 The structure integrity analysis should be carried out for the concrete structure according to the three dimensional system, and the influence of such deformations of structural units as the flexion, the axial, shear and torsion on the structure internal force should be considered.

The simplification analysis shall meet the following requirements:

1 The space structure with regular shape may be analyzed respectively by breaking up into plane structures in different directions along the colonnade or wall axes, but the space teamwork of the plane structure shall be considered;

2 If the influence of the axial, shearing and torsional deformation of members on the analysis of structure internal force is not large, it may be excluded from consideration.

5.2.2 The calculation diagrams of the concrete structure should be determined according to the following methods:

1 The axes of such one-dimension members as beams, columns and rods should take the connecting lines of section geometric centers; the middle axle surfaces of such two-dimension members as walls and slabs should take the plane or curved surface composed by section center lines;

2 The connecting parts of beam-column joints, columns and foundation in cast-in-situ structure and assembled monolithic structure may be adopted with rigid connection; the both ends of the non-integral cast secondary beam and both ends of slab span may be approximately adopted with hinged connection;

3 The calculated span or calculated height of such member bars as beams and columns may be determined according to the clear distance or central distance of the supporting length at both ends and shall be corrected according to the connection rigidity of supporting nodes or the position of supporting resistance;

4 If the rigidity of connection parts between member bars as beams and columns is far larger than that of the mid section of member bars, it may be treated as rigid zone in computation model.

5.2.3 In the structure integrity analysis, for cast-in-situ structures or assembled monolithic structures, their floor system in themselves' plane may be assumed as infinitely rigid. If there is relatively big opening in the floor system or obvious plane deformation may occur in its local, the influence shall be considered in structural analysis.

5.2.4 For cast-in-situ floor system and assembled monolithic floor system, the flange coupled beam rigidity for floor slab and the influence of the bearing capacity should be considered. The effective flange calculated width b_f' in the beam compression zone may be adopted with the minimum value in situations listed in Table 5.2.4; it also may be approximately considered by adopting the beam rigidity enhancement coefficient method, and the rigidity enhancement coefficient shall be determined according to the relative scale of the effective flange dimension of beams and beam section dimensions.

Table 5.2.4 Effective Flange Calculated Width b_f' of Flexural Member in Compressive Zone

Situation	T-shaped and I-shaped section		Inverted L-shaped section
	Ribbed beam (slab)	Isolated beam	Ribbed beam (slab)

1	Considered as calculated span l_0		$l_0/3$	$l_0/3$	$l_0/6$
2	Considered as beam (rib) clear distance s_n		$b+s_n$	—	$b+s_n/2$
3	Considered as flange height h_f'	$h_f' / h_0 \geq 0.1$	—	$b+12 h_f'$	—
		$0.1 > h_f' / h_0 \geq 0.05$	$b+12 h_f'$	$b+6 h_f'$	$b+5 h_f'$
		$h_f' / h_0 < 0.05$	$b+12 h_f'$	b	$b+5 h_f'$

Notes:

- 1 b in this table refers to web thickness of the beam;
- 2 If the transverse rib with spacing less than the spacing of longitudinal rib is set for ribbed beams in the beam span, the requirements of Situation 3 in this table may not be considered;
- 3 For haunched T-shaped, I-shaped and inverted L-shaped sections, if the haunched height h_h in the compressive zone is not less than h'_f and the haunched length b_h is not larger than $3h_h$, then the flange calculated width may be increased by $2b_h$ (for T-shaped and I-shaped sections) and b_h (for inverted L-shaped sections) respectively according to the requirements of Situation 3 in this table.
- 4 If the flange plate of the isolated beam in the compressive zone may produce cracks along the longitudinal rib direction under the action of load through checking and calculation, the calculated width shall take web width b .

5.2.5 If the interaction of the base and structure has an obvious impact on the internal force and deformation of the structure, the influence of the interaction of the base and structure should be considered in structural analysis.

5.3 Elastic Analysis

5.3.1 The elastic analysis method of the structure may be used to the action effect analysis in serviceability limit states and ultimate limit states.

5.3.2 The rigidity of the structural member may be determined according to the following principles:

- 1 The concrete elastic modulus may be adopted according to those specified in Table 4.1.5;
- 2 The moment of inertia of section may be calculated according to the total cross section of homogeneous concrete;
- 3 The influence of the section change on the structural analysis shall be considered for member bars haunched at ends;
- 4 The influence of such factors as the concrete cracking and creeping should be considered for the section rigidity of members in different stressed states to reduce.

5.3.3 The elastic analysis on concrete structure should be adopted with structural mechanics or elastic mechanics analysis methods. The proper simplification analysis method may be adopted for the structure with regular shape according to the action type and characteristic.

5.3.4 If the second-order effect of the structure may bring obvious increase in action effect, the adverse impact of the second-order effect shall be considered in structural analysis.

The gravity second-order effect of the concrete structure may be calculated with the finite element analysis method or the simplified method specified in Appendix B of this code.

The impact of cracking of concrete members on the member rigidity should be considered if the finite element analysis method is adopted.

5.3.5 If the boundary supporting displacement has a relatively big impact on the internal force and deformation of two-way slabs, the influence of vertical deformation and torsion of boundary supporting should be considered in the analysis.

5.4 Analysis on Plastic Redistribution of Internal Forces

5.4.1 The analysis method of plastic redistribution of internal forces may be adopted for concrete continuous beams and continuous one-way slabs.

For the frames, cast-in-situ beams and two-way slabs in the frame-shear wall structure under the action of the gravity load, after their internal forces calculated from the elastic analysis, proper amplitude modulation may be carried out for supports or node moments and corresponding mid-span bending moment may be determined.

5.4.2 For the structure and member designed by considering the analysis method of plastic redistribution of internal forces, the steel reinforcement meeting the requirements of Article 4.2.4 of this code shall be selected, the requirements for serviceability limit states shall be complied with and the effective constructional measures shall be taken.

The analysis method in consideration of the plastic redistribution of internal forces shall not be adopted for members directly bearing the dynamic load and structures with no crack requirement or structures in Class IIIa and Class IIIb environment conditions.

5.4.3 The modulated amplitude of section hogging moment on the reinforced concrete beam support or node edge should not be larger than 25%; the height of beam end section with adjusted bending moment in relative compression zone shall not exceed 0.35 and should not be less than 0.10.

The modulated amplitude of the hogging moment of reinforced concrete slabs should not be larger than 20%.

The modulated amplitude of bending moment of prestressed concrete beams shall meet the requirements of Article 10.1.8 of this code.

5.4.4 For concrete structure members belonging to coordination torsion, the influence of redistribution of internal force should be considered for the torsion moment of supporting beams restricted by adjacent members.

For supporting beams considering the redistribution of internal forces, the bearing capacity shall be calculated as bending, shearing and torsion members.

Note: Other design methods may be adopted if sufficient criteria exist.

5.5 Elastic-Plastic Analysis

5.5.1 For important or stressed complex structures, the elastic-plastic analysis method should be adopted for the checking and calculation on the entire or local structure. The elastic-plastic analysis for the structure should be complied with the following principles:

- 1 The shape, dimension, boundary conditions, material properties and reinforcements of the structure shall be pre-established;
- 2 The performance index of materials should be averaging and should be determined

by test analysis and also may be determined according to the requirements of Appendix C of this code;

3 The adverse impact of the geometrical nonlinearity of the structure should be considered;

4 If the analysis result is used for the bearing capacity design, the resistance of the structure should be adjusted properly by considering the indeterminateness coefficient of the resistance model.

5.5.2 The statical or dynamic analysis methods may be adopted for the elastic-plastic analysis of the concrete structure according to actual conditions. The computation model of basic members of structures should be determined according to the following principles:

1 Such bar members as beams, columns and rods may be reduced to one-dimension unit and the fiber bundle model or plastic hinge model should be adopted;

2 Such members as walls and slabs may be reduced to two-dimension unit, and the film unit, slab unit or shell unit should be adopted;

3 If the elaborate analysis is required to do for complex concrete structures, bulk concrete structures and nodes or local area of structures, the three-dimensional block unit should be adopted.

5.5.3 The stressed -deformation constitutive relations of members, sections or all kinds of calculating units should meet practical stressed conditions. The bond-slip constitutive relations between steel reinforcements and concretes should be considered if the local elaborate analysis is carried out for some members or nodes with larger deformation.

The constitutive relation of steel reinforcements and concrete materials should be determined by test analysis and also may be adopted as those specified in Appendix C of this code.

5.6 Plastic Limit Analysis

5.6.1 For concrete structures without bearing multiple-repeated load action, if there is adequate energy of plastic deformation, the bearing capacity calculation may be carried out by adopting the analysis method of plastic limit theory, and requirements for normal use shall be met simultaneously.

5.6.2 The plastic limit analysis calculation of the integral structure shall meet the following requirements:

1 For conditions that the structural damage mechanism may be predicted, the ultimate bearing capacity of the structure may be analyzed by adopting the plastic limit theory according to set plastic yielding mechanism of the structure;

2 For conditions that the structural damage mechanism is difficult to predict, the ultimate bearing capacity of the structure may be determined by adopting statical or dynamic elastic-plastic analysis methods;

3 For structural members or parts directly bearing the accidental action, the influence of the dynamic effect shall be considered according to dynamic characteristics of the accidental action.

5.6.3 For two-way rectangular slabs supported at edges bearing evenly distributed load, the analysis and design in ultimate limit state may be carried out by adopting such plastic limit

analysis methods as plastic hinge line or strip methods.

5.7 Indirect Action Effect Analysis

5.7.1 If the action effect produced by such indirect actions of concrete as contraction, creeping and temperature variation in the structure may endanger the safety or normal use of the structure, the indirect action effect analysis should be carried out, and corresponding constructional measures and construction measures shall be taken.

5.7.2 For the indirect action effect analysis on the concrete structure, the elastic-plastic analysis method given in Section 5.5 of this code may be adopted; also the approximate analysis may be carried out according to the elastic method by considering the impact of cracks and creeping on the member rigidity.

6 Calculation of Ultimate Limit States

6.1 General

6.1.1 This chapter is applicable to the calculation on the ultimate limit states of reinforced concrete members and prestressed concrete members; the design of plain concrete structural members shall comply with the requirements in Appendix D of this code.

The load-bearing capacity calculation of deep flexural members, brackets and composite members shall meet the relevant requirements in Chapter 9 of this code.

6.1.2 For two-dimensional or three-dimensional non-bar structural members, when obtaining the stress design value distribution of members according to elastic or elastic-plastic analysis method, the quantity of reinforcement may be determined according to the projection of the resultant force of the principal tensile stress design value in the reinforcement direction, the steel reinforcement arrangement shall be determined by the distribution zone of the principal tensile stress, and shall meet the corresponding structural requirements; where the concrete is in compression state, the combined action of the compression steel reinforcement and the concrete shall be considered, and the allocation of compression steel reinforcement shall meet the structural requirements.

6.1.3 When checking the ultimate limit states of concrete structure members with the stress expression, the following requirements shall be met:

1 The strength value of concrete and reinforcement shall be determined according to the condition of design and the performance design objectives of the members.

2 The steel reinforcement stress shall not be greater than the strength value of reinforcement.

3 The concrete stress shall not be greater than the strength value of concrete; under the multi-axial stress condition, the concrete's strength calculation and checking may be carried out according to the relevant requirements specified in Appendix C.4 of this code.

6.2 Calculation of Normal Section Load-bearing Capacity

(I) General Requirements for Calculation of Normal Section Load-bearing Capacity

6.2.1 The load-bearing capacity of normal section shall be calculated according to the following fundamental assumptions:

1 Strain of section remains plane.

2 Tensile strength of concrete is disregarded.

3 The relation curve of compressive stress and strain in concrete shall be taken according to the following provisions:

When $\varepsilon_c \leq \varepsilon_0$

$$\sigma_c = f_c \left[1 - \left(1 - \frac{\varepsilon_c}{\varepsilon_0} \right)^n \right] \quad (6.2.1-1)$$

When $\varepsilon_0 < \varepsilon_c \leq \varepsilon_{cu}$

$$\sigma_c = f_c \quad (6.2.1-2)$$

$$n = 2 - \frac{1}{60}(f_{cu,k} - 50) \quad (6.2.1-3)$$

$$\varepsilon_0 = 0.002 + 0.5(f_{cu,k} - 50) \times 10^{-5} \quad (6.2.1-4)$$

$$\varepsilon_{cu} = 0.0033 - (f_{cu,k} - 50) \times 10^{-5} \quad (6.2.1-5)$$

Where:

——the concrete compressive stress where the concrete compressive strain is ;

——the design value of concrete axial compressive strength shall be adopted according to Table 4.1.4-1 of this code;

ε_0 ——the concrete compressive strain where the concrete compressive stress reaches the f_c value; if $\varepsilon_0 < 0.002$, $\varepsilon_0 = 0.002$ is taken;

ε_{cu} ——the concrete's ultimate compressive strain of normal section; when under non-uniform compression and the calculated value is greater than 0.0033, calculated according to the requirements of Formula (6.2.1-5), $\varepsilon_{cu} = 0.0033$ is taken; when under axial compression, taking ε_0 ;

$f_{cu,k}$ ——the standard value for compressive strength of concrete cube, shall be determined according to Article 4.1.1 of this code;

n ——the coefficient, 2.0 may be taken if the calculated value of n is greater than 2.0.

4 The ultimate tensile strain of longitudinal tension steel reinforcement is to be taken as 0.01.

5 The product of the strain and the elastic modulus of steel reinforcement is taken for the stress of the longitudinal steel reinforcement, but its value shall be in accordance with the following requirements:

$$-f'_y \leq \sigma_{si} \leq f_y \quad (6.2.1-6)$$

$$\sigma_{p0i} - f'_{py} \leq \sigma_{pi} \leq f_{py} \quad (6.2.1-7)$$

Where:

σ_{si} , σ_{pi} ——the stress in longitudinal steel bars, prestressing tendons and/or bars at the i th layer respectively, the tension stress is taken as positive value and the compression stress is taken as negative value;

σ_{p0i} ——the stress of prestressing tendons and/or bars, where the normal stress of concrete at the centroid of the i th layer longitudinal prestressing tendons and/or bars equals zero, calculated according to Formula (10.1.6-3) or Formula (10.1.6-6);

f_y , f_{py} ——the design value of tensile strength of steel bars and prestressing tendons and/or bars respectively, adopted according to Table 4.2.3-1 and Table 4.2.3-2;

f'_y , f'_{py} ——the design value of compressive strength of steel bars and prestressing tendons and/or bars respectively, adopted according to Table 4.2.3-1 and Table 4.2.3-2;

6.2.2 When determining the neutral axis position, for two-way flexural members, the action plane of internal and external bending moment shall superpose mutually; for two-way eccentric tension members, the action point of axial force, the point for resultant of forces of concrete and compression steel reinforcement, and the point for resultant of forces of tension reinforcement shall be in a straight line. If the above-mentioned conditions are not met, the

influence of torsion still shall be considered.

6.2.3 For eccentric compression member with symmetrical section in the action plane of bending moment, where the rod ends bending moment ratio $\frac{M_1}{M_2}$, which is in the same

principal axis direction with the eccentric compression member, is not greater than 0.9, and the axial compression ratio is not greater than 0.9, if the slenderness ratio of members meet the requirements of Formula (6.2.3), the additional bending moment influence of the axial compressive force on flexural member in this direction may not be considered; otherwise the additional bending moment influence of the axial compressive force on flexural member shall be considered respectively based on the two principal axes of section, according to those specified in Article 6.2.4 of this code.

$$l_c / i \leq 34 - 12(M_1 / M_2) \quad (6.2.3)$$

Where M_1, M_2 —The design value of combined bending moment at the same principal axis, determined by the analysis of structural elasticity of the section at both ends of eccentric compression member of which the sides way influence has been considered, the end with a relatively larger absolute value shall be M_2 , the end with a relatively smaller absolute value shall be M_1 ; when the member is bended according to the single curvature, the positive value shall be taken for M_1/M_2 , otherwise the negative value shall be taken;

l_c —The calculated length of members, the distance between the upper and lower supporting points in the direction of corresponding principal axis of the eccentric compression member may be approximately adopted;

i —The sectional radius of gyration in the eccentric direction.

6.2.4 Except the bent structural column, the bending moment design value of control section of other eccentric compression members, after considering the second-order effect of axial compressive force on flexural member, shall be calculated according to the following formulae:

$$M = C_m \eta_{ns} M_2 \quad (6.2.4-1)$$

$$C_m = 0.7 + 0.3 \frac{M_1}{M_2} \quad (6.2.4-2)$$

$$\eta_{ns} = 1 + \frac{1}{1300(M_2 / N + e_a) / h_0} \left(\frac{l_c}{h} \right)^2 \zeta_c \quad (6.2.4-3)$$

$$\zeta_c = \frac{0.5 f_c A}{N} \quad (6.2.4-4)$$

If $C_m \eta_{ns} < 1.0$, taking 1.0; as for shear wall and core-tube wall, $C_m \eta_{ns} = 1.0$ may be taken.

Where C_m —The adjustment coefficient of eccentric distance on the end section of

members; if $C_m < 0.7$, $C_m = 0.7$ shall be adopted;
 η_{ns} ——The amplified coefficient of bending moment;
 N ——The design value of axial compressive force corresponding to design value of bending moment M_2 ;
 e_a ——The additional eccentric distance, determined according to Article 6.2.5 of this code;
 ζ_c ——The section curvature correction coefficient, if the calculated value > 1.0 , $\zeta_c = 1.0$ shall be taken;

h ——The depth of section; as for annular section, the external diameter shall be adopted; for circular section, the diameter shall be adopted;

h_0 ——The effective depth of section; for annular section, taking $h_0 = r_2 + r_s$; for circular section, taking $h_0 = r + r_s$; here r , r_2 and r_s shall be determined according to Article E.0.3 and Article E.0.4 of this code;

A ——The sectional area of members.

6.2.5 The additional eccentric distance e_a of axial compressive force in eccentric direction shall be counted into the calculation of normal section load-bearing capacity of eccentric compression member, the value of 20mm or the value of 1/30 sectional maximum dimension in eccentric direction, whichever is greater, shall be taken.

6.2.6 The stress diagram of the concrete in compression zone may be simplified as an equivalent rectangular stress diagram, when calculating the load-bearing capacity of normal section of the flexural members and the eccentrically stressed members.

The depth x of compression zone in the rectangular stress diagram may take the depth of neutral axis which is determined by an assumption that strain of section remains plane to multiply the coefficient β_1 . Where the concrete strength grade does not exceed C50, $\beta_1 = 0.80$ is taken; where the concrete strength grade is C80, $\beta_1 = 0.74$ is taken. The values between C50 and C80 shall be determined by the linear interpolation method.

The stress value of the rectangular stress diagram may take the design value of axial compressive strength f_c multiplied by the coefficient α_1 . Where the concrete strength grade is not greater than C50, $\alpha_1 = 1.0$ is taken; where the concrete strength grade is C80, $\alpha_1 = 0.94$ is taken. The values between C50 and C80 shall be determined by the linear interpolation method.

6.2.7 Where the yielding of the longitudinal tension steel reinforcement and the failure of concrete in compression zone are occurred simultaneously, the relative depth of limiting compression zone ξ_b shall be calculated according to the following formulae:

1 Reinforced concrete members

As for steel bars with yield points

$$\xi_b = \frac{\beta_1}{1 + \frac{f_y}{E_s \epsilon_{cu}}} \quad (6.2.7-1)$$

As for steel bars without yield points

$$\xi_b = \frac{\beta_1}{1 + \frac{0.002}{\varepsilon_{cu}} + \frac{f_y}{E_s \varepsilon_{cu}}} \quad (6.2.7-2)$$

2 Prestressed concrete members

$$\xi_b = \frac{\beta_1}{1 + \frac{0.002}{\varepsilon_{cu}} + \frac{f_{py} - \sigma_{p0}}{E_s \varepsilon_{cu}}} \quad (6.2.7-3)$$

Where:

ξ_b ——relative depth of limiting compression zone, $\xi_b x_b / h_0$ is taken;

x_b ——the depth of limiting compression zone;

h_0 ——the effective depth of section: the distance from the point for resultant of forces of longitudinal tension steel reinforcement to the compression edge of the section;

E_s ——the elastic modulus of steel reinforcement, adopted according to Table 4.2.5 of this code;

σ_{p0} ——the stress of prestressing tendons and/or bars when normal stress of concrete at the point for resultant of forces of longitudinal prestressing tendons and/or bars in tension zone equals zero, calculated according to the provisions of Formula (10.1.6-3) or Formula (10.1.6-6);

ε_{cu} ——the ultimate compressive strain of the concrete under non-uniform compression, calculated according to requirements in Formula (6.2.1-5) of this code;

β_1 ——the coefficient, calculated according to the requirements specified in the Article 6.2.6 of this code.

Note: When the steel reinforcement, with different kinds or different prestressed values, is allocated in the sectional tension zone, the relative depth of limiting compression zone of flexural members shall be calculated respectively; whichever value is smaller shall be taken.

6.2.8 The stress of longitudinal steel reinforcement shall be in accordance with the following requirements:

1 The stress of longitudinal steel reinforcements should be calculated according to the following formulae:

Steel bars

$$\sigma_{si} = E_s \varepsilon_{cu} \left(\frac{\beta_1 h_{0i}}{x} - 1 \right) \quad (6.2.8-1)$$

Prestressing tendons and/or bars

$$\sigma_{pi} = E_s \varepsilon_{cu} \left(\frac{\beta_1 h_{0i}}{x} - 1 \right) + \sigma_{p0i} \quad (6.2.8-2)$$

2 The stress of longitudinal steel reinforcements may also be calculated according to the following approximate formulae:

Steel bars

$$\sigma_{si} = \frac{f_y}{\xi_b - \beta_1} \left(\frac{x}{h_{0i}} - \beta_1 \right) \quad (6.2.8-3)$$

Prestressing tendons and/or bars

$$\sigma_{pi} = \frac{f_{py} - \sigma_{p0i}}{\xi_b - \beta_1} \left(\frac{x}{h_{0i}} - \beta_1 \right) + \sigma_{p0i} \quad (6.2.8-4)$$

3 The stress of longitudinal steel reinforcements calculated according to Formula ~ Formula , shall meet the relevant requirements specified in the Clause 5 of the Article 6.2.1 of this code.

Where:

h_{0i} ——the distance from the centroid of the i th layer longitudinal steel reinforcement section to the compression edge of the section;

x ——the depth of concrete compression zone of the equivalent rectangular stress diagram;

σ_{si}, σ_{pi} ——the stress of the longitudinal steel bars, prestressing tendons and/or bars at the i th layer respectively, tension stress is taken as positive value and compression stress is taken as negative value;

σ_{p0i} ——the stress of prestressing tendons and/or bars, when the normal stress of concrete at the centroid of the i th layer longitudinal prestressing tendons and/or bars section equals zero, shall be calculated according to Formula (10.1.6-3) or Formula(10.1.6-6) of this code.

6.2.9 The normal section load-bearing capacity of rectangular, I-shaped and T-shaped section members may be calculated according to the provisions of this section; the normal section load-bearing capacity of arbitrary, circular and annular sectional members may be calculated according to the requirements specified in Appendix E of this code.

(II) Calculation of Normal section flexural load-bearing capacity

6.2.10 The inverted T-shaped flexural members with rectangular section or flange situated at tension side, the normal section flexural load-bearing capacity shall meet the following requirements (Figure 6.2.10):

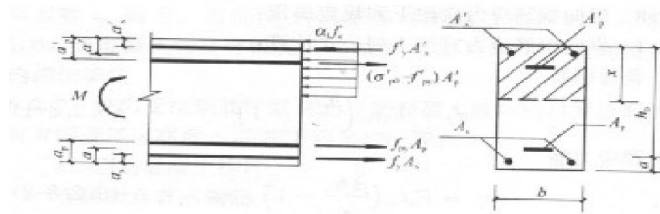


Figure 6.2.10 Calculation of Flexural Load-bearing Capacity with Normal Section of Flexural Members with Rectangular Section

$$M \leq \alpha_1 f_c b x \left(h_0 - \frac{x}{2} \right) + f_y A_s (h_0 - a_s) - (\sigma'_{p0} - f_{py}) A_p (h_0 - a_p) \quad (6.2.10-1)$$

Depth of concrete compression zone shall be determined according to the following formula:

$$\alpha_1 f_c b x = f_y A_s - f_y A_s' + f_{py} A_p + (\sigma'_{p0} - f_{py}) A_p' \quad (6.2.10-2)$$

Depth of concrete compression zone shall meet the following conditions:

$$x \leq \xi_b h_0 \quad (6.2.10-3)$$

$$x \geq 2a' \quad (6.2.10-4)$$

Where:

M ——design value of bending moment;

a_1 ——the coefficient, shall be calculated according to the provisions in the Article 6.2.6 of this code;

f_c ——the design value of axial compressive strength of concrete, adopted according to Table 4.1.4-1 of this code;

A_s, A'_s ——the sectional area of longitudinal steel bars in tension zone, in compression zone respectively;

A_p, A'_p ——the sectional area of longitudinal prestressing tendons and/or bars in tension zone, in compression zone respectively;

σ'_{p0} ——the stress of prestressing tendons and/or bars, when the normal stress of concrete is equal to zero at the point for resultant of forces of longitudinal prestressing tendons and/or bars in compression zone;

b ——the width of rectangular section or width of web for inverted T-shaped section;

h_0 ——the efficient depth of section;

a'_s, a'_p ——the distance from the point for resultant of forces of longitudinal steel bars, prestressing tendons and/or bars in compression zone respectively to the compression edge of the section;

a' ——the distance from the point for resultant of forces of all longitudinal steel reinforcements in compression zone to the compression edge of the section; when longitudinal prestressing tendon and/or bar is not arranged in compression zone or the stress of longitudinal prestressing tendons and/or bars ($\sigma'_{p0} - f'_{py}$) in compression zone is the tension stress, a' in Formula (6.2.10-4) is replaced by a'_s .

6.2.11 As for T-shaped, I-shaped section flexural members with flange situated at compression zone, the calculation of the normal section flexural load-bearing capacity shall meet the following requirements:

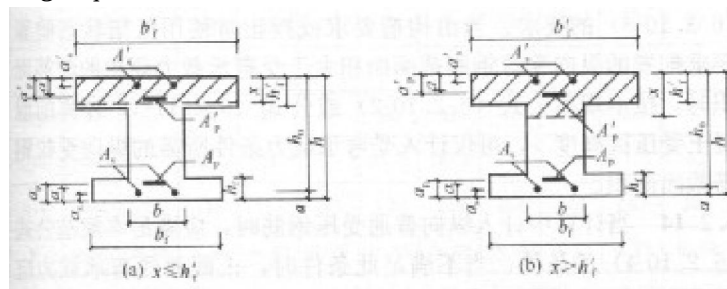


Figure 6.2.11 Location of the Depth in Compression Zone of I-shaped Section Flexural Members

1 If meeting the following conditions, the normal section flexural load-bearing capacity shall be calculated as rectangular section with width of

$$f_y A_s + f_{py} A_p \leq a_1 f_c b_f' h_f' + f_y' A_s' - (\sigma_{p0}' - f_{py}') A_p' \quad (6.2.11-1)$$

2 If not meeting the conditions in Formula (6.2.11-1), it shall be calculated according to the following formula:

$$M \leq a_1 f_c b x \left(h_0 - \frac{x}{2} \right) + a_1 f_c (b_f' - b) h_f' \left(h_0 - \frac{h_f'}{2} \right) + f_y' A_s' (h_0 - a_s') - (\sigma_{p0}' - f_{py}') A_p' (h_0 - a_p') \quad (6.2.11-2)$$

The depth of concrete compression zone shall be determined according to the following formula:

$$a_1 f_c [bx + (b_f' - b) h_f'] = f_y A_s - f_y' A_s' + f_{py} A_p + (\sigma_{p0}' - f_{py}') A_p' \quad (6.2.11-3)$$

Where:

h_f' ——the depth of flange in compression zone of T-shaped, I-shaped section;

b_f' ——the calculated width of flange in compression zone of T-shaped, I-shaped section

shall be determined according to the provisions specified in the Article 6.2.12 of this code.

In the calculation of T-shaped, I-shaped section flexural members using the above formulae, the depth of concrete compression zone still shall conform to the requirements specified in Formula (6.2.10-3) and Formula (6.2.10-4) of this code.

6.2.12 As for the calculated flange width with T-shaped, I-shaped and inverted L-shaped section flexural members situated in compression zone, the minimum values, calculated according to the conditions listed in Table 5.2.4 of this code, may be taken.

6.2.13 The calculation of the normal section flexural load-bearing capacity of flexural members shall meet the requirements specified in Formula of this code. When the sectional area of the longitudinal tension steel reinforcement, distributed according to structural requirements or the checking requirements of serviceability limit states, is larger than the reinforcement area required by flexural load-bearing capacity, the depth of concrete compression zone shall be calculated according to Formula or Formula of this code, only the sectional area of the longitudinal tension steel reinforcements needed by conditions of flexural load-bearing capacity may be counted.

6.2.14 Where the longitudinal ordinary compression steel reinforcements are counted in calculation, the conditions in Formula of this code shall be met; when the conditions are not met, the normal section flexural load-bearing capacity shall meet the following requirements:

$$M \leq f_{py} A_p (h - a_p - a_s') + f_y A_s (h - a_s - a_s') + (\sigma_{p0}' - f_{py}') A_p' (a_p' - a_s') \quad (6.2.14)$$

Where:

a_s, a_p ——the distance from longitudinal steel bars, prestressing tendons and/or bars in the tension zone respectively to the tension edge.

(III) Calculation of Normal Section Compression Load-bearing Capacity

6.2.15 As for reinforced concrete axial compression members, when the stirrups allocated

for members meeting the provisions in Section 9.3 of this code, the normal section compression load-bearing capacity shall meet the following requirements (Figure 6.2.15):

$$N \leq 0.9\phi (f_c A + f_y' A_s') \quad (6.2.15)$$

Where:

N ——the design value of axial compressive force;

Φ ——the stability coefficient of reinforced concrete members, adopted according to Table 6.2.15;

f_c ——the design value of concrete axial compressive strength, adopted according to Table 4.1.4-1 of this code;

A ——the sectional area of members;

A_s' ——the sectional area of all longitudinal steel bars.

If the reinforcement ratio of longitudinal steel bars is greater than 3%, in Formula shall be replaced by .

Table 6.2.15 Stability Coefficient of Reinforced Concrete Axial Compression Member

l_0/b	≤ 8	10	12	14	16	18	20	22	24	26	28
l_0/d	≤ 7	8.5	10.5	12	14	15.5	17	19	21	22.5	24
l_0/i	≤ 28	35	42	48	55	62	69	76	83	90	97
ϕ	1.00	0.98	0.95	0.92	0.87	0.81	0.75	0.70	0.65	0.60	0.56
l_0/b	30	32	34	36	38	40	42	44	46	48	50
l_0/d	26	28	29.5	31	33	34.5	36.5	38	40	41.5	43
l_0/i	104	111	118	125	132	139	146	153	160	167	174
ϕ	0.52	0.48	0.44	0.40	0.36	0.32	0.29	0.26	0.23	0.21	0.19

Notes:

- 1 denotes the calculated length of the member, the length of reinforced concrete column may be taken according to the Article 6.2.20 of this code;
- 2 denotes the short-side size of rectangular section, denotes the diameter of circular section, denotes the minimum radius of gyration of section.

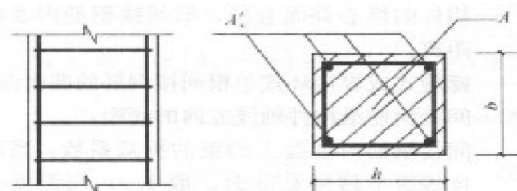


Figure 6.2.15 Reinforced Concrete Axial Compression Member Allocated with Stirrups

6.2.16 As for reinforced concrete axial compression members, when the spiral or welded ring indirect steel reinforcements allocated for members meeting the provisions in Article 9.3.2 of this code, the normal section compression load-bearing capacity shall meet the following requirements (Figure 6.2.16):

$$N \leq 0.9(f_c A_{cor} + f_y' A_s' + 2\alpha f_{yv} A_{ss0}) \quad (6.2.16-1)$$

$$A_{ss0} = \frac{\pi d_{cor} A_{ss1}}{s} \quad (6.2.16-2)$$



Figure 6.2.16 Reinforced Concrete Axial Compression Member Allocated with Spiral Indirect Steel Reinforcement

Where:

f_{yv} —the design value of tensile strength of indirect steel reinforcement, adopted according to those specified in the Article 4.2.3 of this code;

A_{cor} —the area of core section of member, the sectional area of concrete within internal surface range of indirect steel reinforcement shall be taken;

A_{ss0} —the transformed sectional area of spiral or welded ring indirect steel reinforcement;

d_{cor} —the diameter of core section of the member, the distance between internal surface range of indirect steel reinforcement shall be taken;

A_{ss1} —the sectional area of spiral or welded ring single indirect steel reinforcement;

s —space of indirect steel reinforcement along the member axis direction;

a —the reduction coefficient for the limitation of indirect steel reinforcements to the concrete: $a=1.0$ is taken if the concrete strength grade does not exceed C50; where the concrete strength grade is C80, $a=0.85$ is taken; the values between C50 and C80 shall be determined according to linear interpolation method.

Notes:

1 The design value of member compression load-bearing capacity calculated according to Formula shall not be greater than 1.5 times that calculated according to Formula of this code;

2 When one of the following cases is encountered, where the influence of the indirect steel reinforcement shall not be counted, hence the calculation shall be carried out according to those specified in the Article 6.2.15 of this code:

- 1) Where $l_0/d > 12$;
- 2) Where the compression load-bearing capacity calculated according to Formula is less than that calculated according to Formula of this code;
- 3) Where the transformed sectional area of the indirect steel reinforcement is less than 25% of the gross sectional area of the longitudinal steel bars.

6.2.17 The normal section compression load-bearing capacity of the rectangular section eccentric compression members shall meet the following requirements (Figure 6.2.17):

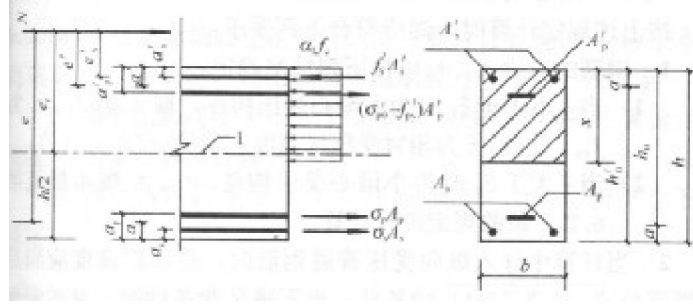


Figure 6.2.17 Calculation of Normal Section Compression Load-bearing Capacity of Eccentric Compression

Member with Rectangular Section

1—the centroid axis of section

$$N \leq \alpha_1 f_c b x + f_y' A_s' - \sigma_s A_s - (\sigma_{p0}' - f_{py}') A_p' - \sigma_p A_p \quad (6.2.17-1)$$

$$Ne \leq \alpha_1 f_c b x \left(h_0 - \frac{x}{2} \right) + f_y' A_s' (h_0 - a_s') - (\sigma_{p0}' - f_{py}') A_p' (h_0 - a_p') \quad (6.2.17-2)$$

$$e = e_i + \frac{h}{2} - a \quad (6.2.17-3)$$

$$e_i = e_0 + e_a \quad (6.2.17-4)$$

Where:

e —the distance from the action point of axial compressive force to the point for resultant of forces of longitudinal steel bar and prestressed tension tendon and/or bar.

σ_s , σ_p —the stress of longitudinal steel bars, prestressed tendons and/or bars at the tension side or smaller compression side;

e_i —the initial eccentric distance;

a —the distance from the point of resultant of forces of longitudinal tension steel bars and prestressed tension tendons and/or bars to the near edge of section;

e_0 —the eccentric distance of axial compressive force with respect to centroid of section $e_0 = M/N$, when the second-order effect needs to be considered, M denotes the design value of bending moment determined according to the Article 5.3.4, the Article 6.2.4 of this code;

e_a —the additional eccentric distance, determined according to the Article 6.2.5 of this code.

When the calculation is as prescribed above, the following requirements still shall be met:

1 The stress of steel reinforcements σ_s and σ_p may be determined according to the following cases:

- 1) Where $\xi \leq \xi_b$ denotes the large eccentric compression members, taking $\sigma_s = f_y$ and $\sigma_p = f_{py}$, here ξ is the relative depth of compression zone, $\xi = x/h_0$;
- 2) Where $\xi > \xi_b$ denotes small eccentric compression members, σ_s and σ_p shall be calculated according to those specified in the Article 6.2.8 of this code.

2 When the longitudinal compression steel bar is counted in calculation, the depth of compression zone shall meet the conditions in Formula (6.2.10-4) of this code; when the condition is not met, the normal section compression load-bearing capacity may be calculated

according to the Article 6.2.14 of this code; where, M in Formula (6.2.14) of this code shall be replaced by. Here e denotes the distance from the action point of axial compressive force to the point for resultant of forces of longitudinal steel bar in the compression zone; the initial eccentric distance shall be determined according to the requirements of Formula (6.2.17-4).

3 As for the small eccentric compression member of non-symmetric reinforcement with rectangular section, where $\xi < \xi_b$, the checking still shall be carried out according to the following formulae:

$$Ne' \leq f_c b h \left(h_0' - \frac{h}{2} \right) + f_y' A_s (h_0' - a_s) - (\sigma_{p0} - f_{yp}') A_p (h_0' - a_p) \quad (6.2.17-5)$$

$$e' = \frac{h}{2} - a' - (e_0 - e_a) \quad (6.2.17-6)$$

Where:

e' ——the distance from the action point of axial compressive force to the point of resultant of forces for the longitudinal steel bar and prestressing tendons and /or bars in compression zone;

h_0' ——the distance from the resultant of forces point of the longitudinal compression steel reinforcement to far-side of section.

4 As for the small eccentric compression member of reinforced concrete of symmetric reinforcement with rectangular section, the sectional area of longitudinal steel bars may also be calculated according to the following approximate formula:

$$A_s' = \frac{Ne - \xi(1 - 0.5\xi)a_1 f_c b h_0^2}{f_y'(h_0 - a_s')} \quad (6.2.17-7)$$

Here, the relative depth of compression zone ξ may be calculated according to the following formula:

$$\xi = \frac{N - \xi_b a_1 f_c b h_0}{Ne - 0.43 a_1 f_c b h_0^2} + \xi_b \quad (6.2.17-8)$$

6.2.18 The calculated compression flange width of the eccentric compression member with I-shaped sections shall be determined in accordance with Article 6.2.12 of this code; the normal section compression load-bearing capacity shall meet the following requirements:

1 Where the depth of compression zone $\xi < \xi_b$, the calculation shall be carried out according to the rectangular section with width as the calculated compression flange width.

2 Where the depth of compression zone $\xi > \xi_b$, the following requirements shall be met:

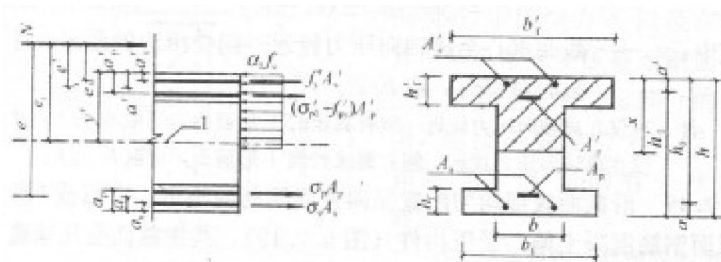


Figure 6.2.18 Calculation of Normal Section Compression Load-bearing Capacity for I-shaped Section Eccentric Compression Member

1—the centroid axis of section

$$N \leq a_1 f_c [bx + (b'_f - b)h'_f] + f'_y A'_s - \sigma_s A_s - (\sigma'_{p0} - f'_{py})A'_p - \sigma_p A_p \quad (6.2.18-1)$$

$$Ne \leq a_1 f_c \left[bx \left(h_0 - \frac{x}{2} \right) + (b'_f - b)h'_f \left(h_0 - \frac{h'_f}{2} \right) \right] + f'_y A'_s (h_0 - a'_s) - (\sigma'_{p0} - f'_{py})A'_p (h_0 - a'_p) \quad (6.2.18-2)$$

The reinforcement stress in formulae, and whether the action of longitudinal compression steel bars being considered, shall all be determined according to the relevant requirements in Article 6.2.17 of this code.

3 If $x > (h - h_f)$, the action of compression parts in flange of smaller compression side shall be counted into the calculation of normal section compression load-bearing capacity; where, the calculated width for the flange of the smaller compression side shall be in accordance with the Article 6.2.12 of this code.

4 As for small eccentric compression member with non-symmetric reinforcement, where $N > N_b$, the checking shall be carried out according to the following formulae:

$$Ne' \leq f_c \left[bh \left(h_0 - \frac{h}{2} \right) + (b_f - b)h_f \left(h_0 - \frac{h_f}{2} \right) + (b'_f - b)h'_f \left(\frac{h'_f}{2} - a' \right) \right] + f'_y A'_s (h_0 - a'_s) - (\sigma'_{p0} - f'_{py})A'_p (h_0 - a'_p) \quad (6.2.18-3)$$

$$e' = y' - a' - (e_0 - e_a) \quad (6.2.18-4)$$

Where:

y' —the distance from centroid of section to compression side relatively nearer to the axial compressive force; when the section is symmetrical, taking $h/2$.

Note: For the T-shaped section with flange at the side relatively nearer to the axial compressive force only, may be taken as b ; for the inverted T-shaped section with flange at the side relatively farther to the axial compressive force, may be taken as b .

6.2.19 As for the eccentric compression member (Figure 6.2.19) of reinforced concrete with rectangular, T-shaped or I-shaped section and longitudinal steel bars allocated uniformly along the web of section, its normal section compression load-bearing capacity should meet the following requirements:

$$N \leq a_1 f_c [\xi b h_0 + (b'_f - b)h'_f] + f'_y A'_s - \sigma_s A_s + N_{sw} \quad (6.2.19-1)$$

$$Ne \leq a_1 f_c \left[\xi (1 - 0.5\xi) b h_0^2 + (b'_f - b)h'_f \left(h_0 - \frac{h'_f}{2} \right) \right] + f'_y A'_s (h_0 - a'_s) + M_{sw} \quad (6.2.19-2)$$

$$N_{sw} = \left(1 + \frac{\xi - \beta_1}{0.5\beta_1 w} \right) f_{yw} A_{sw} \quad (6.2.19-3)$$

$$M_{sw} = \left[0.5 - \left(\frac{\xi - \beta_1}{\beta_1 w} \right)^2 \right] f_{yw} A_{sw} h_{sw} \quad (6.2.19-4)$$

Where:

A_{sw} ——the sectional area of all longitudinal steel bars allocated uniformly along the web of section;

f_{yw} ——the design value of strength of longitudinal steel bar allocated uniformly along the web of section, adopted according to Table 4.2.3-1 of this code;

N_{sw} ——the axial compressive force undertaken by longitudinal steel bar allocated uniformly along the web of section, if is taken in the calculation.

M_{sw} ——the force moment of the internal force of longitudinal steel bar uniformly allocated along the section web to the centroid, if is taken in the calculation.

w ——the depth of longitudinal steel bar segment allocated uniformly, the ratio between h_{sw} and the efficient depth of section h_0 is (h_{sw}/h_0) , $h_{sw}=(h_0-a'_s)$ should be adopted.

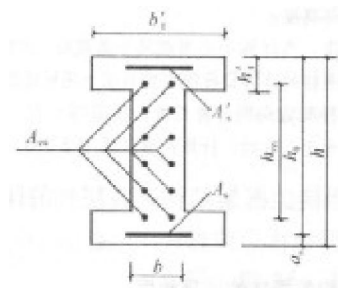


Figure 6.2.19 I-shaped Section with Steel Reinforcement Uniformly Allocated along Web of Section

Note: This article is applicable to the condition that the quantity of longitudinal steel bars allocated uniformly along the web of section at each side is not less than 4.

The stress of steel bar at the tension side or smaller compression side, and whether taking the action of compression steel bar and flange compression parts of the smaller compression side, shall be determined according to the relevant requirements in Article 6.2.17 and Article 6.2.18 of this code.

6.2.20 The calculated length of axial compression and eccentric compression column may be determined according to the following requirements:

1 The calculated length of the bent columns, open-air crane columns and trestle columns of single-storey building with rigid roof may be taken according to Table 6.2.20-1.

Table 6.2.20-1 Calculated Length of Bent Column of Single-storey Building with Rigid Roof, Open-air Crane Column and Trestle Column

Type of column		l_0		
		Bent direction	Vertical bent direction	
			With column bracing	Without column bracing
Housing column with crane	Single-span	$1.5H$	$1.0H$	$1.2H$
	Two-span and multi-span	$1.25H$	$1.0H$	$1.2H$
Housing column without crane	Upper column	$2.0H_u$	$1.25H_u$	$1.5H_u$
	Lower column	$1.0 H_l$	$0.8 H_l$	$1.0H_l$
Open-air crane column and trestle column		$2.0 H_l$	$1.0 H_l$	—

Notes:

- 1 in this Table denotes the overall height calculated from the foundation top; denotes the lower height of the column calculated from foundation top to either the bottom of fabricated crane beam, or the top of cast-in-situ crane beam; denotes the upper height of column calculated from the bottom of fabricated crane beam or the top of cast-in-situ crane beam;

- 2 The calculated length of housing bent column with crane in this Table, when the crane load is neglected in calculation, may be adopted according to the calculated length of housing column without crane, whereas the calculated length of upper column may still adopted according to the buildings with crane;
- 3 The calculated length of upper column on bent direction of housing bent column with crane in this Table, is only applicable to condition that ; if the calculated strength of should be adopted.

2 In general multistoried buildings whose frame structure is made-up with rigid-connected beams and columns, the calculated length of column in each storey may be adopted according to provisions of Table 6.2.20-2.

Table 6.2.20-2 Calculated Length of Column in Each Storey of Frame Structure

Type of floor	Category of column	l_0
Cast-in-situ floor	Ground floor column	$1.0H$
	Columns in rest stories	$1.25H$
Fabricated floor	Ground floor column	$1.25H$
	Columns in rest stories	$1.5H$

Note: in this Table denotes the height of ground floor column from foundation top to the first floor top;

6.2.21 As for reinforced concrete two-way eccentric compression member which has the section with two orthogonal symmetry axes (Figure 6.2.21), either of the following two methods may be adopted for the calculation of the normal section compression load-bearing capacity:

1 The calculation is carried out according to the methods specified in Appendix E of this code; here, in Formula (E.0.1-7) and Formula (E.0.1-8) of Appendix E shall respectively be replaced by , hereof the initial eccentric distance shall be calculated according to the following formulae:

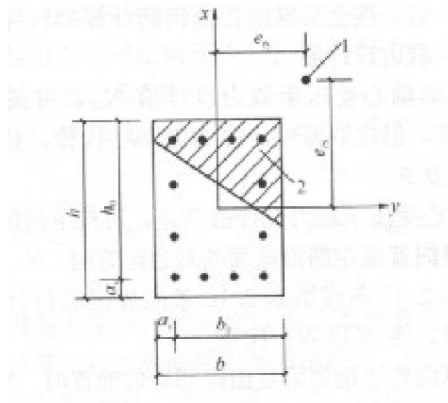


Figure 6.2.21 Section of Two-way Eccentric Compression Member

1—the action point of axial compressive force;

2—the compression zone.

$$e_{ix} = e_{0x} + e_{ax} \quad (6.2.21-1)$$

$$e_{iy} = e_{0y} + e_{ay} \quad (6.2.21-2)$$

Where:

e_{0x} , e_{0y} —the eccentric distance of axial compressive force to the axis, axis crossing the centroid of section respectively, namely M_{0x}/N , M_{0y}/N ; M_{0x} , M_{0y} —the bending moment design value of axial compressive force in axis axis direction respectively denotes the bending moment design value determined according to those specified in Article 5.3.4, Article 6.2.4 of this code;

e_{ax} , e_{ay} —the additional eccentric distance in x axis, y axis direction, determined according to those specified in Article 6.2.5 of this code;

2 The calculation shall be carried out according to the following approximate formula:

$$N \leq \frac{1}{\frac{1}{N_{ux}} + \frac{1}{N_{uy}} - \frac{1}{N_{u0}}} \quad (6.2.21-3)$$

Where:

N_{u0} —the design value of sectional axial compression load-bearing capacity of members;

N_{ux} —the design value of eccentric compression bearing capacity of members calculated according to all longitudinal steel bars after the axial compressive force is acted on x axis and the corresponding calculated eccentric distance e_{ix} is considered;

N_{uy} —the design value of eccentric compression bearing capacity of members calculated according to all longitudinal steel bars, after the axial compressive force is acted on y axis and the corresponding calculated eccentric distance e_{iy} is considered.

The sectional axial compression load-bearing capacity design value of members may be calculated according to the provisions in Formula (6.2.15) of this code; whereas the equal-sign shall be taken, with η replaced by 1, and the stability coefficient η and coefficient 0.9 are disregarded.

The design value of eccentric compression bearing capacity of members may be calculated according to the following conditions:

- 1) When the longitudinal steel bar is allocated along the two opposite side of the section, N_{ux} may be calculated according to those specified in Article 6.2.17 or Article 6.2.18 of this code, whereas the equal-sign shall be taken and with η replaced by 1.
- 2) When the longitudinal steel bar is allocated uniformly along the web of section, N_{ux} may be calculated according to those specified in Article 6.2.19 of this code, whereas the equal-sign shall be taken with η replaced by 1.

The eccentric compression bearing capacity design value of members may be calculated adopting the same method as that of N_{ux} .

(IV) Calculation of Normal Section Tension Load-bearing Capacity

6.2.22 The normal section tension load-bearing capacity of axial tension member shall meet the following requirements:

$$N \leq f_y A_s + f_{py} A_p \quad (6.2.22)$$

Where:

N —the design value of axial tensile force;

A_s , A_p —the gross sectional area of longitudinal steel bars and prestressing tendons and/or bars.

6.2.23 The normal section tension load-bearing capacity of eccentric tension members with rectangular section shall meet the following requirements:

- 1 Small eccentric tension member

When the axial tensile force acts between the points for resultant of forces of and as well as that of and (Figure 6.2.23a):

$$Ne \leq f_y A'_s (h_0 - a'_s) + f_{py} A'_p (h_0 - a'_p) \quad (6.2.23-1)$$

$$Ne' \leq f_y A_s (h_0 - a_s) + f_{py} A_p (h_0 - ap) \quad (6.2.23-2)$$

2 Large eccentric tension member

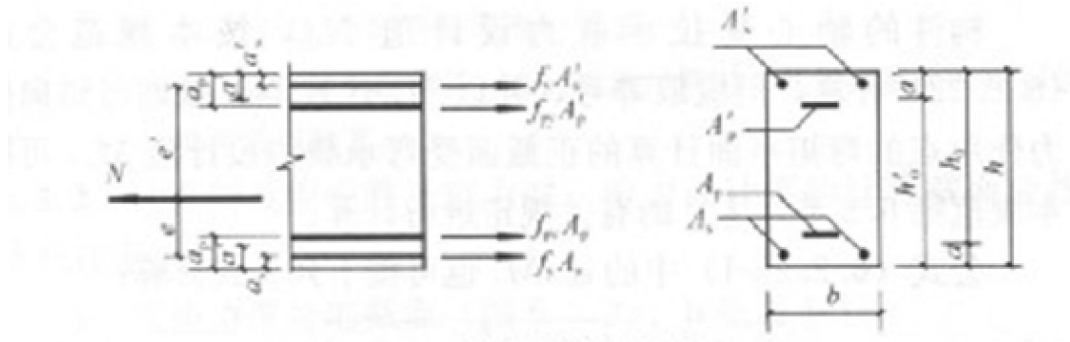
When the axial tensile force does not act between the point for resultant of forces of and , and that of and (Figure 6.2.23b):

$$N \leq f_y A_s + f_{py} A_p - f_y A'_s + (\sigma'_{p0} - f_{py}) A'_p - a_1 f_c b x \quad (6.2.23-3)$$

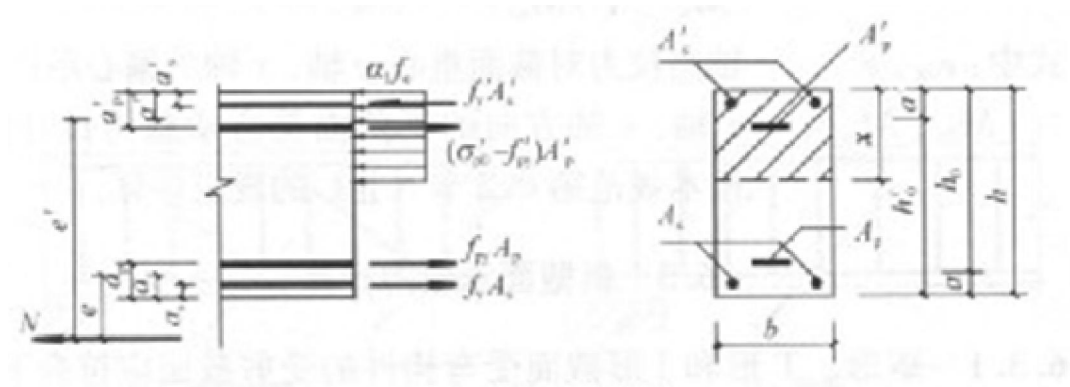
$$Ne \leq a_1 f_c b x \left(h_0 - \frac{x}{2} \right) + f_y A'_s (h_0 - a'_s) - (\sigma'_{p0} - f_{py}) A'_p (h_0 - a'_p) \quad (6.2.23-4)$$

In this case, the depth of concrete compression zone shall meet the requirements in Formula (6.2.10-3) of this code. If the longitudinal compression steel bar is counted in the calculation, the condition in Formula (6.2.10-4) of this code still shall be met; otherwise, may be calculated according to the provisions in Formula (6.2.23-2).

3 As for the rectangular section eccentric tension member of symmetric reinforcement, whether in the condition of large eccentric tension or small eccentric tension, both may be calculated according to the provisions in Formula (6.2.23-2).



(a) The small eccentric tension member



(b) The large eccentric tension member

Figure 6.2.23 Calculation on Normal Section Tension Load-bearing Capacity of Rectangular Section Eccentric Tension Member

6.2.24 As for the rectangular, T-shaped or I-shaped section reinforced concrete eccentric tension member of longitudinal steel bars allocated uniformly along the web of section, the normal section tension load-bearing capacity shall meet those specified in Formula (6.2.25-1) of this code; where the design value of normal section flexural load-bearing capacity may be calculated according to the provisions in Formula (6.2.19-1) and Formula (6.2.19-2) of this code; whereas the equal-sign shall be taken, meanwhile and shall be replaced by respectively.

6.2.25 As for rectangular section reinforced concrete two-way eccentric tension member of symmetric reinforcement, the tension load-bearing capacity of normal section shall meet the following requirements:

$$N \leq \frac{1}{\frac{1}{N_{u0}} + \frac{e_0}{M_u}} \quad (6.2.25-1)$$

Where:

N_{u0} ——the axial tension load-bearing capacity design value of members;

e_0 ——the distance from the action point of axial tensile force and centroid of section;

M_u ——the design value of normal section flexural load-bearing capacity calculated according to the bending moment plane crossing the action point of axial tensile force.

The axial tension load-bearing capacity design value of members, is calculated according to the provisions in Formula (6.2.22) of this code; whereas the equal-sign shall be taken, with replaced by. The design value of normal section flexural load-bearing capacity, calculated according to the bending moment plane crossing the action point of axial tensile force, may be calculated according to the relevant requirements in Section 6.2 of this code.

in Formula (6.2.25-1) may also be calculated according to the following formula:

$$\frac{e_0}{M_u} = \sqrt{\left(\frac{e_{0x}}{M_{ux}}\right)^2 + \left(\frac{e_{0y}}{M_{uy}}\right)^2} \quad (6.2.25-2)$$

Where:

e_{0x}, e_{0y} ——the eccentric distance of axial tensile force to the axis, axis of section centroid

M_{ux}, M_{uy} ——the design value of normal section flexural load-bearing capacity in axis, axis direction, is calculated according to those specified in Section 6.2 of this code.

6.3 Calculation of Inclined Section Load-bearing Capacity

6.3.1 The shear section of flexural members with rectangular, T-shaped and I-shaped section shall meet the following conditions:

When $h_w/b \leq 4$

$$V \leq 0.25\beta_c f_c b h_0 \quad (6.3.1-1)$$

When $h_w/b \geq 6$

$$V \leq 0.2\beta_c f_c b h_0 \quad (6.3.1-2)$$

If $4 < h_w/b < 6$, V is determined according to the linear interpolation method.

Where:

V —the design value of maximum shear force on the inclined section of members;
 β_c —the influence coefficient of concrete strength: if the concrete strength grade does not exceed C50, ; if the concrete strength grade is C80, ; the value between C50 and C80 shall be determined according to the linear interpolation method;

b —the width rectangular section, the web width of T-shaped section or I- shaped section;

h_0 —the effective depth of section;

h_w —the web depth of section: for rectangular section, the effective depth is taken; for T-shaped section, the value of effective depth minus flange depth is taken; for I-shaped section, the net depth of web is taken.

Notes:

- 1 As for simply-supported flexural members with T-shaped or I-shaped section, where practical experience is available, the coefficient in Formula (6.3.1-1) may be changed to 0.3;
- 2 As for members with inclined tension side, where practical experience is available, the controlling requirements for shear section may be appropriately reduced.

6.3.2 In the calculation of shear bearing capacity of inclined section, the calculated section shear force design value shall be adopted according to the following requirements:

- 1 Section at the edge of support (Figure 6.3.2 a, b section 1-1);
- 2 Section at the bent starting point of bent-up steel reinforcement in tension zone(Figure 6.3.2a section 2-2, 3-3);

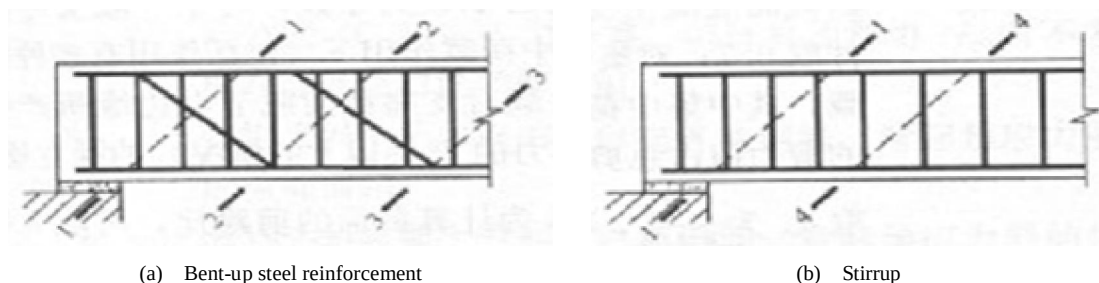


Figure 6.3.2 Calculated Section of Shear Force Design Value of Inclined Section Shear Bearing Capacity

1-1: The inclined section at the edge of support; 2-2, 3-3: The inclined section at the bent starting point of bent-up steel reinforcement in the tension zone; 4-4: The inclined section at the point where the sectional area or spacing of stirrup is changed;

3 The section at the point where the sectional area or spacing of stirrup is changed (Figure 6.3.2b section 4-4);

4 The section at the point where the section dimension is changed.

Notes:

- 1 The section at the point where the depth of beam begins to change, concentrated load acts and other unfavorable section still shall be included for the flexural members with inclined tension side.
- 2 The spacing of stirrups and distance from the bent starting point of front row (with respect to support) to the bent end point of next row, shall meet the structural requirements in Article 9.2.8 and Article 9.2.9 of this code.

6.3.3 As for the common slab flexural members without allocating stirrups and bent-up steel reinforcements, the shear bearing capacity of inclined section shall meet the following requirements:

$$V \leq 0.7\beta_h f_t b h_0 \quad (6.3.3-1)$$

$$\beta_h = \left(\frac{800}{h_0} \right)^{1/4} \quad (6.3.3-2)$$

Where:

β_h ——the influence coefficient of sectional depth: if 800mm, taking 800mm; if 2000mm, taking 2000mm.

6.3.4 If only the stirrup is allocated, the inclined section shear bearing capacity of rectangular T-shaped and I-shaped section flexural members shall meet the following requirements:

$$V \leq V_{cs} + V_p \quad (6.3.4-1)$$

$$V_{cs} = a_{cv} f_t b h_0 + f_{yv} \frac{A_{sv}}{s} h_0 \quad (6.3.4-2)$$

$$V_p = 0.05 N_{p0} \quad (6.3.4-3)$$

Where:

V_{cs} ——the shear bearing capacity design value of concrete and stirrup on the inclined section of members;

V_p ——the design value of shear bearing capacity raised by pre-applied force;

a_{cv} ——the shear bearing capacity coefficient of inclined section concrete; for common flexural members, taking $a_{cv}=0.7$; for isolated beam under concentrated load action (including action with various loads, among which the shear force value produced by concentrated load to the support section or node edge occupies over 75% of the total shear force), taking $a_{cv} = \frac{1.75}{\lambda + 1}$; λ denotes the ratio of shear span to effective depth, $\lambda = a/h_0$ may be taken; if $\lambda < 1.5$, taking $\lambda = 1.5$; if $\lambda > 3$, taking $\lambda = 3$; the distance from the action point of concentrated load to the support section or node edge shall be taken for a ;

A_{sv} ——the gross sectional area of each stirrup limb allocated in the same section, namely ; here denotes the number of stirrup limbs in the same section, denotes the sectional area of single-limb stirrup;

s ——the stirrup spacing along the length direction of member;

f_{yv} ——the design value of tensile strength of stirrup, is adopted according to those specified in Article 4.2.3 of this code;

N_{p0} ——the pre-applied force where the concrete normal prestress on calculated section equals to zero, calculated according to Article 10.1.13 of this code; if $N_{p0} > 0.3 f_c A_0$, taking $N_{p0} = 0.3 f_c A_0$; here A_0 denotes the transformed sectional area of members.

Notes:

- 1 For the condition that the direction of sectional bending moment induced by pre-applied force is the same with that of the outer bending moment, as well as the continuous beam of prestressed concrete and simple-supported beam of prestressed concrete the crack is allowable, shall be taken;
- 2 For pre-tensioned prestressed concrete members, in the calculation of pre-applied force , the influence of transmission length of prestressing tendon and/or bar shall be taken into consideration in accordance with those specified in Article 7.1.9 of this code.

6.3.5 When allocating the stirrup and bent-up steel reinforcement, the inclined section shear bearing capacity of rectangular, T-shaped and I-shaped section flexural members shall meet the following requirements:

$$V \leq V_{cs} + V_p + 0.8f_{yv}A_{sb}\sin\alpha_s + 0.8f_{py}A_{pb}\sin\alpha_p \quad (6.3.5)$$

Where:

V ——the shear force design value at the bent-up point of the allocated steel reinforcement, is adopted according to those specified in Article 6.3.6 of this code;

V_p ——the shear bearing capacity design value of member raised by pre-applied force, is calculated according to the provisions in Formula (6.3.4-3) of this code; whereas the action of prestressing tendon and/or bar is disregarded in the calculation of pre-applied force ;

A_{sb} , A_{pb} ——respectively denotes the sectional area of bent-up steel bar, bent-up prestressing tendon and/or bar in the same plane;

α_s , α_p ——respectively denotes the included angle between bent-up steel bar, the tangent to of bent-up prestressing tendon and/or bar and longitudinal axis of member in the same inclined section.

6.3.6 In the calculation of bent-up steel reinforcement, the design value of section shear force may be adopted according to the following requirements according to the following requirements (Figure 6.3.2a):
1 In the calculation of the bent-up steel reinforcement in the first row (with respect to support), the shear force value at the edge of support is taken;

2 In the calculation of bent-up steel reinforcement in each subsequent row, the shear force value at the bent starting point of bent-up steel reinforcement in one row ahead is taken.

6.3.7 As for rectangular, T-shaped and I-shaped section of ordinary flexural members, if meeting the requirements of the following formula, the shear bearing capacity of inclined section may not be calculated, and the structural requirements of stirrup shall meet the relevant requirements of Article 9.2.9 of this code.

$$V \leq a_{cv}f_t b h_0 + 0.05N_{p0} \quad (6.3.7)$$

Where:

a_{cv} ——shear bearing capacity coefficient of sectional concrete, is adopted according to those specified in Article 6.3.4 of this code.

6.3.8 The flexural members of rectangular, T-shaped and I-shaped section with inclined tension side, its inclined section shear bearing capacity shall meet the following requirements (Figure 6.3.8):

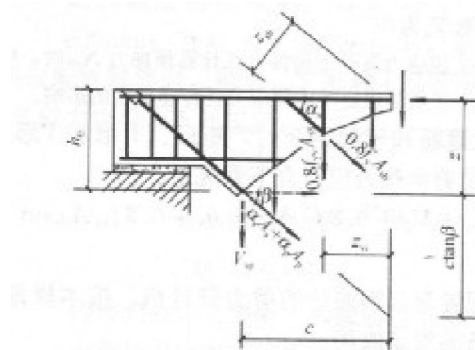


Figure 6.3.8 Calculation on the Inclined Section Shear Bearing Capacity of Flexural Members with Inclined Tension Side

$$V \leq V_{cs} + V_{sp} + 0.8f_s A_{sb} \sin \alpha_s \quad (6.3.8-1)$$

$$V_{sp} = \frac{M - 0.8(\sum f_{yv} A_{sv} z_{sv} + \sum f_y A_{sb} z_{sb})}{z + c \tan \beta} \tan \beta \quad (6.3.8-2)$$

Where:

M ——the bending moment design value at the ending point of inclined section compression zone of members;

V_{cs} ——the shear bearing capacity design value of concrete and stirrup on inclined sections of members, calculated according to the provisions in Formula (6.3.4-2) of this code, where the effective depth of vertical section at the starting end of inclined section tension zone shall be adopted for ;

V_{sp} ——the projection in vertical direction of the resultant force design value of longitudinal non-prestress and prestress tension steel reinforcements with inclined tension side in member section: as for reinforced concrete flexural member, its value shall not be greater than ; as for prestressed concrete flexural member, its value shall not be greater than , and shall not be less than ;

z_{sv} ——the distance from the stirrup resultant of forces in one section to the point of resultant force in inclined section compression zone;

z_{sb} ——the distance from the resultant of forces of bent-up steel bar in one bent-up plane to the point of resultant force in inclined section compression zone;

z ——the distance from the horizontal component of longitudinal tension steel reinforcement resultant of forces at the starting point of inclined section tension zone to the point of resultant force in inclined section compression zone, and may be approximately taken;

β ——the dip angle of longitudinal tension steel reinforcement, inclined at the starting point of inclined section tension zone;

c ——the horizontal projection length of inclined section, may be approximately taken.

Note: At the point where the sectional depth of beam begins to change, the shear bearing capacity of inclined section shall be calculated respectively according to the relevant formulae on beam of uniform section depth and beam of variable section depth, and the stirrup and bent-up steel reinforcement shall be allocated according to the unfavorable one.

6.3.9 The flexural load-bearing capacity of flexural member's inclined section shall meet the following requirements (Figure 6.3.9):

$$M \leq (f_y A_s + f_{py} A_p) z + \sum f_y A_{sb} z_{sb} + \sum f_{py} A_{pb} z_{pb} + \sum f_{yv} A_{sv} z_{sv} \quad (6.3.9-1)$$

Where, the horizontal projection length of inclined section shall be determined according to the following condition:

$$V = \sum f_y A_{sb} \sin \alpha_s + \sum f_{py} A_{pb} \sin \alpha_p + \sum f_{yv} A_{sv} \quad (6.3.9-2)$$

Where:

V ——the shear force design value at the ending point of inclined section compression zone;

z ——the distance from the resultant of forces point of longitudinal tension steel bars and prestressing tendon and/or bar to resultant of forces point in compression zone, and may be approximately taken;

z_{sb}, z_{pb} ——the distance from the resultant of forces point of bent-up steel bar, bent-up prestressing tendon and/or bar in one bent-up plane to the resultant of forces point in inclined section compression zone;

z_{sv} ——the distance from the stirrup resultant of forces point in one inclined section to that of the inclined section compression zone.

When calculating the inclined section flexural load-bearing capacity of anchorage zone

at the end of pre-tensioned prestressed concrete member, in the formulae shall be determined according to the following requirements: the longitudinal prestressed reinforcement design value of tensile strength in the anchorage zone at the anchorage starting point shall take its value as zero, shall be taken when at the anchorage endpoint, and the value between the starting point and the endpoint shall be determined according to linear interpolation. Here, the anchorage length of longitudinal prestressed reinforcement shall be determined according to Article 8.3.1 of this code.

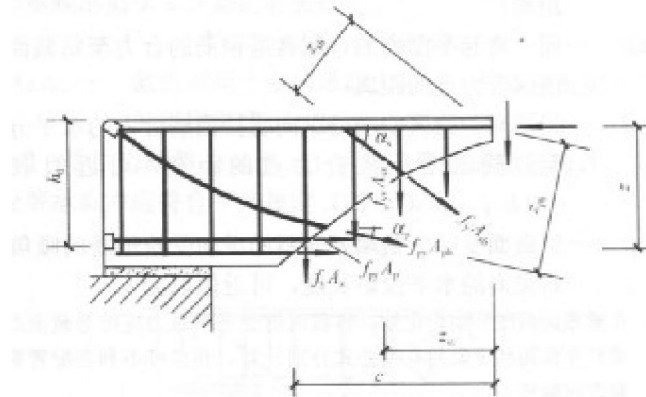


Figure 6.3.9 Calculation on Inclined Section Flexural Load-bearing Capacity of Flexural Members

6.3.10 As for the longitudinal steel reinforcement and stirrup allocated for flexural members, when meeting the structural requirements of Article 8.3.1~Article 8.3.5, Article 9.2.2~Article 9.2.4, Article 9.2.7~Article 9.2.9 of this code, the inclined section flexural load-bearing capacity of member may not be calculated.

6.3.11 As for reinforced concrete eccentric compression members and eccentric tension members with rectangular, T-shaped and I-shaped sections, the shear section shall meet the requirements of Article 6.3.1 of this code.

6.3.12 As for the reinforced concrete eccentric compression members with rectangular, T-shaped and I-shaped sections, the inclined section shear bearing capacity shall meet the following requirements:

$$V \leq \frac{1.75}{\lambda + 1} f_t b h_0 + f_{yv} \frac{A_{sv}}{s} h_0 + 0.07 N \quad (6.3.12)$$

Where:

λ —the ratio of shear span to effective depth of the calculated eccentric compression member section, is taken;

N —the design value of axial compressive force corresponding to the shear force design value; if $N > N_c$ here, N_c denotes the sectional area of member.

The ratio of shear span to effective depth of calculated section shall be adopted according to the following requirements:

1 As for the frame column in the frame structure, where its point of contraflexure is within the range of storey height, λ may be taken. If $\lambda < 1$, taking $\lambda = 1$; if $\lambda > 3$, taking $\lambda = 3$. Here, M denotes the bending moment design value in calculated section, corresponding to shear force design value, h_0 is the net height of column.

2 As for other eccentric compression members, when bearing uniformly distributed load, $\lambda = 1.5$ is taken; when bearing the concentrated load meeting what is stated in Article 6.3.4 of this code, λ is taken, and if $\lambda < 1.5$, $\lambda = 1.5$ is taken; if $\lambda > 3$, $\lambda = 3$ is taken.

6.3.13 As for reinforced concrete eccentric compression member of rectangular T-shaped and I-shaped section, when meeting the following requirements, the inclined section shear bearing capacity may not be calculated, and its stirrup structural requirements shall meet those specified in Article 9.3.2 of this code.

$$V \leq \frac{1.75}{\lambda + 1} f_t b h_0 + 0.07 N \quad (6.3.13)$$

Where:

The ratio of shear span to effective depth and axial compressive force design value shall be determined according to Article 6.3.12 of this code.

6.3.14 As for the reinforced concrete eccentric tension member of rectangular, T-shaped and I-shaped section, its inclined section shear bearing capacity shall meet the following requirements:

$$V \leq \frac{1.75}{\lambda + 1} f_t b h_0 + f_{yv} \frac{A_{sv}}{S} h_0 - 0.2 N \quad (6.3.14)$$

Where:

N ——the axial tensile force design value corresponding to the shear force design value V ;

λ ——the ratio of shear span to effective depth of calculated section, is determined according to Article 6.3.12 of this code.

Where the calculated value at the right side of Formula (6.3.14) is less than, $V=$ and the value of shall not be less than

6.3.15 For the flexural members, eccentric compression and tension members of reinforced concrete with circular sections, the sectional limit condition and inclined section shear bearing capacity may be calculated according to Article 6.3.1~Article 6.3.14 of this code; whereas the sectional width and effective depth of section stated in the above-mentioned provisions and \ formulae shall respectively be substituted by 1.76 and 1.6, here, denotes the radius of circular section. The calculated sectional area of stirrups shall be adopted as that of the circular stirrup.

6.3.16 As for the two-way shear reinforced concrete frame column of rectangular section, its shear section shall be in accordance with the following requirements:

$$V_x \leq 0.25 \beta_c f_c b h_0 \cos \theta \quad (6.3.16-1)$$

$$V_y \leq 0.25 \beta_c f_c h b_0 \sin \theta \quad (6.3.16-2)$$

Where:

V_x ——the shear force design value in direction, of which the corresponding effective depth of section is , and sectional width is ;

V_y ——the shear force design value in y direction, of which the corresponding effective depth of section is , and the sectional width is ;

θ ——the included angle between the action direction of inclined shear force design value and axis, .

6.3.17 As for the two-way shear reinforced concrete frame column of rectangular section, its inclined section shear bearing capacity shall meet the following requirements:

$$V_x \leq \frac{V_{ux}}{\sqrt{1 + \left(\frac{V_{ux} \tan \theta}{V_{uy}} \right)^2}} \quad (6.3.17-1)$$

$$V_y \leq \frac{V_{uy}}{\sqrt{1 + \left(\frac{V_{uy}}{V_{ux} \tan \theta} \right)^2}} \quad (6.3.17-2)$$

The inclined section shear bearing capacity design value in axis, axis direction shall be calculated according to the following formulae:

$$V_{ux} = \frac{1.75}{\lambda_x + 1} f_t b h_0 + f_{yv} \frac{A_{svx}}{s} h_0 + 0.07 N \quad (6.3.17-3)$$

$$V_{uy} = \frac{1.75}{\lambda_y + 1} f_t h b_0 + f_{yv} \frac{A_{svy}}{s} b_0 + 0.07 N \quad (6.3.17-4)$$

Where:

λ_x, λ_y —respectively denotes the calculated ratio of shear span to effective depth in the axis, axis direction of frame column, is determined according to those specified in Article 6.3.12 of this code;

A_{svx}, A_{svy} —respectively denotes the sectional area sum of stirrup's each limb that is parallel to axis, axis and allocated in one section;

N —the axial compressive force corresponding to the inclined direction shear force design value ; if >0.3 =0.3 is taken; here denotes the sectional area of members.

In the calculation of sectional stirrup, may be approximately adopted in Formula and Formula .

6.3.18 For the two-way shear reinforced concrete frame column with rectangular sections, when the following requirements are met, the inclined section shear bearing capacity may not be calculated, and its structural stirrup requirements shall meet those specified in Article 9.3.2 of this code.

$$V_x \leq \left(\frac{1.75}{\lambda_x + 1} f_t b h_0 + 0.07 N \right) \cos \theta \quad (6.3.18-1)$$

$$V_y \leq \left(\frac{1.75}{\lambda_y + 1} f_t h b_0 + 0.07 N \right) \sin \theta \quad (6.3.18-2)$$

6.3.19 As for the two-way shear reinforced concrete frame column of rectangular section, where the included angle between the action direction of inclined shear force design value and the axis is in the range of , the sectional load-bearing capacity may be calculated merely according to the single-direction shear member.

6.3.20 The shear section of reinforced concrete shear wall shall meet the provisions the following condition:

$$V \leq 0.25 \beta_c f_c b h_0 \quad (6.3.20)$$

6.3.21 The inclined section shear bearing capacity of reinforced concrete shear wall in

eccentric compression shall meet the following requirements:

$$V \leq \frac{1}{\lambda - 0.5} (0.5 f_t b h_0 + 0.13 N \frac{A_w}{A}) + f_{yv} \frac{A_{sh}}{s_v} h_0 \quad (6.3.21)$$

Where:

N ——the axial compressive force design value corresponding to the shear force design value N ; $0.2 f_c b h$

A ——the sectional area of shear wall;

A_w ——the sectional area of T-shaped, I-shaped section shear wall web; as for shear wall of rectangular section, is taken;

A_{sh} ——the gross sectional area of horizontally distributed steel reinforcement allocated in one section;

s_v ——the vertical spacing of horizontally distributed steel reinforcement;

λ ——the ratio of shear span to effective depth of calculated section, $\lambda = M/(Vh_0)$ is taken; if $\lambda < 1.5$, $\lambda = 1.5$ is taken; if $\lambda > 2.2$, $\lambda = 2.2$ is taken; here, denotes the bending moment design value corresponding to the shear force design value; when the distance from the calculated section and the wall bottom is less than , λ may be calculated according to bending moment value and shear force value which are at the distance of to the wall bottom.

When the shear force design value is not greater than the first item at the right side of Formula (6.3.21), the horizontally distributed steel reinforcement may be allocated according to the structural requirements of Article 9.4.2, Article 9.4.4 and Article 9.4.6 of this code.

6.3.22 The inclined section shear bearing capacity of reinforced concrete shear wall in eccentric tension shall meet the following requirements:

$$V \leq \frac{1}{\lambda - 0.5} (0.5 f_t b h_0 - 0.13 N \frac{A_w}{A}) + f_{yv} \frac{A_{sh}}{s_v} h_0 \quad (6.3.22)$$

If the calculated value of the right side in the above Formula is less than $f_{yv} \frac{A_{sh}}{s_v} h_0$, $V =$

$f_{yv} \frac{A_{sh}}{s_v} h_0$ is taken.

Where:

N ——the axial tensile force design value corresponding to the shear force design value ;

λ ——the ratio of shear span to effective depth of calculated section, is adopted according to Article 6.3.21 of this code.

6.3.23 The shear section of the connecting beam at the shear wall opening shall meet in the requirements of Article 6.3.1 of this code, its inclined section shear bearing capacity shall meet the following requirements:

$$V \leq 0.7 f_t b h_0 + f_{yv} \frac{A_{sv}}{s} h_0 \quad (6.3.23)$$

6.4 Calculation of Load-bearing Capacity of Distortion Section

6.4.1 Under the co-action of bending moment, shear force and torsional moment, for the rectangular, T-shaped and I-shaped sections with λ no larger than 6 as well as the box-section member with λ no larger than 6 (Figure 6.4.1), the section shall meet the following conditions:

When h_w/b (or h_w/t_w) is not larger than 4

$$\frac{V}{bh_0} + \frac{T}{0.8W_t} \leq 0.25\beta_c f_c \quad (6.4.1-1)$$

When h_w/b (or h_w/t_w) is equal to 6

$$\frac{V}{bh_0} + \frac{T}{0.8W_t} \leq 0.2\beta_c f_c \quad (6.4.1-2)$$

When h_w/b (or h_w/t_w) is larger than 4 but not larger than 6, it shall be determined according to linear interpolation method.

Where:

T ——the design value of torsional moment;

b ——the width of rectangular section, which shall be taken as the web width for T-shaped or I-shaped section and the total thickness of walls on both sides for box section;

W_t ——the moment of plastic resistance of the section of torsion member, which shall be calculated according to those specified in Article 6.4.3 of this code;

h_w ——the web depth of section: for the rectangular section, h_w takes the effective depth; for the T-shaped section, h_w takes the effective depth minus the height of flange; for the I-shaped and box section, h_w takes the net height of web;

t_w ——the wall thickness of the box section, its value shall not be less than $1.5\sqrt{f_c}$, here, b represents the width of the box section.

Note: When λ is larger than 6 or λ is larger than 6, the requirements on the sectional dimension of torsion members and the calculation of the load-bearing capacity of distortion section shall meet the specific provisions.

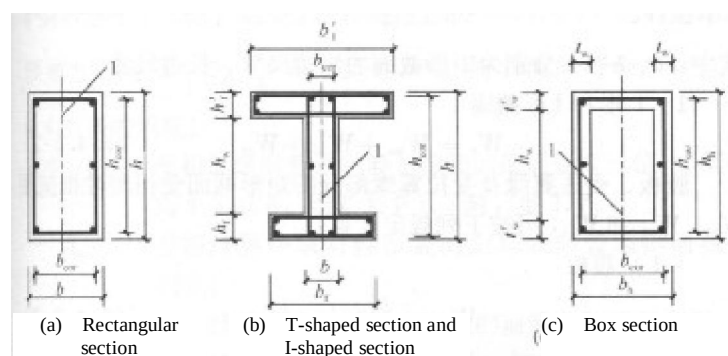


Figure 6.4.1 Section of the Torsion Members

1——the acting plane of the bending moment and shear force

6.4.2 For the members under the co-action of bending moment, shear force and torsional moment, when it meets the following requirements, it may not be carried out with the calculation of shear and torsion load-bearing capacity, however, it shall be configured with constructional longitudinal steel reinforcement and stirrup in accordance with those specified in Article 9.2.5, 9.2.9 and 9.2.10 of this code.

$$\frac{V}{bh_0} + \frac{T}{W_t} \leq 0.7 f_t + 0.05 \frac{N_{p0}}{bh_0} \quad (6.4.2-1)$$

or

$$\frac{V}{bh_0} + \frac{T}{W_t} \leq 0.7 f_t + 0.07 \frac{N}{bh_0} \quad (6.4.2-2)$$

Where:

N_{p0} ——the pre-applied force when the normal prestress of concrete on the calculated section is equal to zero, which shall be calculated according to those specified in Article 10.1.13 of this code, when N is larger than N_{p0} , N takes N_{p0} , hereof, A represents the area of the transformed section of member;

N ——the design value of axial compressive force corresponding to the design values V and T of shear force and torsional moment, when N is larger than N_{p0} , N takes N_{p0} , here, A represents the sectional area of the member.

6.4.3 For the torsion members, the moment of plastic resistance of its section under torsion may be calculated according to the following requirements:

1 Rectangular section

$$W_t = \frac{b^2}{6}(3h - b) \quad (6.4.3-1)$$

Where:

b, h ——respectively the sizes of the short side and long side of the rectangular section.

2 T-shaped and I-shaped sections

$$W_t = W_{tw} + W_{tf}' + W_{tf} \quad (6.4.3-2)$$

The moment of plastic resistance W_{tw}, W_{tf}', W_{tf} of the rectangular section under torsion in the web, compression flange and tension flange may be calculated according to the following requirements:

1) Web

$$W_{tw} = \frac{b^2}{6}(3h - b) \quad (6.4.3-3)$$

2) Compression flange

$$W_{tf}' = \frac{h_f'^2}{2}(b_f' - b) \quad (6.4.3-4)$$

3) Tension flange

$$W_{tf} = \frac{h_f^2}{2}(b_f - b) \quad (6.4.3-5)$$

Where:

b, h ——respectively the web width of the section and the depth of section;

b_f', b_f ——respectively the flange width of the compression zone and the tension zone of

the section;

h'_f, h_f —respectively the flange height of the compression zone and the tension zone of the section.

The adopted flange width adopted in calculation still shall meet the requirements that b'_f is not larger than $b + 6h'_f$ and b_f is not larger than $b + 6h_f$ 3 Box section

$$W_t = \frac{b_h^2}{6}(3h_h - b_h) - \frac{(b_h - 2t_w)^2}{6}[3h_w - (b_h - 2t_w)] \quad (6.4.3-6)$$

Where:

b_h, h_b —Respectively the dimensions of the short side and the long side of the box section.

6.4.4 The torsion load-bearing capacity of the pure torsion members of rectangular section shall meet the following requirements:

$$T \leq 0.35 f_t W_t + 1.2 \sqrt{\xi} f_{yv} \frac{A_{stl} A_{cor}}{s} \quad (6.4.4-1)$$

$$\xi = \frac{f_y A_{stl} s}{f_{yv} A_{stl} u_{cor}} \quad (6.4.4-2)$$

For the prestressed concrete pure torsion members with eccentric distance e_{p0} no larger than $h/6$, when the calculated ξ value is not less than 1.7, ξ shall take 1.7, and the influence item of pre-applied force may be added on the right side of Formula (6.4.4-1), here, the value of N_{p0} shall meet those specified in Article 6.4.2 of this code.

Where:

ξ —the ratio of the strength between the torsion longitudinal steel bars and the reinforcement of stirrups, the ξ value shall not be less than 0.6; when ξ is larger than 1.7, ξ shall take 1.7;

A_{stl} —the sectional areas of all longitudinal steel bars that are regarded as symmetrical arrangement in torsion calculation;

A_{stl} —the single-limb sectional area of the stirrups configured along the perimeter of section in torsion calculation;

f_{yv} —the design value of tensile strength of torsion stirrups, which shall be adopted according to Article 4.2.3 of this code;

A_{cor} —the area of the core part of section, which is taken as $b_{cor}h_{cor}$, here b_{cor}, h_{cor} respectively represent the sizes of the short side and the long side of the section core part within scope of the internal surface of stirrups;

u_{cor} —the perimeter of the core part of section, which shall be taken as $2(b_{cor} + h_{cor})$

Note: When ξ is less than 1.7 or is larger than , the influence item of the pre-applied force shall not be considered, and ξ shall be calculated according to the reinforced concrete pure torsion members.

6.4.5 For the pure torsion members of T-shaped and I-shaped sections, the section of which may be divided into several rectangular sections, and the calculation of torsion load-bearing capacity shall be carried out respectively according to Article 6.4.4 of this code. The design value of torsional moment of each rectangular section may be calculated according to the following requirements:

1 Web

$$T_w = \frac{W_{tw}}{W_t} T \quad (6.4.5-1)$$

2 Compression flange

$$T'_f = \frac{W'_{tf}}{W} T \quad (6.4.5-2)$$

3 Tension flange

$$T_f = \frac{W_{tf}}{W_t} T \quad (6.4.5-3)$$

Where:

T_w ——the design value of torsional moment borne by web;

T'_f, T_t ——respectively the design values of torsional moment borne by the compression

flange and tension flange.

6.4.6 The torsion load-bearing capacity of the reinforced concrete pure torsion members of box section shall meet the following requirements:

$$T \leq 0.35a_h f_t W_t + 1.2\sqrt{\xi} f_{yv} \frac{A_{stl} A_{cor}}{s} \quad (6.4.6-1)$$

$$a_h = 2.5t_w/b_h \quad (6.4.6-2)$$

Where:

a_h ——the influence coefficient of the wall thickness of box section, when a_h is larger than 1.0, a_h takes 1.0.

ξ ——the same as Article 6.4.4 of this code.

6.4.7 For the reinforced concrete member of rectangular section under the co-action of axial compressive force and torsional moment, its torsion load-bearing capacity shall meet the following requirements:

$$T \leq \left(0.35 f_t + 0.07 \frac{N}{A} \right) W_t + 1.2\sqrt{\xi} f_{yv} \frac{A_{stl} A_{cor}}{s} \quad (6.4.7)$$

Where:

N ——the design value of the axial compressive force corresponding to the design value of the torsional moment, when N is larger than , N takes ;

ξ ——the same as Article 6.4.4 of this code.

6.4.8 For the shear and torsion members of rectangular section under the co-action of shear force and torsional moment, their shear and torsion load-bearing capacity shall meet the following requirements:

1 Common shear and torsion members

1) Shear bearing capacity

$$V \leq (1.5 - \beta_t)(0.7 f_t b h_0 + 0.05 N_{p0}) + f_{yv} \frac{A_{sv}}{s} h_0 \quad (6.4.8-1)$$

$$\beta_t + \frac{1.5}{1 + 0.5 \frac{VW_t}{Tbh_0}} \quad (6.4.8-2)$$

Where:

A_{sv} ——the sectional area of stirrups required by the shear bearing capacity;

β_t —— the reduction coefficient of the torsion load-bearing capacity of concrete for common shear and torsion members: when β_t is less than 0.5, β_t takes 0.5; when β_t is larger than 1.0, β_t takes 1.0.

2) Torsion load-bearing capacity

$$T \leq \beta_t (0.35 f_t + 0.05 \frac{N_{p0}}{A_0}) W_t + 1.2 \xi f_{yv} \frac{A_{stl} A_{cor}}{s} \quad (6.4.8-3)$$

Where:

ξ ——the same as Article 6.4.4.

2 Independent shear and torsion members under the action of concentrated load

1) Shear bearing capacity

$$V \leq (1.5 - \beta_t) \left(\frac{1.75}{\lambda + 1} f_t b h_0 + 0.05 N_{p0} \right) + f_{yv} \frac{A_{sv}}{s} h_0 \quad (6.4.8-4)$$

$$\beta_t = \frac{1.5}{1 + 0.2(\lambda + 1) \frac{VW_t}{Tbh_0}} \quad (6.4.8-5)$$

Where:

λ ——the ratio of shear span to effective depth of the calculated section, which shall be adopted according to those specified in Article 6.3.4 of this code;

β_t ——the reduction coefficient of torsion load-bearing capacity of concrete under the action of concentrated load: when λ is less than 0.5, β_t takes 0.5; when λ is larger than 1.0; β_t takes 1.0.

2) Torsion load-bearing capacity

The torsion load-bearing capacity still shall be calculated in accordance with Formula (6.4.8-3), however β_t in the formula shall be calculated in accordance with Formula (6.4.8-5).

6.4.9 The shear and torsion load-bearing capacity of the shear and torsion members of T-shaped or I-shaped section shall meet the following requirements:

1 The shear bearing capacity may be calculated according to Formula (6.4.8-1) and Formula (6.4.8-2) or Formula (6.4.8-4) and Formula (6.4.8-5) of this code, but T and β_t in these formulae shall be respectively replaced by T_f and β_{tf} ;

2 The torsion load-bearing capacity may be calculated by respectively dividing the section into several rectangular sections according to those specified in Article 6.4.5 of this code. In which, the web may be calculated according to Formula (6.4.8-3) and Formula (6.4.8-2) or Formula (6.4.8-3) and Formula (6.4.8-5) of this code, but T and β_t in these formulae shall be replaced by T_f and β_{tf} ; the compression flange and tension flange may be calculated according to those specified in Article 6.4.4 of this code, but T and β_t shall be replaced by T_f and β_{tf} .

and or and .

6.4.10 The shear and torsion load-bearing capacity of the reinforced concrete shear and torsion members of box section shall be calculated according to requirements:

1 Common shear and torsion members

1) Shear bearing capacity

$$V \leq 0.7(1.5 - \beta_t) f_t b h_0 + f_{yv} \frac{A_{sv}}{s} h_0 \quad (6.4.10-1)$$

2) Torsion load-bearing capacity

$$T \leq 0.35 a_h \beta_t f_t W_t + 1.2 \sqrt{\xi} f_{yv} \frac{A_{stl} A_{cor}}{s} \quad (6.4.10-2)$$

Where:

β_t ——calculated according to Formula (6.4.8-2) of this code. But W_t in the formula shall be replaced by ;

a_h ——determined according to those specified in Article 6.4.6 of this code;

ξ ——determined according to those specified in Article 6.4.4 of this code.

2 Independent shear and torsion members under the action of concentrated load

1) Shear bearing capacity

$$V \leq (1.5 - \beta_t) \frac{1.75}{\lambda + 1} f_t b h_0 + f_{yv} \frac{A_{sv}}{s} h_0 \quad (6.4.10-3)$$

Where:

β_t ——calculated according to Formula (6.4.8-5) of this code, but W_t in the formula shall be replaced by $a_h W_t$.

2) Torsion load-bearing capacity

The torsion load-bearing capacity still shall be calculated according to Formula (6.4.10-2) of this code, but in which, β_t shall be calculated according to Formula (6.4.8-5) of this code.

6.4.11 For the reinforced concrete members of rectangular section under the co-action of axial tensile force and torsional moment, its torsion load-bearing capacity shall be calculated according to requirements:

$$T \leq \left(0.35 f_t - 0.2 \frac{N}{A} \right) W_t + 1.2 \sqrt{\xi} f_{yv} \frac{A_{stl} A_{cor}}{s} \quad (6.4.11)$$

Where:

ξ ——determined according to those specified in Article 6.4.4 of this code;

A_{stl} ——the single-limb sectional area of stirrups that are configured along the perimeter of section in the torsion calculation;

A_{stl} ——the sectional areas of all longitudinal steel bars for the torsion of symmetrical layout;

N ——the design value of axial tensile force corresponding to the design value of torsional moment, when N is larger than 1.75 , N takes ;

A_{cor} ——the area of the core part of section, which shall be taken as , here, represent the sizes of short side and long side of the core part of section that are within the scope of internal surface of stirrups;

u_{cor} ——the perimeter of the core part of section, takes

6.4.12 For the shear and torsion flexural members of rectangular, T-shaped, I-shaped and box section under the co-action of bending moment, shear force and torsion moment, the calculation of load-bearing capacity may be carried out according to the following requirements:

1 When V is not larger than V_c or V is not larger than V_{cw} , only the normal section flexural load-bearing capacity of flexural member and the torsion load-bearing capacity of the pure torsion members may be calculated;

2 When T is not larger than T_c or T is not larger than T_{cw} , only the normal section flexural load-bearing capacity of the flexural members and the shear bearing capacity of inclined section may be checked and calculated.

6.4.13 For shear and torsion flexural members of rectangular, T-shaped, I-shaped and box sections, the sectional area of their longitudinal steel reinforcement shall be determined through calculation respectively according to the normal section flexural load-bearing capacity of flexural member and torsion load-bearing capacity of shear and torsion member, and the longitudinal steel reinforcements shall be configured at the corresponding positions; the sectional area of stirrups shall be determined through calculation respectively according to the shear bearing capacity and torsion load-bearing capacity of the shear and torsion member, and the stirrups shall be configured at the corresponding position.

6.4.14 For the reinforced concrete rectangular-section frame column under the co-action of axial compressive force, bending moment, shear force and torsional moment, the shear and torsion load-bearing capacity of which may be calculated according to the following requirements:

1 Shear bearing capacity

$$V \leq (1.5 - \beta_t) \left(\frac{1.75}{\lambda + 1} f_t b h_0 + 0.07 N \right) + f_{yv} \frac{A_{sv}}{s} h_0 \quad (6.4.14-1)$$

2 Torsion load-bearing capacity

$$T \leq \beta_t \left(0.35 f_t + 0.07 \frac{N}{A} \right) W_t + 1.2 \sqrt{\zeta} f_{yv} \frac{A_{stl} A_{cor}}{s} \quad (6.4.14-2)$$

Where:

λ ——the ratio of shear span to effective depth of the calculated section, determined according to Article 6.3.12 of this code;

β_t ——calculated according to Article 6.4.8 of this code and meeting the relevant requirements;

ζ ——adopted according to those specified in Article 6.4.4 of this code.

6.4.15 For the reinforced concrete rectangular-section frame column under the co-action of axial compressive force, bending moment, shear force and torsional moment, when T is not larger than T_c , only the normal section load-bearing capacity and the inclined section shear bearing capacity of the eccentric compression members may be calculated.

6.4.16 For the reinforced concrete rectangular-section frame column under the co-action of axial compressive force, bending moment, shear force and torsional moment, the sectional

area of the longitudinal steel bars of which shall be calculated respectively according to the normal section load-bearing capacity of eccentric compression members and the torsion load-bearing capacity of shear and torsion members, and the longitudinal steel bars shall be configured at the corresponding positions; the sectional area of stirrups shall be calculated respectively according to the shear bearing capacity and the torsion load-bearing capacity of the shear and torsion members, and the stirrups shall be configured at the corresponding position.

6.4.17 For the reinforced concrete rectangular-section frame column under the co-action of axial tensile force, bending moment, shear force and torsional moment, the shear and torsion load-bearing capacity of which shall meet the following requirements:

1 Shear bearing capacity

$$V \leq (1.5 - \beta_t) \left(\frac{1.75}{\lambda + 1} f_t b h_0 - 0.2N \right) + f_{yv} \frac{A_{sv}}{s} h_0 \quad (6.4.17-1)$$

2 Torsion load-bearing capacity

$$T \leq \beta_t \left(0.35 f_t - 0.2 \frac{N}{A} \right) W_t + 1.2 \sqrt{\zeta} f_{yv} \frac{A_{stl} A_{cor}}{s} \quad (6.4.17-2)$$

When the calculated value of the right side of Formula (6.4.17-1) is less than $f_{yv} \frac{A_{sv}}{s} h_0$, it takes $f_{yv} \frac{A_{sv}}{s} h_0$; when calculated value of the right side of Formula (6.4.17-2) is less than $1.2 \sqrt{\zeta} f_{yv} \frac{A_{stl} A_{cor}}{s}$, it takes $1.2 \sqrt{\zeta} f_{yv} \frac{A_{stl} A_{cor}}{s}$.

Where:

λ ——the ratio of shear span to effective depth of the calculated section, which shall be determined according to Article 6.3.12 of this code;

A_{sv} ——the sectional area of stirrups required by the shear bearing capacity;

N ——the design values of the axial tensile force corresponding to the design values V and T of the shear force and torsional moment;

β_t ——calculated according to Article 6.4.8 of this code and meeting the relevant requirements;

ζ ——adopted according to those specified in Article 6.4.4 of this code.

6.4.18 For the reinforced concrete rectangular-section frame column under the co-action of axial tensile force, bending moment, shear force and torsional moment, when $T \leq (0.175 f_t - 0.1 N/A) W_t$, only the normal section load-bearing capacity and the inclined section shear bearing capacity of eccentric tension members may be calculated.

6.4.19 For the reinforced concrete rectangular-section frame column under the co-action of axial compressive force, bending moment, shear force and torsional moment, the sectional area of the longitudinal steel bars of which shall be calculated respectively according to the normal section load-bearing capacity of eccentric tension members and the torsion load-bearing capacity of shear and torsion members, and the longitudinal steel bars shall be configured at the corresponding position; the sectional area of stirrups shall be calculated

respectively according to the shear bearing capacity and the torsion load-bearing capacity of the shear and torsion members, and the stirrups shall be configured at the corresponding position.

6.5 Calculation of Punching Shear Bearing Capacity

6.5.1 Under the action of local load or concentrated reaction, the punching shear bearing capacity of the slab without stirrup or bent-up steel reinforcement shall meet the following requirements (Figure 6.5.1):

$$F_l \leq (0.7\beta_h f_t + 0.25\sigma_{pc,m})\eta\mu_m h_0 \quad (6.5.1-1)$$

The coefficient η in Formula (6.5.1-1) shall be calculated in accordance with the following two formulae and the smaller value shall be taken:

$$\eta_1 = 0.4 + \frac{1.2}{\beta_s} \quad (6.5.1-2)$$

$$\eta_2 = 0.5 + \frac{a_s h_0}{4u_m} \quad (6.5.1-3)$$

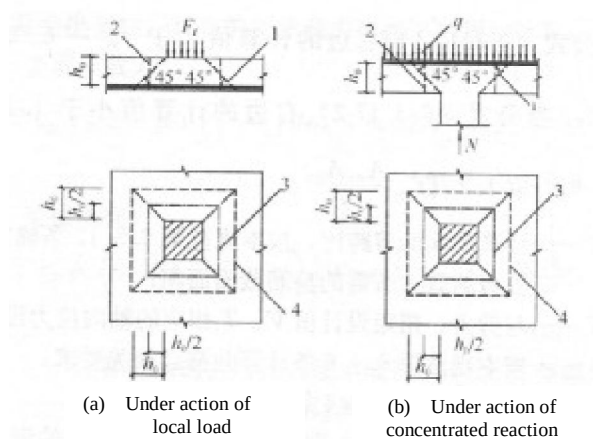


Figure 6.5.1 Calculation of the Punching Shear Bearing Capacity of Slab

1—the inclined section of punching failure cone; 2—the calculated section;

3—the perimeter of calculated section; 4—the bottom surface line of punching failure cone

Where:

F_l —the design value of local load or concentrated reaction; for the slab-column joint, F_l takes the storey difference in the design values of axial compressive force borne by the column minus the design values of loads borne by the slab within the range of punching failure cone at column top; when there are unbalanced moments, F_l shall be determined in accordance with those specified in Article 6.5.6 of this code;

β_h —the influence coefficient of the depth of section: when h is not larger than 800mm, β_h takes 1.0; when h is not less than 2000mm, β_h takes 0.9, when h is between 800mm and 2000mm, β_h is taken according to linear interpolation method;

$\sigma_{pc,m}$ —the weighted mean value of valid compression prestress of concrete according to length in two directions within the calculated sectional perimeter, its value should be controlled between

u_m ——the perimeter of the calculated section, taking the most unfavorable perimeter of the slab's vertical section located at the position of $h_0/2$ from the local load or the concentrated reaction area's perimeter;

h_0 ——the effective depth of section, taking the average value of the effective sectional depth of reinforcements in two directions;

η_1 ——the influence coefficient of the shape of action area of local load or concentrated reaction;

η_2 ——the influence coefficient of the ratio between the calculated sectional perimeter and the effective depth of slab section;

β_s ——the size ratio of the long side and the short side when the action area of local load or concentrated reaction is in shape of rectangle, and should not be larger than 4; when is less than 2, takes 2; for the round punching surface, takes 2;

a_s ——the influence coefficient of column position: for the central column, takes 40; for the border column, takes 30; for the corner column, takes 20.

6.5.2 Where the slab has pore and the distance from the pore to the edge of the action area of local load or concentrated reaction is not larger than , the perimeter of calculated section taken in calculation of punching shear load-bearing capacity shall be deducted by the length between the two tangent lines drawn from the center of the action area of local load or concentrated reaction to the outside edge of pore (Figure 6.5.2).

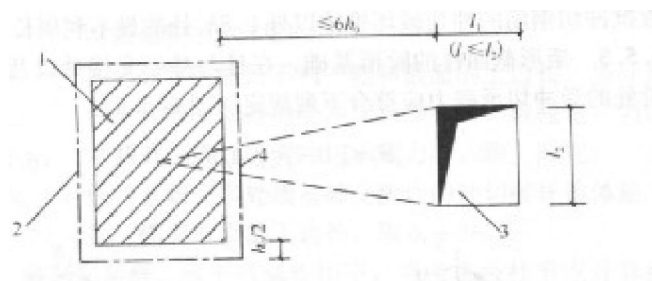


Figure 6.5.2 Perimeter of the Calculated Section Adjacent to Pore

1—the action surface of local load or concentrated reaction; 2—the perimeter of calculated section;

3—the pore; 4—the length shall be subtracted

Note: When in this figure is larger than , the pore side length shall be replaced by $\sqrt{l_1 l_2}$.

6.5.3 Under the action of local load or concentrated reaction, when the punching shear bearing capacity doesn't meet the requirements of Article 6.5.1 of this code and the slab thickness is restricted, stirrup or bent-up steel reinforcement may be configured and shall be in accordance with the detailing requirement stated in Article 9.1.11 of this code. On this occasion, the punching shear section and the punching shear bearing capacity shall meet the following requirements:

1 Punching shear section

$$F_l \leq 1.2 f_t \eta u_m h_0 \quad (6.5.3-1)$$

2 Punching shear bearing capacity with stirrup and bent-up steel reinforcement

$$F_l \leq (0.5 f_t + 0.25 \sigma_{pc,m}) \eta u_m h_0 + 0.8 f_{yv} A_{svu} + 0.8 f_y A_{sbu} \sin \alpha \quad (6.5.3-2)$$

Where:

f_{yv} ——the design value of tensile strength of stirrup, adopted according to the

requirements of Article 4.2.3 of this code;

A_{svu} ——the sectional area of all stirrups which intersect the 45° inclined section of punching failure cone;

A_{sbu} ——the sectional area of all bent-up steel reinforcements which intersect the 45° inclined section of punching failure cone;

α ——the included angle between the bent-up steel reinforcement and the slab bottom.

Note: If the condition permitting, the punching resistance measures, such as configuring stud, section steel cross bridging, may be taken.

6.5.4 For sections, which are beyond the punching failure cone configured with punching resistance steel reinforcement, still shall be carried out with the calculation of punching shear bearing capacity according to those specified in Article 6.5.1 of this code, in that case, u_m shall take the most unfavorable perimeter which is located at the position of $0.5h_0$ beyond the punching failure cone configured with punching resistance steel reinforcement.

6.5.5 As for the stepped foundation of rectangular sectional column, the punching shear bearing capacity at the junction of column and foundation as well as the step changing part of foundation shall meet the following requirements (Figure 6.5.5):

$$F_l \leq 0.7\beta_h f_t b_m h_0 \quad (6.5.5-1)$$

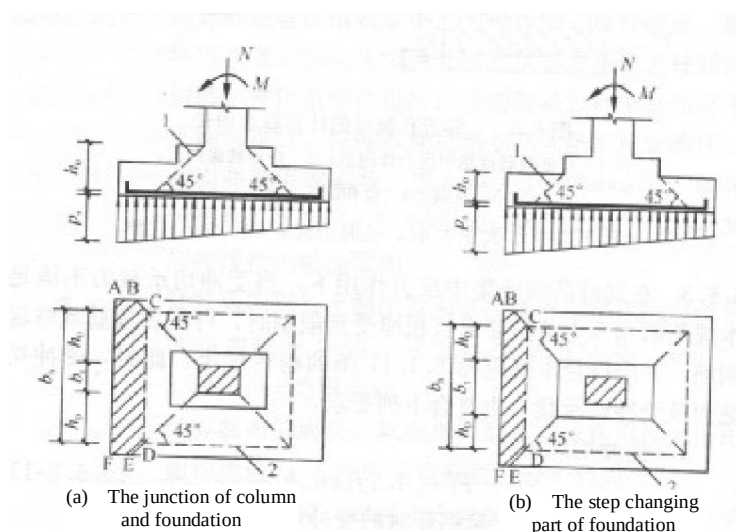


Figure 6.5.5 Section Position for the Calculation of Punching Shear Bearing Capacity of Stepped Foundation

1—the inclined section on the most unfavorable side of the punching failure cone;

2—the bottom surface line of the punching failure cone

$$F_l = p_s A \quad (6.5.5-2)$$

$$b_m = \frac{b_t + b_b}{2} \quad (6.5.5-3)$$

Where:

h_0 ——the effective depth of section at the junction of the column and foundation as well as the step changing part of foundation, taking the average value of the effective section depths of s reinforcements in two directions;

p_s ——the design value of base reaction on the bottom foundation (foundation deadweight and its upper soil weight may be excluded) calculated according to the fundamental combination of load effects and considering the significance coefficient of

structure, when foundation is eccentrically pressed, p_s may take the design value of the maximal base reaction;

A ——the area of polygon taken in consideration of punching load (the shaded area ABCDEF in Figure 6.5.5)

b_t ——the upper side length of inclined section on the most unfavorable side of the punching failure cone: when calculating the punching shear bearing capacity at the junction of column and foundation, b_t takes the column width; when calculating the punching shear bearing capacity at the step changing part of foundation, b_t takes the upper step width;

b_b ——the lower side length of inclined section on the most unfavorable side of the punching failure cone at the junction of column and foundation or the step changing part of foundation, taking

6.5.6 Under the action of vertical load and horizontal load, in consideration that the shear stress on the calculation section of the slab-column joint transfers unbalanced bending moment, the design value of the concentrated reaction shall be replaced by the design value of equivalent concentrated reaction, and may be calculated according to the requirements of Appendix F in this code.

6.6 Calculation of Partial Compression Load-bearing Capacity

6.6.1 For the concrete structural members distributed with indirect steel reinforcement, the sectional dimension of the local compression zone shall meet the following requirements:

$$F_l \leq 1.35\beta_c\beta_l f_c A_{ln} \quad (6.6.1-1)$$

$$\beta_l = \sqrt{\frac{A_b}{A_l}} \quad (6.6.1-2)$$

Where:

——the design value of the local load or local pressure acting on the local compression face;

——the design value of axial compressive strength of concrete; for the checking in the tension stage of post-tensioned prestressed concrete members, the design value hereof may be determined with linear interpolation method according to those specified in Table 4.1.4-1 of this code based on the concrete cube compressive strength value corresponding stage.

——the influence coefficient of concrete strength, which is adopted according to those specified in Article 6.3.1 of this code;

—— the strength improvement coefficient of concrete under local pressure;
the local compression area of concrete;

A_{ln} the net local compression area of concrete; for post-tensioned members, the local compression area of concrete shall be deducted by the area of pore path and groove;

calculated floor area under local pressure, which is determined according to Article 6.6.2 of this code.

6.6.2 The calculated floor area under local pressure may be determined by local compression area and calculated floor area according to the principle of concentric and symmetrical; In common condition, may be adopted according to Figure 6.6.2.

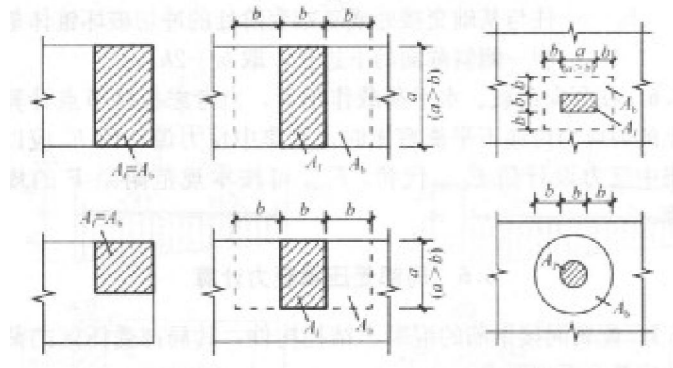


Figure 6.6.2 Calculated Bottom Area under Local Pressure

A_1 local compression area of concrete; A_2 calculated bottom area under local pressure

6.6.3 The partial compression load-bearing capacity, when the partial load-carrying capacity of square grid pattern or spiral pattern indirect steel reinforcements is configured (Figure 6.6.3), shall meet the following requirements:

$$F_1 \leq 0.9(\beta_c \beta_{fc} + 2\alpha_p \beta_{cor} f_{yv}) A_{ln} \quad (6.6.3-1)$$

For reinforcement of square grid pattern (Figure 6.6.3a), the ratio of the sectional area of steel reinforcements in unit length in two directions should not be larger than 1.5; thereof ratio of volumetric reinforcement shall be calculated according to the following formula:

$$\rho_v = \frac{n_1 A_{s1} l_1 + n_2 A_{s2} l_2}{A_{cor} s} \quad (6.6.3-2)$$

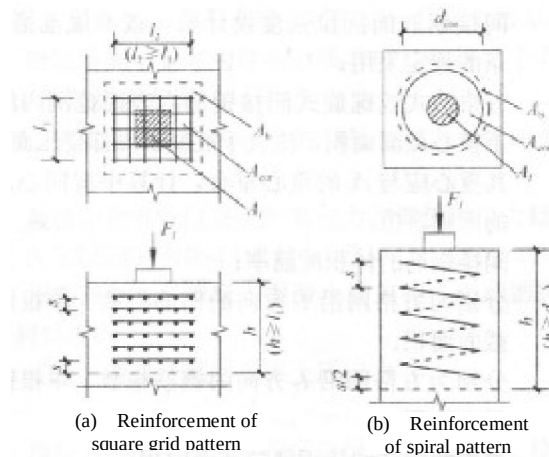


Figure 6.6.3 Indirect Steel Reinforcement in Local Compression Zone

local compression area of concrete; calculated floor area under local pressure;

concrete core area within the internal surface range of square grid pattern or spiral pattern indirect steel reinforcement

For reinforcement of spiral pattern (Figure 6.6.3b), thereof ratio of volumetric reinforcement shall be calculated according to the following formula:

$$\rho_v = \frac{4A_{ss} l}{d_{cor} s} \quad (6.6.3-3)$$

Where:

β_{cor} —the improvement coefficient of partial compression load-bearing capacity when indirect steel reinforcements are configured, which may calculated according to Formula

(6.6.1-2) of this code, but α in this formula shall be replaced by α_{sv} ; when α is larger than 1, takes 1; when α is not larger than 1.25 times of local compression area of concrete, takes 1.0;

α ——the reduction coefficient of constraint of indirect steel reinforcement to concrete, which is adopted according to those specified in Article 6.2.16 of this code;

f_{yv} ——the design value of tensile strength of indirect steel reinforcement, which is adopted according to those specified Article 4.2.3 of this code;

A_{cor} ——the concrete core section area of internal surface range of square grid pattern or spiral pattern indirect reinforcement shall be greater than local compression area of concrete; its gravity center shall be superposed with gravity center of , the value calculated according to the principle of concentric and symmetrical;

ρ_v ——the volumetric reinforcement ratio of indirect reinforcements

n_1, A_{s1} ——respectively the quantity of steel reinforcements in square grid along l_1 direction and the sectional area of single steel reinforcement;

n_2, A_{s2} ——respectively the quantity of steel reinforcements in square grid along l_2 direction and the sectional area of single steel reinforcement;

A_{ss1} ——the sectional area of single spiral indirect steel reinforcements

d_{cor} ——the diameter of concrete section within the internal surface range of spiral pattern indirect steel reinforcements;

s ——the spacing of square grid pattern or spiral pattern indirect steel reinforcements, which should be taken from 30mm to 80mm.

The indirect steel reinforcements shall be configured within the range of height according to those stated in Figure 6.6.3, the steel reinforcements of square grid pattern shall not be less than 4 sheets; the steel reinforcements of spiral pattern shall not be less than 4 circles. For column joints, h shall not be less than $15d$ (d is diameter of column's longitudinal steel reinforcement).

6.7 Checking of Fatigue

6.7.1 In checking of the normal section fatigue stress of flexural members, the following fundamental assumptions may be adopted:

1 Section strain maintains plane;

2 Normal stress pattern of concrete in compression zone is triangle;

3 For reinforced concrete members, without considering tensile strength of concrete in tension zone, the tension is all borne by longitudinal steel reinforcements; for prestressed concrete members with requirements of no appearance of cracks, the normal stress pattern of concrete in compression zone is triangle;

4 Calculation by adopting with transformed section.

6.7.2 In checking of fatigue, loads shall take the standard value; crane load shall be multiplied by the dynamic coefficient and shall meet current national standard "Load Code for the Design of Building Structures" (GB 50009). The crane beam with span no greater than 12m may be taken with the load of the maximum crane.

6.7.3 In checking of fatigue of reinforced concrete flexural members, the concrete stress and the stress amplitude of steel reinforcements at the following positions shall be calculated:

1 The concrete stress of the fiber at edge of compression zone of normal section and

the stress amplitude of steel reinforcements;

2 The shear stress of concrete and the stress amplitude of stirrup at the neutral axis of section.

Note: checking of fatigue may not be carried out for longitudinal compression steel bars.

6.7.4 The normal section fatigue stress of reinforced concrete and prestressed concrete flexural members shall meet the following requirements:

1 The concrete compression stress of fiber at the edge of compressed zone

$$\sigma'_{cc,max} \leq f_c^f \quad (6.7.4-1)$$

2 The concrete tension stress of fiber at the edge of tension zone of prestressed concrete members

$$\sigma'_{ct,max} \leq f_t^f \quad (6.7.4-2)$$

3 The stress amplitude of the longitudinal steel bar in tension zone

$$\Delta\sigma_{si}^f \leq \Delta f_y^f \quad (6.7.4-3)$$

4 The stress amplitude of the longitudinal prestressing tendon and/or bar in tension zone

$$\Delta\sigma_p^f \leq \Delta f_{py}^f \quad (6.7.4-4)$$

Where:

$\sigma'_{cc,max}$ —the concrete compression stress of fiber at the edge of compressed zone in checking of fatigue, which shall be calculated according to Formula (6.7.5-1);

$\sigma'_{ct,max}$ —the concrete tension stress of fiber at the edge of tension zone of prestressed concrete section in checking of fatigue, which shall be calculated according to Article 6.7.11 of this code;

$\Delta\sigma_{si}^f$ —the stress amplitude of i^{th} layer of longitudinal steel reinforcements in the tension zone of section in checking of fatigue, which shall be calculated according to Formula (6.7.5-2);

$\Delta\sigma_p^f$ —the stress amplitude of the outermost layer of longitudinal prestressing tendon and/or bar in the tension zone of section in checking of fatigue, which shall be calculated according to Formula (6.7.11-3) of this code;

f_c^f , f_t^f —respectively the design value of the axial compressive and tensile fatigue strength of concrete, which shall be determined in accordance with Article 4.1.6 of this code;

Δf_y^f —the limit value of fatigue stress amplitude of steel reinforcement, adopted according to Table 4.2.6-1 of this code;

Δf_{py}^f —the limit value of fatigue stress amplitude of prestressed tension tendon and/or

bar, adopted according to Table 4.2.6-2 of this code.

Note: Where the longitudinal tension steel reinforcements are of one same grade, the stress amplitude of the outermost layer of steel reinforcement may be checked.

6.7.5 The concrete compression stress at the normal section of reinforced concrete flexural members as well as the stress amplitude of steel reinforcement shall be calculated according to the following formulae:

1 The concrete compression stress of fiber at edge of compression zone

$$\sigma_{cc,max}^f = \frac{M_{max}^f x_0}{I_0^f} \quad (6.7.5-1)$$

2 The stress amplitude of the longitudinal tension steel reinforcement

$$\Delta\sigma_{si}^f = \sigma_{si,max}^f - \sigma_{si,min}^f \quad (6.7.5-2)$$

$$\sigma_{si,min}^f = a_E^f \frac{M_{min}^f (h_{0i} - x_0)}{I_0^f} \quad (6.7.5-3)$$

$$\sigma_{si,max}^f = a_E^f \frac{M_{max}^f (h_{0i} - x_0)}{I_0^f} \quad (6.7.5-4)$$

Where:

M_{max}^f , M_{min}^f —respectively the maximum and minimum bending moment produced under the combination of corresponding loads in the same section in the checking of fatigue;

$\sigma_{si,min}^f$, $\sigma_{si,max}^f$ —the stress of the i^{th} layer of longitudinal steel reinforcements in the tension zone of corresponding section respectively caused by bending moment M_{max}^f and M_{min}^f ; a_E^f —the ratio value of elastic modulus of steel reinforcement to fatigue deformation modulus of concrete;

I_0^f —the inertia moment of transformed section when the bending moment M_{max}^f and M_{min}^f are in the same direction in checking of fatigue;

x_0 —the depth of the compression zone of transformed section when the bending moment M_{max}^f and M_{min}^f are in the same direction in checking of fatigue;

h_{0i} —the distance from the compression zone edge of the section to the centroid of section of the i^{th} layer longitudinal steel reinforcements in the tension zone when bending moment M_{max}^f and M_{min}^f are in the same direction;

Where the bending moment M_{min}^f and M_{max}^f are opposite in direction, h_{0i} , x_0 and I_0^f

in Formula (6.7.5-3) shall be replaced respectively by h'_{0i} , x'_0 and $I_0^{f'}$ at the opposite position of the section.

6.7.6 In checking of fatigue for reinforced concrete flexural members, the depth of compression zone of transformed section and the inertia moments I_0^f , $I_0^{f'}$ shall be calculated according to the following formulae:

- 1 The rectangular section and the T-shaped section with flange located in tension zone

$$\frac{bx_0^2}{2} + a_E^f A'_s (x_0 - a'_s) - a_E^f A_s (h_0 - x_0) = 0 \quad (6.7.6-1)$$

$$I_0^f = \frac{bx_0^3}{3} + a_E^f A'_s (x_0 - a'_s)^2 + a_E^f A_s (h_0 - x_0)^2 \quad (6.7.6-2)$$

- 2 The I-shaped section and the T-shaped section with flange located in compression zone

- 1) When x_0 is larger than h'_f (Figure 6.7.6)

$$\frac{b'_f x_0^2}{2} - \frac{(b'_f - b)(x_0 - h'_f)^2}{2} + a_E^f A'_s (x_0 - a'_s) - a_E^f A_s (h_0 - x_0) = 0 \quad (6.7.6-3)$$

$$I_0^f = \frac{b'_f x_0^3}{3} - \frac{(b'_f - b)(x_0 - h'_f)^3}{3} + a_E^f A'_s (x_0 - a'_s)^2 + a_E^f A_s (h_0 - x_0)^2 \quad (6.7.6-4)$$

- 2) When x_0 is not larger than h'_f , the calculation shall be carried out according to the rectangular section with width of b'_f .

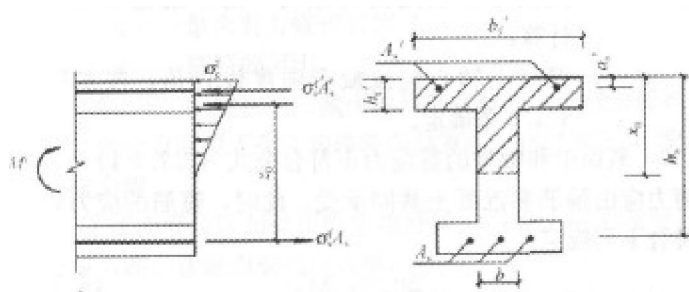


Figure 6.7.6 Calculation of the Normal Section Fatigue Stress of Reinforced Concrete Flexural Member

- 3 The calculation of x'_0 and $I_0^{f'}$, still may adopt the above-mentioned corresponding formulae of x_0 and I_0^f ; hen the bending moment $M_{\min}^f$ and $M_{\max}^f$ are opposite in direction, the compression zones corresponding to are respectively on the downside and upside of this section; when the bending moment $M_{\min}^f$ and $M_{\max}^f$ are in the same

direction, it may adopt $x_0' = x_0$, $I_0^{f'} = I_0^f$.

Notes:

- 1 Where the longitudinal tension steel reinforcements are placed in layers along the depth of their section,

$a_E^f A_s (h_0 - x_0)$ in Formula (6.7.6-1) and (6.7.6-3) may be replaced by

$a_E^f \sum_{i=1}^n A_{si} (h_{0i} - x_0)$; $a_E^f A_s (h_0 - x_0)^2$ in Formula (6.7.6-2) and (6.7.6-4) may be replaced by

$a_E^f \sum_{i=1}^n A_{si} (h_{0i} - x_0)^2$, here, n represents the total layer number of longitudinal tension steel reinforcements,

A_{si} represents the sectional area of all the longitudinal steel reinforcements at the i^{th} layer;

- 2 The stress of the longitudinal compression steel reinforcement shall meet the condition of $a_E^f \sigma_c^f \leq f_y'$; when

$a_E^f \sigma_c^f > f_y'$, $a_E^f A_s'$ in the formulas of this article shall be replaced by $f_y' A_s' / \sigma_c^f$. In which, f_y'

represents the design value of compressive strength of the longitudinal steel reinforcement, σ_c^f represents the concrete stress at the point of resultant force of the longitudinal compression steel reinforcement.

6.7.7 The checking of inclined section fatigue of reinforced concrete flexural members and the distribution of shear force shall meet the following requirements:

1 When the shear stress at the neutral axis of the section meets the following conditions, the shear force in this sector is all borne by concrete, on this occasion, the stirrup may be arranged according to the detailing requirements:

$$\tau^f \leq 0.6 f_t^f \quad (6.7.7-1)$$

Where:

τ^f —the shear stress at the neutral axis of the section, calculated according to Article 6.7.8 of this code:

f_t^f —the design value of concrete axial tensile fatigue strength, determined according to Article 4.1.6 of this code.

2 For the sector in which the shear stress at the neutral axis of the section does not conform to Formula (6.7.7-1), the shear force shall be borne jointly by stirrup and concrete. On this occasion, the stress amplitude $\Delta \sigma_{sv}^f$ of stirrup shall meet the following requirements:

$$\Delta \sigma_{sv}^f \leq \Delta f_{yv}^f \quad (6.7.7-2)$$

Where:

$\Delta \sigma_{sv}^f$ —the stress amplitude of stirrup, calculated according to Formula (6.7.9-1) of this code:

Δf_{yv}^f ——the limit value for fatigue stress amplitude of stirrup, adopted according to

Table 4.2.6-1 of this code.

6.7.8 The shear stress at the neutral axis of reinforced concrete flexural members shall be calculated according to the following formula:

$$\tau^f = \frac{V_{\max}^f}{bz_0} \quad (6.7.8)$$

Where:

$V_{\max}^f$ ——the maximum shear force value of the checked section of such member under the corresponding load combination in the checking of fatigue;

b ——the width of rectangular section, and the web width of T-shaped and I-shaped sections;

z_0 ——the distance from the point of resultant force in compression zone to the point of resultant force of tension steel reinforcements, on this occasion, the depth x_0 of the compression zone calculated according to Formula (6.7.6-1) or Formula (6.7.6-3) of this code.

6.7.9 The stress amplitude of the stirrup on the inclined section of reinforced concrete flexural members shall be calculated according to the following formulae:

$$\Delta \sigma_{sv}^f = \frac{(\Delta V_{\max}^f - 0.1\eta f_t^f b h_0)s}{A_{sv} z_0} \quad (6.7.9-1)$$

$$\Delta V_{\max}^f = V_{\max}^f - V_{\min}^f \quad (6.7.9-2)$$

$$\eta = \Delta V_{\max}^f / V_{\max}^f \quad (6.7.9-3)$$

Where:

$\Delta V_{\max}^f$ ——the maximum shear force amplitude value of the checked section of member in the checking of fatigue;

$V_{\min}^f$ ——the minimum shear force value of the checked section of such member under the corresponding load combination in the checking of fatigue;

η ——the relative value of the maximum shear force amplitude;

s ——the spacing of stirrups;

A_{sv} ——the gross sectional area of all the limbs of stirrups arranged in the same section.

6.7.10 The stress and stress amplitude at the following positions shall be calculated in the checking of fatigue for prestressed concrete flexural members:

1 The concrete stress of fiber at edges of normal section tension zone and compression zone as well as the stress amplitude of the longitudinal prestressing tendon and/or bar and the steel bar in the tension zone;

2 The concrete principal tensile stress at the centroid of section and the position where the width of section changes fiercely.

Notes:

- 1 Checking of fatigue may not be carried out for the longitudinal steel reinforcement in compression zone;
- 2 Checking of fatigue may not be carried out for steel reinforcement for such prestressed concrete members with crack control level 1,

6.7.11 For the prestressed concrete flexural members with requirements no appearance of cracks, the minimum and maximum stress and the stress amplitude of the concrete, longitudinal prestressing tendon and/or bar and steel bar in their normal section shall be calculated according to the following formulae:

- 1 The concrete stress of fibre at edge of tension zone or compression zone

$$\sigma_{c,\min}^f \text{ or } \sigma_{c,\max}^f = \sigma_{pc} + \frac{M_{\min}^f}{I_0} y_0 \quad (6.7.11-1)$$

$$\sigma_{c,\max}^f \text{ or } \sigma_{c,\min}^f = \sigma_{pc} + \frac{M_{\max}^f}{I_0} y_0 \quad (6.7.11-2)$$

- 2 The stress and stress amplitude of the longitudinal prestressing tendon and/or bar in the tension zone

$$\Delta\sigma_p^f = \sigma_{p,\max}^f - \sigma_{p,\min}^f \quad (6.7.11-3)$$

$$\sigma_{p,\min}^f = \sigma_{pe} + a_{pE} \frac{M_{\min}^f}{I_0} y_{0p} \quad (6.7.11-4)$$

$$\sigma_{p,\max}^f = \sigma_{pe} + a_{pE} \frac{M_{\max}^f}{I_0} y_{0p} \quad (6.7.11-5)$$

- 3 The stress and stress amplitude of the longitudinal steel bars in the tension zone

$$\Delta\sigma_s^f = \sigma_{s,\max}^f - \sigma_{s,\min}^f \quad (6.7.11-6)$$

$$\sigma_{s,\min}^f = \sigma_{s0} + a_E \frac{M_{\min}^f}{I_0} y_{0s} \quad (6.7.11-7)$$

$$\sigma_{s,\max}^f = \sigma_{s0} + a_E \frac{M_{\max}^f}{I_0} y_{0s} \quad (6.7.11-8)$$

Where:

$\sigma_{e,\min}^f$, $\sigma_{e,\max}^f$ —respectively the minimum and maximum stress of the fiber concrete

at edge of tension zone or compression zone in checking of fatigue, the minimum and maximum stress shall be distinguished according to their absolute values;

σ_{pc} —the concrete normal stress produced at the edge fiber of tension zone or compression zone due to prestress, after subtracting all prestress loss, calculated according to Formula (10.1.6-1) or Formula (10.1.6-4) of this code;

$M_{\max}^f$, $M_{\min}^f$ —respectively the maximum and minimum bending moment values

generated on the same section under the corresponding load combination in checking of

fatigue;

a_{pE} ——the ratio of the elastic modulus of prestressing tendon and/or bar to the elastic modulus of concrete: $a_{pE}=E_s/E_c$;

I_0 ——the inertia moment of transformed section;

y_0 ——the distance from the edge of tension zone or compression zone to the centroid of transformed section;

$\sigma_{p,min}^f, \sigma_{p,max}^f$ ——respectively the minimum and maximum stress of the outermost layer of prestressing tendon and/or bar in tension zone in checking of fatigue;

$\Delta\sigma_p^f$ ——the stress amplitude of the outermost layer of prestressing tendon and/or bar in tension zone in checking of fatigue;

σ_{pe} ——the effective prestress of the outermost layer of prestressing tendon and/or bar in tension zone after subtracting all the prestress loss, calculated according to Formula (10.1.6-2) or Formula (10.1.6-5) of this code;

y_{0s}, y_{0p} ——respectively the distances from the centroid of the section of the outermost layer steel bars and prestressing tendon and/or bar in tension zone to the centroid of transformed section;

$\sigma_{s,min}^f, \sigma_{s,max}^f$ ——respectively the minimum and maximum stress of the outermost layer of steel bars on in tension zone in checking of fatigue;

$\Delta\sigma_s^f$ ——the stress amplitude of the outermost layer of steel bars in tension zone in checking of fatigue;

σ_{s0} ——the stress generated in the outermost layer of steel bars in tension zone under the action of decompression moment M_{p0} ; in which, M_{p0} is the corresponding bending moment value when the concrete normal prestress at centroid of outermost layer of steel bars in tension zone is equal to zero.

Note: When σ_{s0} in Formula (6.7.11-1) and Formula (6.7.11-2) shall be substituted in positive values when they are tensile stress and substituted in negative values when they compressive stress; σ_{s0} in Formula (6.7.11-7) and Formula (6.7.11-8) shall be substituted in negative value.

6.7.12 The principal tensile stress of concrete in inclined section of prestressed concrete flexural member shall meet the following requirements:

$$\sigma_{tp}^f \leq f_t^f \quad (6.7.12)$$

Where:

σ_{tp}^f ——the concrete principal tensile stress at the fiber for checking of fatigue in the inclined section of prestressed concrete flexural member, which shall be calculated according to Article 7.1.7 of this code; as for crane load, the dynamic coefficient also shall be counted.

7 Checking of Serviceability Limit States

7.1 Checking of Cracks

7.1.1 The checking of the tension edge stress or normal sectional crack width of reinforced concrete and prestressed concrete members shall be carried out according to the following requirements:

1 The tension edge stress of members of Crack Control Level 1 under characteristic combination of loads shall meet the following requirements:

$$\sigma_{ck} - \sigma_{pc} \leq 0 \quad (7.1.1-1)$$

2 The tension edge stress of members of Crack Control Level 2 under characteristic combination of loads shall meet the following requirements:

$$\sigma_{ck} - \sigma_{pc} \leq f_{tk} \quad (7.1.1-2)$$

3 At Crack Control Level 3, the maximum crack width of reinforced concrete members may be calculated according to quasi-permanent combination of loads and by considering the effect of the influence due to long-term action, the maximum crack width of prestressed concrete members may be calculated according to characteristic combination of loads and by considering the effect of the influence due to long-term action. The maximum crack width shall meet the following requirements:

$$w_{max} \leq w_{lim} \quad (7.1.1-3)$$

For prestressed concrete members with Environmental Category II-a under quasi-permanent combination of loads, the tension edge stress shall still meet the following requirements:

$$\sigma_{cq} - \sigma_{pc} \leq f_{tk} \quad (7.1.1-4)$$

Where:

σ_{ck} , σ_{cq} ——the normal stress of concrete at the edge for checking crack resistance under the characteristic combination and the quasi-permanent combination of loads respectively;

σ_{pc} ——the compressive prestress of concrete at the edge for checking crack resistance, after deducting all losses of prestress, which may be calculated by Formula (10.1.6-1) and Formula (10.1.6-4) of this code;

f_{tk} ——the standard value for axial tensile strength of concrete, which may be adopted from Table 4.1.3-2 of this code;

w_{max} ——the maximum crack width calculated according to the characteristic combination or quasi-permanent combination of loads and in consideration of the influence of long-term actions, which shall be calculated according to Article 7.1.2 of this code;

w_{lim} ——the limit value of the maximum crack width, which shall be adopted according to Article 3.4.5 of this code.

7.1.2 In the reinforced concrete tension, flexural and eccentric compression members and the prestressed concrete axial tension and flexural-members with rectangular, T-shaped, inverted T-shaped and I-shaped sections, the maximum crack width, according to the characteristic combination or quasi-permanent combination of loads and in consideration of the influencing of long-term actions, may be calculated according to the following formulae:

$$w_{\max} = a_{cr} \psi \frac{\sigma_s}{E_s} \left(1.9c_s + 0.08 \frac{d_{eq}}{\rho_{te}} \right) \quad (7.1.2-1)$$

$$\psi = 1.1 - 0.65 \frac{f_{tk}}{\rho_{te} \sigma_s} \quad (7.1.2-2)$$

$$d_{eq} = \frac{\sum n_i d_i^2}{\sum n_i v_i d_i} \quad (7.1.2-3)$$

$$\rho_{te} = \frac{A_s + A_p}{A_{te}} \quad (7.1.2-4)$$

Where:

a_{cr} ——the stressed characteristics coefficient of the member, which shall be adopted according to Table 7.1.2-1;

ψ ——the nonuniform coefficient for strain of longitudinal tensile steel reinforcement between cracks; where $\psi < 0.2$, taking $\psi = 0.2$; where $\psi > 1.0$, taking $\psi = 1.0$; for member directly bearing repeated load, taking $\psi = 1.0$;

σ_s ——the stress of longitudinal tension steel bars in reinforced concrete members calculated according to the quasi-permanent combination of loads, or the equivalent stress of longitudinal tension steel reinforcements in prestressed concrete members calculated according to the characteristic combination of loads;

E_s ——the elastic modulus of steel reinforcements, which may be adopted according to Table 4.2.5 of this code;

c_s ——the distance (mm) from the outer edge of longitudinal tension steel reinforcements in the outermost layer to the bottom edge of tension zone; where $c_s < 20$, taking $c_s = 20$; where $c_s > 65$, taking $c_s = 65$;

ρ_{te} ——the ratio of reinforcement for longitudinal tension steel reinforcements calculated according to effective tension concrete section area; for unbounded posttensioning members, the ratio of reinforcement shall be calculated only for longitudinal tension steel bars; in the calculation of maximum crack width, where $\rho_{te} < 0.01$, taking $\rho_{te} = 0.01$;

A_{te} ——the effective tension concrete section area: for axial tension members, taking sectional area of members; for flexural, eccentric compression and eccentric tension members, taking $A_{te} = 0.5bh + (b_f - b)h_f$, here, b_f and h_f refer to the width and depth of tension flanges;

A_s ——the sectional area of longitudinal steel bars in the tension zone;

A_p ——the sectional area of longitudinal prestressing tendon and/or bar in the tension zone;

d_{eq} ——the equivalent diameter (mm) of longitudinal steel reinforcements in tension zone; for unbounded posttensioning members, it is the equivalent diameter (mm) of longitudinal tension steel bars only in tension zone;

d_i ——the nominal diameter of Type i longitudinal steel reinforcements in tension zone; for the diameter of bonded prestressed steel strand bundle, taking $\sqrt{n_1} d_{pl}$; hereinto, d_{pl} refers to the nominal diameter of single steel strand, n_1 refers to the number of single bundle of steel

strand;

n_i —the number of Type i longitudinal steel reinforcements in tension zone; for bonded prestressed steel strand, taking the number of steel strand bundles;

v_i —the relative cohesion characteristic coefficient of Type i longitudinal steel reinforcements in tension zone, which shall be adopted according to Table 7.1.2-2.

Notes:

- 1 For flexural member, which is under crane load but does not need for checking of fatigue, the calculated maximum crack width may be multiplied by coefficient 0.85;
- 2 For beams prepared mesh reinforcement steel fabric in surface layer according to Article 9.2.15 of this code, the maximum crack width calculated according to Formula (7.1.2-1) may be properly reduced and the reduction coefficient may take 0.7;
- 3 For eccentric compression members with $e_0/h_0 \leq 0.55$, the crack width may not be checked and calculated.

Table 7.1.2-1 Stressed Characteristic Coefficient of Members

Type	a_{cr}	
	Reinforced concrete member	Prestressed concrete member
Flexural and eccentric compression	1.9	1.5
Eccentric tension	2.4	—
Axial tension	2.7	2.2

Table 7.1.2-2 Relative Cohesion Characteristic Coefficient of Steel Reinforcements

Classification of steel reinforcements	Steel reinforcements		Pretensioned prestressing steel reinforcements			Post tensioned prestressing steel reinforcements		
	Plain round steel reinforcement	Ribbed steel reinforcement	Ribbed steel reinforcement	Spiral ribbed steel wire	Steel strand	Ribbed steel reinforcement	Steel strand:	Plain steel wire
v_i	0.7	1.0	1.0	0.8	0.6	0.8	0.5	0.4

Note: For ribbed steel reinforcement with epoxy resin coating, the relative cohesion characteristic coefficient shall be adopted by 80% of the coefficient in this table.

7.1.3 Under the quasi-permanent combination or characteristic combination of loads, the compression stress of the concrete at compressive edge of the cracking section of reinforced concrete members and prestressed concrete members, the tension stress of steel reinforcements at different positions and the equivalent stress of the prestressing tendon and/or bar should be calculated according to the following assumptions:

- 1 Keeping section-strain plane;
- 2 Taking triangle for normal stress diagram of concrete in compression zone;
- 3 Without regard to the tensile strength of concrete in tension zone;
- 4 Adopting transformed section.

7.1.4 Under the quasi-permanent combination or characteristic combination of loads, the stress of longitudinal steel bars in tension zone of reinforced concrete members or the equivalent stress of longitudinal steel reinforcements in tension zone of prestressed concrete members also may be calculated according to the following formulae:

- 1 Stress of longitudinal steel bar in tension zone of reinforced concrete member
 - 1) Axial tension member

$$\sigma_{sq} = \frac{N_q}{A_s} \quad (7.1.4-1)$$

2) Eccentric tension member

$$\sigma_{sq} = \frac{N_q e'}{A_s (h_0 - a_s')} \quad (7.1.4-2)$$

3) Flexural member

$$\sigma_{sq} = \frac{M_q}{0.87 h_0 A_s} \quad (7.1.4-3)$$

4) Eccentric compression member

$$\sigma_{sq} = \frac{N_q (e - z)}{A_s z} \quad (7.1.4-4)$$

$$z = \left[0.87 - 0.12(1 - \gamma_f') \left(\frac{h_0}{e} \right)^2 \right] h_0 \quad (7.1.4-5)$$

$$e = \eta_s e_0 + y_s \quad (7.1.4-6)$$

$$\gamma_f' = \frac{(b_f' - b) h_f'}{b h_0} \quad (7.1.4-7)$$

$$\eta_s = 1 + \frac{1}{4000 e_0 / h_0} \left(\frac{l_0}{h} \right)^2 \quad (7.1.4-8)$$

Where:

A_s ——the sectional area of longitudinal steel bars in tension zone: for axial tension members, taking the total sectional area of longitudinal steel bars; for eccentric tension members, taking the sectional area of longitudinal steel bars at the bigger tension side; for flexural and eccentric compression members, taking the sectional area of longitudinal steel bars in tension zone;

N_q, M_q ——the axial force value and the bending moment value calculated according to the quasi-permanent combination of loads;

e' ——the distance from the action point of axial tensile force to the point for resultant forces of the longitudinal steel bars in the compression zone or at the relatively small tension side;

e ——the distance from the action point of axial compressive force to the point for resultant forces of longitudinal tension steel bars;

e_0 ——the original eccentricity under the quasi-permanent combination of loads, taking M_q/N_q ;

z ——the distance from the point for resultant forces of longitudinal tension steel bars to the point for resultant forces of section in compression zone and not larger than $0.87h_0$;

η_s ——the enhancement coefficient of eccentricity for the axial compressive force at

service stage; where l_0/h is not larger than 14, taking 1.0;

y_s ——the distance from the centroid of section to the point for resultant forces of longitudinal tension steel bars;

γ_f' ——the value for ratio of the sectional area of compression flanges to effective sectional area of the web;

b_f', h_f' ——the width and depth of flanges in compression zone respectively; in

Formula (7.1.4-7), where $h_f' > 0.2h_0$, taking $0.2h_0$.

2 Equivalent stress of longitudinal steel reinforcements in tension zone of prestressed concrete member

1) Axial tension member

$$\sigma_{sk} = \frac{N_k - N_{p0}}{A_p + A_s} \quad (7.1.4-9)$$

2) Flexural member

$$\sigma_{sk} = \frac{M_k - N_{p0}(z - e_p)}{(a_1 A_p + A_s)z} \quad (7.1.4-10)$$

$$e = e_p + \frac{M_k}{N_{p0}} \quad (7.1.4-11)$$

$$e_p = y_{ps} - e_{p0} \quad (7.1.4-12)$$

Where:

A_p ——the sectional area of the longitudinal prestressing tendon and/or bar in tension zone: for axial tension members, taking the total sectional area of longitudinal prestressing tendon and/or bar; for flexural members, taking the sectional area of the longitudinal prestressing tendon and/or bar in tension zone;

N_{p0} ——the pre-applied force where the normal direction prestress of concrete on the calculated section is equal to zero, calculated according to the requirements of Article 10.1.13 of this code;

N_k, M_k ——the axial force value and the bending moment value calculated according to the characteristic combination of loads;

z ——the distance from the point for resultant forces of longitudinal steel bars and prestressing tendon and/or bar in tension zone to the point for resultant forces of section compression zone, which shall be calculated according to Formula (7.1.4-5), hereinto, e shall be calculated according to Formula (7.1.4-11);

a_1 ——the equivalent reduction coefficient of the unbonded prestressed tendon, taking a_1 as 0.3; for post-tensioning prestressed tendon of grouting, taking a_1 as 1.0;

e_p ——the distance from the action point of pre-applied force N_{p0} , where the calculated concrete normal direction prestress of the section is zero, to the point for resultant forces of the longitudinal prestressing tendon and/or bar and steel bar in tension zone;

y_{ps} ——the eccentricity of the point for resultant forces of the longitudinal prestressing tendon and/or bar and steel bar in tension zone;

e_{p0} ——the eccentricity of the action point of pre-applied force N_{p0} where the calculated normal direction prestress on the section of concrete is zero, which shall be calculated according to the requirements of Article 10.1.13 of this code;

7.1.5 Under the characteristic combination and quasi-permanent combination of loads, the normal stress of concrete at the section edge in checking of crack resistance shall be calculated according to the following formulae:

1 Axial tension member

$$\sigma_{ck} = \frac{N_k}{A_0} \quad (7.1.5-1)$$

$$\sigma_{cq} = \frac{N_q}{A_0} \quad (7.1.5-2)$$

2 Flexural member

$$\sigma_{ck} = \frac{M_k}{W_0} \quad (7.1.5-3)$$

$$\sigma_{cq} = \frac{M_q}{W_0} \quad (7.1.5-4)$$

3 Eccentric tension and eccentric compression member

$$\sigma_{ck} = \frac{M_k}{W_0} + \frac{N_k}{A_0} \quad (7.1.5-5)$$

$$\sigma_{cq} = \frac{M_q}{W_0} + \frac{N_q}{A_0} \quad (7.1.5-6)$$

Where:

A_0 ——the transformed sectional area of the member;

W_0 ——the moment of elastic resistance for the tension edge of the transformed section of the member.

7.1.6 The principal tensile stress of concrete and the principal compressive stress of concrete in the section of prestressed concrete flexural members shall be checked and calculated respectively:

1 Principal tensile stress of concrete

1) The members of crack control Level 1 shall meet the following requirements:

$$\sigma_{tp} \leq 0.85f_{tk} \quad (7.1.6-1)$$

2) The members of crack control Level 2 shall meet the following requirements:

$$\sigma_{tp} \leq 0.95f_{tk} \quad (7.1.6-2)$$

2 Principal compressive stress of concrete

All the members of crack control Level 1 and 2 shall meet the following requirements:

$$\sigma_{cp} \leq 0.60f_{ck} \quad (7.1.6-3)$$

Where:

σ_{tp} , σ_{cp} —respectively the principal tensile stress and principal compressive stress of concrete, which shall be determined according to Article 7.1.7 of this code.

Here, the section at the unfavorable position in the span shall be selected, and checking shall be carried out at the centroid of the transformed section of the selected section and at the position where the sectional width suddenly changes.

Note: For crane beam in which crack is allowed, the requirements of Formula (7.1.6-2) and Formula (7.1.6-3) shall be conformed in static calculation.

7.1.7 The principal tensile stress and principal compressive stress of concrete shall be calculated according to the following formulae:

$$\left. \begin{array}{l} \sigma_{tp} \\ \sigma_{cp} \end{array} \right\} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau^2} \quad (7.1.7-1)$$

$$\sigma_x = \sigma_{pc} + \frac{M_k y_0}{I_0} \quad (7.1.7-2)$$

$$\tau = \frac{(V_k - \sum \sigma_{pe} A_{pb} \sin a_p) S_0}{I_0 b} \quad (7.1.7-3)$$

Where:

σ_x —the normal stress of concrete at the calculated fiber caused by pre-applied force and bending moment M_k ;

σ_y —the vertical compression stress of concrete caused by standard value for concentrated load F_k ;

τ —the shear stress of concrete at the calculated fiber caused by the shear force V_k and the pre-applied force of bent-up prestressing tendon and/or bar; when the torsion moment is acting on the calculated section, the shear stress induced by torsion moment shall still be counted; for members of post-tensioned prestressed concrete statically indeterminate structure, the secondary shear force caused by pre-applied force shall still be counted when the shear stress is calculated;

σ_{pc} —the normal stress of concrete at the calculated fiber caused by pre-applied force after deduction of total losses of prestress, which shall be calculated according to Formula (10.1.6-1) or Formula (10.1.6-4) of this code:

y_0 —the distance from the centroid of the transformed section to the calculated fiber;

I_0 —the moment of inertia for the transformed section;

V_k —the shear force value calculated according to the characteristic combination of loads;

S_0 —the moment of area for the part of transformed section area above calculated fiber to the centroid of transformed section area of member;

σ_{pe} —the effective prestress of bent-up prestressing tendon and/or bar;

A_{pb} —the sectional area of bent-up prestressing tendon and/or bar in the same bent-up plane on calculated section;

a_p —the included angle between the tangent to bent-up prestressing tendon and/or bar and the longitudinal axes of member on the calculated section.

Note: The values of σ_x , σ_y , σ_{pc} and $M_k V_0 / I_0$ in Formula (7.1.7-1) and Formula (7.1.7-2): the positive value shall be substituted for tension stress; the negative value shall be substituted for compression stress.

7.1.8 For prestressed concrete crane beam in the range of both sides of $0.6h$ length from the action point of concentrated force, the simplified distribution of vertical compression stress and shear stress caused by the standard value of concentrated load F_k may be determined according to Figure 7.1.8 and its maximum value of stress may be calculated according to the following formulae:

$$\sigma_{y,\max} = \frac{0.6F_k}{bh} \quad (7.1.8-1)$$

$$\tau_F = \frac{\tau^l - \tau^r}{2} \quad (7.1.8-2)$$

$$\tau^l = \frac{V_k^l S_0}{I_0 b} \quad (7.1.8-3)$$

$$\tau^r = \frac{V_k^r S_0}{I_0 b} \quad (7.1.8-4)$$

Where:

τ^l , τ^r —respectively the shear stress on the sections at $0.6h$ length from the left and right sides of the action points of standard value of concentrated load F_k ;

τ_F —the shear stress on the acting section of standard value of concentrated load F_k ;

V_k^l , V_k^r —respectively the standard values of shear force of the sections at the left side and right side of the action points for standard value of concentrated load F_k .

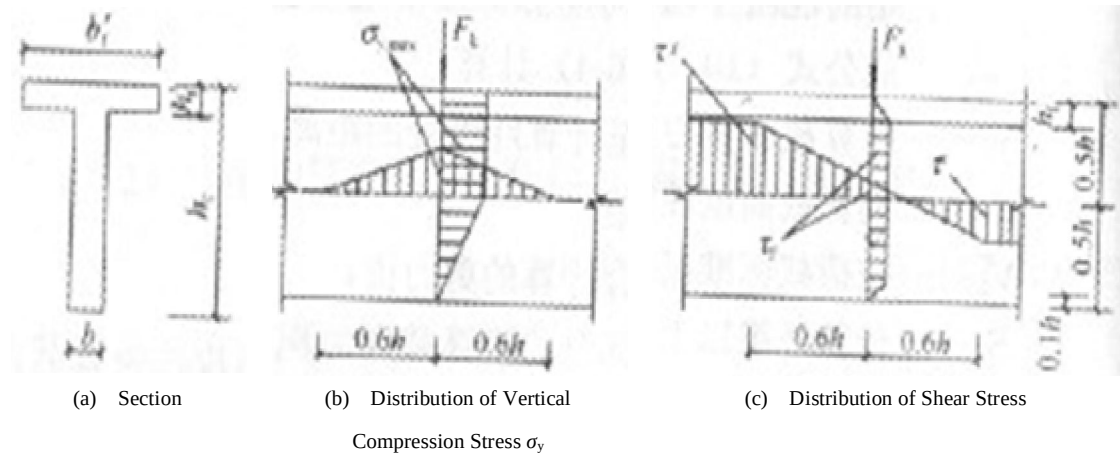


Figure 7.1.8 Stress Distribution Near to Action Point of Concentrated Force for Prestressed Concrete Crane Beam

7.1.9 When doing the crack resistance checking for the normal section and inclined section at the end of pretensioning prestressed concrete members, the variation of the actual stress values within the range of transmission length l_{tr} of prestress for prestressing tendon and/or bar shall be considered. The actual stress of prestressing tendon and/or bar may be considered as linear distribution; at the end of member equals zero; the value of effective prestress σ_{pe} shall be taken at the end of the transmission length of prestress (Figure 7.1.9); the

transmission length of prestress for prestressing tendon and/or bar l_{tr} shall be determined according to Article 10.1.9 of this code.

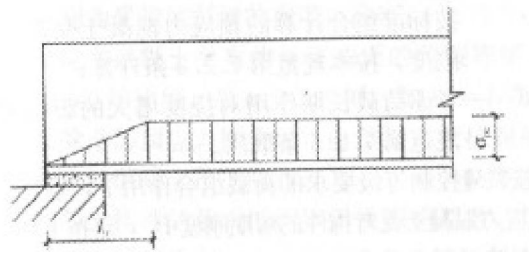


Figure 7.1.9 Variation for the Value of Effective Prestress within the Range of the Transmission Length of Prestress

7.2 Checking of Deflection of Flexural Members

7.2.1 The deflection of reinforced concrete and prestressed concrete flexural members may be calculated according to the structural mechanics method and shall not exceed the limit value specified in Table 3.4.3 of this code.

In members with constant section, it may be assumed that the rigidity is equal in the portion of member with same sign bending moment, and the rigidity at the position with maximum bending moment in that portion shall be adopted. If the section rigidity at support in the calculated span is not larger than twice or it is not less than one half of the section rigidity at mid-span, then the span also may be calculated according to the constant rigidity member, the rigidity of maximum bending moment section may be taken as the rigidity of this member.

7.2.2 The rigidity B of flexural members with rectangular, T-shaped, inverted T-shaped and I-shaped sections considering the long-term action effect by load may be calculated according to the following requirements:

- 1 When adopting characteristic combination of loads,

$$B = \frac{M_k}{M_q(\theta - 1) + M_k} B_s \quad (7.2.2-1)$$

- 2 When adopting quasi-permanent combination of loads,

$$B = \frac{B_s}{\theta} \quad (7.2.2-2)$$

Where:

M_k ——the bending moment calculated according to characteristic combination of loads, taking the maximum value of bending moment within the calculated section;

M_q ——the bending moment calculated according to quasi-permanent combination of loads, taking the maximum value of bending moment within the calculated section;

B_s ——the short-term rigidity of reinforced concrete flexural members calculated according to quasi-permanent combination of loads, or the short-term rigidity of prestressed concrete flexural members calculated according to characteristic combination of loads, which shall be calculated according to Article 7.2.3 of this code;

θ ——the influence coefficient by considering the increase of deflection due to the

long-term action of loads, which shall be taken according to Article 7.2.5 of this code.

7.2.3 The short-term rigidity B_s of reinforced concrete flexural members and prestressed concrete flexural members under the action of load combination required by crack control level may be calculated according to the following formulae:

1 Reinforced concrete flexural member

$$B_s = \frac{E_s A_s h_0^2}{1.15\psi + 0.2 + \frac{6a_E \rho}{1 + 3.5\gamma_f}} \quad (7.2.3-1)$$

2 Prestressed concrete flexural member

1) For member in which crack is not allowed

$$B_s = 0.85 E_c I_0 \quad (7.2.3-2)$$

2) For member in which crack is allowed

$$B_s = \frac{0.85 E_c I_0}{k_{cr} + (1 - k_{cr})w} \quad (7.2.3-3)$$

$$k_{cr} = \frac{M_{cr}}{M_k} \quad (7.2.3-4)$$

$$w = \left(1.0 + \frac{0.21}{a_E \rho} \right) (1 + 0.45\gamma_f) - 0.7 \quad (7.2.3-5)$$

$$M_{cr} = (\sigma_{pc} + \mathcal{M}_{tk}) W_0 \quad (7.2.3-6)$$

$$\gamma_f = \frac{(b_f - b)h_f}{bh_0} \quad (7.2.3-7)$$

Where:

ψ ——the nonuniform coefficient for strain of longitudinal tension steel bars between cracks, which shall be determined according to Article 7.1.2 of this code;

a_E ——the ratio of the elastic modulus of steel reinforcements to the elastic modulus of concrete; namely E_s/E_c ;

ρ ——the ratio of reinforcement of longitudinal tension steel reinforcement: for reinforced concrete flexural members, taking $A_s/(bh_0)$; for prestressed concrete flexural members, taking $(a_1 A_p + A_s)/(bh_0)$, hereof, for grouted post-tensioning prestressed tendon, taking $a_1=1.0$, for unbounded post-tensioning prestressed tendon, taking $a_1=0.3$;

I_0 ——the moment of inertia for the transformed section;

γ_f ——the ratio of the sectional area of tension flanges to the effective sectional area of the web;

b_f, h_f ——respectively the width and depth of flanges in tension zone;

k_{cr} ——the value for the ratio of the cracking bending moment M_{cr} to the bending moment M_k for the normal section of prestressed concrete flexural members, where $K_{cr}>1.0$, taking $K_{cr}=1.0$;

σ_{pc} ——the pre-compression stress in concrete caused by pre-applied force at the edge for

checking of crack resistance after deducting the total losses of prestress;

γ —the plastic influence coefficient for section resistance moment of concrete members, which shall be determined according to Article 7.2.4 of this code.

Note: For members where cracks occur in pretension zone at time of pre-compression, B_s shall be reduced by 10%.

7.2.4 The plastic influence coefficient γ for section resistance moment of concrete members may be calculated according to the following formula:

$$\gamma = \left(0.7 + \frac{120}{h} \right) \gamma_m \quad (7.2.4)$$

Where:

γ_m —the basic value of the plastic influence coefficient for section resistance moment of concrete members, which may be determined on the assumption that the normal section strain may keep plane and taking the concrete stress diagram in tension zone as a trapezoid and the ultimate tensile strain at the tension edge of the concrete is $2f_{tk}/E_c$; for sectional shapes in common use, the value of γ_m may be adopted according to Table 7.2.4;

h —the depth of section (mm): where $h < 400$, taking $h = 400$; where $h > 600$, taking $h = 1,600$; for circular and annular sections, taking $h = 2r$, where r is the radius of circular section or the radius of external ring of annular section.

Table 7.2.4 Basic Value of Plastic Influence Coefficient for Section Resistance Moment γ_m

Item	1	2	3		4		5
Section shape	Rectangular section	T-shaped section with flange located in compression zone	Symmetric I-shaped section or box section		Inverted T-shaped section with flange located in tension zone		Circular and annular section
			$b_f/b \leq 2$ and h_f/h are arbitrary values.	$b_f/b > 2$ and $h_f/h < 0.2$	$b_f/b \leq 2$ and h_f/h are arbitrary values.	$b_f/b > 2$ and $h_f/h < 0.2$	
γ_m	1.55	1.50	1.45	1.35	1.50	1.40	$1.6-0.24r_i/r$

Notes:

1 For I-shaped section $b_f' > b_f$, the value between Item 2 and Item 3 may be adopted; for I-shaped section

$b_f' < b_f$, the value between Item 3 and Item 4 may be adopted;

2 For the box section, b refers to the sum of width of all the ribs;

3 r_i refers to the radius of internal ring of annular section; for circular section, taking $r_i = 0$.

7.2.5 the influence coefficient θ considering the increase of deflection due to the long-term action of loads may be adopted according to the following requirements:

1 Reinforced concrete flexural member

Where $\rho' = 0$, $\theta = 2.0$; where $\rho' = \rho$, $\theta = 1.6$; Where ρ' is the intermediate value, θ may be taken according to the linear interpolation method. Hereof, $\rho' = A_s' / (bh_0)$, $\rho = A_s / (bh_0)$.

For inverted T-shaped section with flange located in tension zone, θ shall be increased 20%.

2 For prestressed concrete flexural members, $\theta = 2.0$.

7.2.6 The inverted camber value due to the pre-applied force in service stage of prestressed concrete flexural members may be calculated using the structural mechanics method according to the rigidity $E_c I_0$, and the influence for long-term action of pre-compression stress shall be considered, the total losses of prestress in the stress of prestressing tendon and/or bar shall be deducted in the calculation. The calculated inverted camber value may be multiplied by the enhancement coefficient 2.0 when adopting simplified calculation.

The long-term inverted camber value of the important or special prestressed concrete flexural members may be determined through the special test analysis or the analysis by adopting calculation methods in consideration of shrinkage and creep influence according to the reinforcement conditions.

7.2.7 For prestressed concrete members, measures shall be taken to control the inverted camber and deflection, and they should meet the following requirements:

1 When considering that the long-term deflection of members calculated after the inverted camber does not conform to the relevant requirements of Article 3.4.3 of this code, the mode of arching in advance in construction may be adopted to control the deflection;

2 For prestressed concrete members with permanent loads relatively small than variable loads, the adverse effect for serviceability due to oversize inverted camber shall be considered, and the corresponding design and construction measures shall be taken.

8 Detailing Requirements

8.1 Expansion Joint

8.1.1 The maximum spacing of expansion joints for reinforced concrete structure may be determined according to those specified in Table 8.1.1.

Table 8.1.1 Maximum Spacing of Expansion Joint for Reinforced Concrete Structure (m)

Structure type		Indoor or in soil	Outdoor
Bent structure	Fabricated	100	70
Frame structure	Fabricated	75	50
	Cast-in-situ	55	35
Shear wall structure	Fabricated	65	40
	Cast-in-situ	45	30
Structure types of retaining wall and basement wall, etc.	Fabricated	40	30
	Cast-in-situ	30	20

Notes:

- 1 The spacing of expansion joint for assembled monolithic structure may adopt the value between fabricated structure and cast-in-situ structure in this table according to specific conditions of structures;
- 2 The spacing of expansion joint for buildings with frame-shear wall structure or frame-core-tube structure may adopt the value between the frame structure and shear wall structure in this table according to specific conditions of structures;
- 3 In case there are no thermal insulation or heat insulation measures on the roof, the spacing of expansion joint for the frame structure and the shear wall structure should adopt according to the value specified in "outdoor" column in this table;
- 4 The spacing of local expansion joint for such exposed structures as cast-in-situ roof overhang and canopy should not be larger than 12m.

8.1.2 Under the following conditions, the maximum spacing of expansion joint in Table 8.1.1 of this code should be properly reduced:

- 1 The bent structure with column height (calculated from top surface of the foundation) less than 8m;
- 2 The bent structure without thermal insulation and heat insulation measures on the roof;
- 3 The structure located in the dry climate area with burning hot weather in summer and frequent rainstorm or the structure under the action of high temperature usually;
- 4 All types of wall structures adopting the slip-form type construction technology;
- 5 The structure with relative big shrinkage of concrete materials and relative long exposed time in the construction period.

8.1.3 The maximum spacing of expansion joint in Table 8.1.1 of this code may be properly increased if there are sufficient criteria under the following conditions:

- 1 Measures for reducing the concrete shrinkage or temperature variation are taken;
- 2 Special measures of prestressing or increasing constructional steel reinforcements are adopted;
- 3 The low shrink concrete materials are adopted, such construction methods as pouring

by separate strips, post-placed strip and control joints are taken, and the construction maintenance is strengthened.

When the spacing of expansion joint is increased, the influence for temperature variation and concrete shrinkage on the structure still shall be considered.

8.1.4 The double column foundation of frame and bent structures may not be broken off when installing the expansion joint.

8.2 Concrete Cover

8.2.1 The thickness of concrete cover for steel bars and the prestressing tendon and/or bar of members shall meet the following requirements.

1 The cover thickness of stressed steel reinforcements of members shall not be less than the nominal diameter d of the steel reinforcement;

2 The cover thickness for the steel reinforcement in outermost layer of the concrete structure with 50-year design life shall be in accordance with those specified in Table 8.2.1; the cover thickness for the steel reinforcement in outermost layer of the concrete structure with 100-year design life shall not be less than 1.4 times value in Table 8.2.1.

Table 8.2.1 Minimum Thickness of Concrete Cover c (mm)

Environmental category	Slab, wall and shell	Beam, column and bar
I	15	20
II-a	20	25
II-b	25	35
III-a	30	40
III-b	40	50

Notes:

- 1 The value of the cover thickness in this table shall be increased by 5mm if the concrete strength grade is not larger than C25;
- 2 The concrete cushion should be arranged for the reinforced concrete foundation; the thickness of concrete cover for the steel reinforcement in the foundation shall be calculated from the top surface of the cushion and shall not be less than 40mm.

8.2.2 The thickness of the concrete cover may be properly reduced if there are sufficient criteria and the following measures are taken:

- 1 There is reliable protective coating on the member surface;
- 2 The factory production is adopted for the precast member;
- 3 Such antirust measures as mixing the corrosion inhibitor in the concrete or adopting the cathodic protection treatment are taken;

4 When reliable building waterproofing or protective measures are taken for the basement wall, the cover thickness for the steel reinforcement contacted with the soil layer may be properly reduced but shall not be less than 25mm.

8.2.3 The effective constructional measures should be taken for the cover if the cover thickness for the longitudinal stressed steel reinforcements in beams, columns and walls is larger than 50mm. The cover thickness for the mesh steel reinforcement shall not be less than 25mm when arranging anti-cracking and anti-peeling mesh reinforcement steel fabric in the cover.

8.3 Anchorage of Steel Reinforcement

8.3.1 When the tensile strength of the steel reinforcement is utilized sufficiently in the calculation, the anchorage of tension steel reinforcements shall meet the following requirements:

1 The basic anchorage length shall be calculated according to the following formulae:

Steel bar

$$l_{ab} = \alpha \frac{f_y}{f_t} d \quad (8.3.1-1)$$

Prestressing tendon and/or bar

$$l_{ab} = \alpha \frac{f_{py}}{f_t} d \quad (8.3.1-2)$$

Where:

l_{ab} ——the basic anchorage length of tension steel reinforcements;

f_y, f_{py} ——the design value of the tensile strength for the steel bar and the prestressing tendon and/or bar;

f_t ——the design value of the axial tensile strength for the concrete, valued as C60 when the concrete strength grade is greater than C60;

d ——the diameter of anchored steel reinforcement;

α ——the appearance coefficient of anchored steel reinforcement, adopted according to Table 8.3.1.

Table 8.3.1 Appearance Coefficient of Anchored Steel Reinforcement α

Type of steel reinforcement	Plain round steel reinforcement	Ribbed steel reinforcement	Spiral ribbed steel wire	Three-skein steel strand	Seven-skein steel strand
α	0.16	0.14	0.13	0.16	0.17

Note: The end of the plain round steel reinforcement shall be made into 180° hook, the length of the straight section after hooking shall not be less than 3d, however, when it is used as compression steel reinforcement, the hook may not be made.

2 The anchorage length of tension steel reinforcements shall be calculated according to the following formulae based on anchorage conditions and shall not be less than 200mm:

$$l_a = \zeta_a l_{ab} \quad (8.3.1-3)$$

Where:

l_a ——the anchorage length of tension steel reinforcements;

ζ_a ——the correction coefficient of the anchorage length; for steel bars, it shall be adopted according to the requirements of Article 8.3.2 of this code; when there are more than one item, it may be calculated as continuous multiply but shall not be less than 0.6; for the prestressing tendon and/or bar, it may be taken as 1.0.

The anchorage requirements of longitudinal tensile steel reinforcements for beam-column joints shall be implemented in accordance with the requirements of Section 9.3 (II) of this code.

3 If the cover thickness for the anchored steel reinforcement is not larger than 5d, the

transverse constructional steel reinforcement shall be arranged within the anchorage length range and the diameter shall not be less than $d/4$; the spacing of such members as beams, columns and diagonal bracing shall not be greater than $5d$; the spacing of such plane members as slabs and walls shall not be larger than $10d$ and all shall not be larger than 100mm, hereof d refers to the diameter of the anchored steel reinforcement.

8.3.2 The correction coefficient ζ_a of anchorage length for longitudinal tension steel bars shall be adopted according to the following requirements:

1 It shall be adopted as 1.10 if the nominal diameter of the ribbed steel reinforcement is larger than 25mm;

2 It shall be adopted as 1.25 for the ribbed steel reinforcement with epoxy coating;

3 It shall be adopted as 1.10 for the steel reinforcement easily disturbed in the construction process;

4 When the practical reinforcement area of longitudinal stressed steel reinforcements is larger than the design calculation area, the ratio value of the design calculation area to the practical reinforcement area shall be adopted as the correction coefficient; however, for structural members directly bearing dynamic loads with seismic protection requirements, this correction shall not be considered;

5 If the cover thickness of anchored steel reinforcement is $3d$, the correction coefficient may be taken as 0.80; if the cover thickness is $5d$, the correction coefficient may be taken as 0.70; the intermediate shall be valued as interpolation, When the hook or mechanical anchoring measures are adopted at the end of the longitudinal tension steel bars, the anchorage length (projected length) including the hook or anchor end may be taken as 60% of the basic anchorage length l_{ab} . The type (Figure 8.3.3) and technical requirements for hooks and mechanical anchoring shall be in accordance with those specified in Table 8.3.3.

Table 8.3.3 Type and Technical Requirements for Hook and Mechanical Anchoring of Steel Reinforcement

Anchoring type	Technical requirements
90° hook	End with 90° hook, the internal diameter of the hook is $4d$, its straight reach length after bending is $12d$
135° hook	End with 135° hook, the internal diameter of the hook is $4d$, its straight reach length after bending is $5d$
Fillet welding with anchor bar at one side	Fillet welding $5d$ -long steel reinforcements with the same diameter at one side of the end
Fillet welding with anchor bar at both sides	Fillet welding $3d$ -long steel reinforcements with the same diameter at both sides of the end
Anchor slab at welded end	Plug welding with perforation for d -thick anchor slab at end
Bolt anchor head	Screwing in bolt anchor head at end

Notes:

- 1 The weld and thread length shall meet the requirements of load-bearing capacity;
- 2 The pressure-bearing net area of bolt anchor head and welded anchor slab shall not be less than 4 times of sectional area of the anchored steel reinforcement;
- 3 The specification for the bolt anchor head shall meet the requirements of relevant standards;
- 4 The clear spacing of steel reinforcements between the bolt anchor head and welded anchor slabs should not be less than $4d$, otherwise the adverse effect of group anchor shall be considered;
- 5 The hook at corner of the section and the reinforced direction for fillet welding anchor bar at one side should be offset

toward the inner side of the section.

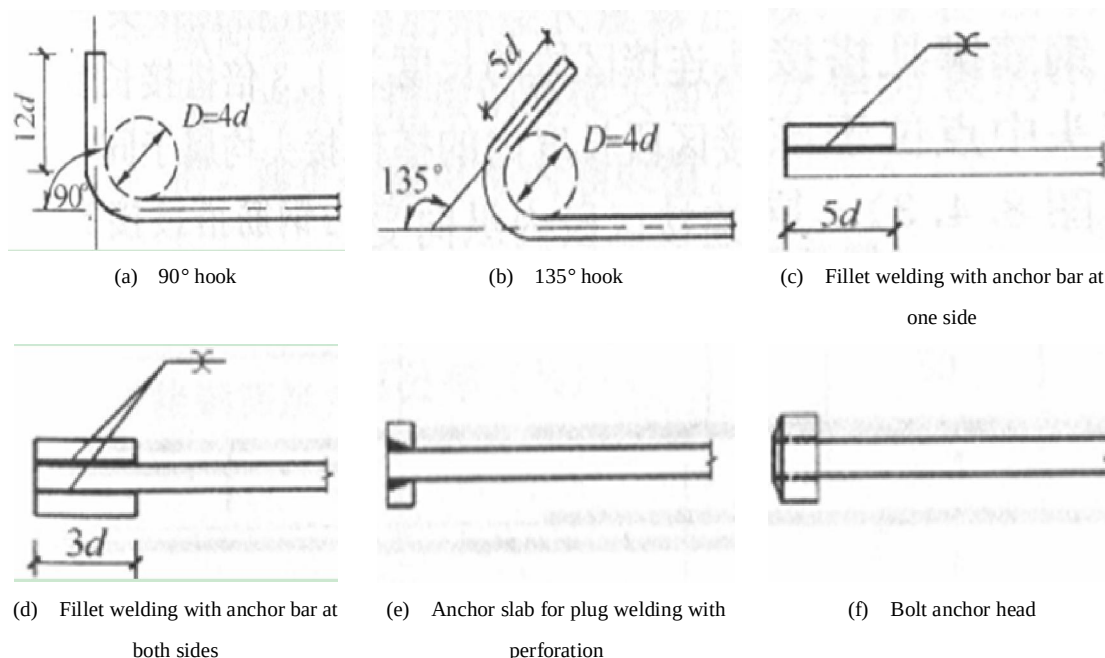


Figure 8.3.3 Type and Technical Requirements for Hook and Mechanical Anchoring of Steel Reinforcement

8.3.4 When the compression strength of longitudinal compressive steel reinforcement in the concrete structure is sufficiently utilized in calculation, its anchorage length shall not be less than 70% of the corresponding tension anchorage length.

Such anchoring measures as end hooks and fillet welding anchor bar at one side shall not be adopted for the compression steel reinforcement.

The transverse constructional steel reinforcement within the range of the anchorage length of the compression steel reinforcement shall meet the relevant requirements of Article 8.3.1 of this code.

8.3.5 For precast members under dynamic loads, the end of the longitudinal stressed steel bars shall be welded on the steel plate or angle steel, which shall be reliably anchored in the concrete. The dimension of the steel plate or angle steel shall be determined according to calculation and the thickness should not be less than 10mm.

The anchor at the end of stressed steel bars in other members also may be realized by welding steel plates or section steel.

8.4 Splices of Steel Reinforcement

8.4.1 The splice of steel reinforcement may adopt binding and splicing, mechanical connection or welding. The types and quality of mechanical connection joints and welded joints shall comply with current relevant standards of China.

The connection joint of stressed steel reinforcements in the concrete structure should be arranged at the place where the stressed force is relative small. The joints on a same stressed steel reinforcement should be arranged as less as possible. The connection joints should not be arranged for longitudinal stressed steel reinforcements in important members and key force transmission positions of the structure.

8.4.2 The binding and splicing shall not be adopted for longitudinal stressed steel reinforcements in the axial tension and small eccentric tension members; the diameter of tension steel reinforcements should not be larger than 25mm and the diameter of compression steel reinforcements should not be larger than 28mm if the binding and splicing is adopted for steel reinforcements in other members.

8.4.3 The binding and splicing joints of adjacent longitudinal stressed steel reinforcements in the same member should be mutually staggered. The length of the joints for binding and splicing of adjacent longitudinal stressed steel reinforcements in a same member should be mutually staggered. The length of connection sector of joints for binding and splicing of steel reinforcements is 1.3 times the overlapping length, all the overlapping joints of which midpoints are located in the length of this connection sector belong to the same connection sector (Figure 8.4.3). In the same connection sector, the overlapping joint area percentage of longitudinal stressed steel reinforcements is the value for ratio of the sectional area of longitudinal stressed steel reinforcements with overlapping joints to the sectional area of all the longitudinal stressed steel reinforcements. When steel reinforcements are overlapping with different diameters, it shall be calculated according to the steel reinforcement with relative small diameter.

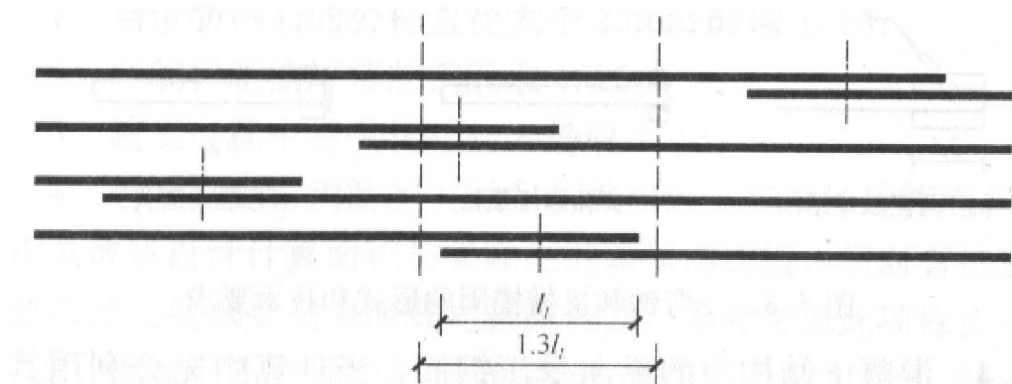


Figure 8.4.3 Binding and Splicing Joint of Longitudinal Tension Steel Reinforcement in a Same Connection Sector

Note: There are two sticks of steel reinforcements with overlapping joints in the same connection sector as shown in the Figure; the overlapping joint area percentage of the steel reinforcements is 50% if the diameters of the steel reinforcements are same.

The overlapping joint area percentage of tension steel reinforcements in the same connection sector: should not be larger than 25% for members of beam, slab and wall types; should not be larger than 50% for members of column types. If it is really necessary to increase the overlapping joint area percentage of tension steel reinforcements, it should not be larger than 50% for members of beam types; and it may be relaxed according to actual conditions for splicing points of slabs, walls, columns and precast members.

When the binding and splicing connection is adopted for twin bars, the connection shall be carried out with the mode of staggered-overlapping each single bar. The joint area percentage shall be calculated according to all single bars in the same connection sector. The overlapping length of steel reinforcements in twin bars shall be calculated according to the single bar respectively.

8.4.4 The overlapping length of joints for binding and splicing of longitudinal tension steel

reinforcements shall be calculated basing on the overlapping joint area percentage of steel reinforcements in the same connection sector according to the following formula and shall not be less than 300mm.

$$l_l = \xi_l l_a \quad (8.4.4)$$

Where:

l_l ——the overlapping length of longitudinal tension steel reinforcements;

ξ_l ——the correction coefficient of the overlapping length for the longitudinal tension steel reinforcement, which shall be adopted according to Table 8.4.4. If the joint area percentage of longitudinal overlapping steel reinforcements is the medium value in this table, the correction coefficient may be valued according to interpolation.

Table 8.4.4 Correction Coefficient of Overlapping Length for Longitudinal Tension Steel Reinforcement

Joint area percentage of longitudinal overlapping steel reinforcement (%)	≤25	50	100
ξ_l	1.2	1.4	1.6

8.4.5 When the overlapping connection is adopted for the longitudinal compression steel reinforcement in members, the compression overlapping length shall not be less than 70% of the overlapping length for longitudinal tension steel reinforcements specified in Article 8.4.4 of this code and shall not be less than 200mm.

8.4.6 The transverse constructional steel reinforcement within the range of overlapping length for longitudinal stressed steel reinforcements of members of beam and column types shall meet the requirements of Article 8.3.1 of this code; if the diameter of the compression steel reinforcement is larger than 25mm, two pieces of stirrup still shall be arranged respectively within the range of 100mm outside two end surfaces of the overlapping joint.

8.4.7 The mechanical connection joints of longitudinal stressed steel reinforcements should be mutually staggered. The length of the mechanical connection sector of steel reinforcements is $35d$; d refers to the relative small diameter of the connection steel reinforcement. All the mechanical connection joints of which midpoints of joints are located in the length of this connection sector belong to the same connection sector.

The joint area percentage of longitudinal tension steel reinforcements in the same connection sector should not be larger than 50%; but it may be relaxed according to actual conditions for splicing points of slabs, walls, columns and precast members. The joint percentage of longitudinal compression steel reinforcements may be free from restriction.

The cover thickness of the mechanical connection sleeve should meet the relevant requirements for minimum cover thickness of steel reinforcements. The transverse clear spacing of the mechanical connection sleeve should not be less than 25mm; the spacing of stirrup at the sleeve still shall meet the corresponding constructional requirements.

The mechanical connection joints in structural members directly under dynamic loads shall not only meet the fatigue resistance required in the design, the joint area percentage of longitudinal stressed steel reinforcements in the same connection sector shall not be larger than 50%.

8.4.8 The welding for fine grain hot-rolled ribbed steel reinforcement and the ribbed steel reinforcement with diameter larger than 28mm shall be determined after tests; the steel reinforcement with waste heat management should not be welded.

The mechanical connection joints of longitudinal stressed steel reinforcements should be mutually staggered. The length of the connection sector of welded joints of steel

reinforcements is $35d$ and shall not be less than 500mm; d refers to the relative small diameter of the connection steel reinforcement. All the welded joints of which midpoints of joints are located in the length of this connection sector belong to the same connection sector.

The joint area percentage of longitudinal tension steel reinforcements should not be larger than 50%; but it may be relaxed according to actual conditions for splicing points of precast members. The joint percentage of longitudinal compression steel reinforcements may be free from restriction.

8.4.9 For the members in need of checking of fatigue, the binding and splicing joints shall not be adopted for longitudinal tension steel reinforcements and the welded joints should not be adopted neither; It is prohibited to weld ant accessories (except end anchorage) on the steel reinforcements.

The adoption of welded joints for longitudinal tension steel reinforcements in reinforced concrete crane beams, roof beams and bottom chord of roof truss directly under crane loads shall meet the following requirements:

- 1 The flash butt welding shall be adopted and the burrs and hemming shall be removed;
- 2 The area percentage of welded joints of longitudinal tension steel reinforcements in the same connection sector shall not be larger than 25%, the length of the connection sector of welded joints shall be taken as $45d$, d refers to the relative big diameter of the longitudinal stressed steel reinforcements;
- 3 In checking of fatigue, the welded joints shall meet the requirements of limit value for fatigue stress amplitude specified in Article 4.2.6 of this code.

8.5 Minimum Ratio of Reinforcement for Longitudinal Stressed Steel Reinforcement

8.5.1 The ratio of reinforcement $\rho_{\min}$ for longitudinal stressed steel reinforcements in reinforced concrete structure members shall not be less than the value specified in Table 8.5.1.

Table 8.5.1 Minimum Ratio of Reinforcement for Longitudinal Stressed Steel Reinforcement $\rho_{\min}$ (%)

Stressed type			Minimum ratio of reinforcement
Compression member	All longitudinal steel reinforcements	Strength grade 500MPa	0.50
		Strength grade 400MPa	0.55
		Strength grade 300MPa and 335MPa	0.60
	One side of longitudinal steel reinforcements		0.20
Tension steel reinforcements at one side of flexural members, eccentric tension and axial tension members			Relative larger value between 0.20 and $45f_t/f_y$

Notes:

- 1 For the minimum ratio of reinforcement for all longitudinal steel reinforcements in compression members: when the concrete with C60 above strength grade is adopted, it shall be increased 0.10 according to those specified in this table;
- 2 For tension steel reinforcements in flexural members of slab type: when the steel reinforcements of strength grade 400MPa and 500MPa are adopted, the minimum ratio of reinforcement shall be allowed to adopt from the larger value between 0.15 and $45f_t/f_y$;
- 3 The compression steel reinforcement in the eccentric tension member shall be considered according to the

longitudinal steel reinforcement at one side of the compression member;

- 4 The ratio of reinforcement of the longitudinal steel reinforcement at one side and all longitudinal steel reinforcements of compression member as well as the ratio of reinforcement of tension steel reinforcements at one side of axial tension members and small eccentric tension members shall be all calculated according to the total sectional area of members;
- 5 The ratio of reinforcement of tension steel reinforcements at one side of flexural members and large eccentric tension members shall be calculated according to the sectional area after deducting the compression flange area $(b'_f - b)h'_f$ from the total sectional area;
- 6 When the steel reinforcement is arranged along the member section perimeter, "longitudinal steel reinforcement at one side" refers to the longitudinal steel reinforcement arranged along one of two opposite sides in the forced direction.

8.5.2 For concrete slabs horizontally bedded on the base, the minimum ratio of reinforcement for tension steel reinforcements in slabs may be properly reduced but shall not be less than 0.15%.

8.5.3 For secondary reinforced concrete flexural members in structures, if the depth of section required in the construction is far larger than the load bearing demands, the ratio of reinforcement for longitudinal tension steel reinforcements may be calculated according to the following formulae:

$$\rho_s \geq \frac{h_{cr}}{h} \rho_{\min} \quad (8.5.3-1)$$

$$h_{cr} = 1.05 \sqrt{\frac{M}{\rho_{\min} f_y b}} \quad (8.5.3-2)$$

Where:

ρ_s ——the ratio of reinforcement for longitudinal tension steel reinforcements in members calculated as total section;

$\rho_{\min}$ ——the minimum ratio of reinforcement for longitudinal stressed steel reinforcements, which shall be adopted according to Article 8.5.1 of this code;

h_{cr} ——the critical depth of the section of members, $h/2$ shall be taken where it is less than $h/2$;

h ——the section depth of the member;

b ——the section width of the member;

M ——the design value of the bend bearing capacity of normal section of the member.

9 Fundamental Requirements for Structural Members

9.1 Slabs

(I) Fundamental Requirements

9.1.1 The concrete slabs shall be calculated in accordance with the following principles:

- 1 The slab supported by two opposite sides shall be calculated according to one-way slab;
- 2 The slab supported by four sides shall be calculated according to the following provisions:
 - 1) When the length ratio of the long side to the short side is not larger than 2.0, the slab shall be calculated according to two-way slab;
 - 2) When the length ratio of the long side to the short side is larger than 2.0 but less than 3.0, the slab should be calculated according to two-way slab;
 - 3) When the length ratio of the long side to the short side is not less than 3.0, the slab should be calculated according to the one-way slab that is stressed along the direction of short side.

9.1.2 The dimensions of cast-in-situ concrete slabs should meet the following requirements:

1 The span-to-thickness ratio of slab: that of reinforced concrete one-way slab is not larger than 30 and that of reinforced concrete two-way slab is not larger than 40; that of the slab without beam support but with column cap is not larger than 35 and that of the slab without beam support and column cap is not larger than 30. The span-to-thickness ratio of the prestressed slab may be increased appropriately; if the load and span of the slab are relatively large, the span-to-thickness ratio should be reduced appropriately.

2 The thickness of cast-in-situ reinforced concrete slab shall not be less than the values specified Table 9.1.2.

Table 9.1.2 Minimum Thickness of Cast-in-situ Reinforced Concrete Slabs (mm)

Slab type		Minimum thickness
One-way slab	Roof slab	60
	Floor slab of civil architecture	60
	Floor slab of industrial architecture	70
	Floor slab under carriage way	80
Two-way slab		80
Ribbed floor	Faceplate	50
	Rib height	250
Cantilever slab (at root)	The cantilever length is not larger than 500mm	60
	The cantilever length is 1200mm	100
Beamless floor slab		150
Cast-in-situ hollow floor		200

9.1.3 The spacing of stressed steel reinforcements in the slab should not be larger than 200mm in case the slab thickness is not larger than 150mm and should not be larger than 1.5 times of the slab thickness or 250mm in case the slab thickness is larger than 150mm.

9.1.4 As for the multi-span slabs adopting the separate type reinforcements, all the steel reinforcements at the slab bottom should be extended into the supports; the length of the negative reinforcements extended into the span shall be determined according to the negative bending moment diagram and also shall meet the anchorage requirements of steel reinforcements.

The anchorage length of longitudinal stressed steel reinforcements in the lower portion of simply supported slab or continuous slab extending into the support shall not be less than 5 times of the diameter of steel reinforcement, and these steel reinforcements shall extended beyond the center line of the support. Where the temperature and shrinkage stress in the continuous slab are relatively large, the length extended into the support should be increased appropriately.

9.1.5 The volumetric void ratio of cast-in-situ concrete hollow floor slab should not be larger than 50%.

Where the box-type inner bore is adopted, the thickness of the top slab shall not be less than 1/15 of the clear distance between ribs or 50mm. Where the bottom slab is set with stressed steel reinforcements, the slab thickness shall not be less than 50mm. The ratio of the rib width between inner bores to the height of inner bore should not be less than 1/4 and the shall not be less than 60mm; as for the prestressed slabs, the rib width shall not be less than 80mm.

Where the tubular inner bore is adopted, the thickness of the slabs both at the top and bottom of the bore shall not be less than 40mm, the ratio of the rib width to the inner bore diameter should not be less than 1/5 and the rib width shall not be less than 50mm; as for the prestressed slabs, the rib width shall not be less than 60mm.

(II) Constructional Reinforcement

9.1.6 For the cast-in-situ concrete slabs designed according to simply supported edge or non-stressed edge, if the slab is integrally casted with the concrete beam and wall or is fixed in masonry wall, hence constructional steel reinforcements shall be applied for the slab surface and the following requirements shall be met:

1 For the steel reinforcements, the diameter should not be less than 8mm, the spacing should not be larger than 200mm, and the area of reinforcement in unit width should not be less than 1/3 sectional area of steel reinforcements at the mid-span slab bottom in the corresponding direction. For the one-way slab integrally casted with the concrete beam and concrete wall, the sectional area of steel reinforcements in the non-stressed direction should not be less than 1/3 sectional area of steel reinforcements at the mid-span slab bottom in the stressed direction.

2 The length of the steel reinforcement extended into the slab from the concrete beam side, column side or wall side should not be less than $l_0/4$, the length of steel reinforcement extended into the slab edge at the support of masonry wall should not be less than $l_0/7$, therein, the calculated span l_0 shall be considered according to the stressed direction in relation to the one-way slab and be considered according to the direction of short side in relation to the two-way slab.

3 At the corners of floor slabs, the additional steel reinforcements should be applied

perpendicularly along two directions, parallel along inclined direction or in radial pattern.

4 The steel reinforcements shall be reliably anchored in beams, walls or columns.

9.1.7 When designed according to one-way slab, distribution steel reinforcements shall be arranged perpendicular to the stressed direction, the reinforcements in unit width should not be less than 15% of the stressed steel reinforcements in unit width, and the ratio of reinforcement should not be less than 0.15%; the diameter of distribution steel reinforcement should not be less than 6mm and the spacing between them should not be greater than 250mm; in case of greater concentrated load, the area of distribution steel reinforcements still shall be increased and their spacing should not be larger than 200mm.

Where practical experiences or reliable measures are available, the distribution steel reinforcements for precast one-way slab may not be restricted by this clause.

9.1.8 Within the range of cast-in-situ slab region where the temperature and shrinkage stress are large, anti-cracking constructional steel reinforcements shall be applied on the slab surface in two ways. The ratio of reinforcement all should not be less than 0.10% and the spacing should not be greater than 200mm. The anti-cracking constructional steel reinforcements may be arranged by penetrating depending on the former steel reinforcements or may be set with other steel reinforcements to overlap with the former steel reinforcements according to the requirements of tension steel reinforcements or to anchor in the peripheral members.

At the neck position of slab plane, the slab thickness and the reinforcements should be increased appropriately. Along the hole edges and concave corners of the slabs, additional anti-cracking constructional steel reinforcements should be applied and reliable anchoring measures should be taken.

9.1.9 For the thick concrete slab as well as the foundational raft slab that is horizontally laid on the subgrade, when the slab thickness is larger than 2m, other than the longitudinal and transverse steel reinforcements shall be applied along the upper and lower surfaces of the slab, the constructional steel mesh reinforcements parallel to the slab surface shall be applied along the slab thickness within a range of no larger than 1m, the diameter of the steel mesh reinforcement should not be less than 12mm and the spacing of the steel reinforcements both in longitudinal and transverse directions should not be larger than 300mm.

9.1.10 Where the thickness of concrete slab is not less than 150mm, U-shaped constructional steel reinforcements should be applied at the end of the free edge of slab to overlap with the steel reinforcements at the top and bottom of slab, the overlapping length should not be less than 15 times of the diameter of U-shaped constructional steel reinforcement or 200mm; or the steel reinforcements may also be bent up and down respectively at the slab surface and slab bottom.

(III) Slab-column Structure

9.11.11 When the anti-punching stirrups or bent-up steel reinforcements are applied in the concrete slabs, the following construction requirements shall be met:

1 The slab thickness shall not be less than 150mm;

2 Stirrups and corresponding erection steel reinforcements that are required in calculation shall be applied within the range intersecting with the 45° punching failure conical surface, their distribution length outward from the acting surface of concentrated load or from

the edge of column section shall not be less than $1.5h_0$ (Figure 9.1.11a); the diameter of stirrup shall not be less than 6mm and the stirrups shall be made into closed type, their spacing shall not be larger than $h_0/3$ or 100mm;

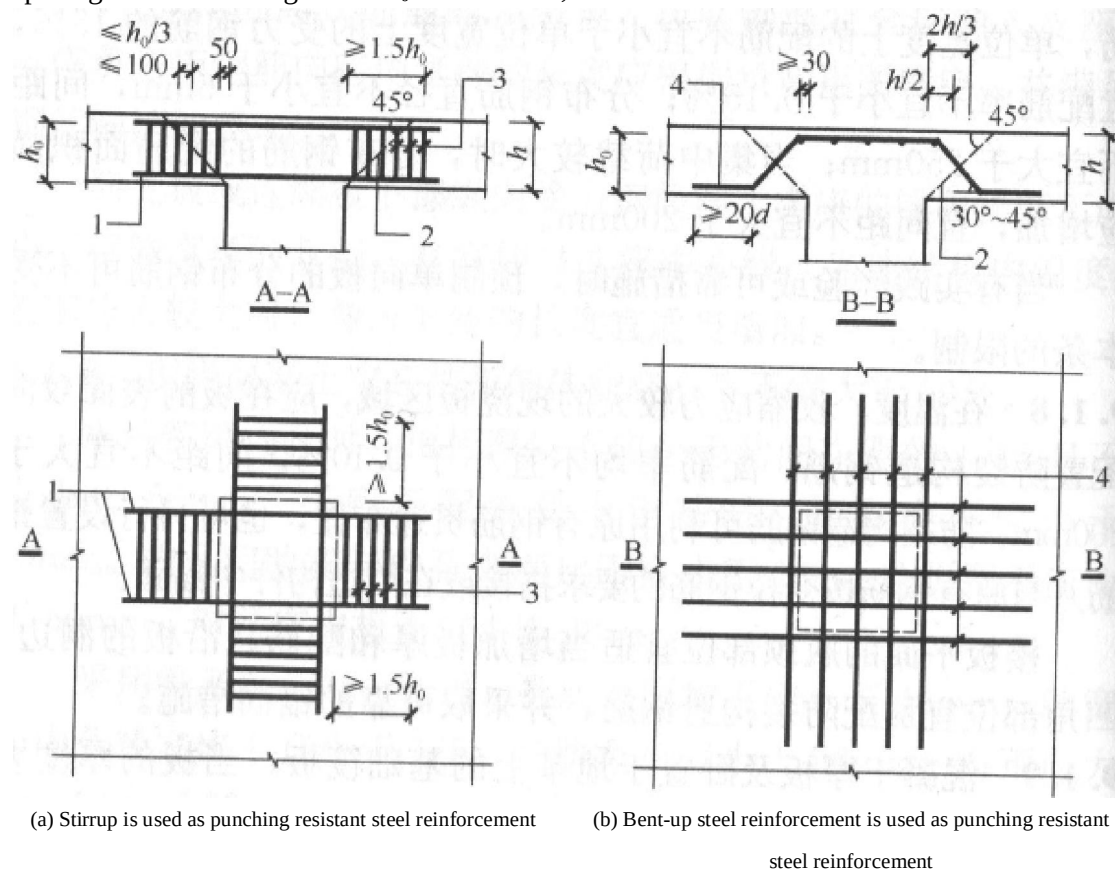


Figure 9.1.11 Arrangement of Punching Resistant Steel Reinforcements in Slab

Note: Dimensional unit in the Figure is mm.

1—Erection steel reinforcement; 2—Punching failure conical surface; 3—Stirrup; 4—Bent-up steel reinforcement

3 The bent-up angle of the required bent-up steel reinforcement according to calculation may be selected between 30° and 45° based on the slab thickness; the inclined part of bent-up steel reinforcement shall be intersected with the conical surface destroyed by punching (Figure 9.1.11b), the intersection point shall be at the scope of $(1/2 \sim 2/3)h$ outside the acting surface or concentrated load or outside the column section. The diameter of the bent-up steel reinforcement should not be less than 12mm and at least three bent-up steel reinforcements shall be applied in each direction.

9.1.12 The slab-column joints may adopt the structural form of column cap or supporting plate. The shape and dimension of slab-column joint shall contain a 45° punching failure cone and shall meet requirements on punching bearing capacity.

The height of column cap shall not be less than the slab thickness h , and the thickness of supporting plate shall not be less than $h/4$. The dimensions of column cap or supporting plate in two directions in the plane all should not be less than the sum of width b of column section in the same direction with $4h$ (Figure 9.1.12).

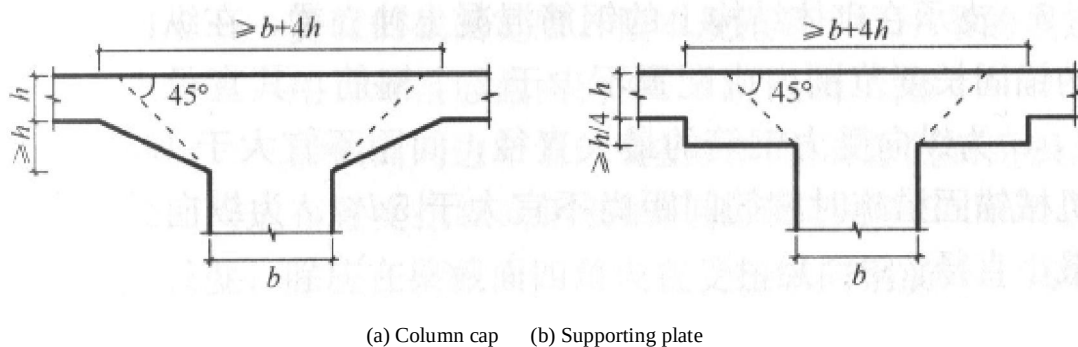


Figure 9.1.12 Slab-column Structure with Column Cap or Supporting Plate

9.2 Beams

(I) Longitudinal Reinforcement

9.2.1 The longitudinal stressed reinforcements of beam shall meet the following requirements:

1 The quantity of the steel reinforcements extended into the scope of beam support shall not be less than 2.

2 If the beam height is not less than 300mm, the diameter of steel reinforcement shall not be less than 10mm; otherwise, the diameter of steel reinforcement shall not be less than 8mm.

3 The clear spacing between the steel reinforcements in the horizontal direction at top of beam shall not be less than 30mm or $1.5d$; while the clear spacing between the steel reinforcements in the horizontal direction at bottom of beam shall not be less than 25mm or d . Where over two layers of steel reinforcements are applied at the bottom, the center-to-center distance in the horizontal direction between those steel reinforcements above two layers shall be larger than that between those at the lower two layers by one times; the clear spacing between the steel reinforcements at each layer shall not be less than 25mm or d , here, d is the maximum diameter of steel reinforcement.

4 In the reinforcement compact area of beam, the reinforcement form of twin bars should be adopted.

9.2.2 For the longitudinal stressed reinforcements at the lower portion of reinforced concrete simply supported beam and simply supported end of continuous beam, the anchorage length extended into the support from the support edge shall meet the following requirements:

1 When V is not larger than $0.7f_tbh_0$, the anchorage length shall not be less than $5d$; when V is larger than $0.7f_tbh_0$, the anchorage length shall not be less than $12d$ for ribbed steel reinforcements and shall not be less than $15d$ for plain round steel reinforcements, here, d is the maximum diameter of steel reinforcement;

2 If the anchorage length of the longitudinal stressed reinforcement extended into the range of beam support fails to meet the requirements stated in Clause 1 of this article, hook or mechanical anchoring measures may be taken, and other effective anchoring measures meeting those specified in Article 8.3.3 of this code shall be taken;

3 For the reinforced concrete independent beam supported on the masonry structure, at

least two stirrups shall be applied within the range of the anchorage length of longitudinal stressed steel reinforcements, the diameter of stirrup should not be less than $d/4$, d is the maximum diameter of longitudinal stressed steel reinforcement; the spacing should not be larger than $10d$, and the stirrup spacing still should not be larger than $5d$ when mechanical anchoring measures are taken, d is the minimum diameter of longitudinal stressed steel reinforcement.

Note: For the simply supported beam and the simply supported end of continuous beam, of which the concrete strength grade is C25 or below, if there is concentrated load acting within a scope of $1.5h$ to the support edge and V is larger than $0.7f_tbh_0$, the ribbed steel reinforcements should be taken with effective anchoring measures or their anchorage length may be less than $15d$, here, d is the diameter of anchored steel reinforcement.

9.2.3 The longitudinal tension steel reinforcements with negative bending moment in the support section of reinforced concrete beam should not be cut off in the tension zone, if they have to be cut off, the following requirements shall be met:

1 When V is not larger than $0.7f_tbh_0$, the longitudinal tension steel reinforcement shall be cut off after it has been extended by no less than $20d$ outside the section in which this steel reinforcement is unneeded according to the calculation of flexural load-bearing capacity of normal section, and the length of this steel reinforcement extended out of the section in which the reinforcement strength is fully used shall not be less than $1.2l_a$;

2 When V is larger than $0.7f_tbh_0$, the longitudinal tension steel reinforcement shall be extended by no less than h_0 outside the section in which this steel reinforcement is unneeded according to the calculation of flexural load-bearing capacity of normal section and shall be cut off at the place no less than $20d$ outside that section, and length of this steel reinforcement extended out of the section in which the reinforcement strength is fully used shall not be less than the sum of $1.2l_a$ and h_0 ;

3 If the cut points determined according to Clause 1 and 2 of this article are still located in the corresponding tension zone of negative bending moment, then the longitudinal tension steel reinforcement shall be extended by no less than $1.3h_0$ outside the section in which this steel reinforcement is unneeded according to the calculation of flexural load-bearing capacity of normal section and shall be cut off at the place no less than $20d$ outside that section, and length of this steel reinforcement extended out of the section in which the reinforcement strength is fully used shall not be less than the sum of $1.2l_a$ and $1.7h_0$;

9.2.4 In the reinforced concrete cantilever beam, at least two upper steel reinforcements shall be extended to the outer end of cantilever beam and shall be bent downward by no less than $12d$; the rest steel reinforcements shall not be cut off at the upper portion of the beam but shall be bent downward at the bent starting points as specified in Article 9.2.8 of this code and be anchored at the lower side of the beam according to those specified in Article 9.2.7 of this code.

9.2.5 The minimum ratio of reinforcement $\rho_{tl,min}$ of the longitudinal torsion steel reinforcements in the beam shall meet the following requirements:

$$\rho_{tl,min} = 0.6 \sqrt{\frac{T}{V_b} \frac{f_t}{f_y}} \quad (9.2.5)$$

When $T/(Vb) > 2.0$, $T/(Vb) = 2.0$.

Where:

$\rho_{tl,min}$ ——the minimum ratio of reinforcement of longitudinal torsion steel reinforcements, which equals to $A_{stl}/(bh)$;

b ——the width of the shear section, which shall be taken according to those specified in Article 6.4.1 of this code and b shall be replaced by b_h for the box-section members;

A_{stl} ——the gross sectional area of those longitudinal torsion steel reinforcements arranged along the periphery of section.

The spacing between the longitudinal torsion steel reinforcements arranged along the periphery of section shall not be larger than 200mm or the length of the short side of beam section; in addition to applying longitudinal torsion steel reinforcements at the four corners of the beam section, the rest longitudinal torsion steel reinforcements should be arranged uniformly and symmetrically along the periphery of section. The longitudinal torsion steel reinforcements shall be anchored in the support as tension steel reinforcements.

In the flexural members with shear and torsion, the sectional area of the longitudinal stressed steel reinforcements arranged at the flexural tension side of the section shall not be less than the sum of the sectional area of steel reinforcements calculated according to the minimum ratio of reinforcement of tension steel reinforcements in flexural member as specified in Article 8.5.1 of this code and the sectional area of steel reinforcements calculated according to the ratio of reinforcement of longitudinal torsion steel reinforcements specified in this article and distributed to the flexural tension side.

9.2.6 The longitudinal constructional steel reinforcements at the upper portion of the beam shall meet the following requirements:

1 When the beam end is calculated according to simply supported beam but is actually restrained partially, the longitudinal constructional steel reinforcements shall be applied in the upper portion of the support zone, and their sectional area shall not be less than a quarter of the sectional area required in the calculation of longitudinal stressed steel reinforcements that are applied in mid-span lower portion of beam, and their quantity shall not be less than 2. The length of this longitudinal constructional steel reinforcement extruded from the edge of support into the span shall not be less than $l_0/5$, here, l_0 is the calculated span of beam.

2 For the erection steel reinforcement, its diameter should not be less than 8mm when the beam span is less than 4m, shall not be less than 10mm when the beam span is between 4m and 6m and should not be less than 12mm when the beam span is larger than 6m.

(II) Transverse Reinforcement

9.2.7 The concrete beam should be adopted with stirrups as the steel reinforcements to bear the shear force.

If the bent-up steel reinforcements are adopted, the bent-up angle should be 45° or 60°; proper anchorage length parallel to the axial direction of beam shall be reserved beyond the bent end point, and the anchorage length shall not be less than $20d$ in the tension zone and shall not be less than $10d$ in the compression zone, here, d is the diameter of bent-up steel reinforcement; the steel reinforcements at corners among the bottom layer steel reinforcements of beam shall not be bent up and those at corners among the top layer steel reinforcements shall not be bent downward.

9.2.8 In the tension zone of concrete beam, the bent starting point of bent-up steel reinforcement may be set in front of the section in which this steel reinforcement is unneeded according to the calculation of flexural load-bearing capacity of normal section, however, the intersection point of this bent-up steel reinforcement with the center line of beam shall be located outside such section in which this steel reinforcement is unneeded (Figure 9.2.8); meanwhile, the distance between the bent starting point and the calculated section in which this steel reinforcement is fully used shall not be less than $h_0/2$.

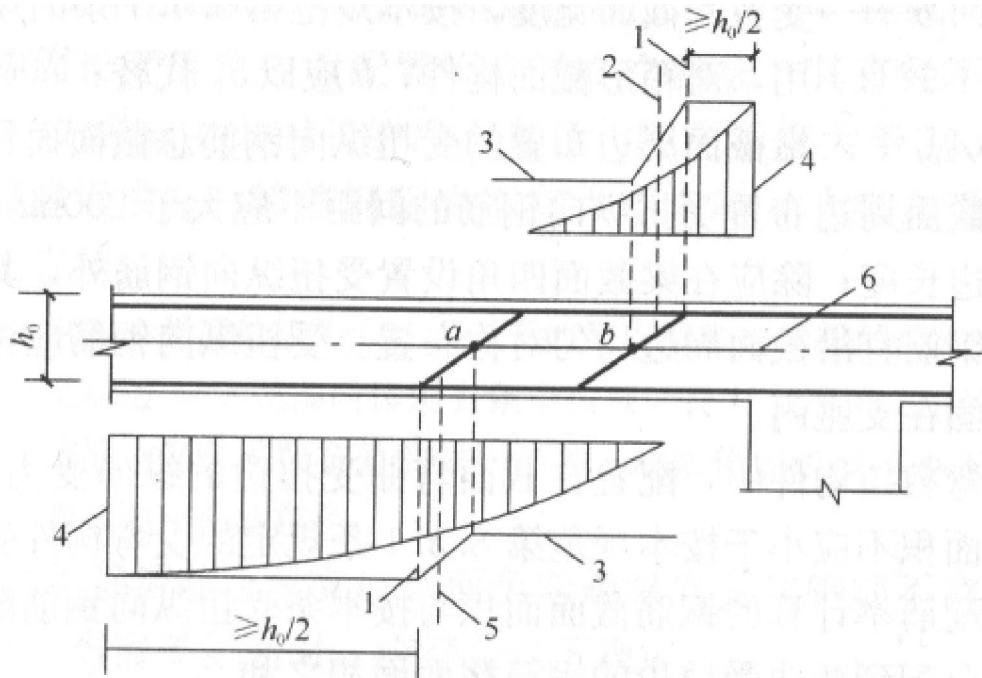


Figure 9.2.8 Relationship Between Bent Starting Point of Bent-up Steel Reinforcement and Bending Moment Diagram

1—Bent starting point of tension zone; 2—Section needing no steel reinforcement "b" according to calculation; 3—Diagram of flexural load-bearing capacity of normal section; 4—Section fully using the strength of steel reinforcement "a" or "b" according to calculation; 5—Section needing no steel reinforcement "a" according to calculation; 6—Center line of beam

Where providing the bent-up steel reinforcements as required according to the calculation, the distance from the bent starting point of the front row starting from the support to the bent end point of the next row shall not be larger than the maximum spacing of stirrups with " $V > 0.7f_t b h_0 + 0.05N_{p0}$ " as stated in Table 9.2.9 of this code. The bent-up steel reinforcement shall not adopt the form of floating steel reinforcement.

9.2.9 Arrangement of stirrups in the beam shall meet the following requirements:

1 For the beams needing no stirrup according to the calculation of load-bearing capacity, if the depth of section is larger than 300mm, the constructional stirrups shall be applied along the full length of beam; if the depth of section $h = 150\text{mm} \sim 300\text{mm}$, the constructional stirrups may only be applied within a scope of $l_0/4$ at the ends of member, here, l_0 is the span. However, if there concentrated loads acting within a scope of $l_0/2$ at the middle part of member, the stirrups shall be applied along the full length of beam. If the depth of section is less than 150mm, no stirrup may be applied.

2 The diameter of stirrup should not be less than 8mm for such beam with depth of section larger than 800mm and should not be less than 6mm for such beam with depth of

section no larger than 800mm. If longitudinal compression steel reinforcements required in calculation are applied in beam, the diameter of stirrup still shall not be less than $d/4$, and d is the maximum diameter of compression steel reinforcement.

3 The maximum spacing of stirrups in the beam should be in accordance with those specified in Table 9.2.9; if V is larger than $0.7f_tbh_0+0.05N_{p0}$, the ratio of reinforcement of stirrups ρ_{sv} [$\rho_{sv}=A_{sv}/(bs)$] still shall not be less than $0.24f_t/f_{yv}$.

Table 9.2.9 Maximum Spacing of Stirrups in Beam (mm)

Depth of beam h	$V>0.7f_tbh_0+0.05N_{p0}$	$V\leq 0.7f_tbh_0+0.05N_{p0}$
$150<h\leq 300$	150	200
$300<h\leq 500$	200	300
$500<h\leq 800$	250	350
$h>800$	300	400

4 Where the longitudinal compression steel reinforcement required according to calculation are applied in the beam, the stirrups shall meet the following requirements:

- 1) The stirrup shall be made into closed type and the length of the straight section of hook shall not be less than $5d$ (d is the diameter of stirrup).
- 2) The spacing of stirrups shall not be larger than $15d$ or 400mm. Where there are more than five longitudinal compression steel reinforcements in one same layer and the diameter of steel reinforcement is larger than 18mm, the stirrup spacing shall not be larger than $10d$ (d is the minimum diameter of longitudinal compression steel reinforcement).
- 3) If the width of beam is larger than 400mm and there are over three longitudinal compression steel reinforcements in one same layer or if the width of beam is not larger than 400mm but there are over four longitudinal compression steel reinforcements in one same layer, the compound stirrup shall be arranged.

9.2.10 In flexural member with shear and torsion, the ratio of reinforcement of stirrups (ρ_{sv}) shall not be less than $0.28f_t/f_{yv}$.

The stirrup spacing shall meet those specified in Table 9.2.9 of this code, therein, the stirrups required due to torsion shall be made into closed type and shall be arranged along the periphery of section. If compound stirrup is adopted, the stirrups located in the section shall not be included in the area of stirrups required for torsion. The tail end of the stirrup required due to torsion shall made into a hook of 135° and the length of the straight section at the end of this hook shall not be less than $10d$ (d is the diameter of stirrup).

In hyperstatic structures, in consideration of the stirrups applied for coordinating torsion, the stirrup spacing should not be larger than $0.75b$, here, b shall be adopted according to those specified in Article 6.4.1 of this code, however, for the box-section members, b shall all be replaced by b_h .

(III) Partial Reinforcement

9.2.11 The concentrated loads acting in the lower portion of beam or within the scope of the depth of section of beam shall all be bear by additional transverse reinforcements, and the additional transverse reinforcements should adopt stirrups.

The stirrups shall be arranged within a length scope of $(2h_1+3b)$ (Figure 9.2.11). When

suspenders are adopted, the bent-up segment shall be extended up to the top edge of beam and the length of the horizontal segment at tail end shall not be less than those specified in Article 9.2.7 of this code.

The gross sectional area of the required additional transverse reinforcements shall meet the following requirements:

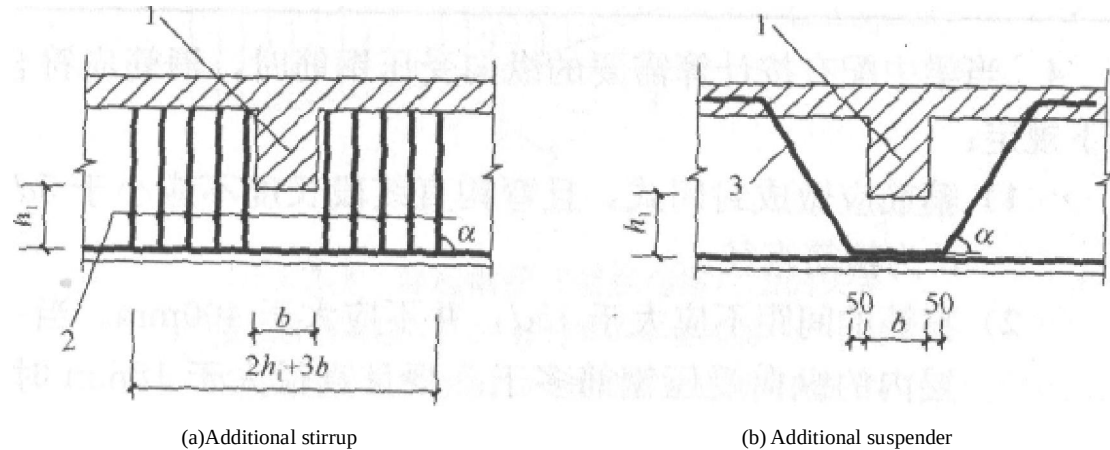


Figure 9.2.11 Arrangement of Additional Transverse Reinforcements When concentrated Loads act in Scope of Beam Section Depth

Note: Dimensional unit in the Figure is mm.

1—Position transferring concentrated loads; 2—Additional stirrup; 3—Additional suspender

$$A_{sv} \geq \frac{F}{f_{yv} \sin \alpha} \quad (9.2.11)$$

Where:

A_{sv} —the gross sectional area of additional transverse reinforcements required for bearing the concentrated loads; if additional suspenders are adopted, A_{sv} shall be the sum of the sectional area of the left and right bent-up segments;

F —the design value of concentrated loads acting at the bottom of beam or in the scope of beam section depth;

α —the included angle between additional transverse reinforcements and the axial line of beam.

9.2.12 The folded beam shall be added with stirrups (Figure 9.2.12) at its inside corner. These stirrups shall be able to bear the resultant force of the longitudinal tension steel reinforcements that are not anchored in compression zone and shall be able to bear no less than 35% of the resultant force of all the longitudinal steel reinforcements in any case.

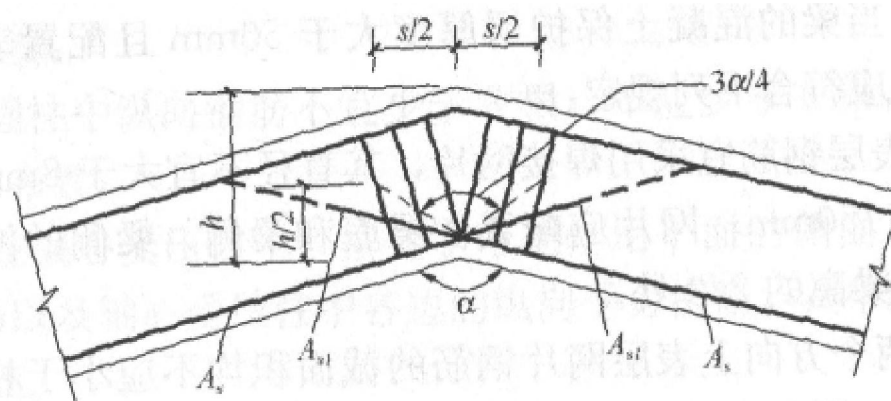


Figure 9.2.12 Reinforcements at the Inside Corner of Folded Beam

The resultant force of those longitudinal tension steel reinforcements that are bear by stirrups shall be calculated according to the following formula:

The resultant force of longitudinal tension steel reinforcements that are not anchored in the compression zone is:

$$N_{s1} = 2 f_y A_{s1} \cos \frac{\alpha}{2} \quad (9.2.12-1)$$

35% of the resultant force of all the longitudinal tension steel reinforcements is:

$$N_{s2} = 0.7 f_y A_s \cos \frac{\alpha}{2} \quad (9.2.12-2)$$

Where:

A_s ——the sectional area of all the longitudinal tension steel reinforcements;

A_{s1} ——the sectional area of longitudinal tension steel reinforcements that are not anchored in the compression zone;

α ——the inside corner of member.

The stirrups calculated according to the aforesaid conditions shall be arranged within in the scope of length s , $s = h \tan(3\alpha/8)$.

9.2.13 Where the depth of beam web is not less than 450mm, the longitudinal constructional steel reinforcements shall be arranged along the depth on both side faces of the beam. The spacing of the longitudinal constructional steel reinforcements (excluding the stressed steel reinforcements at the upper and lower portions of beam as well as the erection steel reinforcements) on each side should not be larger than 200mm, and the sectional area shall not be less than 0.1% of the sectional area of web (bh_w), however, this requirement may be reduced if the beam width is relatively large. Hereof, the depth of web (h_w) shall be adopted according to those specified in Article 6.3.1 of this code.

9.2.14 For thin webbed girder or for reinforced concrete beam requiring checking of fatigue, the longitudinal constructional steel reinforcements in diameter of 8mm~14mm shall be applied with the spacing of 100mm~150mm along both sides in the web that is at the lower one half of the beam depth and shall be arranged in form of lower thick and upper disperse. In the web at the upper one half of beam depth, the longitudinal constructional steel reinforcements may be applied according to those specified in Article 9.2.13 of this code.

9.2.15 Where the thickness of concrete cover for beam is larger than 50mm and the surface layer steel mesh reinforcements are applied, the following requirements shall be met:

- 1 The surface layer steel reinforcements should adopt the welded mesh, the diameter of

steel reinforcement should not be larger than 8mm and the spacing of them shall not be larger than 150mm; the meshes shall be set at the bottom and sides of the beam, those steel mesh reinforcements at beam sides shall be extended up to $2/3$ of the beam depth.

2 The sectional area of the surface layer steel mesh reinforcements in two directions both shall not be less than 1% of the area of corresponding concrete cover (the shaded area in Figure 9.2.15).

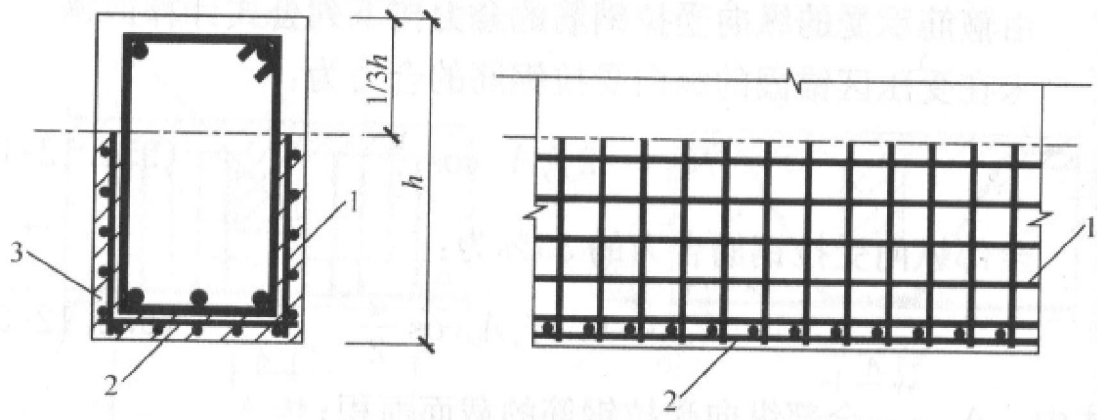


Figure 9.2.15 Detailing Requirements for Applying Surface Layer Steel Mesh Reinforcements

1—Surface layer steel mesh reinforcements at beam side; 2—Surface layer steel mesh reinforcements at beam bottom; 3—Area in which steel mesh reinforcements are applied

9.2.16 The design of deep flexural member shall comply with those specified in Appendix G of this code.

9.3 Columns, Beam-column Joints and Brackets

(I) Columns

9.3.1 Arrangement of longitudinal steel reinforcements in column shall meet the following requirements:

1 The diameter of longitudinal stressed steel reinforcement should not be less than 12mm; the ratio of reinforcement of all the longitudinal steel reinforcements should not be larger than 5%;

2 The clear spacing of longitudinal steel reinforcements in column shall not be less than 50mm and should not be larger than 300mm;

3 Where the depth of section of the eccentric compression column is not less than 600mm, the longitudinal constructional steel reinforcements in diameter no less than 10mm shall be applied on the side face of column and also shall be set with compound stirrups or tie bars correspondingly;

4 The quantity of longitudinal steel reinforcements applied in circular column should not be less than 8 but shall not be less than 6, and these steel reinforcements should be uniformly arranged along the periphery of the column;

5 In eccentric compression column, the center-to-center distance of these longitudinal stressed steel reinforcements on the side face perpendicular to the acting plane of bending moment as well as of these longitudinal stressed steel reinforcements on each side in the axial

compression column should not be larger than 300mm.

Note: In the horizontally casted precast column, the minimum clear spacing of longitudinal steel reinforcements may be adopted in accordance with the relevant provisions on beam as stated in Article 9.2.1 of this code.

9.3.2 Stirrups in column shall meet the following requirements:

1 The diameter of stirrup shall not be less than $d/4$ (d is the maximum diameter of longitudinal steel reinforcement) or 6mm;

2 The stirrup spacing shall not be larger than 400mm or the size of the short side of member section and also shall not be larger than $15d$ (d is the minimum diameter of longitudinal steel reinforcement);

3 The peripheral stirrups in column and other compression members shall be made into closed type; for the stirrups in circular column, their overlapping length shall not be less than the anchorage length stated in Article 8.3.1 of this code, and their tail ends shall be made into 135° hooks, the length of the straight section at the tail end of hook shall not be less than $5d$ (d is the diameter of stirrup);

4 Compound stirrups shall be arranged in case the size of the short side of column section is larger than 400mm and the quantity of longitudinal steel reinforcements on each side is larger than 3 or the size of the short side of column section is not larger than 400mm but the quantity of longitudinal steel reinforcements on each side is larger than 4;

5 Where the ratio of reinforcement of all longitudinal stressed steel reinforcements is larger than 3%, the diameter of stirrup shall not be less than 8mm, the stirrup spacing shall not be larger than $10d$ or 200mm. The tail end of stirrup shall be made into 135° hook and the length of the straight section at tail end of this hook shall not be less than $10d$ (d is the minimum diameter of longitudinal stressed steel reinforcement);

6 In column in which spiral stirrups or welding ring stirrups are applied, if the action of indirect steel reinforcements is considered in the calculation of the compression load-bearing capacity of normal section, the stirrup spacing shall not be larger than 80mm or $d_{cor}/5$ (d_{cor} is the diameter of core section determined according to the internal surface of stirrup) and should not be less than 40mm.

9.3.3 For I-shaped section column, the flange thickness should not be less than 120mm and the web thickness should not be less than 100mm. If the web has opening, 2~3 strengthening steel reinforcements in diameter no less than 8mm should be applied at the periphery of the opening, the sectional area of strengthening steel reinforcements in each direction should not be less than the sectional area of those steel reinforcements cut in this direction.

For the I-shaped section column having opening in its web, if the transverse dimension of the opening is less than one half of the depth of column section and the vertical dimension of the opening is less than the clear spacing between two adjacent openings, the rigidity of column may be calculated according to the solid web I-shaped section column, however, the weak part of the opening shall be deducted in the calculation of load-bearing capacity. If the opening size exceeds the above-mentioned requirements, the rigidity and load-bearing capacity of column shall be calculated according to double-limb column.

(II) Beam-column Joints

9.3.4 The anchorage of longitudinal steel reinforcements of beam at the end nodes at middle

layer of frame shall meet the following requirements:

- 1 Anchorage of the upper longitudinal steel reinforcements of beam into the node:
 - 1) Where the linear anchoring form is adopted, the anchorage length shall not be less than l_a and the steel reinforcement shall be extended over the center line of column, the length extending over the center line should not be less than $5d$ (d is the diameter of the upper longitudinal steel reinforcement of beam).
 - 2) Where the sectional dimension of column fails to meet the requirements of linear anchoring, the upper longitudinal steel reinforcements of beam may be adopted with the anchorage mode by adding mechanical anchor heads at the ends of steel reinforcements as stated in Article 8.3.3 of this code. The upper longitudinal steel reinforcements of beam should be extended into the inner side of those outer longitudinal steel reinforcements of column, the anchorage length of their horizontal projection, including the mechanical anchor heads, shall not be less than $0.4l_{ab}$ (Figure 9.3.4a).
 - 3) The upper longitudinal steel reinforcements of beam may also adopt the anchorage mode of 90° bending, by which, they shall be extended into the inner side of the longitudinal steel reinforcements outside the beam and shall be bent inwards the node, their horizontal projection length, including the arc, shall not be less than $0.4l_{ab}$, and the length of the projection of bent steel reinforcement (including the arc segment) in the bending plane shall not be less than $15d$ (Figure 9.3.4b).

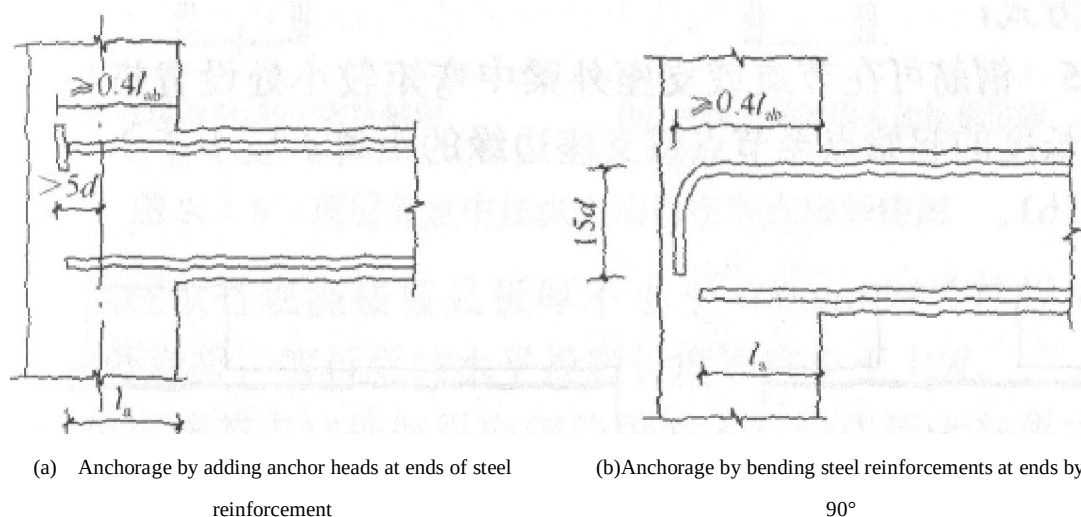


Figure 9.3.4 Anchorage of Upper Longitudinal Steel Reinforcements of Beam in End Nodes at Middle Layer

- 2 Anchorage of the lower longitudinal steel reinforcements of beam into the end node:
 - 1) Where the tensile strength of steel reinforcements is fully used in calculation, the anchorage mode and length of steel reinforcements shall be same as the requirements for upper steel reinforcements.
 - 2) Where the strength of this steel reinforcement is not used or only its compressive strength is used in calculation, the anchorage length extending into the node shall meet the requirements on anchorage of lower longitudinal steel reinforcements of beam at the intermediate nodes as stated in Article 9.3.5

of this code.

9.3.5 For the intermediate nodes at middle layer of frame or the intermediate support of continuous beam, the upper longitudinal steel reinforcements of beam shall penetrate through the node or support. The lower longitudinal steel reinforcements of beam should penetrate through the node or point. If they must be anchored, the following anchorage requirements shall be met:

1 Where the strength of this steel reinforcement is not used in calculation, the anchorage length of this steel reinforcement extended into the node or support shall not be less than $12d$ for ribbed steel reinforcement and not be less than $15d$ for plain steel reinforcement (d is the maximum diameter of steel reinforcement);

2 Where the compressive strength of steel reinforcements is fully used in calculation, the steel reinforcements shall be anchored in the intermediate nodes or intermediate supports according to compression steel reinforcement, and their linear anchorage length shall not be less than $0.7l_a$;

3 Where the tensile strength of steel reinforcements is fully used in calculation, the steel reinforcements may be anchored in the nodes or supports in linear anchoring mode, their anchorage length shall not be less than the tension anchorage length l_a of steel reinforcement (Figure 9.3.5a);

4 Where the sectional dimension of column is insufficient, the mechanical anchoring measure by adding anchor heads at the ends of steel reinforcements should be adopted in accordance with those specified in Clause 1 of Article 9.3.4 of this code or the anchorage mode of 90° bending may also be adopted;

5 The steel reinforcements may be set with overlapping joints outside the node or support and at places where bending moment in the beam is small, the distance from the starting point of their overlapping length to the edge of node or support shall not be less than $1.5h_0$ (Figure 9.3.5b).

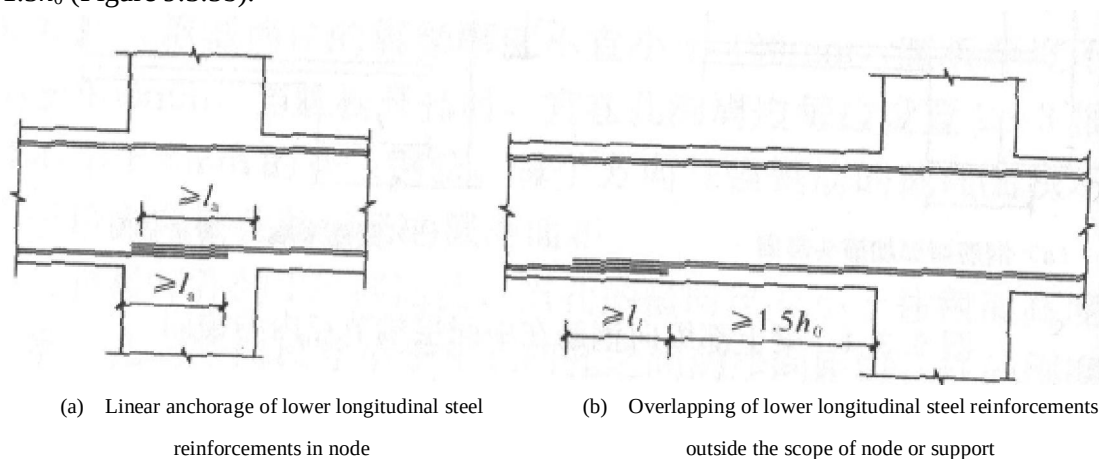


Figure 9.3.5 Anchorage and Overlapping of Lower Longitudinal Steel Reinforcements of Beam within Scope of Intermediate Node or Intermediate Support

9.3.6 The longitudinal steel reinforcements of column shall penetrate through the intermediate node at middle layer or the end node, and the joint shall be set outside the node zone.

The anchorage of longitudinal steel reinforcements of column at the intermediate node of top layer shall meet the following requirements:

1 The longitudinal steel reinforcements of column shall be extended up to the column top, and the anchorage length, counted from the beam bottom, shall not be less than l_a .

2 If the sectional dimension fails to meet the requirements on linear anchorage, the anchoring measure of 90° bending may be adopted. Here, the anchorage length by the vertical projection of steel reinforcement (including the arc) shall not be less than $0.5l_{ab}$, and the horizontal projection length in the bending plane (including the arc segment) should not be less than $12d$ (Figure 9.3.6a).

3 If the sectional dimension is insufficient, the mechanical anchoring measure with anchor head may also be adopted, in this case, the vertical anchorage length with anchor head included shall not be less than $0.5l_{ab}$ (Figure 9.3.6b).

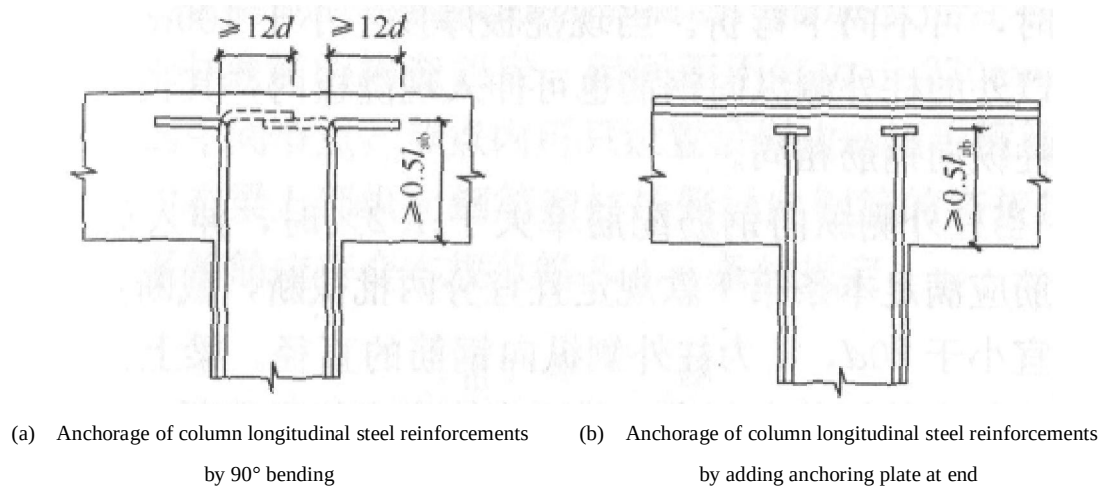


Figure 9.3.6 Anchorage of Column Longitudinal Steel Reinforcements in the Top Layer of Nodes

4 Where cast-in-situ floor slab is applied at the column top and the slab thickness is not less than 100mm, the longitudinal steel reinforcements of column may also be bent outward and their horizontal projection length after bending should not be less than $12d$.

9.3.7 The outer longitudinal steel reinforcements of column at the top layer of end node may be bent into the beam to act as the upper longitudinal steel reinforcements of the beam; and the upper longitudinal steel reinforcements of beam may also be overlapped with the outer longitudinal steel reinforcements of column at the nodes and the nearby positions, the following overlapping manners may be adopted:

1 The overlapping joints may be arranged along the outer side of the top layer end node and the top of beam end, and the overlapping length shall not be less than $1.5l_{ab}$ (Figure 9.3.7a). Therein, the sectional area of those outer steel reinforcements of column extended into the beam should not be less than 65% of the total area; the outer steel reinforcements of column beyond the scope of beam width should be anchored along the top of node till the inner side of column. Where the outer longitudinal steel reinforcements of column are located at the first layer of column top, they should, after being extended into the inner side of column, should be bent down by no less than $8d$ (d is the diameter of column longitudinal steel reinforcement) and be cut off (Figure 9.3.7a); where they are located at the second layer of column top, they may not be bent down. If the thickness of cast-in-situ slab is not less than 100mm, the outer longitudinal steel reinforcements of column beyond the beam width may also be inserted into the cast-in-situ slab and the length is same as that of the column longitudinal steel reinforcements extended into the beam.

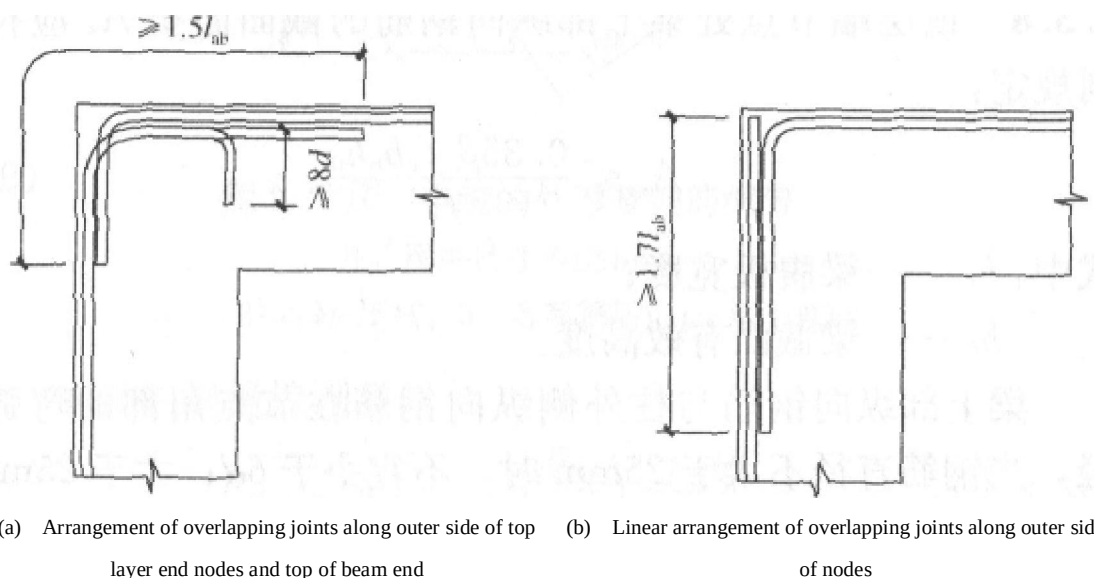


Figure 9.3.7 In-node Anchorage and Overlapping of Longitudinal Steel Reinforcements of Beam and Column at Top Layer of End Nodes

2 In case the ratio of reinforcement of the outer longitudinal steel reinforcements of column is larger than 1.2%, the longitudinal steel reinforcements of column extended into the beam shall comply with those specified in Clause 1 of this article and should be cut off in two batches, the distance between cut points should not be less than $20d$ (d is the diameter of outer longitudinal steel reinforcement of column). The upper longitudinal steel reinforcements of beam shall be extended to the outer side of node, bent down and cut off at the depth position of the lower edge of beam.

3 The overlapping joints of longitudinal steel reinforcements may also be arranged linearly along the outer side of the column top at the node (Figure 9.3.7b), in this case, the overlapping length, counted from the column top, shall not be less than $1.7l_{ab}$. If the ratio of reinforcement of the upper longitudinal steel reinforcements of beam is larger than 1.2%, the upper longitudinal steel reinforcements of beam extended into the outer side of column shall comply with the overlapping length specified in Clause 1 of this article and should be cut off in two batches, the distance between their cut points should not be less than $20d$ (d is the diameter of upper longitudinal steel reinforcement of beam).

4 If the depth of beam section is relatively large, the longitudinal steel reinforcements of beam and column are relatively small, and the linear overlapping length counted from the beam bottom has already met the requirement on $1.5l_{ab}$ before extending up to the column top, thus the overlapping length shall be extended up to the column top and shall meet the requirement on the overlapping length of $1.7l_{ab}$; or, if the overlapping length by bending, counted from the beam bottom, has already met the requirement on $1.5l_{ab}$ before extending to the inner edge of column, thus the length of the horizontal segment of bent steel reinforcement, with the arc included, shall not be less than $15d$ (d is the diameter of longitudinal steel reinforcement of column).

5 The anchorage of inner longitudinal steel reinforcements of column shall meet the provisions on top layer intermediate nodes as given in Article 9.3.6 of this code.

9.3.8 The sectional area A_s of upper longitudinal steel reinforcements of beam on the top layer end node shall meet the following requirements:

$$A_s \leq \frac{0.35\beta_c f_c b_b h_0}{f_y} \quad (9.3.8)$$

Where:

b_b ——the width of the beam web;

h_0 ——the effective depth of beam section.

If the diameter of steel reinforcement is not larger than 25mm, the internal radius of the arc of upper longitudinal steel reinforcements of beam or outer longitudinal steel reinforcements of column at the corner of the node should not be less than $6d$; otherwise, this internal radius should not be less than $8d$. In the concrete outside the arc of steel reinforcement, anti-cracking and anti-peeling constructional steel reinforcements shall be distributed.

9.3.9 Horizontal stirrups shall be distributed in the nodes of frame, these stirrups shall meet the detailing requirements on stirrups in column as stated in Article 9.3.2 of this code, however, their spacing should not be larger than 250mm. As for the intermediate node with beams at four sides, only the rectangular stirrups may be applied along the periphery in this node. In case the overlapping joints of upper longitudinal steel reinforcements of beam with outer longitudinal steel reinforcements of column exist in the top layer end node, the horizontal stirrups in the node shall meet those specified in Article 8.4.6 of this code.

(III) Brackets

9.3.10 The sectional dimension of the column bracket (Figure 9.3.10) with $a \leq h_0$ shall meet the following requirements:

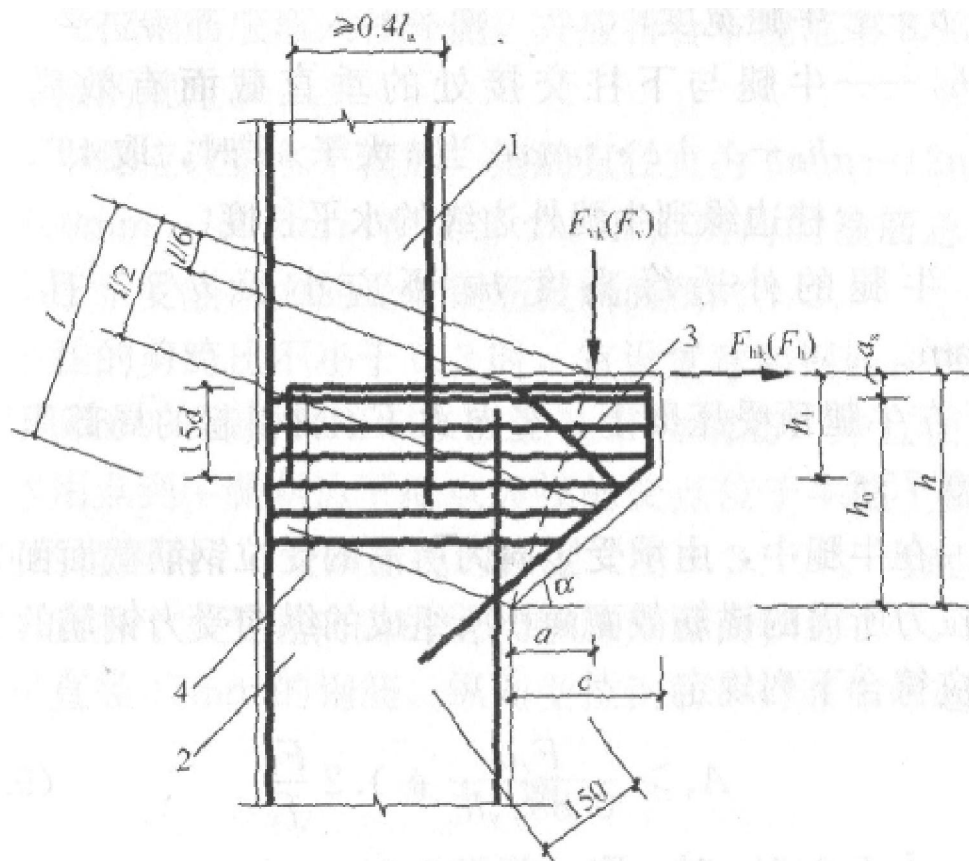


Figure 9.3.10 Appearance and Steel Reinforcement Arrangement of Bracket

Note: Dimensional unit in the Figure is mm.

1—Upper column; 2—Lower column; 3—Bent-up steel reinforcement; 4—Horizontal stirrup

1 Requirements for crack control of bracket

$$F_{vk} \leq \beta \left(1 - 0.5 \frac{F_{hk}}{F_{vk}} \right) \frac{f_{tk} b h_0}{0.5 + \frac{a}{h_0}} \quad (9.3.10)$$

Where:

F_{vk} ——the value of vertical force acting on top of the bracket and calculated according to the characteristic combination of load effects;

F_{hk} ——the value of horizontal tensile force acting on top of the bracket and calculated according to the characteristic combination of load effects;

β ——the crack control coefficient, which is 0.65 for the bracket supporting the crane beam and is 0.80 for other brackets;

a ——the horizontal distance from the action point of vertical force to the edge of lower column, for which the installation deviation of 20mm shall be considered; a equals to zero in case the action point of vertical force with consideration of installation deviation is still in the section of lower column;

b ——the width of bracket;

h_0 ——the effective depth of vertical section at the connecting position of bracket and lower column, $h_0 = h_1 - a_s + c \cdot \tan \alpha$, in which, α will be taken as 45° if $a > 45^\circ$, and c is the horizontal length from the edge of lower column to the outer edge of bracket.

2 The depth h_1 of the outer edge of bracket shall not be less than $h/3$ or 200mm.

3 The local compression stress caused by vertical force F_{vk} on the top compression face of bracket shall not exceed $0.75f_c$.

9.3.11 In the bracket, the gross sectional area of longitudinal stressed steel reinforcements, which is composed of the sectional area of the tension steel reinforcements required to bear the vertical force and the sectional area of the anchor bars required for bearing the horizontal tensile force, shall meet the following requirements:

$$A_s \geq \frac{F_v a}{0.85 f_y h_0} + 1.2 \frac{F_h}{f_y} \quad (9.3.11)$$

Here, a will be taken as $0.3h_0$ in case $a < 0.3h_0$.

Where:

F_v ——the design value of vertical force acting at top of bracket;

F_h ——the design value of horizontal tensile force acting at top of bracket.

9.3.12 The longitudinal stressed steel reinforcements distributed along the top of bracket should adopt the hot-rolled ribbed steel bars of Grade HRB400 or Grade HRB500. All the longitudinal stressed steel reinforcements and bent-up steel reinforcements should be cut off after being extended down into the lower column by 150mm along the outer edge of bracket.

The anchorage length of longitudinal stressed steel reinforcements and bent-up steel reinforcements extended into the upper column shall not be less than the anchorage length l_a of tension steel reinforcements as specified in Article 8.3.1 of this code if the linear anchorage mode is adopted; in case the dimension of upper column is insufficient, the anchorage of steel reinforcements shall comply with the provisions in Article 9.3.4 of this code on the anchorage by 90° bending of upper steel reinforcements of beam in the middle layer end nodes of frame. In this case, the anchorage length shall be counted from the inner side of upper column.

The ratio of reinforcement of those longitudinal stressed steel reinforcements required for bearing the vertical force shall not be less than 0.20% or $0.45f_t/f_y$ and still should be larger than 0.60%, the quantity of steel reinforcements should not be less than 4 and the diameter of steel reinforcement shall be 12mm.

When the bracket is set at the top of upper column, the outer longitudinal stressed steel reinforcements of column at opposite side to the bracket may be horizontally bent into the bracket along the column top to act the longitudinal tension steel reinforcements of bracket. When the top longitudinal tension steel reinforcements of bracket are applied separately from the outer longitudinal steel reinforcements of column at opposite side to the bracket, they shall be bent into the outer side of the column and also shall meet provisions in Article 8.4.4 of this code on the overlapping of steel reinforcements.

9.3.13 The bracket shall be distributed with horizontal stirrups, the diameter of stirrup should be from 6mm to 12mm and the spacing of stirrups should be from 100mm to 150mm; the gross sectional area of those stirrups within a scope of $2h_0/3$ at the upper portion should not be less than 1/2 of the sectional area of those tension steel reinforcements bearing the vertical force.

Where the ratio of shear span to effective depth of bracket is not less than 0.3, proper number of bent-up steel reinforcements should be distributed. The bent-up steel reinforcements should adopt the hot-rolled ribbed steel bars of Grade HRB400 or Grade

HRB500, their intersection point with the connecting line from the action point of concentrated load to the lower extreme point of sloped edge of bracket should be located in a scope of $l/6 \sim l/2$ at the upper portion of bracket (l is the length of this connecting line) (Figure 9.3.10). The sectional area of bent-up steel reinforcements should not be less than 1/2 of the sectional area of the tension steel reinforcements bearing vertical force or that of two steel reinforcements with diameter of 12mm. The longitudinal tension steel reinforcements shall not be used as bent-up steel reinforcements.

9.4 Walls

9.4.1 When the ratio between the long side and short side (thickness) of the section of vertical member is larger than 4, this member should be designed according to the requirements of wall.

The thickness of the wall supporting precast floor (roof) slab should not be less than 140mm; for the shear wall structure, the thickness still should not be less than 1/25 of the storey height; and for the frame-shear wall structure still should not be less than 1/20 of the storey height.

Where the precast slab is adopted, the thickness of supporting wall shall meet the requirements on the penetration of vertical steel reinforcements in wall.

9.4.2 The wall with thickness larger than 160mm shall be arranged with double rows of distribution steel mesh reinforcements; for the shear wall at important position in the structure, if its thickness is not larger than 160mm, it also shall be arranged with double rows of distribution steel mesh reinforcements.

These double rows of distribution steel mesh reinforcements shall be arranged along two side faces of the wall and shall be connected by adopting with tie bars; the diameter of tie bar should not be less than 6mm and the spacing of tie bars should not be larger than 600mm.

9.4.3 Under the action of horizontal loads and vertical loads that are parallel to the wall surface, the wall shall be carried out with the calculation of load-bearing capacity of normal section respectively according to eccentric compression or eccentric tension based on the internal force obtained through structural analysis as well as the relevant provisions in Section 6.2 of this code, and also shall be carried out with the calculation of shear bearing capacity of inclined section according to the relevant provisions in Section 6.3 of this code. At the acting place of concentrated load, the calculation of partial load-carrying capacity still shall be carried out in accordance with Section 6.6 of this code.

In calculation of load-bearing capacity, the calculated width of the flange of shear wall may take the minimum value among the spacing of shear walls, the width of wing wall between doors and windows, the thickness of shear wall plus six times of the thickness of wing walls on both sides as well as the 1/10 of the total height of shear wall columns.

9.4.4 For the horizontally and vertically distributed steel reinforcements of wall, their diameter should not be less than 8mm and their spacing should not be larger than 300mm. The steel reinforcements may be distributed by using welded steel mesh reinforcement.

In the wall, both the ratio of reinforcement $\rho_{sh} \left(\frac{A_{sh}}{bs_v} \right)$, s_v is the spacing of horizontally

distributed steel reinforcements) of the horizontally distributed steel reinforcements in wall and the ratio of reinforcement ρ_{sv} ($\frac{A_{sv}}{bs_h}$, s_h is the spacing of vertically distributed steel

reinforcements) of vertically distributed steel reinforcements should not be less than 0.20%; in the walls at important positions, the ratio of reinforcement of the horizontally and vertically distributed steel reinforcements should be increased appropriately.

At the positions with large temperature and shrinkage stress in the wall, the ratio of reinforcement of horizontally distributed steel reinforcements should be increased appropriately.

9.4.5 For the wall with height of building no larger than 10m and the storey number no larger than 3, the thickness of its section shall not be less than 120mm, and the ratio of reinforcement of both the horizontally and vertically distributed steel reinforcements in it should not be less than 0.15%.

9.4.6 The reinforcement detailing in wall shall meet the following requirements:

1 The vertically distributed steel reinforcements of the wall may be overlapped at the same height, and the overlapping length shall not be less than $1.2l_a$.

2 The overlapping length of the horizontally distributed steel reinforcements of wall shall not be less than $1.2l_a$. The clear spacing along the horizontal direction between the overlapping joints of the same row of horizontally distributed steel reinforcements as well as that between the overlapping joints of the up and down adjacent horizontally distributed steel reinforcements should not be less than 500mm.

3 The horizontally distributed steel reinforcements in the wall shall be extended to the wall ends and be horizontally bent inward by $10d$ (d is the diameter of steel reinforcement).

4 For with wall having wing wall or corner at its end, the horizontally distributed steel reinforcements on both sides of the inner wall and on inner side of the outer wall shall be extended beyond the wing wall or corner and shall be horizontally bent by $15d$ towards both sides respectively. At the corner wall, the horizontally distributed steel reinforcements on outer side of outer wall shall be bent into the wing wall at the exterior angle of wall end and be overlapped with the horizontally distributed steel reinforcements on outer side of wing wall.

5 For the wall with frame, the horizontally and vertically distributed steel reinforcements should penetrate through the column and beam respectively or be anchored in the column and beam.

9.4.7 The connecting beam at wall opening shall be arranged with stirrups along the full length, the diameter of stirrup shall not be less than 6mm and spacing of stirrups should not be larger than 150mm. In the scope of the anchorage length of the longitudinal steel reinforcements for connecting beam at top layer opening extended into the wall, the stirrups shall be distributed with a spacing no larger than 150mm, and the diameter of stirrup should be same as the diameter of those stirrups in the span. Meanwhile, the vertical steel reinforcements at the edges of the door and window openings shall meet the requirements on anchorage length of tension steel reinforcements.

The upper and lower horizontal steel reinforcements on both sides of wall opening shall meet the requirements on flexural load-bearing capacity of normal section of the connecting

beam for opening, and at least two steel reinforcements in diameter no less than 12mm shall be distributed. As for the openings that are negligible in calculation and analysis, the sectional area of steel reinforcements on edges of these openings should respectively not be less than one half of the gross sectional area of those horizontally distributed steel reinforcements cut off at the opening. The length of the longitudinal steel reinforcement extended from the opening edge into the wall shall not be less than the anchorage length of tension steel reinforcements.

9.4.8 Vertical stressed steel reinforcements shall be applied on both ends of the column of shear wall and be used for the calculation for the flexural load-bearing capacity of normal section of wall together with the vertically distributed steel reinforcements in the wall. The vertical stressed steel reinforcements at each end should not be less than four steel reinforcements in diameter of 12mm or 2 steel reinforcements in diameter 16mm, and should also distributed with stirrups or tie bars in diameter no less than 6mm by a spacing of 250mm along the direction of this vertical steel reinforcement.

9.5 Composite Members

(I) Horizontal Composite Members

9.5.1 The horizontal composite flexural members formed in Phase II shall be taken with reliable support during the construction stage if the height of precast member is less than 40% of the depth of total section.

The composite flexural member having reliable support during the construction stage may be designed and calculated according to entire flexural member, however, the shear bearing capacity of its inclined section and the shear bearing capacity of its composite surface shall be calculated in accordance with Appendix H of this code.

For the composite flexural member having no support during the construction stage, the bottom precast members and concreted composite members shall be carried out with loading calculation of Phase II according to the requirements stated in Appendix H of this code.

9.5.2 The concrete composite beam and slab shall meet the following requirements:

1 The thickness of composite layer concrete of composite beam should not be less than 100mm and the concrete strength grade should not be less than C30. All the stirrups of precast beam shall be extended into the composite layer, the length of the straight segment of each limb extending into the composite layer should not be less than $10d$ (d is the diameter of stirrup). The top surface of precast beam shall be built into such a rough surface with unevenness no less than 6mm.

2 The thickness of composite layer concrete of composite slab shall not be less than 40mm and the concrete strength grade should not be less than C25. The surface of precast slab shall be made into such a rough surface with unevenness no less than 4mm. For the composite slab bearing large load and the prestressed composite slab, constructional steel reinforcements extending into the composite layer shall be distributed on the precast bottom slab.

9.5.3 The flexural members concreted on the floor slabs and roofs of existing structures shall meet those specified in Article 9.5.2 of this code and shall be carried out with the calculation of construction stage and service stage in accordance with relevant provisions

stated in Section 3.3 and Section 3.7 of this code.

(II) Vertical Composite Members

9.5.4 Composite columns and walls formed with precast members and post-cast concrete shall be respectively carried out with the calculation of precast member and integral structure according to the working conditions during construction stage and service stage.

9.5.5 For the vertical composite members form by concreting at the periphery of column or on side face of wall in the existing structures, their load bearing history and the roof supporting conditions in construction shall be taken into account, and the calculation of load-bearing capacity for construction stage and service stage shall be carried out in accordance with the principles stated in Section 3.3 and Section 3.7 of this code.

9.5.6 Relying on the vertical composite column of existing structure, the calculation of load-bearing capacity of wall during the service stage shall take the influence of the changes in geometric parameters of existing members in accordance with the measured results.

The design values of strength of concrete and steel reinforcement for vertical composite column and existing members of wall shall be determined in accordance with Article 3.7.3 of this code; the strength of concrete and steel reinforcement in part of post-cast concrete shall determined by multiplying the reduction coefficient of strength utilization in accordance with those specified in Chapter 4 of this code and should be adjusted appropriately by considering the actual roof supporting conditions in construction.

9.5.7 The thickness of the secondary concrete layer over the column shall not be less than 60mm and the concrete strength grade shall not be less than the strength of existing columns. The unevenness of rough joint surface shall not be less than 6mm, and the constructional steel reinforcements shall be distributed in the interface by such methods as bar planting and welding. The diameter of the longitudinal stressed steel reinforcements in the post-cast layer shall not be less than 14mm; the diameter of stirrup shall not be less than 8mm or the diameter of corresponding stirrup in column, and the stirrup spacing shall be same as those in the column.

The thickness of the secondary concrete layer over the wall shall not be less than 50mm and the concrete strength grade shall not be less than the strength of existing walls. The unevenness of rough joint surface shall not be less than 4mm, and the constructional steel reinforcements shall be distributed in the interface by such methods as bar planting and welding. The diameter of vertical and horizontal steel reinforcements in the post-cast layer should not be less than 8mm and shall not be less than the diameter of corresponding steel reinforcement in wall.

9.6 Precast Concrete Structures

9.6.1 All the precast members and connection constructions in the precast concrete structures and assembled monolithic concrete structures shall be designed in accordance with the following principles:

1 The arrangement and connecting mode of precast members shall be determined in the structural scheme and force transmission route, based on which the analysis of integral

structure and the design of members and connection shall be carried out;

2 The design of precast members shall meet the functions of building and meet the standardization requirements;

3 The connection of precast members should be set at the part with small stress in the structure and should be convenient for construction; the connection construction between structural members shall meet the requirements on internal force transmission of structure;

4 Various precast members and their connection constructions shall be checked in accordance with the unfavorable working conditions that may be produced during the production, construction and service process, and those for the precast non-load-bearing members still shall meet those specified in Article 9.6.8 of this code.

9.6.2 During the production and construction process, the precast concrete members shall be carried out with checking of construction stage in accordance with the loads, calculation diagrams and strength of solid concrete of actual working conditions. In the checking, the deadweight of member shall be multiplied by the corresponding dynamic coefficient: this dynamic coefficient may be taken as 1.5 during demoulding, turning, hoisting and transportation and taken as 1.2 for temporary fixation.

Note: The dynamic coefficient may also be increased or decreased appropriately according to specific conditions.

9.6.3 The connection construction of various precast members in precast concrete structures and assembled monolithic concrete structures shall be convenient for the installation and monolithic assembly. For the connection without consideration of internal force transmission in the calculation also shall have reliable fixation measures.

9.6.4 The longitudinal stressed steel reinforcements of frame beam in assembled monolithic structure as well as the vertical stressed steel reinforcements in columns and walls should adopt such forms as mechanical connection and welding; the stressed steel reinforcements in such members as slabs and walls may adopt the overlapping connection form; the concrete joint surface shall be carried out with roughing treatment or made into groove; the splicing points shall be grouted with such concrete with strength grade no less than that of precast member.

At the beam-column joints in assembled monolithic structures, the longitudinal steel reinforcements of columns shall penetrate through the node; the longitudinal steel reinforcements of beams shall meet the anchorage requirements stated in Section 9.3 of this code.

Where the columns are adopted with assembled joggle joints, the axial compression load-bearing capacity of the section in the sector near to the joint should be 1.3~1.5 times of the load-bearing capacity required in the calculation of this section. On this occasion, measures, like adding transverse steel mesh reinforcements in the concrete of joints and the sectors nearby, improving the post-placed concrete strength grade and distributing additional longitudinal steel reinforcements, shall be taken.

9.6.5 The assembled monolithic floor and roof adopting with precast slab shall be taken with the following constructional measures:

1 Precast slabs shall be made into double-geared edges; the upper opening width of the edge joint shall not be less than 30mm; plugs shall be set in the end holes of hollow slabs and the depth of plug in the hole should not be less than 60mm; and the edge joint shall be grouted with fine stone concrete with strength grade no less than C30;

2 The ends of precast slabs should be extended out and be connected with the anchored steel reinforcements, and also should be spliced with the steel reinforcements extended out of the supporting structures (ring beam, beam top or wall crown) of slabs and the full-length steel reinforcements distributed in the edge joints at slab ends.

9.6.6 For the assembled monolithic floor and roof having high requirements on integrity, the form of precast members with cast-in-situ composite layers shall be adopted; or, the reinforced concrete post-placed strip shall be arranged one side of precast slab, negative moment steel reinforcements shall be applied at the slab ends and steel tie bars shall be distributed along the edge joint at periphery of slab to connect with the support.

9.6.7 In assembled monolithic structures, the connecting steel reinforcement distributed along the periphery of precast load-bearing wall slab shall be connected with the supporting structure and the adjacent wall slab and also shall be concreted to form into one entirety with the peripheral floors and walls.

9.6.8 The design of non-load-bearing precast members shall meet the following requirements:

1 The non-load-bearing precast members should adopt flexible connection with the supporting structures;

2 When inlaying the non-load-bearing precast members in frames or adopting welded connection, their influence on the lateral stiffness of frame shall be taken in account;

3 The connection construction between cladding panel and the major structure shall have a certain deformation adaptability.

9.7 Embedded Parts and Connecting Pieces

9.7.1 The anchoring plates of stressed embedded parts should adopt steel of Grade Q235 or Q345, the thickness of anchoring plate shall be determined through calculation according to the stressing condition and should not be less than 60% of the diameter of anchor bar; the thickness of the anchoring plates of tension or flexural embedded parts still should be larger than $b/8$ (b is the spacing of anchor bars).

The anchor bars of stressed embedded parts shall adopt the HRB400 or HPB300 steel reinforcements but shall not adopt the cold-processed steel reinforcements.

Straight anchor bars and anchoring plates shall adopt the T-type welding. If the diameter of anchor bar is not larger than 20mm, the submerged arc pressure welding shall be adopted; otherwise, the perforated plug welding should be adopted. Where the manual welding is adopted, the weld bead height should not be less than 6mm, and also should not be less than $0.5d$ for steel reinforcements of Grade 300MPa and should not be less than $0.6d$ for other steel reinforcements (d is the diameter of anchor bar).

9.7.2 For the stressed embedded parts composed of anchoring plates and symmetrically arranged straight anchor bars (Figure 9.7.2), the gross sectional area A_s of anchor bars shall meet the following requirements:

1 When shear force, normal tension and bending moment coact, the gross sectional area shall be calculated according to the following two formulae and be taken as the larger value among the calculated ones:

$$A_s \geq \frac{V}{\alpha_r \alpha_v f_y} + \frac{N}{0.8 \alpha_b f_y} + \frac{M}{1.3 \alpha_r \alpha_b f_y z} \quad (9.7.2-1)$$

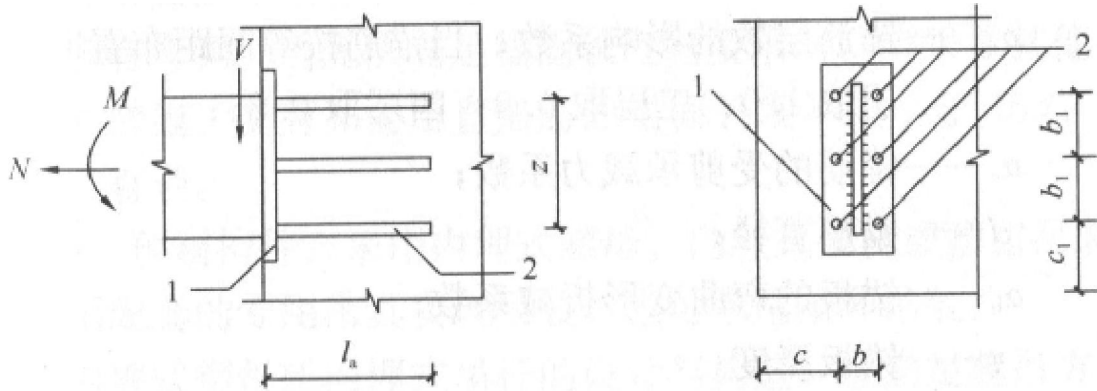


Figure 9.7.2 Embedded Parts Composed of Anchoring Plates and Straight Anchor Bars

1—Anchoring plate; 2—Straight anchor bar

$$A_s \geq \frac{N}{0.8 \alpha_b f_y} + \frac{M}{0.4 \alpha_r \alpha_b f_y z} \quad (9.7.2-2)$$

2 When shear force, normal pressure and bending moment coact, the gross sectional area shall be calculated according to the following two formulae and be taken as the larger value among the calculated ones:

$$A_s \geq \frac{V - 0.3N}{\alpha_r \alpha_v f_y} + \frac{M - 0.4Nz}{1.3 \alpha_r \alpha_b f_y z} \quad (9.7.2-3)$$

$$A_s \geq \frac{M - 0.4Nz}{0.4 \alpha_r \alpha_b f_y z} \quad (9.7.2-4)$$

When $M < 0.4Nz$, $M = 0.4Nz$.

Coefficients α_v and α_b in the above-mentioned formulae shall be calculated according to the following formulae:

$$\alpha_v = (4.0 - 0.08d) \sqrt{\frac{f_c}{f_y}} \quad (9.7.2-5)$$

$$\alpha_b = 0.6 + 0.25 \frac{t}{d} \quad (9.7.2-6)$$

When $\alpha_v > 0.7$, $\alpha_v = 0.7$; and $\alpha_b = 1.0$ if the measures for preventing the flexural deformation of anchoring plates are taken.

Where:

f_y —The design value of tensile strength of anchor bar, which shall be adopted in accordance with Section 4.2 of this code and shall not be larger than 300N/mm^2 ;

V —the design value of shear force;

N —the design value of normal tension or normal pressure, the design value of normal pressure shall not be larger than $0.5f_c A$, hereof, A is the area of anchoring plate;

M —the design value of bending moment;

α_r ——the influence coefficient of the number of layers of anchor bars; when the anchor bars are arranged with equal spacing: $\alpha_r=1.0$ for two layers; $\alpha_r=0.9$ for three layers; $\alpha_r=0.85$ for four layers;

α_v ——the shear bearing capacity coefficient of anchor bar;

d ——the diameter of anchor bar;

α_b ——the reduction coefficient for flexural deformation of anchoring plate;

t ——the thickness of anchoring plate;

z ——the distance between center lines of the anchor bars at the outermost layer along the acting direction of shear force.

9.7.3 For the embedded parts of which the anchoring plate, symmetrically arranged bent anchor bar and blind anchor bar bear the shear force jointly (Figure 9.7.3), the sectional area A_{sb} of the bent anchor bar shall meet the following requirements:

$$A_{sb} \geq 1.4 \frac{V}{f_y} - 1.25 \alpha_v A_s \quad (9.7.3)$$

Where, coefficient α_v is adopted in accordance with Article 9.7.2 of this code. When the straight anchor bar is distributed according to the detailing requirements, A_s shall be taken as 0.

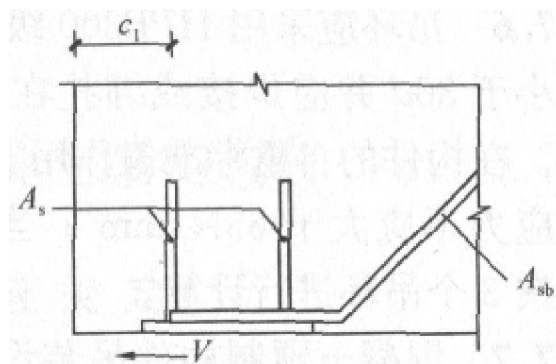


Figure 9.7.3 Embedded Parts Composed of Anchoring Plates, Bent Anchor Bars and Straight Anchor Bars

Note: The included angle between bent anchor bar and steel plate should not be less than 15° but not larger than 45°.

9.7.4 For the embedded parts, the distance from the center of anchor bar to the edge of anchoring plate shall not be less than $2d$ or 20mm. The embedded part shall be located with its anchor bar being at the inner side of the outer layer main steel reinforcements of the member.

The diameter of the stressed straight anchor bars for embedded parts should not be less than 8mm but not be larger than 25mm. No more than four straight anchor bars, namely, no more than four rows of straight anchor bars, should be adopted; and two straight anchor bars may be adopted for the shear embedded parts.

As for the tension and flexural embedded parts (Figure 9.7.2), the spacing b and b_1 of their anchor bars and the distance c and c_1 from the anchor bar to the edge of member all shall not be less than $3d$ or 45mm.

For shear embedded parts (Figure 9.7.2), the spacing b and b_1 of their anchor bars shall not be larger than 300mm, and b_1 shall not be less than $6d$ or 70mm; the distance c_1 from the anchor bar to the edge of member shall not be less than $6d$ or 70mm, and both b and c shall not be less than $3d$ or 45mm.

The anchorage length of tension straight anchor bar and bent anchor bar shall not be less

than the anchorage length of tension steel reinforcements at stated in Article 8.3.1 of this code; when the anchor bars adopt the steel reinforcements of Grade HPB300, their tail ends still shall be made into hooks. If the requirements for anchorage length can not be met, other effective anchoring measures shall be taken. The anchorage length of shear and compression straight anchor bars shall not be less than $15d$ (d is the diameter of anchor bar).

9.7.5 The precast members should be adopted with embedded nuts, embedded hanger rod or prepared lifting hole as well as the mating special lifting tools to realize the hoisting, they may also be adopted with hanging rings for the hoisting.

The design and construction of embedded nut or embedded hanger rod shall meet the requirements on convenience and safety of hoisting. The special embedded nuts or embedded hanger rods and the mating lifting tools shall be selected in accordance with the corresponding product standards and the requirements of application technologies.

9.7.6 The hanging rings shall be manufactured with HPB300 steel reinforcement, it shall be anchored into concrete by a depth no less than $30d$ and shall be welded or banded onto the framework of steel reinforcement, here, d is the diameter of steel reinforcement for hanging ring. Under the action of the standard deadweight value of the member, the reinforcement stress of each hanging ring calculated according to two sections shall not be larger than 65N/mm^2 ; where four hanging rings are designed on one member, the reinforcement stress shall be calculated according to three hanging rings.

9.7.7 The position of the hoisting facilities for precast concrete member shall be able to ensure the steady stressing of member during the hoisting and transportation process. When arranging embedded parts, hanging rings, lifting holes and various embedded type prepared lifting tools, the effect of hoisting loads bear by the member at these positions shall be carried out with the checking of load-bearing capacity and corresponding constructional measures also shall be taken so as to avoid the partial failure of concrete at the lifting point.

10 Prestressed Concrete Structural Members

10.1 General

10.1.1 For prestressed concrete structural members, the calculation of load-bearing capacity and checking of serviceability limit states shall not only be carried out according to the design condition, the checking in construction stage shall also be carried out.

10.1.2 The prestress action effect shall be included in the design of prestressed concrete structure; for hyperstatic structure, the corresponding secondary bending moment, secondary shear force and secondary axial force shall be included in the combination calculation.

For ultimate limit states, if the prestress action effect is favourable for the structure, the partial coefficient of the prestress action γ_p shall take 1.0, if it is unfavorable, γ_p shall take 1.2; for serviceability limit states, the partial coefficient of the prestress action γ_p shall take 1.0.

For the item included in the combination of prestress action effect, if the prestress action effect is favourable for the load-bearing capacity, the significance coefficient of structure γ_0 shall take 1.0; if the prestress action effect is unfavorable for the load-bearing capacity, the significance coefficient of structure γ_0 shall be determined according to Article 3.3.2 of this code.

10.1.3 The tension control stress of the prestressing tendon and/or bar σ_{con} shall meet the following requirements:

1 Stress-relieved steel wire and steel strand

$$\sigma_{con} \leq 0.75f_{ptk} \quad (10.1.3-1)$$

2 Medium-strength prestressed wire

$$\sigma_{con} \leq 0.70f_{ptk} \quad (10.1.3-2)$$

3 Prestressed twisted steel reinforcement

$$\sigma_{con} \leq 0.85f_{pyk} \quad (10.1.3-3)$$

Where:

f_{ptk} ——the standard value of ultimate strength for the prestressing tendon and/or bar;

f_{pyk} ——the standard value of yield strength for the prestressed twisted steel reinforcement.

The value for the tension control stress of the stress-relieved steel wire, steel strand and the medium-strength prestressed wire shall not be less than $0.4f_{ptk}$; the value for the tension stress control of the prestressed twisted steel reinforcement should not be less than $0.5f_{pyk}$.

The limit value of the above-mentioned tension control stress may be proportionately increased $0.05f_{ptk}$ or $0.05f_{pyk}$ if one of the following conditions is met:

- 1) The prestressing tendon and/or bar are required to arrange in the compression zone at service stage in order to improve the crack resistance of members in construction stage;
- 2) The loss of prestress caused by such factors as the stress relaxation, friction, the tension of the steel reinforcement in batches and the temperature difference between the prestressing tendon and/or bar and the stretching bed is required to counteract partially.

10.1.4 When applying the prestress, the required concrete cube compressive strength shall be determined after calculation, but it should not be less than 75% of the designed concrete strength grade value.

Note: When the tensioning of the prestressing tendon and/or bar is to prevent the shrinkage crack on the concrete appeared in the early stage, it may be free from the above-mentioned restriction, but it shall meet the requirements of partial compression load-bearing capacity.

10.1.5 For the hyperstatic structure in post-tensioned prestressed concrete, the elastic theory may be adopted for the analysis of the internal force and deformation caused by the prestress and the following requirements should be met:

1 The secondary bending moment M_2 should be calculated according to the following formulae if the calculation is based on the elastic analysis:

$$M_2 = M_r - M_1 \quad (10.1.5-1)$$

$$M_1 = N_p e_{pn} \quad (10.1.5-2)$$

Where:

N_p ——The pre-applied force on the post-tensioned prestressed concrete member shall be calculated according to Formula (10.1.7-3) of this code;

e_{pn} ——the distance from the centroid of the net section to the action point of the pre-applied force, calculated according to Formula (10.1.7-4) of this code;

M_1 ——the value for the bending moment of the eccentricity of the net section centroid caused by the pre-applied force N_p ;

M_r ——the value for the bending moment on the section of the structural member caused by the equivalent load of the pre-applied force N_p .

The secondary shear force may be calculated according to the distribution analysis for the secondary bending moment of the member, and the secondary axial force should be calculated according to the constraint condition of the structure.

2 Measures should be taken in the design to avoid or reduce the adverse effect of such constraint members as supports, columns and walls on the prestress action effect of beams and slabs.

10.1.6 The concrete normal stress caused by the pre-applied force and the stress of the prestressing tendon and/or bar in the corresponding stage may be calculated according to the following formulae respectively:

1 Pre-tensioned member

The concrete normal stress caused by the pre-applied force

$$\sigma_{pc} = \frac{N_{p0}}{A_0} \pm \frac{N_{p0} e_{p0}}{I_0} y_0 \quad (10.1.6-1)$$

The effective prestress of the prestressing tendon and/or bar at corresponding stage

$$\sigma_{pe} = \sigma_{con} - \sigma_l - \alpha_E \sigma_{pc} \quad (10.1.6-2)$$

When the concrete normal stress at the point of the resultant forces of the prestressing tendon and/or bar is zero, the stress of the prestressing tendon and/or bar, is

$$\sigma_{p0} = \sigma_{con} - \sigma_l \quad (10.1.6-3)$$

2 Post-tensioned member

The concrete normal stress caused by the pre-applied force is

$$\sigma_{pc} = \frac{N_p}{A_n} \pm \frac{N_p e_{pn}}{I_n} y_n + \sigma_{p2} \quad (10.1.6-4)$$

The effective prestress of the prestressing tendon and/or bar at corresponding stage is

$$\sigma_{pe} = \sigma_{con} - \sigma_1 \quad (10.1.6-5)$$

When the concrete normal stress at the point for resultant forces of the prestressing tendon and/or bar is zero, the stress of the prestressing tendon and/or bar is

$$\sigma_{p0} = \sigma_{con} - \sigma_1 + \alpha_E \sigma_{pc} \quad (10.1.6-6)$$

Where:

A_n ——the area of the net section, namely the sum of the total sectional area of the concrete by deducting the area of such weakened parts as the pore path and the groove and the sectional area of the concrete converted from the sectional area of the longitudinal non-prestressing tendon and/or bar; for the section composed of different concrete strength grades, its area shall be converted to the sectional area of the same concrete strength grade according to the ratio of the elastic modulus of the concrete;

A_0 ——the transformed sectional area: including the net section area and the sectional area of concrete converted from the sectional area of all the longitudinal prestressing tendon and/or bar;

I_0, I_n ——the moment of inertia for the transformed section, the moment of inertia for the net section;

e_{p0}, e_{pn} ——respectively the distance from the transformed section centroid and the net section centroid to the action point of the pre-applied force, calculated according to the requirements of Article 10.1.7 of this code;

y_0, y_n ——the distance from the transformed section centroid and the net section centroid to the calculated fiber respectively;

σ_1 ——the value for prestress loss at the corresponding stage, calculated according to the requirements of Article 10.2.1~Article 10.2.7 of this code;

α_E ——the ratio of the elastic modulus of the steel reinforcement to the elastic modulus of the concrete; $\alpha_E = E_s / E_c$, hereof, E_s shall be adopted according to Table 4.2.5 of this code, and E_c shall be adopted according to Table 4.1.5 of this code;

N_{p0}, N_p ——the pre-applied force of the pre-tensioned member and the post-tensioned member respectively, calculated according to Article 10.1.7 of this code;

σ_{p2} ——the normal stress of the concrete section caused by the secondary internal force of the prestress.

Note: In Formula (10.1.6-1) and Formula (10.1.6-4), if the stress direction of the second item on the right and first item is same, taking the plus sign, if not, taking the minus sign; Formula (10.1.6-2) and Formula (10.1.6-6) are applicable to the condition that σ_{pc} is the compression stress, when σ_{pc} is the tension stress, the negative value shall be substituted in.

10.1.7 The pre-applied force and its eccentric distance of the action point (Figure 10.1.7) should be calculated according to the following formulae:

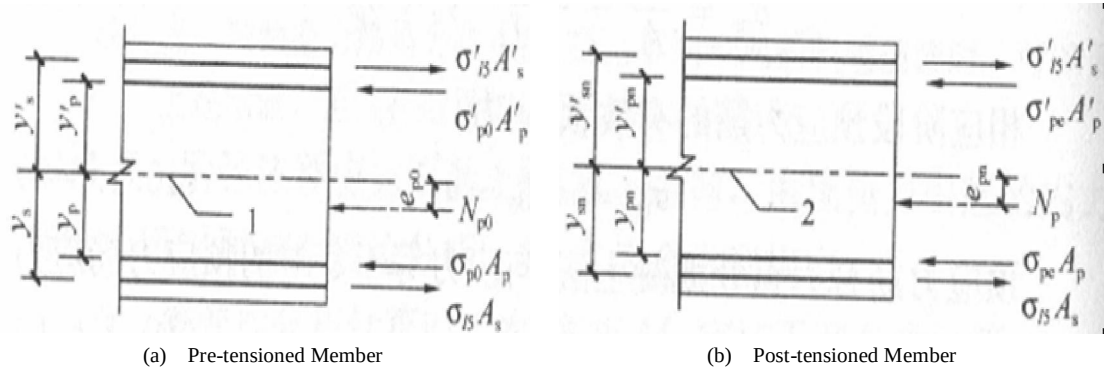


Figure 10.1.7 Position of Action Point for Pre-applied Force

1—the gravity axis of the transformed section; 2—the gravity axis of the net section

1 Pre-tensioned member

$$N_{p0} = \sigma_{p0} A_p + \sigma'_{p0} A'_p - \sigma_{l5} A_s - \sigma'_{l5} A'_s \quad (10.1.7-1)$$

$$e_{p0} = \frac{\sigma_{p0} A_p y_p - \sigma'_{p0} A'_p y'_p - \sigma_{l5} A_s y_s + \sigma'_{l5} A'_s y'_s}{\sigma_{p0} A_p + \sigma'_{p0} A'_p - \sigma_{l5} A_s - \sigma'_{l5} A'_s} \quad (10.1.7-2)$$

2 Post-tensioned member

$$N_p = \sigma_{pe} A_p + \sigma'_{pe} A'_p - \sigma_{l5} A_s - \sigma'_{l5} A'_s \quad (10.1.7-3)$$

$$e_{pn} = \frac{\sigma_{pe} A_p y_{pn} - \sigma'_{pe} A'_p y'_{pn} - \sigma_{l5} A_s y_{sn} + \sigma'_{l5} A'_s y'_{sn}}{\sigma_{pe} A_p + \sigma'_{pe} A'_p - \sigma_{l5} A_s - \sigma'_{l5} A'_s} \quad (10.1.7-4)$$

Where:

σ_{p0} , σ'_{p0} —respectively the stress of the prestressing tendon and/or bar in tension zone and compression zone under the condition that the concrete normal stress at the point of the resultant forces of the prestressing tendon and/or bar is zero;

σ_{pe} , σ'_{pe} —the effective prestress of the prestressing tendon and/or bar in tension zone and compression zone respectively;

A_p , A'_p —the sectional area of the longitudinal prestressing tendon and/or bar in tension zone and in compression zone respectively;

A_s , A'_s —the sectional area of longitudinal steel bars in tension zone and in compression zone respectively;

y_p , y'_p —the distance from the point for resultant forces of prestress to the transformed section centroid in tension zone and in compression zone respectively;

y_s , y'_s —the distance from the steel bar centroid to the transformed section centroid in tension zone and in compression zone respectively;

σ_{l5} , σ'_{l5} —the value for the prestressing loss due to concrete shrinkage and creep at

the point for resultant forces of the prestressing tendon and/or bar in tension zone and in compression zone respectively, calculated according to the requirements of Article 10.2.5 of this code;

y_{pn}, y'_{pn} ——the distance from the point for resultant forces of prestress to the net section centroid in tension zone and in compression zone respectively;

y_{sn}, y'_{sn} ——the distance from the steel bar centroid to the net section centroid in tension zone and in compression zone respectively.

Notes:

- 1 In Formula (10.1.7-1) ~ Formula (10.1.7-4), where $A'_p = 0$, then $\sigma'_{l5} = 0$;
- 2 When calculating the secondary internal force, σ_{l5} and σ'_{l5} in Formula (10.1.7-3) and Formula (10.1.7-4) may approximately adopt zero.

10.1.8 For post-tensioned bonded prestressed concrete frame beams and continuous beams allowed to appear cracks, when the calculating is according to ultimate limit states under the action of the gravity load, the redistribution of the internal force may be considered and the requirements for checking of serviceability limit states shall be satisfied. If the depth of the section relative to the compression zone ξ is not less than 0.1 and not larger than 0.3, the design value for the maximum negative bending moment on the section of the support in any span may be determined according to the following formulae:

$$M = (1-\beta)(M_{GQ}+M_2) \quad (10.1.8-1)$$

$$\beta = 0.2(1-2.5\xi) \quad (10.1.8-2)$$

And the modulated amplitude should not exceed 20% of the design value for the bending moment under the gravity load.

Where:

M ——the design value for the bending moment on the control section of the support;

M_{GQ} ——the design value for the bending moment on the control section under the gravity load calculated according to the elastic analysis;

ξ ——the depth of the section relative to the compression zone, calculated according to the requirements of Chapter 6 of this code;

β ——the amplitude modulation coefficient for the bending moment.

10.1.9 The transfer length of prestress l_{tr} for the prestressing tendon and/or bar in the pre-tensioned member shall be calculated according to the following formula:

$$l_{tr} = \alpha \frac{\sigma_{pe}}{f'_{tk}} d \quad (10.1.9)$$

Where:

σ_{pe} ——the effective prestress of the prestressing tendon and/or bar in extension;

d ——the nominal diameter of the prestressing tendon and/or bar, adopted according to Appendix A of this code;

α ——the appearance coefficient of the prestressing tendon and/or bar, adopted according to Table 8.3.1 of this code;

f'_{tk} ——the standard value of axial tensile strength corresponding to the concrete cube

compressive strength f'_{cu} in extension, determined according to the linear interpolation method specified in Table 4.1.3-2 of this code.

When the construction technology of sudden extension of the prestress is adopted, for the prestressed wire with smooth surface, the starting point of l_{tr} shall be calculated from the place where $l_{tr}/4$ away from the tail end of the member.

10.1.10 When calculating the bend bearing capacity of normal section sum inclined section in the end anchorage zone of pre-tensioned prestressed concrete members, the design value of tensile strength of the prestressing tendon and/or bar within the range of the anchorage length shall be taken as zero at the starting point of the anchorage, it shall be taken as f_{py} at the end point of the anchorage, and it shall be determined according to linear interpolation method between these two points. The anchorage length l_a of the prestressing tendon and/or bar shall be determined according to Article 8.3.1 of this code.

When the construction technology of sudden extension of the prestress is adopted, for the prestressed wire with smooth surface, the anchorage length shall be calculated from the place where $l_{tr}/4$ away from the tail end of the member.

10.1.11 For members allowed to appear tension stress in pretension zone at such construction stages as manufacture, transportation and installation stages, or members with total section compressed in prepressing, the normal stress of the concrete at the edge of the section under the action of the pre-applied force, deadweight and construction loads (the dynamic coefficient shall be considered if necessary) should meet the following requirements (Figure 10.1.11):

$$\sigma_{ct} \leq f'_{tk} \quad (10.1.11-1)$$

$$\sigma_{cc} \leq 0.8 f'_{ck} \quad (10.1.11-2)$$

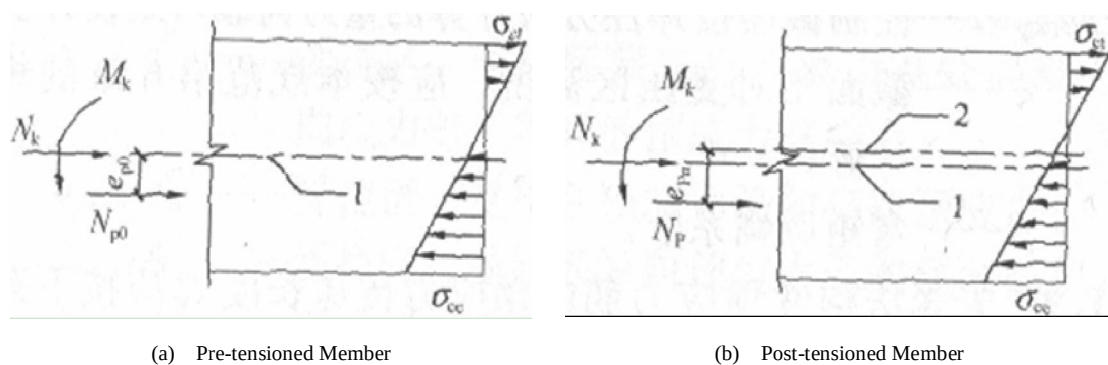


Figure 10.1.11 Checking of Prestressed Concrete Member at Construction Stage

1—the gravity axis of the transformed section; 2—the gravity axis of the net section

The concrete tension stress of the fiber at the edge of the section in pretension zone at the end sector for simply-supported members is allowed to larger than f'_{tk} but shall not be larger than $1.2 f'_{tk}$.

The concrete normal stress at the edge of the section may be calculated according to the following formula:

$$\sigma_{cc} \text{ or } \sigma_{ct} = \sigma_{pc} + \frac{N_k}{A_0} \pm \frac{M_k}{W_0} \quad (10.1.11-3)$$

Where:

σ_{ct} ——the concrete tension stress of the fiber at the edge of calculated section in pretension zone at corresponding construction stage;

σ_{cc} ——the concrete compression stress of the fiber at the edge of calculated section in precompression zone at corresponding construction stage;

f'_{tk} , f'_{ck} ——the standard value for the tensile strength and the standard value for the

compressive strength corresponding to the concrete cube compressive strength f'_{cu} at each construction stage respectively, determined according to the linear interpolation method specified in Table 4.1.3-2 and Table 4.1.3-1 of this code respectively;

N_k , M_k ——the values for the axial force and the bending moment on the calculated section caused by the characteristic combination of the member deadweight and construction loads respectively;

W_0 ——the resistance moment of the elasticity on the transformed section at the checking edge.

Notes:

- 1 The pretension zone and precompression zone refer to the section tension stress zone and compression stress zone formed when applying the prestress respectively;
- 2 In Formula (10.1.11-3), if σ_{pc} is the compression stress, the positive value shall be adopted; if σ_{pc} is the tension stress, the negative value shall be adopted; if N_k is the axial compressive force, the positive value shall be adopted; if N_k is the axial tensile force, the negative value shall be adopted; if the external fibre stress caused by M_k is the compression stress, the plus sign shall be adopted for the symbol in the formula; if it is the tension stress, the minus sign shall be adopted for the symbol in the formula;
- 3 If there is reliable engineering experience, the normal tensile stress of the concrete in pretension zone for composite flexural members may be controlled basing on that σ_{ct} is not larger than $2f'_{tk}$.

10.1.12 For members allowed to appear tension stress in pretension zone at construction stage, the ratio of reinforcement $(A'_s + A'_p)/A$ of the longitudinal steel reinforcement in pretension zone should not be less than 0.15%, A'_p shall not be counted in for post-tensioned members, hereinto, A is the sectional area of member. The diameter of longitudinal steel bars in pretension zone should not be larger than 14mm, and they shall be arranged uniformly along the external edge of the member in pretension zone.

Note: For members of slab type not allowed to appear cracks in pretension zone at construction stage, the reinforcement in longitudinal steel reinforcements in pretension zone may be determined according to specific conditions and practical experience.

10.1.13 For members of pre-tensioned and post-tensioned prestressed concrete structure, in the calculation of load-bearing capacity and crack width, when the prestress in normal direction for used concrete is zero, the pre-applied force N_{p0} and the eccentric distance of

action point e_{p0} shall be all calculated according to Formula (10.1.7-1) and Formula (10.1.7-2) of this code. Here, the stress σ_{p0} and σ'_{p0} of the prestressing tendon and/or bar in pre-tensioned and post-tensioned members shall be all calculated according to the requirements of Article 10.1.6 of this code.

10.1.14 For unbounded prestressed flexural members with rectangular sections, in calculation of load-bearing capacity of normal section, the stress design value σ_{pu} of the unbonded prestressed tendon should be calculated according to the following formulae:

$$\sigma_{pu} = \sigma_{pe} + \Delta\sigma_p \quad (10.1.14-1)$$

$$\Delta\sigma_p = (240 - 335\xi_p) \left(0.45 + 5.5 \frac{h}{l_0} \right) \frac{l_2}{l_1} \quad (10.1.14-2)$$

$$\xi_p = \frac{\sigma_{pe} A_p + f_y A_s}{f_c b h_p} \quad (10.1.14-3)$$

For continuous beams, continuous one-way slabs and continuous two-way slabs with span number no less than three; $\Delta\sigma_p$ value shall not be less than 50N/mm^2 .

The stress design value σ_{pu} of unbonded prestressed tendon shall still meet the following conditions:

$$\sigma_{pu} \leq f_{py} \quad (10.1.14-4)$$

Where:

σ_{pe} ——the effective prestress in unbonded prestressed tendon after deducting all losses of prestress (N/mm^2);

$\Delta\sigma_p$ ——the increment of stress in unbonded prestressed tendon (N/mm^2);

ξ_p ——the characteristic value of composite reinforcement, which should not be larger than 0.4; for continuous beams and slabs, the mean value for the characteristic value of composite reinforcements in supports in each span and in mid-span section shall be adopted;

h ——the depth of section in flexural members;

h_p ——the distance from the point for resultant forces of unbonded prestressed tendon to the compressive edge of section;

l_1 ——the total length between two anchored ends of continuous unbonded prestressed tendon;

l_2 ——the sum of span length of loads related with l_1 and determined by the most unfavorable arrangement diagram of live load.

For flexural members of T-shaped and I-shaped sections with flanges located in compression zone, if the depth of compression zone is larger than the flange depth, the characteristic value ξ_p of composite reinforcement may be calculated according to following formula:

$$\xi_p = \frac{\sigma_{pe} A_p + f_y A_s - f_c (b'_f - b) h'_f}{f_c b h_p} \quad (10.1.14-5)$$

Where:

h'_f ——the depth of flange on T-shaped and I-shaped sections in compression zone;

b'_f ——the calculated width for the flange on T-shaped and I-shaped sections in compression zone.

10.1.15 In tension zone of unbounded prestressed concrete flexural members, the arrangement for the sectional area A_s of longitudinal steel bars and the distribution shall meet the following requirements:

1 One-way slab

$$A_s \geq 0.002bh \quad (10.1.15-1)$$

Where:

b ——the width of section;

h ——the depth of section.

the diameter of longitudinal steel bars shall not be less than 8mm and the spacing shall not be larger than 200mm.

2 Beam

The larger value from calculated results of the following two formulae shall be adopted for A_s :

$$A_s \geq \frac{1}{3} \left(\frac{\sigma_{pu} h_p}{f_y h_s} \right) A_p \quad (10.1.15-2)$$

$$A_s \geq 0.003bh \quad (10.1.15-3)$$

Where:

h_s ——the distance from the point for resultant forces of the longitudinal tension steel bars to the compressive edge of section.

The diameter of longitudinal tension steel bars should not be less than 14mm and they shall be distributed uniformly at the tension edge of beams.

For beams designed according to Crack Control Level 1, when no less than 75% of the design value for bending moment is bearing by unbonded prestressed tendon, the area of longitudinal tension steel bars shall meet the requirements of the calculation of load-bearing capacity and Formula (10.1.15-3).

10.1.16 For two-way flat slabs in slab-column structure of unbounded prestressed concrete, the sectional area A_s of longitudinal steel bars and the distribution shall meet the following requirements:

1 In hogging moment area of the column side, the sectional area of longitudinal steel bars in each direction shall meet the following requirements:

$$A_s \geq 0.00075hl \quad (10.1.16-1)$$

Where:

l ——the span of slab parallel to the direction for calculating longitudinal stressed steel reinforcements;

h ——the slab thickness.

The longitudinal steel bars determined by the above formula shall be distributed within the range of slab width (namely 1.5h away from each column side). At least four sticks of steel reinforcements with diameter no less than 16mm shall be arranged in each direction. The spacing of longitudinal steel reinforcements shall not be larger than 300mm; the length stretching out of the column side shall be at least 1/6 of clear span at each side of the support.

If the action of longitudinal steel bars is considered in calculation of load-bearing capacity, the length stretching out of the column side shall be determined according to the calculation and shall meet the requirements for anchorage length specified in Section 8.3 of this code.

2 Under characteristic combination of loads, the longitudinal steel bars may be arranged according to the construction only in positive bending moment zone if the normal tensile stress of concrete at crack resistance checking edge in each direction in positive bending moment zone meets the following requirements:

$$\sigma_{ck} - \sigma_{pc} \leq 0.4f_{tk} \quad (10.1.16-2)$$

3 Under characteristic combination of loads, when the normal tensile stress of concrete at crack resistance checking edge in each direction in positive bending moment zone exceeds $0.4f_{tk}$ and is not larger than $1.0f_{tk}$, the sectional area of longitudinal steel bars shall meet the following requirements:

$$A_s \geq \frac{N_{tk}}{0.5f_y} \quad (10.1.16-3)$$

Where:

N_{tk} ——the resultant force in tension zone of non-cracking section of concrete of members under characteristic combination of loads;

f_y ——the design value of tensile strength on steel reinforcements, if f_y is larger than 360N/mm^2 , it shall adopt 360N/mm^2 .

The longitudinal steel bars shall be distributed uniformly in tension zone of slabs and shall be arranged in full length near by the tension edge.

4 Embedded ring-beams or reinforced concrete boundary beams shall be arranged at the edge and corner of flat slabs. The diameter of longitudinal steel reinforcements in embedded ring-beams shall not be less than 12mm and the number of longitudinal steel reinforcements shall not be less than four; the diameter of stirrup shall not be less than 6mm and the spacing shall not be larger than 150mm.

Note: In the area of cast-in-situ two-way flat slabs with higher temperature and larger shrinkage stress, the ordinary constructional steel mesh reinforcement shall be arranged according to Article 9.1.8 of this code.

10.1.17 The design value for normal section flexural load-bearing capacity of prestressed concrete flexural members shall meet the following requirements:

$$M_u \geq M_{cr} \quad (10.1.17)$$

Where:

M_u ——the design value for normal section flexural load-bearing capacity of members, calculated according to Formula (6.2.10-1), Formula (6.2.11-2) or Formula (6.2.14) of this code, but the equal sign shall be adopted and M shall be replaced by M_u ;

M_{cr} ——the value for normal section cracking bending moment of members, calculated according to Formula (7.2.3-6) of this code.

10.2 Calculation of Value for Loss of Prestress

10.2.1 The value for loss of prestress in the prestressing tendon and/or bar may be calculated according to those specified in Table 10.2.1.

If the calculated values for total losses of prestress are less than the following values, the

following values shall be adopted:

Pre-tensioned member 100N/mm²;
Post-tensioned member 80N/mm².

Table 10.2.1 Value for Loss of Prestress (N/mm²)

Factor for losses		Symbol	Pre-tensioned member	Post-tensioned member
Deformation of tensioning end anchorage and inner shrinkage of prestressing tendon and/or bar		σ_{11}	Calculated according to the requirements of Article 10.2.2 of this code	Calculated according to the requirements of Article 10.2.2 and Article 10.2.3 of this code
Friction of prestressing tendon and/or bar	Friction with pore path wall	σ_{12}	-	Calculated according to the requirements of Article 10.2.4 of this code
	Friction of anchor at tensioning end		Determined according to actual measured values or data provided by manufacturer	
	Friction at steering device		Determined according to actual conditions	
The temperature difference between the prestressing tendon and/or bar and the equipment bearing tension in the heating maintenance of concrete		σ_{13}	$2\Delta t$	-
Stress relaxation of prestressing tendon and/or bar		σ_{14}	Stress-relieved steel wire and steel strand Ordinary relaxation: $0.4 \left(\frac{\sigma_{con}}{f_{ptk}} - 0.5 \right) \sigma_{con}$ Underrelaxation: Where $\sigma_{con} \leq 0.7f_{ptk}$, $0.125 \left(\frac{\sigma_{con}}{f_{ptk}} - 0.5 \right) \sigma_{con}$ Where $0.7f_{ptk} < \sigma_{con} \leq 0.8f_{ptk}$, $0.2 \left(\frac{\sigma_{con}}{f_{ptk}} - 0.575 \right) \sigma_{con}$ Medium-strength prestressed wire: $0.08\sigma_{con}$ Prestressed twisted steel reinforcement: $0.03\sigma_{con}$	
Shrinkage and creep of concrete		σ_{15}	Calculated according to the requirements of Article 10.2.5 of this code	
Partial extrusion of concrete (for ring-shaped members with reinforcements made of prestressing tendon and/or bar and of which the diameter d is not larger than 3m)		σ_{16}	-	30

Notes:

- 1 Δt in this table refers to the temperature difference (°C) between the prestressing tendons and/or bar and the equipment bearing tension in the heating maintenance of concrete;
- 2 Where $\sigma_{con}/f_{ptk} \leq 0.5$, The value for the loss of stress relaxation of the prestressing tendon and/or bar may be taken as zero.

10.2.2 For linear prestressing tendon and/or bar, the value for loss of prestress σ_{l1} caused by anchorage deformation and inner shrinkage of the prestressing tendon and/or bar shall be calculated according to the following formula:

$$\sigma_{l1} = \frac{a}{l} E_s \quad (10.2.2)$$

Where:

a ——the value for the deformation of tensioning end anchorage and inner shrinkage of the prestressing tendon and/or bar (mm), may be adopted according to those specified in Table 10.2.2;

l ——the distance from the tensioning end to anchored end (mm).

Table 10.2.2 Value a for Anchorage Deformation and Inner Shrinkage of Prestressing Tendon and/or Bar (mm)

Type of anchorage		a
Supporting-type anchorage (steel tendon stamping-end anchorage)	Gap of nut cap	1
	Gap of each post-added subplate	1
Clip-type anchorage	With top pressure	5
	Without top pressure	6~8

Notes:

1 The value for anchorage deformation and inner shrinkage of the prestressing tendon and/or bar in this table may also be determined according to actual measured data;

2 The value for anchorage deformation and inner shrinkage of other types shall be determined according to actual measured data.

For the structure assembled of blocks, the precompression deformation of crack filling between blocks still shall be included in the loss of prestress. When the concrete or mortar is adopted as crack filler, the precompression deformation value for each crack filling may be taken as 1mm.

10.2.3 The value for loss of prestress σ_{l1} of curve or broken-line prestressing tendon and/or bar in post-tensioned members, caused by anchorage deformation and inner shrinkage of the prestressing tendon and/or bar, shall be determined according to the condition that the deformation value for the prestressing tendon and/or bar within the range of the reverse friction influence length l_f between the curve or broken-line prestressing tendon or bar and pore path walls equals the value for the anchorage deformation and inner shrinkage of the prestressing tendon and/or bar; and the reverse friction coefficient may be selected according to those values specified in Table 10.2.4.

The reverse friction influence length l_f and the value for loss of prestress σ_{l1} of post-tensioning prestressed tendon of common tendon shapes within the range of reverse friction influence length l_f may be calculated according to Appendix J of this code.

10.2.4 The value for loss of prestress σ_{l2} due to friction between the prestressing tendons and/or bar and pore path wall should be calculated according to the following formula:

$$\sigma_{l2} = \sigma_{con} \left(1 - \frac{1}{e^{kx} + \mu\theta} \right) \quad (10.2.4-1)$$

Where $(kx + \mu\theta)$ is not larger than 0.3, σ_{l2} may be calculated according to the following approximate formula:

$$\sigma_{l2} = (kx + \mu\theta) \sigma_{\text{con}} \quad (10.2.4-2)$$

Note: When the clip-type group anchor system is adopted, the anchor friction loss should be deducted in σ_{con} .

Where:

x ——the length from the tensioning end to the pore path of calculated section, which may take the projected length (m) of the pore path at this sector on the longitudinal axis approximately;

θ ——the sum of included angles for tangent lines at each part of curve pore path from tensioning end to calculated section (rad);

k ——the friction coefficient in consideration of partial deviation in each meter length of the pore path, adopted according to those specified in Table 10.2.4;

μ ——the friction coefficient between the prestressing tendon and/or bar and the pore path walls, adopted according to those specified in Table 10.2.4.

Table 10.2.4 Friction Coefficient

Shaping mode of pore path	k	μ	
		Steel strand and steel tendon	Prestressed twisted steel reinforcement
Embedded metal bellows	0.0015	0.25	0.50
Embedded plastic bellows	0.0015	0.15	-
Embedded steel pipe	0.0010	0.30	-
Shaping by loose core	0.0014	0.55	0.60
Unbonded prestressed tendon	0.0040	0.09	-

Note: The friction coefficient also may be determined according to actual measured data.

In Formula (10.2.4-1), for the space curve that changed in parabola and arc curve as well as the generalized space curve that may be superposed after segmenting, the sum of included angles θ may be calculated according to the following approximate formulae:

Parabola and arc curve:

$$\theta = \sqrt{\alpha_v^2 + \alpha_h^2} \quad (10.2.4-3)$$

Generalized space curve:

$$\theta = \sum \sqrt{\Delta\alpha_v^2 + \Delta\alpha_h^2} \quad (10.2.4-4)$$

Where:

α_v, α_h ——the bending angle of the parabola and arc curve shaped by vertical and horizontal projection of the space curve prestressing tendon and/or bar changed with the parabola and arc curve respectively;

$\Delta\alpha_v, \Delta\alpha_h$ ——the bending angle increment at segmented curve shaped by vertical and horizontal projection of generalized space curve prestressing tendon and/or bar respectively.

10.2.5 The value for loss of prestress σ_{l5} and σ'_{l5} for longitudinal prestressing tendon and/or bar in tension zone and compression zone due to concrete shrinkage and creep may be determined according to the following methods:

1 General situation

Pre-tensioned member

$$\sigma_{l5} = \frac{60 + 340 \frac{\sigma_{pc}}{f'_{cu}}}{1 + 15\rho} \quad (10.2.5-1)$$

$$\sigma'_{l5} = \frac{60 + 340 \frac{\sigma'_{pc}}{f'_{cu}}}{1 + 15\rho'} \quad (10.2.5-2)$$

Post-tensioned member

$$\sigma_{l5} = \frac{55 + 300 \frac{\sigma_{pc}}{f'_{cu}}}{1 + 15\rho} \quad (10.2.5-3)$$

$$\sigma'_{l5} = \frac{55 + 300 \frac{\sigma'_{pc}}{f'_{cu}}}{1 + 15\rho'} \quad (10.2.5-4)$$

Where:

σ_{pc} , σ'_{pc} ——the concrete direct compressive stress at the point for resultant forces of the prestressing tendon and/or bar in tension zone and compression zone respectively;

f'_{cu} ——the concrete cube compressive strength when applying prestress;

ρ , ρ' ——respectively the ratio of reinforcement for the prestressing tendon and/or bar and steel bars in tension zone and compression zone: for pre-tensioned members, $\rho = (A_p + A_s)/A_0$, $\rho' = (A'_p + A'_s)/A_0$; for post-tensioned members, $\rho = (A_p + A_s)/A_n$, $\rho' = (A'_p + A'_s)/A_n$; For members with symmetrically arranged prestressing tendon and/or bar and steel bars, the ratio of reinforcement ρ and ρ' shall be calculated according to half of gross sectional area for steel reinforcements.

The concrete normal compression stress at the point for resultant forces of prestressing tendon and/or bar in tension zone and compression zone σ_{pc} and σ'_{pc} shall be calculated according to the requirements of Article 10.1.6 and Article 10.1.7 of this code. Here, the loss of concrete before the precompression (in the first batch) shall only be considered for the value for loss of prestress, the stress σ_{l5} and σ'_{l5} in steel bars shall be taken as zero; σ_{pc} and σ'_{pc} shall not be larger than $0.5f'_{cu}$; when σ'_{pc} is the tension stress, σ'_{pc} in Formula (10.2.5-2) and Formula (10.2.5-4) shall be taken as zero. The influence of deadweight may be considered according to manufacture conditions of members when calculating the concrete normal stress σ_{pc} and σ'_{pc} .

While the structure is under the environment with annual average relative humidity less than 40%, values for σ_{i5} and σ'_{15} shall be increased 30%.

2 For important structural members, when time-dependent concrete shrinkage and creep as well as the stress relaxation of the prestressing tendon and/or bar is required to consider for the value for loss of prestress, it should be calculated according to Appendix K of this code.

10.2.6 When tensioning in batches is adopted for the prestressing tendon and/or bar of post-tensioned members, the influence of concrete elastic compression or elongation caused by the prestressing tendon and/or bar tensioned in the latter batch on the prestressing tendon and/or bar tensioned in the former batch shall be considered; the tension control stress value σ_{con} for the prestressing tendon and/or bar tensioned in the former batch may be increased or $\alpha_E \sigma_{pci}$ may be reduced. Hereof, σ_{pci} is the concrete normal stress at the centroid of prestressing tendon and/or bar tensioned in the former batch caused by the prestressing tendon and/or bar tensioned in the latter batch.

10.2.7 The combination of the value for loss of prestress for prestressed concrete members in each stage should be carried out according to those specified in Table 10.2.7.

Table 10.2.7 Combination of Value for Loss of Prestress at Each Stage

Combination of value for loss of prestress	Pre-tensioned member	Post-tensioned member
Loss before concrete precompression (in the first batch)	$\sigma_{i1} + \sigma_{i2} + \sigma_{i3} + \sigma_{i4}$	$\sigma_{i1} + \sigma_{i2}$
Loss before concrete precompression (in the second batch)	σ_{i5}	$\sigma_{i4} + \sigma_{i5} + \sigma_{i6}$

Note: For pre-tensioned members, if the distinction of the proportion, in the loss in the first batch and second batch, for the loss value σ_{i4} due to the stress relaxation of the prestressing tendon and/or bar may be determined according to actual conditions.

10.3 Detailing of Prestressed Concrete Members

10.3.1 The clear spacing between pre-tensioned prestressing tendon and/or bar should not be less than 2.5 times of nominal diameter or 1.25 times of maximum particle diameter of the concrete coarse aggregate and shall meet the following requirements: for prestressed wire, it shall not be less than 15mm; for three-skein steel strand, it shall not be less than 20mm; for seven-skein steel strand, it shall not be less than 25mm. If there is reliable guaranty for concrete vibro compactness, clear spacing may be broadened to 1.0 times of maximum particle diameter of coarse aggregate.

10.3.2 The end of pre-tensioned prestressed concrete members should be taken with the following constructional measures:

1 For the prestressing tendon and/or bar arranged in single stick, its end should be arranged with spiral reinforcements;

2 For multi-stick of prestressing tendon and/or bar dispersedly-distributed, if the length range is within $10d$ from member end and is not less than 100mm, 3~5 sheets of steel mesh reinforcement perpendicular to the prestressing tendon and/or bar should be arranged, hereof d is the nominal diameter of the prestressing tendon and/or bar;

3 For thin slabs adopted reinforcements with prestressed wires, the transverse steel reinforcement should be properly thickened within the length range of 100mm from the slab end;

4 For members of trough slab type, additional transverse reinforcements shall be arranged within the length range of 100mm from the member end, and the number shall not be less than two.

10.3.3 The transverse ribs for strengthening the integrity and transverse stiffness should be arranged for precast ribbed slabs. The stressed steel reinforcements with end transverse ribs shall be bent into the longitudinal rib. When the pretension long line method is adopted for manufacturing prestressed concrete ribbed slabs with end transverse ribs, effective measures for preventing cracks from appearing on the end transverse rib in extension of the prestress shall be taken in the design and manufacture.

10.3.4 For parts of such members as prestressed concrete roof beams and crane beams with larger inclined principal tensile stress near by the support, a part of prestressing tendon and/or bar should be bent-up-arranged.

10.3.5 For flexural members with all prestressing tendon and/or bar bent up at member end or pre-tensioned members with linear reinforcements, when the member end is welded with bottom supporting structure, the adverse effect caused by concrete shrinkage, creep and temperature variation shall be considered, and the longitudinal constructional steel reinforcement should be arranged at parts where may produce cracks at the member end.

10.3.6 The type and quality of the anchorage, clamp and connector used for the post-tensioned prestressing tendon and/or bar shall comply with current relevant standards of China.

10.3.7 The arrangement of post-tensioned prestressing tendon and/or bar and prepared pore path shall meet the following detailing requirements:

1 The horizontal clear spacing between the prepared pore path in precast members should not be less than 50mm and should not be less than 1.25 times of the particle diameter of coarse aggregate; the clear spacing from the pore path to the member edge should not be less than 30mm and should not be less than 50% of the diameter of pore path;

2 The clear spacing of prepared pore path in cast-in-place concrete beams in vertical direction shall not be less than the external diameter of pore path; the clear spacing in horizontal direction should not be less than 1.5 times of external diameter of pore path and shall not be less than 1.25 times of particle diameter of coarse aggregate. The clear spacing from outer wall of pore path to the member edge: it should not be less than 50mm for the beam bottom, it should not be less than 40mm for the beam side; for the beam of Crack Control Level 3, and it should not be less than 60mm or 50mm for the beam bottom and beam side respectively.

3 The internal diameter of prepared pore path should be 6mm~15mm larger than the external diameter of prestressed bundle and the external diameter of the connector in need of crossing the pore path, and the sectional area of pore path should be 3.0 ~4.0 times of the sectional area penetrating into the prestressed bundle.

4 When there is reliable experience and the quality of concrete pouring is able to ensure, the prepared pore path may be arranged closely and parallelly in horizontal, but the quantity for parallelly prepared pore path shall not exceed two bundles.

5 When the flat anchorage system is adopted in cast-in-situ floor slabs, the number of the prestressing tendon and/or bar crossed each prepared pore path should be 3~5; under conventional loading, the clear spacing of pore path in horizontal direction shall not exceed the larger value between 8 times of slab thickness and 1.5m.

6 The spacing of single unbonded prestressed tendon in slabs should not be larger than 6 times of slab thickness and should not be larger than 1m; the number of the unbonded prestressed tendon in strip-shaped bundles should not exceed 5, the spacing of strip-shaped bundles should not be larger than 12 times of slab thickness and should not be larger than 2.4m.

7 The unbonded prestressed tendon in bundling arrangement in beams, the horizontal clear spacing of bundling should not be less than 50mm; the clear distance from the bundle to the member edge should not be less than 40 mm.

10.3.8 In the end anchorage zone of post-tensioned prestressed concrete members, the indirect steel reinforcement shall be arranged according to the following requirements:

1 When the ordinary subplate is adopted, the calculation for partial compression load-bearing capacity shall be carried out according to the requirements of Section 6.6 of this code, and the indirect steel reinforcement shall be arranged, the ratio of volumetric reinforcement shall not be less than 0.5%, and the rigidity spread angle of the subplate shall take 45°;

2 In calculation for partial compression load-bearing capacity, the partial pressure design value for bonded prestressed concrete member shall take 1.2 times of tensioning control force; the larger value between 1.2 times of tensioning control force and $(f_{ptk}A_p)$ shall be taken for unbonded prestressed concrete;

3 When the monoblock casting subplate is adopted, the design of partial compression zone shall meet the requirements of relevant standards;

4 Outside the arranged zone for partial compression indirect steel reinforcements, within the range of additional reinforcement zone where the length of member end l is not less than 3 times of the distance e from the point for resultant forces of prestressing tendon and/or bar at upper part or lower part of the gravity center line to the adjacent edge but is not larger than 1.2 times of depth of section h at member end, and the height is $2e$, the additional anti-splitting stirrup or meshes (Figure 10.3.8) shall be arranged uniformly, and the area of reinforcement may be calculated according to the following formula:

$$A_{sb} \geq 0.18 \left(1 - \frac{l_1}{l_b} \right) \frac{P}{f_{yv}} \quad (10.3.8-1)$$

And the ratio of volumetric reinforcement shall not be less than 0.5%.

Where:

P ——the design value for the resultant force acted on the prestressing tendon and/or bar at upper part or lower part of the section gravity center line at member end, which may be determined according to the requirements of Clause 2 of this article;

l_1, l_b ——the side length or diameter along the height direction A_1 and A_b of member respectively, A_1 and A_b shall be determined according to Article 6.6.2 of this code;

f_{yv} ——the design value of tensile strength on additional anti-splitting steel reinforcements, adopted according to the requirements of Article 4.2.3 of this code.

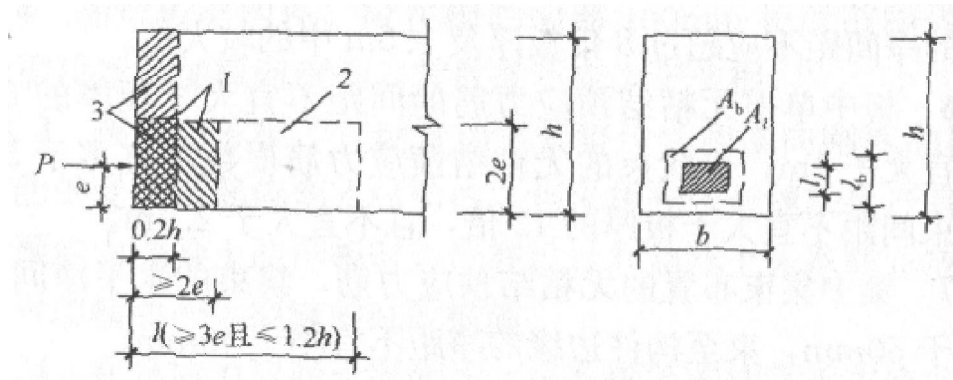


Figure 10.3.8 Range of Reinforcement Preventing End Crack

1—the arrangement zone for partial compression indirect steel reinforcement; 2—the additional anti-splitting reinforcement zone;

3—the additional anti-end-crack reinforcement zone

5 When the prestressing tendon and/or bar at member end is required to intensively arrange at the lower part of the section or at the upper part and lower part of the section, the additional vertical anti-end-crack constructional steel reinforcement shall be arranged within the range of $0.2h$ of the member end (Figure 10.3.8), and the sectional area shall meet the requirements of the following formulae:

$$A_{sv} \geq \frac{T_s}{f_{yv}} \quad (10.3.8-2)$$

$$T_s = \left(0.25 - \frac{e}{h} \right) P \quad (10.3.8-3)$$

Where:

T_s ——the tension of anchor end face;

P ——the design value for the resultant force acted on the prestressing tendon and/or bar at upper part or lower part of the section gravity center line at member end, which may be determined according to the requirements of Clause 2 of this article;

e ——the distance from the point for resultant forces of prestressing tendon and/or bar at upper part or lower part of section gravity center line to the near edge of section;

h ——the depth of section at member end;

Where e is larger than $0.2h$, the constructional steel reinforcement may be properly arranged according to actual conditions. The vertical anti-end-crack steel reinforcement should be arranged nearby the end face, the welded steel fabric, closed stirrup or other types may be adopted; and ribbed steel reinforcements should be adopted.

When there is prestressing tendon and/or bar at both upper part and lower part of end section, the larger value calculated according to the resultant force of prestress at upper part and lower part respectively shall be adopted for the gross sectional area of additional vertical steel reinforcements.

The anti-end-crack steel reinforcement also shall be calculated according to above-mentioned methods in the transverse of end face of member, and the arrangement of mesh reinforcement shall be formed with above-mentioned vertical steel reinforcements.

10.3.9 When there is partial concave bow for members at end, broken-line constructional

steel reinforcements (Figure 10.3.9) or other effective constructional steel reinforcements shall be added.

10.3.10 In post-tensioned prestressed concrete members, when the curve prestressed bundle is adopted, the radius of curvature r_p should be determined according to the following formula, but it should not be less than 4m.

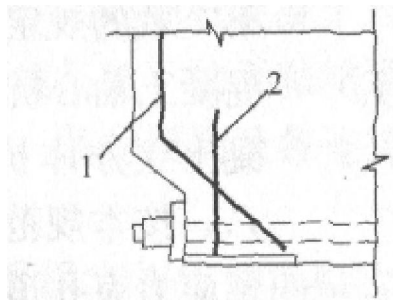


Figure 10.3.9 Constructional Steel Reinforcement at Concave Bow of End

1—the broken line constructional steel reinforcement; 2—the vertical constructional steel reinforcement

$$r_p \geq \frac{P}{0.35 f_c d_p} \quad (10.3.10)$$

Where:

P —the design value for the resultant force of prestressed bundle, which may be determined according to the requirements of Clause 2 of Article 10.3.8 in this code;

r_p —the radius of curvature for prestressed bundle (m);

d_p —the external diameter of pore path in prestressed bundle;

f_c —the design value for axial compressive strength of concrete; in checking of radius of curvature at tensioning stage, the design value of compressive strength f'_c corresponding to the concrete cube compressive strength f'_{cu} at construction stage may be taken, and it shall be determined according to linear interpolation method and Table 4.1.4-1 of this code.

For members with broken line reinforcements, the radius of curvature at the bending place of prestressed bundle may be properly reduced. When the radius of curvature r_p dissatisfies the above-mentioned requirements, the steel mesh reinforcement or spiral reinforcement may be arranged at inner side of the bending place of curve prestressed bundle.

10.3.11 In prestressed concrete structures, when the curve prestressed bundle is arranged along the concave of the member (Figure 10.3.11); the anti-cracking design shall be carried out. When the radius of curvature r_p meets the following formula requirements, the constructional U-shaped steel dowel may be arranged only.

$$r_p \geq \frac{P}{f_t (0.5d_p + c_p)} \quad (10.3.11-1)$$

When it is not met, the sectional area of each single-limb U-shaped steel dowel shall be determined according to the following formula:

$$A_{sv1} \geq \frac{Ps_v}{2r_p f_{yv}} \quad (10.3.11-2)$$

Where:

P ——the design value for the resultant force of prestressed bundle, which may be determined according to the requirements of Clause 2 of Article 10.3.8 in this code;

f_t ——the design value for axial compressive strength of concrete; or the design value of tensile strength f_t' corresponding to the concrete cube compressive strength f_{cu}' at construction tensioning stage, determined according to linear interpolation method and Table 4.1.4-2 of this code;

c_p ——the net thickness of concrete cover in pore path of prestressed bundle;

A_{sv1} ——the sectional area of each single-limb steel dowel;

s_v ——the spacing of U-shaped steel dowel;

f_{yv} ——the design value of tensile strength on U-shaped steel dowel, adopted according to Table 4.2.3-1 of this code, where it is larger than 360N/mm^2 , take 360N/mm^2 .

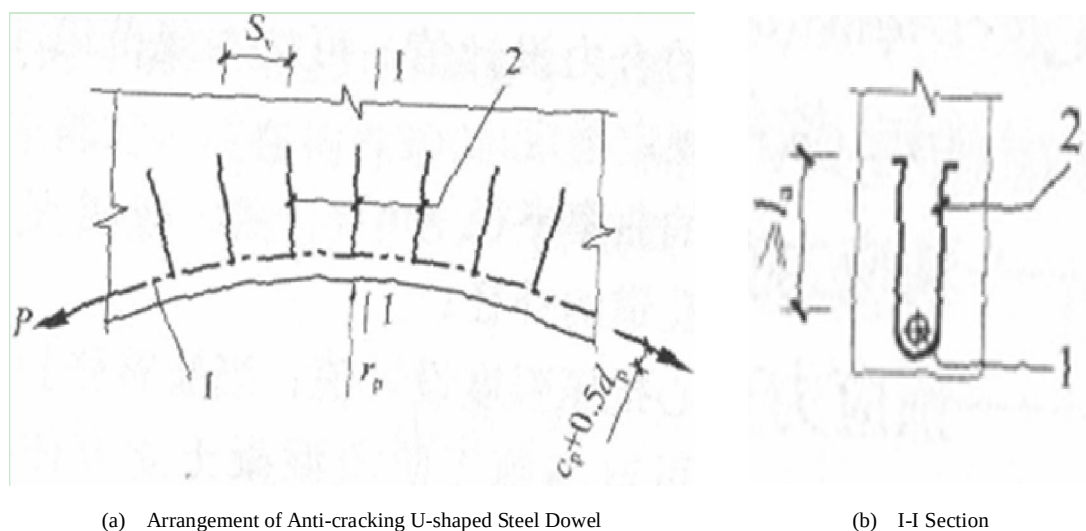


Figure 10.3.11 Constructional Schematic of Anti-cracking U-shaped Steel Dowel

1—the prestressed bundle; 2—U-shaped steel dowel arranged uniformly along the curve prestressed bundle

The anchorage length of U-shaped steel dowel shall not be less than l_a ; when the practical anchorage length l_e is less than l_a , the sectional area of each single-limb U-shaped steel dowel may be valued according to A_{sv1}/k . Hereinto, k shall take the smaller value between $l_e/15d$ and $l_e/200$, and k shall not be larger than 1.0.

When there are several parallel pore paths and the center to center distance is not larger than $2d_p$, the design value for the resultant force of prestressing tendon and/or bar shall be determined according to the prestressing tendon and/or bar in all adjacent pore paths.

10.3.12 The anchorage arrangement, dimension of tensioning equipments and requirements of partial compression shall be considered for the dimension of member end, which shall be properly increased if necessary.

10.3.13 For exposed metallic anchorage of post-tensioning prestressed concrete, reliable anticorrosion and fire precaution measures shall be taken and the following requirements shall be met:

1 The sealed anchorage end with plastic cap filled with sufficient anti-corrosive grease shall be adopted for exposed anchorage of unbonded prestressed tendon, and non-shrink mortar or fine stone concrete shall be adopted for sealing;

2 For unbounded prestressed anchorage system under Environment Category II-b, III-a and III-b, totally enclosed anti-corrosion system shall be adopted, the anchor end for sealing and each connecting part shall be able to bear 10kPa of hydrostatic pressure and shall not be pervious to water.

3 When the concrete is adopted for sealing, the strength grade should be consistent with the concrete strength grade of members and shall not be less than C30. The sealed anchor concrete and member concrete shall be bonded reliably, for instance, the concrete interface around the anchorage shall be roughened and rinsed out before sealing, and 1~2 sheets of steel mesh reinforcements should be arranged, the steel mesh reinforcement shall be tied with member concrete.

4 When non-shrink mortar or concrete are adopted for sealing protection, the cover thickness of the end at the anchorage and prestressing tendon and/or bar shall not be less than: 20mm for Environmental Category I; 50mm for Environmental Category II-a and II-b; 80mm for Environmental Category III-a and III-b.

11 Seismic Design of Reinforced Concrete Structural Members

11.1 General

11.1.1 The concrete structure of seismic protection shall not only meet the requirements of Chapter 1~ Chapter 10 of this code, but also the seismic design of structural members shall be carried out according to the requirements of this chapter basing on the seismic design principle specified in the current national standard "Code for Seismic Design of Buildings" (GB 50011).

11.1.2 For concrete buildings of seismic protection, the seismic protection category and corresponding seismic protection criterion shall be determined according to the requirements of the current national standard "Standard for Classification of Seismic Protection of Building Constructions" (GB 50223).

Note: Category A, Category B and Category C buildings in this chapter are short for buildings of particular fortification category, major fortification category and standard fortification category respectively specified in "Standard for Classification of Seismic Protection of Building Constructions" (GB 50223).

11.1.3 The seismic design of buildings with concrete structure members shall be adopted with different seismic grades according to the protection category, intensity, structure type and the height of buildings, and shall meet the requirements of the corresponding calculation and constructional measures. The seismic grade of Category C buildings shall be determined according to those specified in Table 11.1.3.

Table 11.1.3 Seismic Grade of Concrete Structure

Structure type			Protection intensity									
			6		7		8		9			
Frame structure	Height(m)		≤24	>24	≤24	>24	≤24	>24	≤24			
	Ordinary frame		IV	III	III	II	II	I	I			
	Large-span frame		III		II		I		I			
Frame-shear wall structure	Height (m)		≤60	>60	≤24	>24 and ≤60	≤24	>24 and ≤60	>60	≤24	>24 and ≤50	
	Frame		IV	III	IV	III	II	III	II	I	II	I
	Shear wall		III		III	II		II	I		I	
Shear wall structure	Height (m)		≤80	>80	≤24	>24 and ≤80	>80	≤24	>24 and ≤80	>80	≤24	24~60
	Shear wall		IV	III	IV	III	II	III	II	I	II	I
Partial frame-supporting shear wall structure	Height (m)		≤80	>80	≤24	>24 and ≤80	>80	≤24	>24 and ≤80	-	-	
	Shear wall	General position	IV	III	IV	III	II	III	II			
		Reinforced position	III	II	III	II	I	II	I			

	Frame of frame-supported storey		II		II	I	I		
Tube structure	Frame-core-tube	Frame	III		II		I		I
		Core tube	II		II		I		I
	Tube-in-tube	Inner tube	III		II		I		I
		Outer tube	III		II		I		I
Slab-column-shear wall structure	Height (m)		≤34	>35	≤35	>35	≤35	>35	
	Slab column and perimeter frame		III	II	II	II	I		-
	Shear wall		II	II	II	I	II	I	
Structure of single-story plant	Hinged bent		IV		III		II		I

Notes:

- 1 If the building site is of Category I, except for the requirements of Protection Intensity 6, it shall be allowable to adopt details of seismic design according to the corresponding seismic grade (one grade lower than the ones listed in the above table). However, the corresponding calculation requirements shall not be reduced;
- 2 Where it approximates or equals to the height boundary, the seismic grade shall be allowed to determine according to the combination of the degree of irregularity of buildings as well as conditions of site and base;
- 3 The large-span frame refers to the frame with span no less than 18m;
- 4 The frames stated in this table exclude special-shaped column frames;
- 5 When the frame-core tube structure with the height of building no larger than 60m is designed according to the requirements of frame-shear wall structure, the seismic grade shall be determined according to the requirements of the frame-shear wall structure stated in this table.

11.1.4 The determination of the seismic grade of structural members in reinforced concrete buildings still shall meet the following requirements:

1 For the frame-shear wall structure, under the action of specified horizontal seismic force, if the overturning moment borne by the frame bottom is larger than 50% of total overturning moment at structure bottom, the seismic grade of the frame shall be determined according to frame structure.

2 For the podium connected with the main building, the seismic grade shall be determined according to the podium itself and the seismic grade of relevant range shall not be less than that of the main building. And details of seismic design for the main building structure shall be properly strengthened at the adjacent up and down storey corresponding to the top slab of the podium. If the podium and the main building are separated, the seismic grade shall be determined according to the podium itself.

3 When the top-slab of the basement is used as the building-in location of superstructure, the seismic grade of the first storey underground shall be the same as that of the superstructure, the seismic grade of the details of seismic design for other storey lower than the first storey underground may be reduced by one grade storey by storey but shall not be less than Grade 4. For the part of the basement without superstructure, the seismic grade of the details of seismic design may be taken as Grade 3 or 4 according to specific conditions.

4 When the seismic grade of Category A and B buildings is determined by increasing one grade as required, if the height exceeds the maximum applicable height of corresponding buildings, then much more efficient details of seismic design than that of corresponding

seismic grade shall be taken.

11.1.5 The range of bottom reinforced part of the shear wall shall meet the following requirements:

1 The height of the bottom reinforced part shall be counted from the top-slab of basement.

2 For the shear wall of part frame-support shear wall structures, the height of the bottom reinforced part may be taken from the larger value between the height of the frame-supported storey and the two storeys above this frame-supported storey and 1/10 of the total height of the shear wall linked with the ground. As for the shear wall of other structures, if the height of the building is larger than 24m, the height of its bottom reinforced part may be taken from the larger value between the height of these two bottom storeys and 1/10 of the total height of wall-column; if the height of the building is not larger than 24m, the height of bottom storey may be taken as that of the bottom reinforced part.

3 When the calculated building-in end of the structure is located on or below the base plate of first storey underground, the range of bottom reinforced part determined according to Clause 1 and 2 of this article still shall be extended downward to the calculated building-in end.

11.1.6 When the load-bearing capacity of concrete structure members is checked by considering the seismic combination, all the adjustment shall be carried out according to the seismic adjustment coefficient of load-bearing capacity γ_{RE} , which shall be adopted according to those specified in Table 11.1.6.

The seismic load-bearing capacity of normal section shall be calculated according to the requirements of Section 6.2 of this code, but the right item in relevant calculation formulae shall divide by corresponding seismic adjustment coefficient of load-bearing capacity γ_{RE} .

If only the vertical earthquake action is calculated, the seismic adjustment coefficient of load-bearing capacity γ_{RE} of all structural members shall be taken as 1.0.

Table 11.1.6 Seismic Adjustment Coefficient of Load-bearing Capacity

Type of structural member	Calculation of normal section load-bearing capacity					Calculation of inclined section load-bearing capacity	Calculation of punching shear bearing capacity	Calculation of partial compression load-bearing capacity
	Flexural member	Eccentric compression column		Eccentric tension member	Shear wall	All types of member and frame node		
		Axial compression ratio less than 0.15	Axial compression ratio no less than 0.15					
γ_{RE}	0.75	0.75	0.8	0.85	0.85	0.85	0.85	1.0

Note: The seismic adjustment coefficient of load-bearing capacity γ_{RE} calculated on the anchor bar section of embedded parts shall be taken as 1.0.

11.1.7 The anchorage and connection of longitudinal stressed steel reinforcements in concrete structural members shall not only meet the relevant requirements of Section 8.3 and Section 8.4 but also shall meet the following requirements:

1 The seismic anchorage length l_{aE} of longitudinal tension steel reinforcements shall be calculated according to the following formula:

$$l_{aE} = \xi_{aE} l_a \quad (11.1.7-1)$$

Where:

ξ_{aE} ——the correction coefficient of seismic anchorage length for longitudinal tension steel reinforcements, taking 1.15 for Grade 1 and 2 of seismic grade, 1.05 for Grade 3 and 1.00 for Grade 4;

l_a ——the anchorage length of longitudinal tension steel reinforcements, determined according to Article 8.3.1 of this code.

2 When the overlapping connection is adopted, the seismic overlapping length l_{lE} of longitudinal tension steel reinforcements shall be calculated according to the following formula:

$$l_{lE} = \xi_l l_{aE} \quad (11.1.7-2)$$

Where:

ξ_l ——the correction coefficient of overlapping length for longitudinal tension steel reinforcements, determined according to Article 8.4.4 of this code.

3 The connection of longitudinal stressed steel reinforcements may be adopted with binding and splicing, mechanical connection or welding.

4 The connected position of longitudinal stressed steel reinforcements should keep clear of the stirrup densified area at the beam end and column end; the mechanical connection or welding shall be adopted if the connection must be done here.

5 For concrete members, the joint area percentage of longitudinal stressed steel reinforcements located in the same connection sector should not exceed 50%.

11.1.8 The stirrup should adopt welding sealed stirrup, continuous spiral stirrup or continuous compound spiral stirrup. When non-welding sealed stirrup is adopted, the tail end shall be made into hook of 135° , the length of straight sector of hook terminal shall not be less than 10 times of diameter of stirrup; the stirrup spacing within the range of overlapping length of longitudinal steel reinforcements shall not be larger than 5 times of smaller diameter of overlapping steel reinforcements and should not be larger than 100mm.

11.1.9 The embedded parts in consideration of the earthquake action shall meet the following requirements:

1 The sectional area of straight anchor steel reinforcement may be calculated and increased by 25% according to the relevant requirements of Chapter 9 of this code, and the anchor slab thickness shall be properly increased.

2 The anchorage length of anchor bar shall meet the relevant requirements of Section 9.7 of this code and shall be increased 10%; if not, the effective measure shall be taken. One stick of sealed stirrup with diameter no less than 10mm should be arranged at the place where nearby the anchor slab.

3 The embedded parts should not be arranged in plastic hinge zone; otherwise, effective measures shall be taken.

11.2 Materials

11.2.1 The concrete strength grade of concrete structure shall meet the following

requirements:

1 The shear wall should not exceed C60; for other members, they should not exceed C60 at Intensity 9 or C70 at Intensity 8.

2 The frame-supported beam, frame-supported column and Seismic Grade 1 frame beam, column and node shall not be less than C30; other kinds of structural members shall not be less than C20.

11.2.2 In the beam, column, support and shear wall boundary members, the stressed steel reinforcement should adopt hot-rolled ribbed bars; when the hot-rolled ribbed steel reinforcement with designation "E" specified in the current national standard "Steel for the Reinforcement of Concrete—Part 2: Hot Rolled Ribbed Bars"(GB 1499.2), the intensity and elastic modulus shall be adopted according to related requirements of hot-rolled ribbed bars specified in Section 4.2 of this code.

11.2.3 For the frame and diagonal bracing members designed according to Seismic Grade 1, 2 and 3, the longitudinal stressed steel bar shall meet the following requirements:

1 The ratio of the actual measured value of tensile strength to actual measured value of yield strength of steel reinforcements shall not be less than 1.25;

2 The ratio of the actual measured value of yield strength to standard value of yield strength of steel reinforcements shall not be larger than 1.30;

3 The actual measured value of overall elongation under the maximum tension of steel reinforcements shall not be less than 9%.

11.3 Frame Beams

11.3.1 In calculation of normal section flexural load-bearing capacity of beams, the depth of concrete compression zone at beam end counted in longitudinal compression steel reinforcements shall meet the following requirements:

Seismic Grade 1

$$x \leq 0.25h_0 \quad (11.3.1-1)$$

Seismic Grade 2 and 3

$$x \leq 0.35h_0 \quad (11.3.1-2)$$

Where:

x ——the depth of concrete compression zone;

h_0 ——the effective depth of section.

11.3.2 The design value of shear force V_b at frame beam end in consideration of seismic combination shall be calculated according to the following requirements:

1 Frame structure of Seismic Grade 1 and Frame of Seismic Grade 1 at Protection Intensity 9

$$V_b = 1.1 \frac{(M_{bua}^l + M_{bua}^r)}{l_n} V_{Gb} \quad (11.3.2-1)$$

2 Other Conditions

Seismic Grade 1

$$V_b = 1.3 \frac{(M_b^l + M_b^r)}{l_n} V_{Gb} \quad (11.3.2-2)$$

Seismic Grade 2

$$V_b = 1.2 \frac{(M_b^l + M_b^r)}{l_n} V_{Gb} \quad (11.3.2-3)$$

Seismic Grade 3

$$V_b = 1.1 \frac{(M_b^l + M_b^r)}{l_n} V_{Gb} \quad (11.3.2-4)$$

Seismic Grade 4, taking the design value of shear force under seismic combination.

Where:

M_{bua}^l , M_{bua}^r ——respectively the bending moment value at left and right end of frame

beams corresponding to the normal sectional seismic flexural load-bearing capacity, which is calculated according to the sectional area of actual arranged steel reinforcements (including the compression steel reinforcement and the steel reinforcement in floor slab within the range of effective flange width of beams) and the standard value for material strength and in consideration of seismic adjustment coefficient of load-bearing capacity;

M_b^l , M_b^r ——the design value of bending moment at left and right end of frame beam

in consideration of seismic combination;

V_{Gb} ——the design value of shear force caused by the representative value of gravity load in consideration of seismic combination, which may be determined by calculating according to simple-supported beams;

l_n ——the clear span of beam.

In formula (11.3.2-1), the sum of M_{bua}^l and M_{bua}^r shall be calculated accordingly in clockwise and counter-clockwise directions respectively and the larger value shall be adopted.

In Formula (11.3.2-2) ~Formula (11.3.2-4), the sum of M_b^l and M_b^r shall adopt the larger value for the sum of the design value for bending moment at both end in consideration of seismic combination calculated in clockwise and counter-clockwise directions respectively; at Seismic Grade 1, when the bending moment at both end is all hogging moment, the bending moment value with smaller absolute value shall adopt zero.

11.3.3 For frame beams with rectangular, T-shaped and I-shaped sections in consideration of seismic combination, when the span-height ratio is larger than 2.5, the shear section shall meet the following conditions:

$$V_b \leq \frac{1}{\gamma_{RE}} (0.20 \beta_c f_c b h_0) \quad (11.3.3-1)$$

When the span-height ratio is not larger than 2.5, the shear section shall meet the following conditions:

$$V_b \leq \frac{1}{\gamma_{RE}} (0.15 \beta_c f_c b h_0) \quad (11.3.3-2)$$

11.3.4 For frame beams with rectangular, T-shaped and I-shaped sections in consideration of seismic combination, the shear bearing capacity of inclined section shall meet the following requirements:

$$V_b = \frac{1}{\gamma_{RE}} \left[0.6 \alpha_{cv} f_t b h_0 + f_{yv} \frac{A_{sv}}{s} h_0 \right] \quad (11.3.4)$$

Where:

α_{cv} ——the concrete shear bearing capacity coefficient of section, valued according to Article 6.3.4 of this code.

11.3.5 The section dimension of frame beam shall meet the following requirements:

- 1 The sectional width should not be less than 200mm;
- 2 The ratio of depth of section to width should not be larger than 4;
- 3 The ratio of the clear span to the depth of section should not be less than 4.

11.3.6 The arrangement of steel reinforcements for frame beams shall meet the following requirements:

1 The ratio of reinforcement for longitudinal tension steel reinforcements shall not be less than the value specified in Table 11.3.6-1;

Table 11.3.6-1 Minimum Ratio of Reinforcement for Longitudinal Tension Steel Reinforcement of Frame Beam (%)

Seismic grade	Position in beam	
	Lug support	Mid-span
Grade 1	The larger value between 0.40 and $80f_t/f_y$	The larger value between 0.30 and $65f_t/f_y$
Grade 2	The larger value between 0.30 and $65f_t/f_y$	The larger value between 0.25 and $55f_t/f_y$
Grade 3 and 4	The larger value between 0.25 and $55f_t/f_y$	The larger value between 0.20 and $45f_t/f_y$

2 The ratio of sectional area of longitudinal stressed steel reinforcements at bottom and top of section at beam end of frame beams shall not only be determined according to calculation, the ratio at Seismic Grade 1 shall not be less than 0.5 and the ratio at Seismic Grade 2 and 3 shall not be less than 0.3;

3 The length of the stirrup densified area at the beam end, the maximum spacing between stirrups and the minimum diameter of stirrup shall be adopted according to those specified in Table 11.3.6-2; if the ratio of reinforcement for longitudinal tension steel reinforcements at beam end is larger than 2%, the values of the minimum diameter of stirrup shall be increased by 2mm in this table.

Table 11.3.6-2 Constructional Requirements of Stirrup Densified Area at Beam End of Frame Beam

Seismic grade	Length of densified area (mm)	Maximum spacing of stirrups (mm)	Minimum diameter (mm)
Grade 1	The larger value between 2 times	The minimum value in 6 times of diameter of	10

	of beam depth and 500	longitudinal steel reinforcement, 1/4 of beam depth and 100	
Grade 2	The larger value between 1.5 times of beam depth and 500	The minimum value in 8 times of diameter of longitudinal steel reinforcement, 1/4 of beam depth and 100	8
Grade 3		The minimum value in 8 times of diameter of longitudinal steel reinforcement, 1/4 of beam depth and 150	8
Grade 4		The minimum value in 8 times of diameter of longitudinal steel reinforcement, 1/4 of beam depth and 150	6

Note: If the diameter of the stirrup is larger than 12mm, the quantity is no less than 4 and the spacing of stirrup legs is not larger than 150mm, the maximum spacing of stirrups for Grade 1 and 2 shall be allowed to broaden properly but shall not be larger than 150mm.

11.3.7 The ratio of reinforcement for longitudinal tension steel reinforcements at beam end should not be larger than 2.5%. At least two sticks of longitudinal steel reinforcements with full length shall be arranged at top surface and bottom surface along full length of beam respectively; for Seismic Grade 1 and 2, the diameter of steel reinforcement shall not be less than 14mm and shall not be less than 1/4 of relatively big sectional area in longitudinal stressed steel reinforcements at top surface and bottom surface at both end of beams respectively; for Seismic Grade 3 and 4, the diameter of steel reinforcement shall not be less than 12mm.

11.3.8 For the spacing of stirrup legs within the beam stirrup densified area length: for Seismic Grade 1, it should not be larger than the larger value between 200mm and 20 times of diameter of stirrup; for Seismic Grade 2 and 3, it should not be larger than the larger value between 250mm and 20 times of diameter of stirrup; all should not be larger than 300mm for each seismic grade.

11.3.9 The distance from the first stirrup arranged at beam end to the edge of node of frame shall not be larger than 50mm. The spacing of stirrup in non-densified area should not be larger than 2 times of spacing of stirrup in densified area. The area ratio of reinforcement ρ_{sv} for stirrups along full length of beam shall meet the following requirements:

Seismic Grade 1

$$\rho_{sv} \geq 0.30 \frac{f_t}{f_{yv}} \quad (11.3.9-1)$$

Seismic Grade 2

$$\rho_{sv} \geq 0.28 \frac{f_t}{f_{yv}} \quad (11.3.9-2)$$

Seismic Grade 3 and 4

$$\rho_{sv} \geq 0.26 \frac{f_t}{f_{yv}} \quad (11.3.9-3)$$

11.4 Frame Columns and Frame-supported Columns

11.4.1 Except the nodes of the frame top column, the nodes of the column which axial-compression-ratio is less than 0.15, and the nodes of and frame-supported beams and frame-supported columns, the design values of sectional bending moment at the upper and lower end of the frame column node and the upper and lower end of the node in intermediate storey shall meet the following requirements:

1 Frame structure of Seismic Grade 1 and frame of Seismic Grade 1 at Protection Intensity 9

$$\sum M_c = 1.2 \sum M_{bua} \quad (11.4.1-1)$$

2 Frame structure
Seismic Grade 2

$$\sum M_c = 1.5 \sum M_b \quad (11.4.1-2)$$

Seismic Grade 3

$$\sum M_c = 1.3 \sum M_b \quad (11.4.1-3)$$

Seismic Grade 4

$$\sum M_c = 1.2 \sum M_b \quad (11.4.1-4)$$

3 Other conditions
Seismic Grade 1

$$\sum M_c = 1.4 \sum M_b \quad (11.4.1-5)$$

Seismic Grade 2

$$\sum M_c = 1.2 \sum M_b \quad (11.4.1-6)$$

Seismic Grade 3 and 4

$$\sum M_c = 1.1 \sum M_b \quad (11.4.1-7)$$

Where:

$\sum M_c$ ——the sum of the design values of the bending moment at the upper and lower column end of the node in consideration of seismic combination; for the determination of design value of the column end bending moment, generally, the sum of bending moment calculated in Formula (11.4.1-1) ~ Formula (11.4.1-5) may be distributed according to the bending moment ratio gained from elasticity analysis at the upper and lower column end in consideration of the seismic combination;

$\sum M_{bua}$ ——the larger value of the sum of bending moment values corresponding to the bend bearing capacity of normal section at the left and right beam ends of the same joint, calculated according to adopting the actual arranged steel reinforcements and standard value of the material strength in clockwise and counterclockwise directions and in consideration of the seismic adjustment coefficient of load-bearing capacity. When there are cast-in-situ slabs, the actual arranged steel reinforcement at the beam end shall include the longitudinal steel reinforcement of the floor-slab within the range of the beam's effective flange width;

$\sum M_b$ ——the larger value of the sum of the bending moment design values at the left and right beam end of the same joint, calculated in clockwise and counterclockwise directions in consideration of seismic combination ; for Seismic Grade 1, when the bending moments at both ends are hogging moments, the bending moment value with smaller absolute value shall take zero.

11.4.2 For the bottom storey of the frame structure of Seismic Grade 1, 2, 3 and 4, the

bending moment design value of the section combination at the column lower end shall respectively multiply amplified coefficients 1.7, 1.5, 1.3 and 1.2. The column longitudinal steel reinforcements at the bottom storey shall be arranged in accordance with the unfavorable situation at upper and lower end of the column.

Note: The bottom storey refers to the first storey above the foundation without the basement or above the basement.

11.4.3 The design value of shear force frame V_c of the frame column and frame-supported column shall be calculated according to the following formulae:

1 Frame structure of Seismic Grade 1 and frame of Seismic Grade 1 at Protection Intensity 9

$$V_c = 1.2 \frac{(M_{cua}^t + M_{cua}^b)}{H_n} \quad (11.4.3-1)$$

2 Frame structure
Seismic Grade 2

$$V_c = 1.3 \frac{(M_c^t + M_c^b)}{H_n} \quad (11.4.3-2)$$

Seismic Grade 3

$$V_c = 1.2 \frac{(M_c^t + M_c^b)}{H_n} \quad (11.4.3-3)$$

Seismic Grade 4

$$V_c = 1.1 \frac{(M_c^t + M_c^b)}{H_n} \quad (11.4.3-4)$$

3 Other conditions
Seismic Grade 1

$$V_c = 1.4 \frac{(M_c^t + M_c^b)}{H_n} \quad (11.4.3-5)$$

Seismic Grade 2

$$V_c = 1.2 \frac{(M_c^t + M_c^b)}{H_n} \quad (11.4.3-6)$$

Seismic Grade 3 and 4

$$V_c = 1.1 \frac{(M_c^t + M_c^b)}{H_n} \quad (11.4.3-7)$$

Where:

M_{cua}^t , M_{cua}^b —respectively the bending moment value at the upper and lower end of the frame column corresponding to the normal sectional seismic bearing capacity, calculated in accordance with the sectional area of actual arranged reinforcing steel and the material strength standard value in consideration of the seismic adjustment coefficient of load-bearing

capacity;

M_c^t , M_c^b —respectively the bending moment design value of at the upper and lower end of the frame column after adjustment in consideration of the seismic combination; the net height of column.

In Formula (11.4.3-1), the sum of M_{cua}^t and M_{cua}^b shall be respectively calculated in clockwise and counterclockwise directions, and the larger value shall be taken; N may take the design value of the axial compressive force generated by the representative value of the gravity load.

In Formula (11.4.3-2) ~ Formula (11.4.3-5), the sum of M_c^t and M_c^b shall be respectively calculated in clockwise and counterclockwise directions, and the larger value shall be taken. The values of M_c^t and M_c^b shall meet the requirements of Article 11.4.1 and Article 11.4.2 of this code.

11.4.4 For the frame-supported column of Seismic Grade 1 and 2, the additional axial force caused by seismic action shall respectively multiply amplified coefficients 1.5 and 1.2; in calculation of axial compression ratio, the amplified coefficient may not be considered.

11.4.5 For corner columns of frames of all seismic grades, the design values of bending moment and shear force shall multiply the amplified coefficient no less than 1.1 based on the adjustment according to Article 11.4.1 ~ Article 11.4.3 of this code.

11.4.6 For the frame columns and frame-supported columns with rectangular sections in consideration of seismic combination, the shear section shall meet the following conditions:

Frame column with ratio of shear span to effective depth λ larger than 2

$$V_c \leq \frac{1}{\gamma_{RE}} (0.2\beta_c f_c b h_0) \quad (11.4.6-1)$$

Frame column with ratio of shear span to effective depth λ no larger than 2 and frame-supported column

$$V_c \leq \frac{1}{\gamma_{RE}} (0.15\beta_c f_c b h_0) \quad (11.4.6-2)$$

Where:

——the calculated ratio of shear span to effective depth of frame columns and frame-supported columns, taking ; hereof, M should take the larger one of bending moment design values at the upper and lower ends of columns in consideration of seismic combination, V takes the design value of shear force corresponding to M , represents the effective depth of column section; when the inflection point of the frame column in the frame structure is within the range of column storey height, it may be taken as $\lambda =$, here, represents the net height of the column.

11.4.7 For frame columns and frame-supported columns with rectangular sections in consideration of seismic combination, the shear bearing capacity of inclined section shall meet the following requirements:

$$V_c \leq \frac{1}{\gamma_{RE}} \left[\frac{1.05}{\lambda + 1} f_t b h_0 + f_{yv} \frac{A_{sv}}{s} h_0 + 0.056N \right] \quad (11.4.7)$$

Where:

λ ——the calculated ratio of shear span to effective depth of the frame column and frame-supported column; when λ is less than 1.0, it takes 1.0; when λ is larger than 3.0, it takes 3.0;

N ——the design value of axial compressive force of the frame column and frame-supported column in consideration of seismic combination, when N is larger than , it takes .

11.4.8 For the frame column and frame-supported column with rectangular sections in consideration of seismic combination, when there is tensile force, the seismic shear bearing capacity of the inclined section shall meet the following requirements:

$$V_c \leq \frac{1}{\gamma_{RE}} \left[\frac{1.05}{\lambda + 1} f_t b h_0 + f_{yv} \frac{A_{sv}}{s} h_0 + 0.2N \right] \quad (11.4.8)$$

Where:

N ——the design value of axial tensile force of the frame column in consideration of seismic combination.

When the calculated value in the right bracket of the above Formula is less than $f_{yv} \frac{A_{sv}}{s} h_0$, it takes $f_{yv} \frac{A_{sv}}{s} h_0$, and the value of $f_{yv} \frac{A_{sv}}{s} h_0$ shall not be less than .

11.4.9 For the reinforced concrete frame column under two-way shear force on the rectangular section in consideration of seismic combination, the shear section shall meet the following conditions;

$$V_x \leq \frac{1}{\gamma_{RE}} 0.2\beta_c f_c b h_0 \cos \theta \quad (11.4.9-1)$$

$$V_y \leq \frac{1}{\gamma_{RE}} 0.2\beta_c f_c b h_0 \sin \theta \quad (11.4.9-2)$$

Where:

V_x ——the design value of shear force along x axis direction, the corresponding effective depth of section is , the section width is b ;

V_y ——the design value of shear force in the y axis direction, the corresponding effective depth of section is , the section width is b ;

θ ——the included angle between the action directions of inclined directional design value V of shear force and x axis, takes

11.4.10 For the reinforced concrete frame column under two-way shear force on the rectangular section, the shear bearing capacity of inclined section shall meet the following conditions:

$$V_x \leq \frac{V_{ux}}{1 + \left(\frac{V_{ux} \tan \theta}{V_{uy}} \right)^2} \quad (11.4.10-1)$$

$$V_y \leq \frac{V_{uy}}{1 + \left(\frac{V_{uy}}{V_{ux} \tan \theta} \right)^2} \quad (11.4.10-2)$$

$$V_{ux} = \frac{1}{\gamma_{RE}} \left[\frac{1.05}{\lambda_x + 1} f_t b h_0 + f_{yv} \frac{A_{svx}}{s_x} h_0 + 0.056 N \right] \quad (11.4.10-3)$$

$$V_{uy} = \frac{1}{\gamma_{RE}} \left[\frac{1.05}{\lambda_y + 1} f_t b h_0 + f_{yv} \frac{A_{svy}}{s_y} b_0 + 0.056 N \right] \quad (11.4.10-4)$$

Where:

λ_x, λ_y ——the calculated ratio of shear span to effective depth of the frame column, determined according the requirements of Article 6.3.12 of this code;

A_{svx}, A_{svy} ——the total sectional area of limbs of stirrups parallel to x axis and y axis arranged on the same section;

N ——the design value of axial compressive force corresponding to the design value of shear force V in inclined direction, when N is larger than , it takes ; hereof, A represents the sectional area of members.

In calculation of sectional stirrup, it may approximately take $\gamma_{RE} = 1$ for the calculation in Formula (11.4.10-1) and Formula (11.4.10-2).

11.4.11 The section dimension of frame column shall meet the following requirements:

1 For the column with rectangular section, when it is at Seismic Grade 4 or the number of storey is not larger than 2, the minimum section dimension should not be less than 300mm, when it is at Seismic Grade 1, 2, 3 and the number of storey exceeds 2, it should not be less than 400mm; for the section diameter of round columns, when it is at Seismic Grade 4 or the number of storey does not exceed 2, it should not be less than 350mm, when it is at Seismic Grade 1, 2, 3 and the number of storey exceeds 2, it should not be less than 450mm.

2 The ratio of shear span to effective depth of the column should be larger than 2;

3 The side ratio of the long side to the short side of column section should not be larger than 3.

11.4.12 The arrangement of steel reinforcement for frame columns and frame-supported columns shall meet the following requirements:

1 The percentage of reinforcement of all the longitudinal stressed steel reinforcements in frame columns and frame-supported columns shall not be less than the value specified in Table 11.4.12-1, meanwhile, the percentage of reinforcement at each side shall not be less than 0.2; for the higher high-rise buildings in site of Category IV, the minimum percentage of reinforcement shall be increased by 0.1;

Table 11.4.12-1 Minimum Percentage of Reinforcement of all the Longitudinal Stressed Steel Reinforcements of Columns (%)

Column type	Seismic grade			
	Grade 1	Grade 2	Grade 3	Grade 4
Central column, border column	0.9(1.0)	0.7(0.8)	0.6(0.7)	0.5(0.6)
Corner column, frame-supported column	1.1	0.9	0.8	0.7

Notes:

- 1 The value in the bracket of this table is used for the column of frame structure;
- 2 When longitudinal stressed steel reinforcements of Grade 335MPa and Grade 400MPa are adopted, the values in the table shall be respectively adopted by increasing 0.1 and 0.05;
- 3 When the concrete strength grade is above C60, the value in the table shall be adopted by increasing 0.1.

2 The stirrup at the upper and lower ends of frame columns and frame-supported columns shall be densified; the largest spacing and the minimum diameter of stirrup in densified area shall be in accordance with the requirements of those specified in Table 11.4.12-2;

Table 11.4.12-2 Constructional Requirements of Stirrup at Column End in Densified Area

Seismic grade	The largest spacing of stirrup (mm)	The minimum diameter of stirrup (mm)
Grade 1	The smaller value between 6 times of diameter of longitudinal steel reinforcement and 100	10
Grade 2	The smaller value between 8 times of diameter of longitudinal steel reinforcement and 100	8
Grade 3	The smaller value between 8 times of diameter of longitudinal steel reinforcement and 150 (for column root, 100)	8
Grade 3	The smaller value between 8 times of diameter of longitudinal steel reinforcement and 150 (for column root, 100)	6 (for column root, 8)

Note: The column root refers to the range of densified area of stirrup at the lower end of bottom storey column.

3 For frame-supported column and frame columns with ratio of shear span to effective depth no larger than 2, stirrup shall be densified within the range of column total height, and the stirrup spacing shall meet the requirements of Seismic Grade 1 specified in Clause 2 of this article;

4 On the condition that the diameter of stirrup in the frame column of Seismic Grade 1 is larger than 12mm and the spacing of stirrup legs is not larger than 150mm, the diameter of frame column of Seismic Grade 2 is not less than 10mm and the spacing of stirrup legs is not larger than 200mm, the stirrup spacing shall be allowed to adopt 150mm except the lower end of the bottom storey column; if the ratio of shear span to effective depth of frame column of Seismic Grade 4 is not larger than 2, the diameter of stirrup shall not be less than 8mm.

11.4.13 When the border column and corner column of frame as well as the end column of shear wall are in small eccentric tension under the seismic combination, the gross sectional area of longitudinal stressed steel reinforcements in column shall be increased by 25% than calculated value.

The ratio of reinforcement of all the longitudinal stressed steel reinforcements in frame column and frame-supported column shall not be greater than 5%. The longitudinal steel reinforcement of column should be in symmetrical arrangement. For the column with section

dimension larger than 400mm, the spacing of longitudinal steel reinforcement should not be larger than 200mm. When it is designed according to Seismic Grade 1, and the ratio of shear span to effective depth of column is not larger than 2, the ratio of reinforcement of longitudinal steel reinforcement at each side of column should not be larger than 1.2%.

11.4.14 The length in densified area of stirrup of frame column shall take the maximum values among the long side dimension of the column section (or circular section diameter) and 1/6 of the column net height and 500mm; for the corner column of Seismic Grade 1 and 2, the stirrup shall be densified along the column's total height. The length in densified area of stirrup of column root at bottom storey shall take the value no less than 1/3 of the net height of column at this storey; when there are rigid ground, besides the densified area of stirrup at column end, the stirrup shall still be densified within the range of 500mm height from upper and lower of the rigid ground.

11.4.15 For the spacing of stirrup legs within the range of densified area of column stirrup: it should not be larger than 200mm at Seismic Grade 1; it should not be larger than the larger value between 250mm and 20 times of the diameter of stirrup at Seismic Grade 2 and 3; it should not be larger than 300mm at Seismic Grade 4. Every other longitudinal steel reinforcement should be restrained in two directions with the stirrup or tie bar; when tie bar is adopted and there is binding between stirrups and longitudinal steel reinforcements, the tie bar should be abut against longitudinal steel reinforcements and hooked with stirrups.

11.4.16 For every kind of structural frame columns and frame-supported columns of Seismic Grade 1, 2, 3 and 4, the axial compression ratio should not be larger than the limit value specified in Table 11.4.16. For the higher high-rise buildings in site of Category IV, the limit value of axial compression ratio of columns shall be properly reduced.

Table 11.4.16 Axial Compression Ratio of Column

Structural system	Seismic grade			
	Grade 1	Grade 2	Grade 3	Grade 4
Frame structure	0.65	0.75	0.85	0.90
Frame-shear wall structure, tube structure	0.75	0.85	0.90	0.95
Part frame-supported shear wall structure	0.60	0.70	-	

Notes:

- 1 The axial compression ratio refers to the ratio of the design value of axial compressive force on the column under seismic action combination to the product of the total section area of column and the design value of concrete axial compressive strength ;
- 2 When the concrete strength grade is C65 and C70, the limit value of axial compression ratio should be reduced by 0.05 according to those values specified in this table; when the concrete strength grade is C75 and C80, the limit value of axial compression ratio should be reduced by 0.10 according to those values specified in this table;
- 3 The limit value in this table is applicable to the column with ratio of shear span to effective depth larger than 2 and concrete strength no greater than C60; for the column with ratio of shear span to effective depth no larger than 2, the limit value of axial compression ratio shall be reduced by 0.05; for the column with ratio of shear span to effective depth less than 1.5, the limit value of axial compression ratio shall be specially studied and special constructional measures shall be taken;
- 4 On the condition that cross compound stirrup is adopted along total column height, and spacing of stirrup is not larger than 100mm, spacing of stirrup legs is not larger than 200mm, diameter is not less than 12mm, or compound spiral stirrup is adopted along total column height, and thread pitch is not larger than 100mm, spacing of stirrup legs is not

larger than 200mm, diameter is not less than 12mm; or continuous compound rectangular spiral stirrup is adopted along total column height, and spiral clear distance is not larger than 80mm, spacing of stirrup legs is not larger than 200mm, diameter is not less than 10mm, the limit value of axial compression ratio may all be increased by 0.10 according to the values in this table;

- 5 When the core column formed by additional longitudinal steel reinforcement is arranged in column sectional central part, and the gross sectional area of additional longitudinal steel reinforcement is not less than 0.8% of the column sectional area, the limit value of axial compression ratio may be increased by 0.05 according to those values specified in this table; when this measure is adopted with the measure of Note 4 at the same time, the limit value of axial compression ratio may be increased by 0.15 according to those values specified in this table, but the stirrup characteristic value λ_v of stirrup still shall be determined in accordance with the requirements of increasing 0.10 for axial compression ratio;
- 6 The adjusted limit value of axial compression ratio shall not be greater than 1.05.

11.4.17 The reinforcement ratio per unit volume of stirrup in stirrup densified area shall meet the following requirements:

- 1 The ratio of volumetric reinforcement for stirrups in densified area of column stirrup shall meet the following requirements:

$$\rho_v \geq \lambda_v \frac{f_c}{f_{yv}} \quad (11.4.17)$$

Where:

ρ_v ——the ratio of volumetric reinforcement in densified area of column stirrup, calculated according to the requirements of Article 6.6.3 of this code, the stirrup volume in overlap section shall be subtracted in calculation;

f_{yv} ——the design value of tensile strength of stirrup;

f_c ——the design value of axial compressive strength of concrete; when the strength grade is less than C35, the value is taken as C35;

λ_v ——the minimum stirrup characteristic value, adopted according to the requirements specified in Table 11.4.17;.

Table 11.4.17 Minimum Stirrup Characteristic Value λ_v of Stirrup in Densified Area of Column Stirrup

Seismic grade	Stirrup type	Axial compression ratio								
		≤0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.05
Grade 1	Ordinary stirrup, compound stirrup	0.10	0.11	0.13	0.15	0.17	0.20	0.23	-	-
	Spiral stirrup, compound or continuous compound rectangular spiral stirrup	0.08	0.09	0.11	0.13	0.15	0.18	0.21	-	-
Grade 2	Ordinary stirrup, compound stirrup	0.08	0.09	0.11	0.13	0.15	0.17	0.19	0.22	0.24
	Spiral stirrup, compound or continuous compound rectangular spiral stirrup	0.06	0.07	0.09	0.11	0.13	0.15	0.17	0.20	0.22
Grade 3, 4	Ordinary stirrup, compound stirrup	0.06	0.07	0.09	0.11	0.13	0.15	0.17	0.20	0.22
	Spiral stirrup, compound or continuous compound rectangular spiral stirrup	0.05	0.06	0.07	0.09	0.11	0.13	0.15	0.18	0.20

Notes:

- 1 The ordinary stirrup refers to a single rectangular stirrup or a single round stirrup; the spiral stirrup refers to a single spiral stirrup; the compound stirrup refers to the stirrups composed of rectangular, polygon, round stirrups or tie bars; compound spiral stirrup refers to the stirrups composed of spiral stirrups and rectangular, polygon, round stirrups or

tie bars; continuous compound rectangular spiral stirrup refers to the stirrup in which all the spiral stirrups are processed by the same steel reinforcement;

- 2 In calculation of ratio of volumetric reinforcement for the compound spiral stirrup, hereinto the volume of non- spiral stirrup shall be multiplied by coefficient 0.8;
- 3 When the concrete strength grade is greater than C60, the stirrup should be adopted with compound stirrup, compound spiral stirrup or continuous-compound-rectangular-spiral stirrup, when the axial compression ratio is not larger than 0.6, the minimum stirrup characteristic value in the densified area should be increased by 0.02 according to those values specified in this table; when the axial compression ratio is larger than 0.6, it should be increased by 0.03 according to those values specified in this table.

2 For the column of Seismic Grade 1, 2, 3 and 4, the ratio of volumetric reinforcement for stirrups in stirrup densified area shall not be less than 0.8%, 0.6%, 0.4% and 0.4% respectively;

3 Frame-supported columns should be adopted with compound spiral stirrups or cross compound stirrups, the minimum stirrup characteristic value shall be increased by 0.02 according to those values specified in Table 11.4.17 and the ratio of volumetric reinforcement shall not be less than 1.5%;

4 When the ratio of shear span to effective depth λ is not larger than 2, compound spiral stirrups or cross compound stirrups should be adopted, and the ratio of volumetric reinforcement for stirrups shall not be less than 1.2%; at Protection Intensity 9 and Seismic Grade 1, the ratio of volumetric reinforcement shall not be less than 1.5%.

11.4.18 Outside of the densified area of stirrup, the ratio of volumetric reinforcement for stirrups should not be less than half of the ratio of reinforcement in densified area; for Seismic Grade 1 and 2, the stirrup spacing shall not be larger than $10d$; for Seismic Grade 3 and 4, the stirrup spacing shall not be larger than $15d$, here, d represents the diameter of longitudinal steel reinforcements.

11.5 Column of Hinged Bent

11.5.1 The longitudinal stressed steel reinforcements and stirrups of columns of hinged bent shall be determined by calculating according to the design value of bending moment and design value of shear force under seismic combination as well as the relevant requirements of Section 11.4 of this code; the construction shall not only meet the relevant requirements of this section but also shall meet the relevant requirements of Chapter 8, Chapter 9, Section 11.1 and Section 11.2 of this code.

11.5.2 The stirrup densified area of columns of hinged bent shall meet the following requirements:

- 1 Length of stirrup densified area:
 - 1) For the column top sector, it shall take 500mm below the column top and shall not be less than the depth of section at the column top;
 - 2) For the crane beam sector, it shall take 300mm above the upper column root to the top surface of the crane beam;
 - 3) For the column root sector, it shall take 500mm above the foundation top to the indoor terrace;
 - 4) For the bracket sector, it shall take the total height of bracket;

- 5) For connection nodes of inter-column support and columns as well as the parts where the column displacement is restrained, it shall take 300mm above and below the node respectively.

2 In the stirrup densified area, the maximum spacing of stirrups shall be 100mm; the diameter of stirrups shall be in accordance with those specified in Table 11.5.2.

Table 11.5.2 Minimum Diameter of Stirrup in Stirrup Densified Area of Column of Hinged Bent (mm)

Sector in densified area	Seismic grade and site category					
	Grade 1	Grade 2	Grade 2	Grade 3	Grade 3	Grade 4
	All category of sites	Site Category III and IV	Site Category I and II	Site Category III and IV	Site Category I and II	All category of sites
Ordinary column top and column root sector	8(10)		8		6	
Corner column top	10		10		8	
Crane beam and bracket sector Column root sector with support	10		8		8	
Column top sector with support Part where the columnar deflection is restrained	10		10		8	

Note: The value in the parentheses in this table is used for the column root.

11.5.3 When the hinged bent in side direction is restrained and the height from the restrained point to the column top is not larger than 2 times of side length of the column section in this direction, the construction of embedded steel plates at the column top and the stirrup densified area at the column top still shall meet the following requirements:

1 The length of embedded steel plates at the column top along the plane direction of bent should adopt the depth of section h of the column top, but in any case it shall not be less than $h/2$ or 300mm;

2 When the eccentric distance e_0 of the axial force on the column top in the bent plane is within range of $h/6 \sim h/4$, the ratio of volumetric reinforcement for stirrups in the stirrup densified area at the column top: should not be less than 1.2% for Seismic Grade 1; should not be less than 1.0% for Seismic Grade 2; should not be less than 0.8% for Seismic Grade 3 and 4.

11.5.4 Under the action of vertical force and horizontal tensile force of seismic combination, the calculation and arrangement of reinforcement of the column bracket for supporting such roof structures as the roof beam and the roof truss of low span in the plants with unequal heights shall not only be carried out according to the requirements of Section 9.3 of this code but also shall meet the following requirements:

1 Anchor bar under horizontal tensile force: it shall not be less than two sticks of steel reinforcements in diameter of 16mm for Seismic Grade 1, it shall not be less than two sticks of steel reinforcements in diameter of 14mm for Seismic Grade 2, it shall not be less than two sticks of steel reinforcements in diameter of 12mm for Seismic Grade 3 and 4;

2 The anchoring measures and anchorage length of longitudinal tension steel

reinforcements and anchor bars in the bracket shall meet the relevant requirements of Article 9.3.12 of this code, but the anchorage length l_a of tension steel reinforcements therein shall be replaced by l_{aE} ;

3 The minimum diameter of horizontal stirrup of the bracket shall be 8mm and the maximum spacing shall be 100mm.

11.5.5 The straight anchor bars in embedded parts at the column top of columns of hinged bent shall not only meet the requirements of Article 11.1.9 of this code but also shall meet the following requirements:

1 At Seismic Grade 1, it shall not be less than four sticks of straight anchor bars in diameter of 16mm;

2 At Seismic Grade 2, it shall not be less than four sticks of straight anchor bars in diameter of 14mm;

3 For columns with inter-column support, the embedded parts at the column top shall be added with shear resistant steel plates.

11.6 Nodes of Frame Beam Column

11.6.1 The frame of Seismic Grade 1, 2 and 3 shall be carried out with the checking of the seismic shear bearing capacity in node core zone; the node of frame of Seismic Grade 4 may not be calculated, but it shall meet the requirements of details of seismic design. The checking methods of seismic shear bearing capacity and details of seismic design on the nodes of intermediate storey of frame-supported storey are the same as that on nodes of middle storey of frames.

11.6.2 The design value of shear force V_j in the node core zone of the frame beam column of Seismic Grade 1, 2 and 3 shall be calculated according to the following requirements:

1 Intermediate node and end node at top storey

1) Frame structure of Seismic Grade 1 and frame of Seismic Grade 1 at Protection Intensity 9:

$$V_j = \frac{1.15 \sum M_{bua}}{h_{b0} - a'_s} \quad (11.6.2-1)$$

2) Other conditions:

$$V_j = \frac{\eta_{jb} \sum M_b}{h_{b0} - a'_s} \quad (11.6.2-2)$$

2 Intermediate node and end node at other storey

1) Frame structure of Seismic Grade 1 and frame of Seismic Grade 1 at Protection Intensity 9:

$$V_j = \frac{1.15 \sum M_{bua}}{h_{b0} - a'_s} \left(1 - \frac{h_{b0} - a'_s}{H_c - h_b} \right) \quad (11.6.2-3)$$

2) Other conditions:

$$V_j = \frac{\eta_{jb} \sum M_b}{h_{b0} - a'_s} \left(1 - \frac{h_{b0} - a'_s}{H_c - h_b} \right) \quad (11.6.2-4)$$

Where:

$\sum M_{\text{bua}}$ ——the sum of bending moment values corresponding to the actual arranged normal section seismic flexural load-bearing capacity in counterclockwise or clockwise directions of the beam end at left and right sides of nodes; it may be determined according to the actual arranged area of steel reinforcements (including longitudinal compression steel reinforcements) and the standard value of material strength;

$\sum M_b$ ——the sum of combined bending moment design values in counterclockwise or clockwise directions of the beam end at left and right sides of nodes; the bending moment with smaller absolute value shall take zero where the beam end at left and right sides of nodes of Seismic Grade 1 frames are all hogging moment;

η_{jb} ——the amplified coefficient of the shear force on the node, for the frame structure, it shall take 1.50 for Grade 1, 1.35 for Grade 2 and 1.20 for Grade 3; for frames in other structures, it shall take 1.35 for Grade 1, 1.20 for Grade 2 and 1.10 for Grade 3;

h_{b0}, h_b ——the respectively effective depth of section and the depth of section for beams, when the beam depth at both sides of nodes are different, the mean value shall be taken;

H_c ——the distance between inflection points of upper columns and lower columns of nodes;

a'_s ——the distance from the point for resultant forces of longitudinal compression steel reinforcements to near-side of the section.

11.6.3 The shear horizontal section in node core zone of frame beam columns shall meet the following conditions:

$$V_j \leq \frac{1}{\gamma_{\text{RE}}} (0.3\eta_j \beta_c f_c b_j h_j) \quad (11.6.3)$$

Where:

h_j ——the depth of section in node core zone of frames, which may be taken as the column section depth h_c in the checking direction;

b_j ——the effective checking width of the section in node core zone of frames, where b_b is not less than $b_c/2$, it may take b_c ; where b_b is less than $b_c/2$, it may take the smaller value between $(b_b+0.5h_c)$ and b_c ; when the center lines of beams and columns are not in the same line and the eccentric distance e_0 is not larger than $b_c/4$, it may take the smallest value among $(b_b+0.5h_c)$, $(0.5b_b+0.5b_c+0.25h_c-e_0)$ and b_c . Here, b_b is the sectional width of beams in checking directions; b_c is the sectional width of this lateral column;

η_j ——the restrained influence coefficient of orthogonal beams for nodes: when the floor slab is cast-in-situ, the center lines of beams and columns coincide, the respectively sectional width of beams at four sides is not less than 1/2 of the sectional width of this lateral column, and the depth of beam in orthogonal directions is not less than 3/4 of height of higher frame beams, η_j may be taken as 1.50, but η_j should be taken as 1.25 at Protection Intensity 9; η_j shall be taken as 1.00 when above-mentioned conditions are not met.

11.6.4 The seismic shear bearing capacity of frame beam-column nodes shall meet the following requirements:

1 Frame of Seismic Grade 1 at Protection Intensity 9

$$V_j \leq \frac{1}{\gamma_{RE}} \left(0.9\eta_j f_t b_j h_j + f_{yv} A_{svj} \frac{h_{b0} - a'_s}{s} \right) \quad (11.6.4-1)$$

2 Other conditions

$$V_j \leq \frac{1}{\gamma_{RE}} \left(1.1\eta_j f_t b_j h_j + 0.05\eta_j N \frac{b_j}{b_c} + f_{yv} A_{svj} \frac{h_{b0} - a'_s}{s} \right) \quad (11.6.4-2)$$

Where:

N ——the design value of axial force on the bottom of upper columns of nodes corresponding to the design value of shear force in consideration of seismic combination; where N is the compression, it shall take the smaller value of the design values of axial compressive force, and where N is larger than $0.5f_c b_c h_c$, it shall take $0.5f_c b_c h_c$; where N is the tension, it shall take 0;

A_{svj} ——the total sectional area of stirrups at the same section checking direction within the range of effective checking width in core zone;

h_{b0} ——the effective depth of section of frame beams, which may take mean value where the depth of section of beams at both sides of nodes is not same.

11.6.5 For beam-column joints of circular column frames, when the beam center line and column center line coincide, the shear horizontal section shall meet the following conditions:

$$V_j \leq \frac{1}{\gamma_{RE}} (0.3\eta_j \beta_c f_c A_j) \quad (11.6.5)$$

Where:

A_j ——the effective sectional area in node core zone: where the beam width $b_b \geq 0.5D$, $A_j = 0.8D^2$; where $0.4D \leq b_b < 0.5D$, $A_j = 0.8D(b_b + 0.5D)$;

D ——the diameter of circular column section;

b_b ——the sectional width of beams;

η_j ——the restrained influence coefficient of orthogonal beams for nodes, which may be adopted according to Article 11.6.3 of this code.

11.6.6 For beam-column joints of circular column frames, when the beam center line and column center line coincide, the seismic shear bearing capacity shall meet the following requirements:

1 Frame of Seismic Grade 1 at Protection Intensity 9

$$V_j \leq \frac{1}{\gamma_{RE}} \left(1.2\eta_j f_t A_j + 1.57 f_{yv} A_{sh} \frac{h_{b0} - a'_s}{s} + f_{yv} A_{svj} \frac{h_{b0} - a'_s}{s} \right) \quad (11.6.6-1)$$

2 Other conditions

$$V_j \leq \frac{1}{\gamma_{RE}} \left(1.5\eta_j f_t A_j + 0.05\eta_j \frac{N}{D^2} A_j + 1.57 f_{yv} A_{sh} \frac{h_{b0} - a'_s}{s} + f_{yv} A_{svj} \frac{h_{b0} - a'_s}{s} \right) \quad (11.6.6-2)$$

Where:

h_{b0} ——the effective depth of beam section;

A_{sh} ——the sectional area of single circular stirrup;

A_{svj} ——the total sectional area of tie bars and non-circular stirrups at the same section checking direction.

11.6.7 The anchorage and overlapping for longitudinal stressed steel reinforcements of frame beams and frame columns in frame node zone shall meet the following requirements:

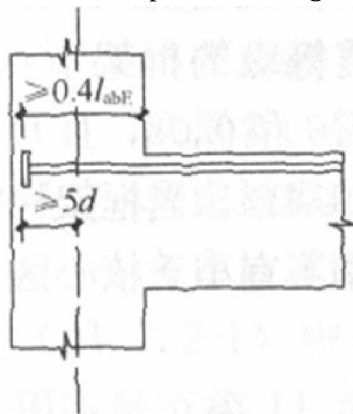
1 At intermediate node of middle storey of frames, the upper longitudinal steel reinforcements of frame beams shall be penetrated into the intermediate nodes. The diameter of each beam longitudinal steel reinforcement penetrated central columns, for all classes of frames at Protection Intensity 9 and frame structures of Seismic Grade 1, when the column is in rectangular sections, it should not be larger than 1/25 of the sectional dimension of the column in this direction, when the column is in circular sections, it should not be larger than 1/25 of the chord length of the column section where the longitudinal steel reinforcement is located; at Seismic Grade 1, 2 and 3, when the column is in rectangular sections, it should not be larger than 1/20 of the sectional dimension of the column in this direction, for circular column sections, it should not be larger than 1/20 of the chord length of the column section where the longitudinal steel reinforcement is located.

2 For intermediate nodes at intermediate storey of frames, end nodes at intermediate storey, intermediate nodes at top storey and end nodes at top storey, the anchorage and overlapping of longitudinal steel reinforcements in beams and columns at node parts shall meet relevant detailing requirements of Figure 11.6.7. l_{IE} in the figure shall be adopted according to the requirements of Article 11.1.7 of this code, and l_{abE} shall be adopted according to the following formula:

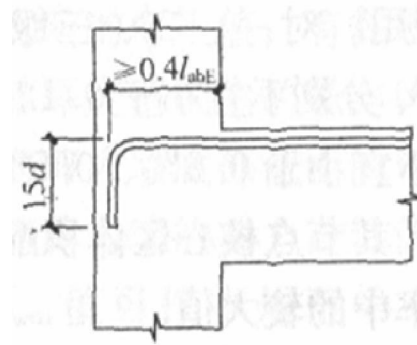
$$l_{abE} = \xi_{aE} l_{ab} \quad (11.6.7)$$

Where:

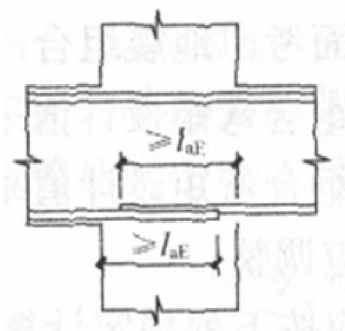
ξ_{aE} ——the correction coefficient of anchorage length of longitudinal tension steel reinforcements, adopted according to the requirements of Article 11.1.7.



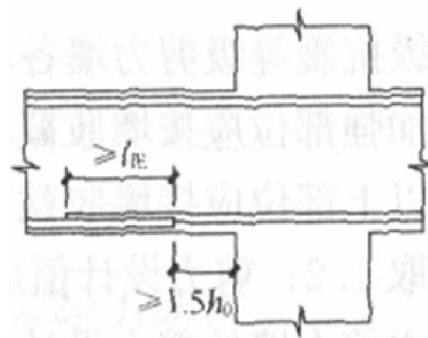
(a) Anchorage of Beam Reinforcement at End Node in Intermediate Storey plus Anchor Head (Anchor Slab)



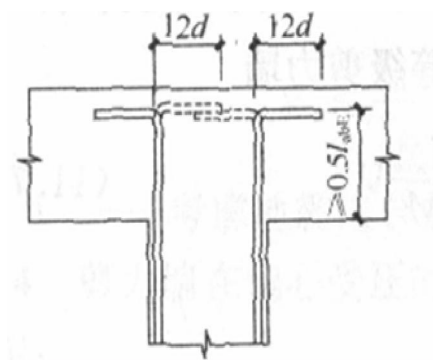
(b) 90° Bending Anchorage of Beam Reinforcement at End Node in Intermediate Storey



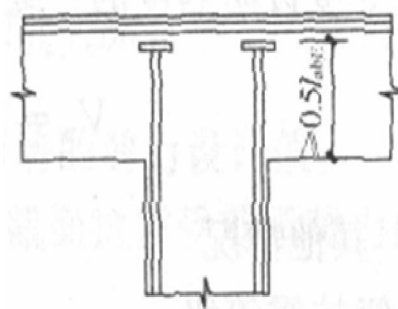
(c) Straight Anchorage of Beam Reinforcement at Intermediate Node in Intermediate Storey inside Nodes



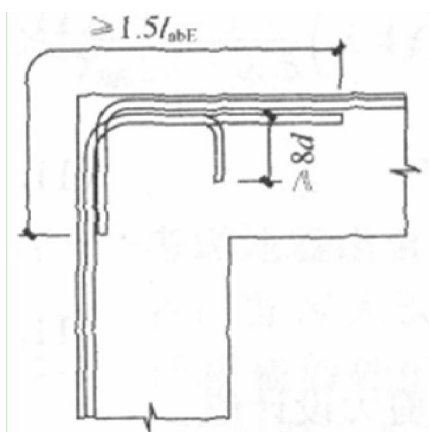
(d) Overlapping of Beam Reinforcement at Intermediate Node in Intermediate Storey outside Nodes



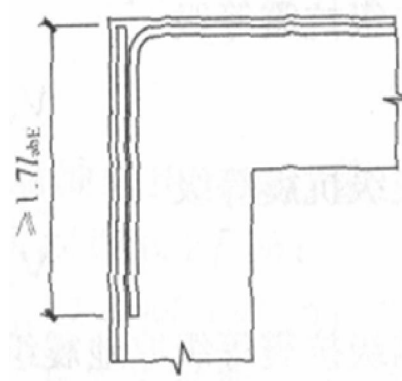
(e) 90° Bending Anchorage of Column Reinforcement at Intermediate Node at Top Storey



(f) Anchorage of Column Reinforcement at Intermediate Node at Top Storey plus Anchor Head (Anchor Slab)



(g) Bending Overlapping of Steel Reinforcement at Outer Side of End Node at Top Storey and at Top of Beam End



(h) Linear Overlapping of Steel Reinforcement at Outer Side of End Node at Top Storey

Figure 11.6.7 Anchorage and Overlapping of Longitudinal Stressed Steel Reinforcements of Beams and Columns in Node Zone

11.6.8 The maximum spacing and minimum diameter of the stirrup in the frame node zone should be adopted according to those specified in Table 11.4.12-2 of this code. For the frame node core zone of Seismic Grade 1, 2 and 3, the stirrup characteristic value λ_v should not be less than 0.12, 0.10 and 0.08 respectively and the ratio of volumetric reinforcement for stirrups should not be less than 0.6%, 0.5% and 0.4% respectively. When the ratio of shear span to effective depth for the frame column is not larger than 2, the volumetric stirrup ratio in the node core zone should not be less than the larger value of the volumetric stirrup ratio at

upper and lower column ends in the core zone.

11.7 Shear Walls and Connecting Beams

11.7.1 For the design value for bending moment at each wall-column section on the shear wall of Seismic Grade 1 in consideration of seismic combination, it shall be adopted according to the design value for bending moment in seismic combination at wall-column section for bottom reinforced part, and it shall be adopted by using the design value for bending moment in seismic combination at wall-column section multiplying the amplified coefficient for the part above the bottom reinforced part and the value may be taken as 1.2; the design value of shear force shall be adjusted correspondingly.

11.7.2 The design value of shear force V_w in consideration of the shear wall shall be calculated according to the following requirements:

1 Bottom reinforced part

1) Shear wall of Seismic Grade 1 at Protection Intensity 9

$$V_w = 1.1 \frac{M_{wua}}{M} V \quad (11.7.2-1)$$

2) Other conditions

Seismic Grade 1

$$V_w = 1.6V \quad (11.7.2-2)$$

Seismic Grade 2

$$V_w = 1.4V \quad (11.7.2-3)$$

Seismic Grade 3

$$V_w = 1.2V \quad (11.7.2-4)$$

The design value of shear force under seismic combination shall be taken for Seismic Grade 4.

2 Other positions

$$V_w = V \quad (11.7.2-5)$$

Where:

M_{wua} ——the bending moment of the bottom section of the shear wall, which is corresponding to the seismic load-bearing capacity of normal section and is calculated according to the sectional area of actual arranged steel reinforcements and the standard value of material strength in consideration of seismic adjustment coefficient of load-bearing capacity; if any wing wall is built, the longitudinal steel reinforcements within the range of one times of the wing wall thickness at both sides of the wall shall be included;

M ——the design value for bending moment at bottom section on the shear wall in consideration of seismic combination;

V ——the design value of shear force on the shear wall in consideration of seismic combination.

In Formula (11.7.2-1), the M_{wua} value may be determined by adopting related and same method for calculating the M_{cua} value at frame column end specified in Article 11.4.3 of this code according to the requirements of Article 6.2.19 of this code, but the γ_{RE} value shall take the seismic adjustment coefficient of normal section load-bearing capacity of the shear wall.

11.7.3 The shear section of the shear wall shall meet the following requirements:

When the ratio of shear span to effective depth is larger than 2.5,

$$V_w \leq \frac{1}{\gamma_{RE}} (0.2 \beta_c f_c b h_0) \quad (11.7.3-1)$$

When the ratio of shear span to effective depth is not larger than 2.5,

$$V_w \leq \frac{1}{\gamma_{RE}} (0.15 \beta_c f_c b h_0) \quad (11.7.3-2)$$

Where:

V_w ——the design value of shear force on the shear wall in consideration of seismic combination.

11.7.4 The seismic shear bearing capacity of inclined section on the shear wall in eccentric compression shall meet the following requirements:

$$V_w \leq \frac{1}{\gamma_{RE}} \left[\frac{1}{\lambda - 0.5} \left(0.4 f_t b h_0 + 0.1 N \frac{A_w}{A} \right) + 0.8 f_{yv} \frac{A_{sh}}{s} h_0 \right] \quad (11.7.4)$$

Where:

N ——the smaller one of the design value for axial compressive force on the shear wall in consideration of seismic combination; where N is larger than $0.2 f_c b h$, it shall take $0.2 f_c b h$;

λ ——the ratio of shear span to effective depth at calculated section, taking $\lambda = M/(V h_0)$; where λ is less than 1.5, it shall take 1.5; where λ is larger than 2.2, it shall take 2.2; hereof, M is the design value for bending moment corresponding to the design shear force value V ; when the distance between the calculated section and the wall bottom is less than $h_0/2$, it shall be calculated according to the design value of shear force and design value of bending moment at the place where is $h_0/2$ away from the wall bottom.

11.7.5 The seismic shear bearing capacity of inclined section on the shear wall in eccentric tension shall meet the following requirements:

$$V_w \leq \frac{1}{\gamma_{RE}} \left[\frac{1}{\lambda - 0.5} \left(0.4 f_t b h_0 - 0.1 N \frac{A_w}{A} \right) + 0.8 f_{yv} \frac{A_{sh}}{s} h_0 \right] \quad (11.7.5)$$

Where:

N ——the larger one of the design value for axial tensile force on the shear wall in consideration of seismic combination.

When the calculated value in the right square brackets of Formula (11.7.5) is less than $0.8 f_{yv} \frac{A_{sh}}{s} h_0$, it shall be taken as equivalent to $0.8 f_{yv} \frac{A_{sh}}{s} h_0$.

11.7.6 For shear wall of Seismic Grade 1, the shear bearing capacity at horizontal construction joint shall meet the following requirements:

$$V_w \leq \frac{1}{\gamma_{RE}} (0.6 f_y A_s + 0.8 N) \quad (11.7.6)$$

Where:

N ——the design value of axial force at horizontal construction joint in consideration of seismic combination, the positive value shall be adopted for compression, and the negative value shall be adopted for tension;

A_s ——the sectional area of all vertical steel reinforcements at horizontal construction joint of the shear wall, including the gross sectional area of vertically distributed steel reinforcements, additional vertical steel dowel and longitudinal steel reinforcements in boundary members (excluding wing walls at both sides).

11.7.7 For the connecting beam in tube and shear wall opening, when the symmetric reinforcement is adopted, the normal section flexural load-bearing capacity shall meet the following requirements:

$$M_b \leq \frac{1}{\gamma_{RE}} [f_y A_s (h_0 - a'_s) + f_{yd} A_{sd} z_{sd} \cos \alpha] \quad (11.7.7)$$

Where:

M_b ——the design value for bending moment at connecting beam end on the shear wall in consideration of seismic combination;

f_y ——the design value of tensile strength on longitudinal steel reinforcements;

f_{yd} ——the design value of tensile strength on diagonal bars;

A_s ——the sectional area of longitudinal steel reinforcements in single-side tension;

A_{sd} ——the sectional area of single-direction diagonal bars, which shall be taken as 0 if there is no diagonal bars;

z_{sd} ——the distance from the diagonal bars at calculated section to the point for resultant forces of section in compression zone;

α ——the included angle of diagonal bars and beam longitudinal axis;

h_0 ——the effective depth of connecting beam section.

11.7.8 The design value of shear force V_{wb} of the connecting beam at the tube and shear wall opening shall be calculated according to the following requirements:

1 Frame of Seismic Grade 1 at Protection Intensity 9

$$V_{wb} = 1.1 \frac{M_{bua}^l + M_{bua}^r}{l_n} + V_{Gb} \quad (11.7.8-1)$$

2 Other conditions

$$V_{wb} = \eta_{vb} \frac{M_b^l + M_b^r}{l_n} + V_{Gb} \quad (11.7.8-2)$$

Where:

M_{bua}^l, M_{bua}^r ——respectively the bending moment value corresponding to actual arranged flexural load-bearing capacity in clockwise or counterclockwise direction at left and right ends of connecting beam, which shall be calculated according to actual arranged area of steel reinforcement (including compression steel reinforcements) and the standard value for material strength and in consideration of seismic adjustment coefficient of load-bearing capacity;

M_b^l, M_b^r ——respectively the design value for bending moment at left and right

connecting beam end of the shear wall and tube in consideration of seismic combination. The sum of M_b^l and M_b^r shall be calculated according to clockwise and counterclockwise directions respectively and the larger value shall be taken. For Seismic Grade 1, when the bending moment at both end are all hogging moment, the bending moment value with smaller absolute value shall be taken as zero.

l_n ——the clear span of connecting beam;

V_{Gb} ——the design value of shear force caused by the representative value of gravity load in consideration of seismic combination, which may be determined by calculating according to simple-supported beams;

η_{vb} ——the amplified coefficient of the shear force on connecting beam. For connecting beams of ordinary stirrups, it shall be taken as 1.3 at Seismic Grade 1, 1.2 at Seismic Grade 2, 1.1 at Seismic Grade 3 and 1.0 at Seismic Grade 4; η_{vb} of connecting beam arranged with diagonal bars shall be taken as 1.0.

11.7.9 For the connecting beam of the shear wall and tube opening at each seismic grade, when it is arranged with ordinary stirrups, the section limit condition and shear bearing capacity of inclined section shall meet the following requirements:

1 When span-depth ratio is larger than 2.5,

1) The shear section shall meet the following requirements:

$$V_{wb} \leq \frac{1}{\gamma_{RE}} (0.20 \beta_c f_c b h_0) \quad (11.7.9-1)$$

2) The shear bearing capacity of inclined section on connecting beams shall meet the following requirements:

$$V_{wb} \leq \frac{1}{\gamma_{RE}} \left(0.42 f_t b h_0 + \frac{A_{sv}}{s} f_{yv} h_0 \right) \quad (11.7.9-2)$$

2 When the span-depth ratio is not larger than 2.5,

1) The shear section shall meet the following requirements:

$$V_{wb} \leq \frac{1}{\gamma_{RE}} (0.15 \beta_c f_c b h_0) \quad (11.7.9-3)$$

2) The shear bearing capacity of inclined section on connecting beams shall meet the following requirements:

$$V_{wb} \leq \frac{1}{\gamma_{RE}} \left(0.38 f_t b h_0 + 0.9 \frac{A_{sv}}{s} f_{yv} h_0 \right) \quad (11.7.9-4)$$

Where:

f_t ——the design value of concrete tensile strength;

f_{yv} ——the design value of tensile strength on stirrups;

A_{sv} ——the sectional area of stirrups being arranged in the same section.

11.7.10 For the connecting beam of Seismic Grade 1 and 2, when the span-depth ratio is not larger than 2.5, besides ordinary stirrups, the intersecting steel reinforcements in inclined direction should be arranged additionally, the section limit condition and shear bearing

capacity of inclined section may be calculated according to the following requirements:

1 When the sectional width of connecting beam of the opening is not less than 250mm, the intersecting diagonal bar and reinforcement may be adopted (Figure 11.7.10-1); the section limit condition and shear bearing capacity of inclined section shall meet the following requirements:

1) The shear section shall meet the following requirements:

$$V_{wb} \leq \frac{1}{\gamma_{RE}} (0.25 \beta_c f_c b h_0) \quad (11.7.10-1)$$

2) The shear bearing capacity of inclined section shall meet the following requirements:

$$V_{wb} \leq \frac{1}{\gamma_{RE}} [0.4 f_t b h_0 + (2.0 \sin \alpha + 0.6 \eta) f_{yd} A_{sd}] \quad (11.7.10-2)$$

$$\eta = (f_{sv} A_{sv} h_0) / (s f_{yd} A_{yd}) \quad (11.7.10-3)$$

Where:

η ——the reinforcement strength ratio of stirrups and diagonal bars, it shall be taken as 0.6 where it is less than 0.6 and it shall be taken as 1.2 where it is larger than 1.2;

α ——the included angle of diagonal bars and beam longitudinal axis;

f_{yd} ——the design value of tensile strength on diagonal bars;

A_{sd} ——the sectional area of single-direction diagonal bars;

A_{sv} ——The total sectional area of each limb of stirrups arranged in the same section.

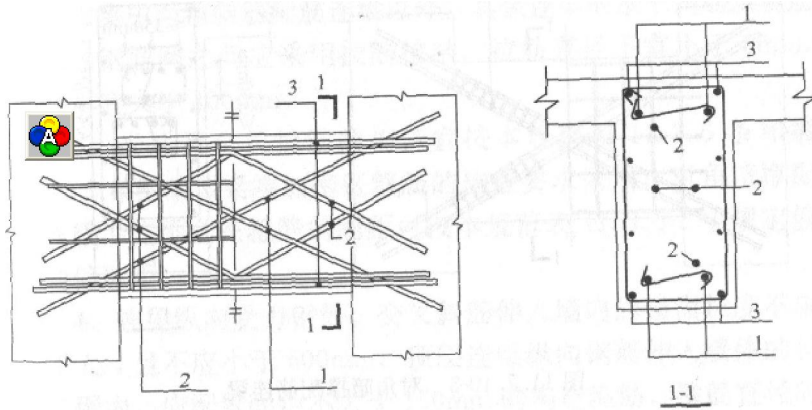


Figure 11.7.10-1 Connecting Beam of Intersecting Diagonal Bar and Reinforcement

1—the diagonal bar; 2—the broken-line reinforcement; 3—the longitudinal steel reinforcement

2 When the sectional width of connecting beam is not less than 400mm, the intensive diagonal reinforcement (Figure 11.7.10-2) or diagonal concealed bracing reinforcement (Figure 11.7.10-3) may be adopted, the section limit condition and shear bearing capacity of inclined section shall meet the following requirements:

1) The shear section shall meet the requirements of Formula (11.7.10-1).

2) The shear bearing capacity of inclined section shall meet the following requirements:

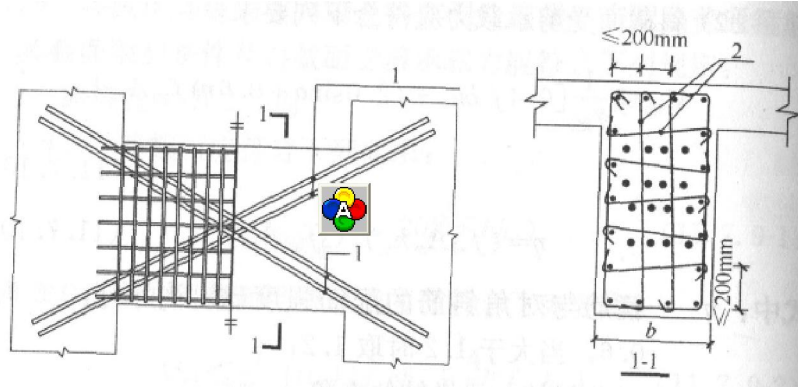


Figure 11.7.10-2 Connecting Beam of Intensive Diagonal Reinforcement

1—the diagonal bars; 2—the tie bar

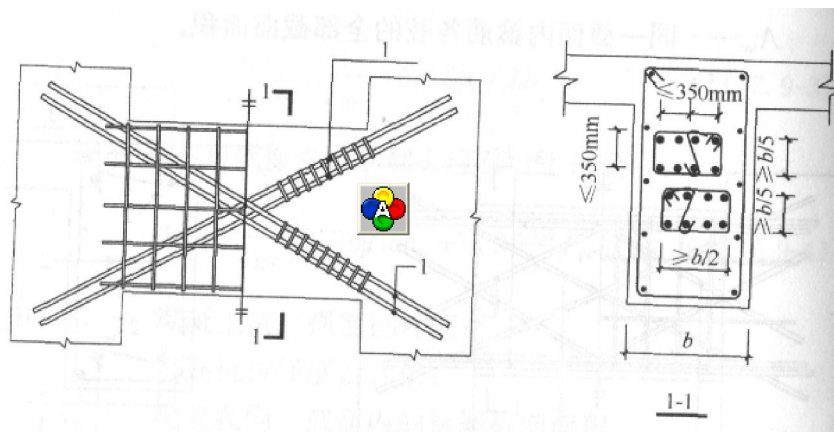


Figure 11.7.10-3 Connecting Beam of Diagonal Concealed Bracing Reinforcement

1—diagonal concealed bracing

$$V_{wb} \leq \frac{2}{\gamma_{RE}} f_{yd} A_{sd} \sin \alpha \quad (11.7.10-4)$$

11.7.11 The construction of longitudinal steel reinforcements, diagonal bars and stirrups for connecting beams on the shear wall and tube opening shall meet the following requirements:

1 The minimum ratio of reinforcement of single-side longitudinal steel reinforcements on connecting beams along upper and lower edge shall not be less than 0.15%, and the reinforcement should not be less than 2φ12; the single-direction diagonal bars on connecting beams with intersecting diagonal reinforcements should not be less than 2φ12, the sectional area of single group broken-line reinforcements may be taken as half of the sectional area of single-direction diagonal bars, and the diameter should not be less than 12mm; each group of diagonal bars on connecting beams with intensive diagonal reinforcements and connecting beams with diagonal concealed bracing shall be at least composed of four sticks of steel reinforcements with diameters no less than 14mm.

2 The diagonal bars on connecting beams with intersecting diagonal reinforcements shall be arranged with no less than three sticks of tie bars at beam end part, the spacing of tie bars shall not be larger than the smaller value of connecting beam width and 200mm, and the diameter shall not be less than 6mm; the connecting beam with intensive diagonal reinforcements shall be arranged with two-way tie bars along horizontal direction and vertical

direction within the beam section, the tie bar shall be hooked with outer longitudinal steel reinforcements, the spacing shall not be larger than 200mm, and the diameter shall not be less than 8mm; in connecting beams with diagonal concealed bracing reinforcements, the outer-edge of concealed bracing stirrups along the beam sectional width direction should not be less than half of beam width, it should not be less than 1/5 of beam width in the other direction; the diagonal concealed bracing restrains that the spacing of stirrups should not be larger than 6 times of diameter of steel reinforcements with concealed bracing, it may be taken as 100mm when the calculated spacing is less than 100mm, and the spacing of stirrup legs shall not be larger than 350mm.

Except connecting beams with intensive diagonal reinforcements, the steel mesh reinforcements formed by horizontal steel reinforcements and stirrups of the rest connecting beams shall be tied by adopting tie bars, the diameter of the tie bar should not be less than 6mm, and the spacing should not be larger than 400mm.

3 The construction of stirrups along the full length of connecting beams should be adopted according to constructional requirements of stirrups in frame beam end densified area specified in Article 11.3.6 and Article 11.3.8 of this code; for connecting beams with diagonal concealed bracing reinforcements, the spacing of stirrups along the full length of connecting beams may be taken as twice of specified value in Table 11.3.6-2 of this code.

4 The anchorage length for longitudinal stressed steel reinforcements and intersecting diagonal bars of connecting beams stretching into the wall shall not be less than l_{aE} and shall not be less than 600mm; within the range of length for longitudinal steel reinforcements of connecting beams at top storey stretching into the wall, the constructional stirrup with spacing no larger than 150mm shall be arranged, the diameter of stirrup shall be the same as the diameter of stirrup of this connecting beam.

5 The horizontally distributed steel reinforcement of the shear wall may be adopted as the longitudinal constructional steel reinforcement of connecting beams to penetrate within range of connecting beams. When the depth of beam web h_w is not less than 450mm, the diameter of longitudinal constructional steel reinforcements arranged along the beam depth range at both sides shall not be less than 10mm, and the spacing shall not be larger than 200mm; for connecting beams with span-depth ratio no larger than 2.5, the area ratio of reinforcement for longitudinal constructional steel reinforcements at both sides of beams still shall not be less than 0.3%.

11.7.12 The sectional thickness of wall-column on the shear wall shall meet the following requirements:

1 Shear wall structure: at Seismic Grade 1 and 2, at general position, it shall not be less than 160mm and should not be less than storey height or 1/20 of non-support part length; at Seismic Grade 3 and 4, it shall not be less than 140mm and should not be less than storey height or 1/25 of non-support part length. At bottom reinforced part of Seismic Grade 1 and 2, it shall not be less than 200mm and should not be less than storey height or 1/16 of non-support part length, when there are no end columns or wing walls at wall end, the wall thickness should not be less than storey height or 1/12 of non-support part length.

2 Frame-shear wall structure: at general position, it shall not be less than 160mm and should not be less than storey height or 1/20 of non-support part length; at bottom reinforced part, it shall not be less than 200mm and should not be less than storey height or 1/16 of

non-support part length.

3 Frame-core tube structure and tube-in-tube structure: at general position, it shall not be less than 160mm and should not be less than storey height or 1/20 of non-support part length; at bottom reinforced part, it shall not be less than 200mm and should not be less than storey height or 1/16 of non-support part length. The wall thickness should not be changed for the bottom reinforced part of the tube and its upper one storey.

11.7.13 When the shear wall thickness is larger than 140mm, the distribution steel reinforcement in vertical and horizontal directions shall not be less than the double row arrangement.

11.7.14 The reinforcement for horizontal and vertical distribution steel reinforcements on the shear wall shall meet the following requirements:

1 The ratio of reinforcement for horizontal and vertical distribution steel reinforcements on the shear wall of Seismic Grade 1, 2 and 3 all shall not be less than 0.25%; it shall not be less than 0.2% for the shear wall of Seismic Grade 4;

2 For the bottom reinforced part of the shear wall in partial frame-support shear wall structures, the ratio of reinforcement for horizontal and vertical distribution steel reinforcements shall not be less than 0.3%.

Note: For the shear wall of Seismic Grade 4 with height less than 24m and very small shear pressure ratio, the minimum ratio of reinforcement for vertical distribution reinforcements shall be allowed to adopt as 0.15%.

11.7.15 The spacing of horizontal and vertical distribution steel reinforcements on the shear wall should not be larger than 300mm, the diameter should not be larger than 1/10 of wall thickness and shall not be less than 8mm; the diameter of vertical distribution steel reinforcements should not be less than 10mm.

For the bottom reinforced part in partial frame-support shear wall structure, the spacing of horizontal and vertical distribution steel reinforcements on the shear wall should not be larger than 200mm.

11.7.16 For the shear wall of Seismic Grade 1, 2 and 3, the axial compression ratio of wall-column at bottom reinforced part should not exceed the limit value specified in Table 11.7.16.

Table 11.7.16 Limit Value of Axial Compression Ratio on Shear Wall

Seismic grade (protection intensity)	Grade 1 (Intensity 9)	Grade 1 (Intensity 7 and 8)	Grade 2 and 3
Limit value of axial compression ratio	0.4	0.5	0.6

Note: The axial compression ratio of the shear wall column refers to the ratio of the design value of the wall axial pressure under the action of the representative value of the gravity load to the product of the total sectional area of the wall and the design value of axial compressive strength of concrete.

11.7.17 The boundary member shall be arranged at both ends of the shear wall and both sides of the opening and should meet the following requirements:

1 For the shear wall of Seismic Grade 1, 2 and 3, under the action of the representative value of the gravity load, when the axial compression ratio at bottom section of the wall-column is larger than those specified in Table 11.7.17, the restraining boundary members shall be arranged according to the requirements of Article 11.7.18 of this code for the wall-column at bottom reinforced part and its upper one story; when the axial compression

ratio of the wall-column is not larger than those specified in Table 11.7.17, the constructional boundary members may be arranged according to the requirements of Article 11.7.19 of this code;

Table 11.7.17 Maximum Axial Compression Ratio of Constructional Boundary Members Arranged for Shear Walls

Seismic grade (protection intensity)	Grade 1 (Intensity 9)	Grade 1 (Intensity 7 and 8)	Grade 2 and 3
Axial compression ratio	0.1	0.2	0.3

2 In partial frame-support shear wall structure, at bottom reinforced part of the shear wall linked with the ground at Seismic Grade 1, 2 and 3 and both ends of the wall-column at its upper one storey, the wing walls or end columns should be arranged and the restraining boundary members shall be arranged according to the requirements of Article 11.7.18 of this code; for the shear wall not linked with the ground, the restraining boundary members shall be arranged at bottom reinforced part and both ends of wall-column of the shear wall at the upper one storey;

3 The shear wall at general position for the shear wall of Seismic Grade 1, 2 and 3 as well as the shear wall of Seismic Grade 4 shall be arranged with constructional boundary members according to Article 11.7.19 of this code;

4 For the frame-core-tube structure, the boundary member of the core tube corner wall of Seismic Grade 1, 2 and 3 still shall be strengthened according to the following requirements: at bottom reinforced part, the length of restraining boundary members for the wall-column should take 1/4 of depth of section at wall-column, and the stirrup should be adopted completely within range of restraining boundary members; above the bottom reinforced part, the restraining boundary members should be arranged according to the requirements of Figure 11.7.18 of this code.

11.7.18 The restraining boundary members (embedded columns, end columns, wing walls and corner walls) arranged at the shear wall end shall meet the following requirements (Figure 11.7.18):

1 The length l_c of restraining boundary members along the wall-column and the stirrup characteristic value λ_v should meet those specified in Table 11.7.18, the arranged range of stirrups and the area of corresponding stirrup characteristic value λ_v and $\lambda_v/2$ are as shown in Figure 11.7.18, the ratio of volumetric reinforcement ρ_v shall meet the following requirement:

$$\rho_v \geq \lambda_v \frac{f_c}{f_{yv}} \quad (11.7.18)$$

Where:

λ_v ——the stirrup characteristic value, which may be included in tie bar in calculation.

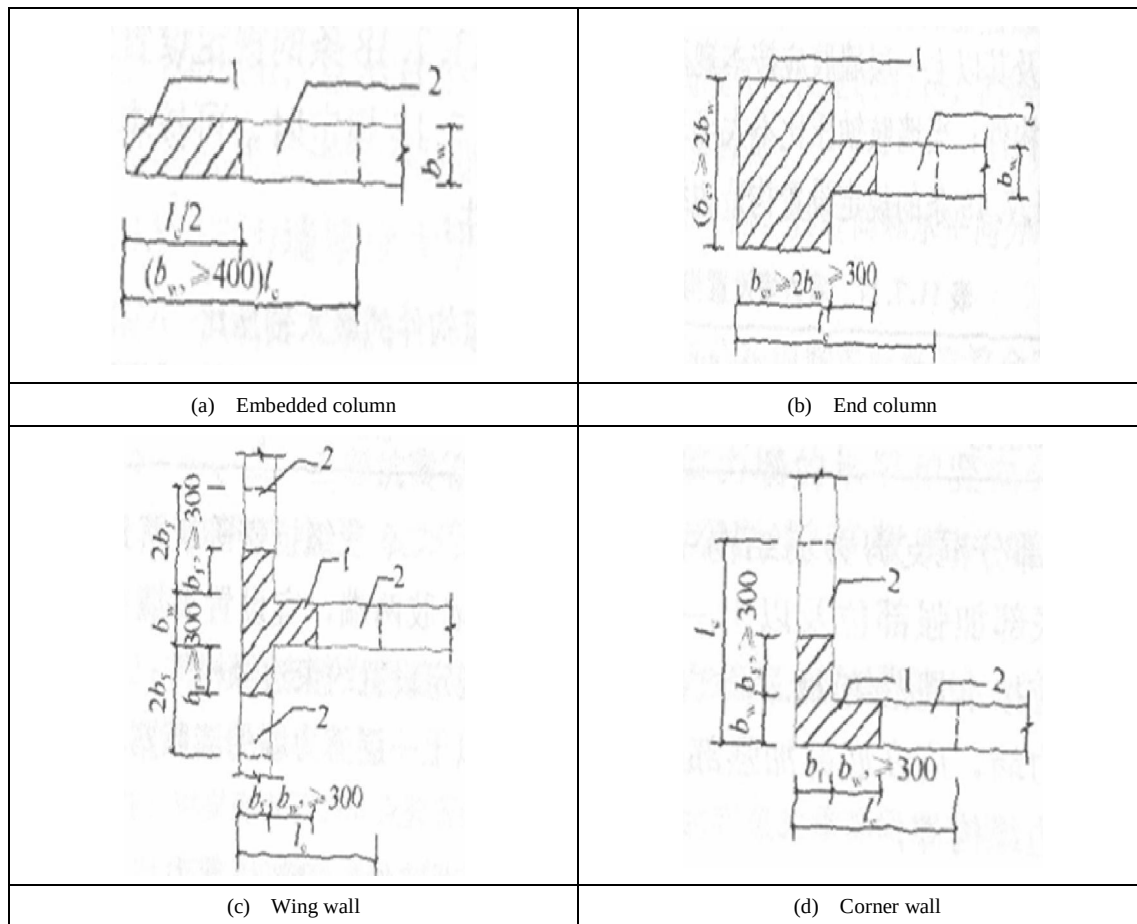


Figure 11.7.18 Restraining Boundary Members of Shear Wall

Note: The dimensional unit in this figure is mm.

1—the area where the stirrup characteristic value is λ_v ; 2—the area where the stirrup characteristic value is $\lambda_v/2$

When calculating the volumetric stirrup ratio, the sectional area of horizontal distribution steel reinforcements meeting constructional requirements and with reliable anchorage at wall end may be properly included.

2 The sectional area of longitudinal steel reinforcements in restraining boundary members of the shear wall of Seismic Grade 1, 2 and 3 for the embedded columns, end columns, wing walls and corner walls shown in Figure 11.7.18 shall not be less than 1.2%, 1.0% and 1.0% of shaded area in this figure respectively.

3 For the vertical spacing of stirrups or tie bars in restraining boundary members, it should not be larger than 100mm for Seismic Grade 1; it should not be larger than 150mm for Seismic Grade 2 and 3.

Table 11.7.18 Length l_c along Wall-column and Its Stirrup Characteristic Value λ_v for Restraining Boundary Members

Seismic grade (protection intensity)		Grade 1 (Intensity 9)		Grade 1 (Intensity 7 and 8)		Grade 2 and 3	
Axial compression ratio		≤ 0.2	> 0.2	≤ 0.3	> 0.3	≤ 0.4	> 0.4
λ_v		0.12	0.20	0.12	0.20	0.12	0.20
l_c (mm)	Embedded column	0.20 h_w	0.25 h_w	0.15 h_w	0.20 h_w	0.15 h_w	0.20 h_w
	End column, wing wall or corner wall	0.15 h_w	0.20 h_w	0.10 h_w	0.15 h_w	0.10 h_w	0.15 h_w

Notes:

1 When the length of the wing wall at both sides is less than 3 times of the thickness, it shall be deemed as the shear wall

without wing walls; when the side length of the end column section is less than 2 times of wall thickness, it shall be deemed as the shear wall without end columns;

- 2 The length l_c along wall-column of restraining boundary members shall not only meet those specified in Table 11.7.18, but also should not be less than the wall thickness and 400mm; when there are end columns, wing walls or corner walls, it still shall not be less than the wing wall thickness or the depth of section at the end column along the wall-column direction plus 300mm;
- 3 h_w is the depth of section at wall-column of the shear wall.

11.7.19 The range of constructional boundary members (embedded columns, end columns, wing walls and corner walls) arranged at the shear wall end shall be determined according to Figure 11.7.19, the longitudinal steel reinforcements in constructional boundary members shall not only meet the calculation requirements but also shall be in accordance with those specified in Table 11.7.19.

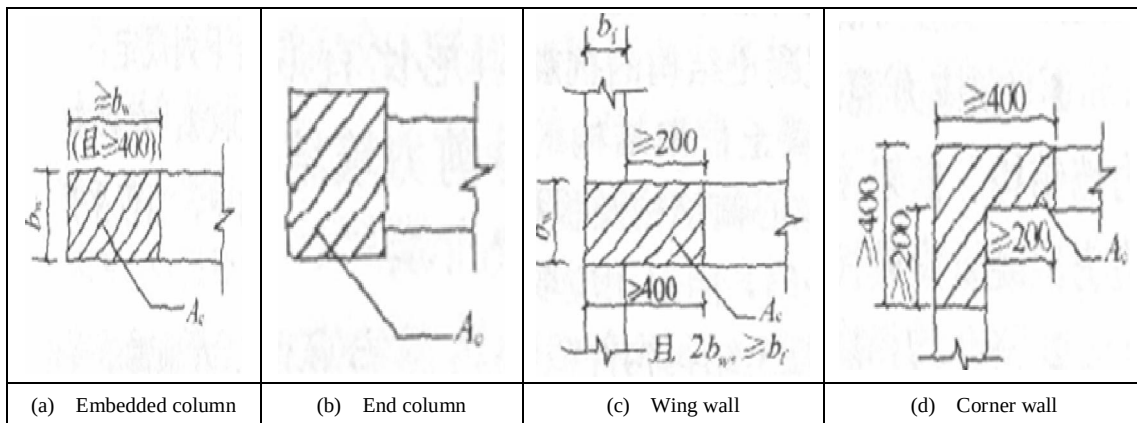


Figure 11.7.19 Constructional Boundary Members of Shear Wall

Note: The dimensional unit in this figure is mm.

Table 11.7.19 Constructional Reinforcement Requirements for Constructional Boundary Members

Seismic grade	Bottom reinforced part			Other parts		
	Minimum reinforcement amount of longitudinal steel reinforcement (taking the larger value)	Stirrup and tie bar		Minimum reinforcement amount of longitudinal steel reinforcement (taking the larger value)	Stirrup and tie bar	
		Minimum diameter (mm)	Maximum spacing (mm)		Minimum diameter (mm)	Maximum spacing (mm)
1	$0.01A_c$, $6\phi 16$	8	100	$0.008A_c$, $6\phi 14$	8	150
2	$0.008A_c$, $6\phi 14$	8	150	$0.006A_c$, $6\phi 12$	8	200
3	$0.006A_c$, $6\phi 12$	6	150	$0.005A_c$, $4\phi 12$	6	200
4	$0.005A_c$, $4\phi 12$	6	200	$0.004A_c$, $4\phi 12$	6	250

Notes:

- 1 A_c is the shaded area as shown in Figure 11.7.19;
- 2 For other parts, the level spacing of tie bars shall not be greater than twice of spacing of longitudinal steel reinforcements, the stirrups should be arranged at corner;
- 3 When the end column is under the concentrated load, it shall meet the reinforcement requirements for the frame column.

11.8 Prestressed Concrete Structural Members

11.8.1 The prestressed concrete structure may be used in zones of Seismic Protection Intensity 6, 7 and 8, when the Intensity 9 zone is needed to adopt the prestressed concrete structure, the sufficient criteria shall be provided and reliable measures shall be taken.

The seismic design of unbonded prestressed concrete structure shall meet specialized requirements.

11.8.2 In the seismic design, the transfer girder of post-tensioned prestressed frames, portals and transfer storey should be adopted with bonded prestressed tendons. The prestressed tension member of load-bearing structure and prestress frame of Seismic Grade 1 shall be adopted with unbonded prestressed tendons.

11.8.3 The seismic calculation for the prestressed concrete structure shall meet the following requirements:

1 The damping ratio of prestressed concrete frame structures should take 0.03; in the frame-shear wall structure, frame-core-tube structure and slab column-shear wall structure, when the prestressed concrete beam or slab is adopted only, the damping ratio shall take 0.05;

2 In the section seismic checking of prestressed concrete structural members, in seismic combination, for the partial coefficient of prestress action, when the prestress action effect is favourable for the load-bearing capacity of the member, it shall take 1.0, if not, it shall take 1.2;

3 When the prestressing tendon and/or bar is crossing the core zone of node of frame, the checking of the section seismic shear bearing capacity of node core zone shall be carried out according to the relevant requirements of Section 11.6 of this code, and the favourable influence of effective pre-applied force may be considered.

11.8.4 The seismic construction of prestressed concrete frame shall not only meet the requirements of the reinforced concrete structure but also shall meet the following requirements:

1 For the end section of prestressed concrete frame beam, the depth of concrete compression zone included in longitudinal compression steel reinforcements shall meet the requirements of Article 11.3.1 of this code; the ratio of reinforcement for all longitudinal tension steel reinforcements converted according to the design value of tensile strength on steel bars should not be larger than 2.5%.

2 In prestressed concrete frame beams, the mixing reinforcement mode of prestressing tendons and steel bars shall be adopted; the reinforcement at the beam end section should meet the following requirements:

$$A_s \geq \frac{1}{3} \left(\frac{f_{pv} h_p}{f_y h_s} \right) A_p \quad (11.8.4)$$

Note: For post-tensioned bonded prestressed concrete frames in frame-shear wall and frame-core-tube structures of Seismic Grade 2 and 3, the coefficient 1/3 in right items of Formula (11.8.4) may be changed to 1/4.

3 The sectional area ratio of the bottom longitudinal steel bars to the top longitudinal stressed steel reinforcements at the beam end section of prestressed concrete frame beams shall meet the requirements of Clause 2 of Article 11.3.6 in this code. When the sectional area of the top longitudinal stressed steel reinforcements is calculated, the prestressing tendon and/or bar shall be converted to the sectional area of steel bars according to the design value of tensile strength.

The ratio of reinforcement for longitudinal steel bars at bottom surface of the frame beam end still shall not be less than 0.2%.

4 When the axial compression ratio of prestressed concrete frame columns is calculated, the design value of axial compressive force shall take the design value of axial compressive force on the column combination plus the design value of effective pre-applied force on prestressing tendons, the axial compression ratio shall meet corresponding requirements of Article 11.4.16 in this code.

5 The stirrups of prestressed concrete frame columns should be densified in total height. The large-span frame side column may be adopted with non-symmetric reinforcement mode (the mixing reinforcements of prestressing tendons and steel bars are arranged at one side where the section tension is larger and steel bars are arranged only at the other side).

11.8.5 For post-tensioning prestressed concrete slab column-shear wall structures, the end section of the column strip on the slab column shall meet the requirements of the depth of compression zone specified in Article 11.8.4 of this code and the requirements of section reinforcements stated in Formula (11.8.4).

The slab-column joints shall meet the requirements of Section 11.9 of this code.

11.8.6 The anchorage and connector of post-tensioning prestressed tendons should not be arranged in the core zone of beam-column joints.

11.9 Slab-column Joints

11.9.1 For slab-column joints of Seismic Grade 1, 2 and 3, the checking of seismic punching shear bearing capacity shall be carried out according to Article 11.9.3 and Appendix F of this code.

11.9.2 At Protection Intensity 8, the slab-column joints with supporting boards or column caps should be adopted, and the physical dimension of column caps and supporting boards shall meet the requirements of Article 9.1.10 of this code. Simultaneously, the root thickness (including board thickness) of supporting boards or column caps shall not be less than 16 times of the diameter of column longitudinal steel reinforcements, and the side length of supporting boards or column caps shall not be less than the sum of 4 times of board thickness and the corresponding side length of column sections.

11.9.3 Under seismic combination, when the shear stress on the critical section of slab-column joints transferring unbalanced bending moment is considered, the design value $F_{l,eq}$ of equivalent concentrated reaction in consideration of seismic grade may be calculated according to the requirements of Appendix F of this code, here, F_l is the design value of vertical force borne by the critical section of slab-column joints. The design value of equivalent concentrated reaction caused by the unbalanced bending moment of seismic combination at slab-column joints shall multiply the amplified coefficient, for nodes in slab-column structures of Seismic Grade 1, 2 and 3, this amplified coefficient may be taken as 1.7, 1.5 and 1.3 respectively.

11.9.4 Under seismic combination, for the slab-column joints arranged with stirrups or male pins, the punching shear section and punching shear bearing capacity shall meet the following requirements:

1 Punching shear section

$$F_{l,eq} \leq \frac{1}{\gamma_{RE}} (1.2 f_t \eta u_m h_0) \quad (11.9.4-1)$$

2 Punching shear bearing capacity

$$F_{l,eq} \leq \frac{1}{\gamma_{RE}} [(0.3 f_t + 0.15 \sigma_{pc,m}) \eta u_m h_0 + 0.8 f_{yv} A_{svu}] \quad (11.9.4-2)$$

3 For sections outside the punching failure cone arranged with punching resistant steel reinforcements, the checking of punching shear bearing capacity still shall be carried out according to the following formula:

$$F_{l,eq} \leq \frac{1}{\gamma_{RE}} (0.42 f_t + 0.15 \sigma_{pc,m}) \eta u_m h_0 \quad (11.9.4-3)$$

Where:

u_m ——the perimeter of critical sections, u_m in Formula (11.9.4-1) and Formula (11.9.4-2) shall be adopted according to the requirements of Article 6.5.1 of this code; u_m in Formula (11.9.4-3) shall take the most unfavorable perimeter where is $0.5h_0$ outside the periphery of the punching resistant steel reinforcements in the outermost row.

11.9.5 The constructional concealed beams should be arranged in the column strip for flat slabs without column caps, the concealed beam width may be taken as the column width plus no larger than 1.5 times of slab thickness at both sides of columns respectively. The upper longitudinal steel reinforcements on the lug support of concealed beams shall not be less than 1/2 of the sectional area of longitudinal steel reinforcements on the column strip; the lower longitudinal steel reinforcements in concealed beams should not be less than 1/2 of the sectional area of upper longitudinal steel reinforcements.

The diameter of stirrups in concealed beams shall not be less than 8mm, the spacing should not be larger than 3/4 times of slab thickness, the spacing of stirrup legs should not be larger than 2 times of slab thickness; the length of stirrup densified area for concealed beams at lug support shall not be less than 3 times of slab thickness, the stirrup spacing should not be larger than 100mm, and the spacing of stirrup legs should not be larger than 250mm.

11.9.6 The continuous prestressing tendons and longitudinal steel bars at slab bottom penetrating the node column section along two principal axis directions shall meet the following requirements:

1 The gross sectional area of continuous steel reinforcements penetrating the node column section along two principal axis directions shall meet the requirements of the following formula:

$$f_{py} A_p + f_y A_s \geq N_G \quad (11.9.6)$$

Where:

A_s ——the sectional area of longitudinal steel bars at slab bottom penetrating the column section; for steel bars that one end is in stretch-bending anchorage at the opposite side of the column section, the sectional area shall be calculated according to the half;

A_p ——the sectional area of continuous prestressing tendons penetrating the column section; for prestressing tendon and/or bar that one end is in the anchorage at the opposite side of the column section, the sectional area shall be calculated according to the half;

f_{py} ——the design value of tensile strength on prestressing tendon and/or bar, for unbonded prestressed tendons, it shall adopt the stress design value σ_{pu} of unbonded prestressed tendons according to Article 10.1.14 of this code;

N_G ——the design value of axial compressive force on the column under the action of the representative value of the gravity load in the floor slab of this storey.

2 The continuous prestressing tendon and/or bar shall be arranged at the upper part of slab-column joints, entering into the slab mid-span in inferior-fovea way.

3 The connection positions of longitudinal steel bars at the slab bottom should be outside the larger value of the distance between the column surface l_{aE} and 2 times of slab thickness, and shall keep clear of the range of tension zone at the slab bottom.

Appendix A Nominal Diameter, Nominal Sectional Area and Theoretical Weight of Steel Reinforcement

Table A.0.1 Nominal Diameter, Nominal Sectional Area and Theoretical Weight of Steel Reinforcement

Nominal diameter (mm)	Nominal sectional area of different number of steel reinforcement (mm ²)									Theoretical weight of single steel reinforcement (kg/m)
	1	2	3	4	5	6	7	8	9	
6	28.3	57	85	113	142	170	198	226	255	0.222
8	50.3	101	151	201	252	302	352	402	453	0.395
10	78.5	157	236	314	393	471	550	628	707	0.617
12	113.1	226	339	452	565	678	791	904	1017	0.888
14	153.9	308	461	615	769	923	1077	1231	1385	1.21
16	201.1	402	603	804	1005	1206	1407	1608	1809	1.58
18	254.5	509	763	1017	1272	1527	1781	2036	2290	2.00(2.11)
20	314.2	628	942	1256	1570	1884	2199	2513	2827	2.47
22	380.1	760	1140	1520	1900	2281	2661	3041	3421	2.98
25	490.9	982	1473	1964	2454	2945	3436	3927	4418	3.85(4.10)
28	615.8	1232	1847	2463	3079	3695	4310	4926	5542	4.83
32	804.2	1609	2413	3217	4021	4826	5630	6434	7238	6.31(6.65)
36	1017.9	2036	3054	4072	5089	6107	7125	8143	9161	7.99
40	1256.6	2513	3770	5027	6283	7540	8796	10053	11310	9.87(10.34)
50	1963.5	3928	5892	7856	9820	11784	13748	15712	17676	15.42(16.28)

Note: Figures in parentheses are the values of prestressed twisted steel reinforcements.

Table A.0.2 Nominal Diameter, Nominal Sectional Area and Theoretical Weight of Steel Strands

Type	Nominal diameter (mm)	Nominal sectional area (mm ²)	Theoretical weight (kg/m)
1×3	8.6	37.7	0.296
	10.8	58.9	0.462
	12.9	84.8	0.666
1×7 Standard type	9.5	54.8	0.430
	12.7	98.7	0.775
	15.2	140	1.101
	17.8	191	1.500
	21.6	285	2.237

Table A.0.3 Nominal Diameter, Nominal Sectional Area and Theoretical Weight of Steel Wires

Nominal diameter (mm)	Nominal sectional area(mm ²)	Theoretical weight (kg/m)
5.0	19.63	0.154
7.0	38.48	0.302
9.0	63.62	0.499

Appendix B Amplified Coefficient Method for Approximate Calculation of Sway Second-order Effect of Eccentric Compression

Members

B.0.1 In the frame structures, shear wall structures, frame-shear wall structures and tube structures, if the amplified coefficient method is adopted for the approximate calculation of the second-order effect ($P-\Delta$ effect) of structure due to sway, thus the bending moment at end of column and wall column, the bending moment at beam end and the storey drift, obtained from the first-order elasticity analysis without consideration of $P-\Delta$ effect, shall be multiplied by the amplified coefficient η_s respectively according to Formula (B.0.1-1) and Formula (B.0.1-2):

$$M = M_{ns} + \eta_s M_s \quad (\text{B.0.1-1})$$

$$\Delta = \eta_s \Delta_1 \quad (\text{B.0.1-2})$$

Where:

M_s ——the design value of bending moment produced at end of first-order elasticity analysis member due to load or action that causes the sway of structure;

M_{ns} ——the design value of bending moment produced at end of first-order elasticity analysis member due to load that does not cause the sway of structure;

Δ_1 ——the storey drift in first-order elasticity analysis;

η_s ——the amplified coefficient of $P-\Delta$ effect, which shall be determined in accordance with Article B.0.2 or Article B.0.3 of this code, therein, η_s at beam end shall be taken as the mean value of η_s values at the upper and lower column ends or upper and lower wall column ends at the corresponding node.

B.0.2 In frame structures, η_s of columns in the calculated storey may be calculated according to the following formula:

$$\eta_s = \frac{1}{1 - \frac{\sum N_j}{DH_0}} \quad (\text{B.0.2})$$

Where:

D ——the lateral stiffness of the calculated storey. When calculating the amplified coefficients for bending moment of structural members and for displacement of structures, the stiffness of the structural members shall be adopted respectively in accordance with those specified in Article B.0.5 of this code;

N_j ——the design value of axial force of the j^{th} row columns at the calculated storey;

H_0 ——the storey height of the calculated storey;

B.0.3 η_s in shear wall structures, frame-shear wall structures and tube structures may be calculated according to the following formula:

$$\eta_s = \frac{1}{1 - 0.14 \frac{H^2 \sum G}{E_c J_d}} \quad (\text{B.0.3})$$

Where:

$\sum G$ ——the sum of design values of the gravity loads of all storeys;

$E_c J_d$ ——the flexural stiffness of the vertical uniform cantilever flexural member that is equivalent to the designed structure, which may be calculated according to the principle that the top displacement of this cantilever flexural member under the action of horizontal loads distributed in inverted triangle is same as that of the designed structure. When calculating the amplified coefficients for bending moment of structural members and for displacement of structures, the stiffness of the structural members shall be adopted respectively in accordance with those specified in Article B.0.5 of this code;

H ——the total height of structure.

B.0.4 The design value of bending moment of the bent structural columns, in consideration of second-order effect, may be calculated according to the following formulae:

$$M = \eta_s M_0 \quad (\text{B.0.4-1})$$

$$\eta_s = 1 + \frac{1}{1500 e_i / h_0} \left(\frac{l_0}{h} \right)^2 \zeta \quad (\text{B.0.4-2})$$

$$\zeta_c = \frac{0.5 f_c A}{N} \quad (\text{B.0.4-3})$$

$$e_i = e_0 + e_a \quad (\text{B.0.4-4})$$

Where:

ζ_c ——the correction coefficient for sectional curvature; $\zeta = 1.0$ if $\zeta > 1.0$.

e_i ——the initial eccentric distance;

M_0 ——the design value of column end bending moment in first-order elasticity analysis;

e_0 ——the eccentric distance of axial pressure to centroid of section, and $e_0 = M_0 / N$;

e_a ——the additional eccentric distance, which is determined according to those specified in Article 6.2.5 of this code;

l_0 ——the calculated length of bent frame column, which is adopted in accordance with Table 6.2.20-1 of this code;

h, h_0 ——respectively the depth of section and effective depth of section of the column in the considered bending direction;

A ——the sectional area of column. For I-shaped section: $A = bh + 2(b_f - b)h_f'$.

B.0.5 Where the amplified coefficient η_s for bending moment in various structures is calculated by adopting with Article B.0.2 and Article B.0.3 of this code, the elastic flexural stiffness $E_c I$ of members should be multiplied by the reduction coefficient; for beams, the reduction coefficient shall be 0.4; for columns, the reduction coefficient shall be 0.6; and for shear wall columns and corewall columns, the reduction coefficient shall 0.45; in the calculation of the amplified coefficient η_s for displacement in various structures, the stiffness may not be reduced.

Note: Where the checking shows that the control sections of shear wall column or corewall column do not crack, the stiffness reduction coefficient may be taken as 0.7 for the calculation of amplified coefficient η_s for bending moment.

Appendix C Constitutive Relations for Steel Reinforcement and Concrete and the Multi-axial Strength Criterion for Concrete

C.1 Constitutive Relation of Steel Reinforcement

C.1.1 The mean values f_{ym} and f_{stm} of yield strength and ultimate strength of steel bars may be calculated according to the following formulae:

$$f_{ym} = f_{yk} / (1 - 1.645\delta_s) \quad (C.1.1-1)$$

$$f_{stm} = f_{stk} / (1 - 1.645\delta_s) \quad (C.1.1-2)$$

Where:

f_{yk} , f_{ym} —respectively the standard value and mean value of yield strength of steel reinforcement;

f_{stk} , f_{stm} —respectively the standard value and mean value of ultimate strength of steel reinforcement;

δ_s —the variation coefficient for strength of steel reinforcement, which should be determined in accordance with the experimental statistics.

C.1.2 The stress-strain constitutive relation curve of the monotonic loading steel reinforcement (Figure C.1.2) may be determined according to the following requirements.

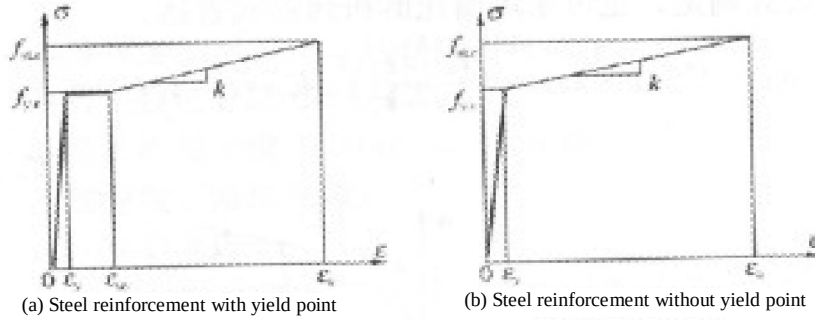


Figure C.1.2 Stress-strain Constitutive Relation Curve of the Monotonic Loading Steel Reinforcement

1 Steel reinforcement with yield point

$$\sigma_s = \begin{cases} E_s \varepsilon_s & \varepsilon_s \leq \varepsilon_y \\ f_{y,r} & \varepsilon_y < \varepsilon_s \leq \varepsilon_{uy} \\ f_{y,r} + k(\varepsilon_s - \varepsilon_{uy}) & \varepsilon_{uy} < \varepsilon_s \leq \varepsilon_u \\ 0 & \varepsilon_s > \varepsilon_u \end{cases} \quad (C.1.2-1)$$

2 Steel reinforcement with yield point

$$\sigma_p = \begin{cases} E_s \varepsilon_s & \varepsilon_s \leq \varepsilon_y \\ f_{y,r} + k(\varepsilon_s - \varepsilon_y) & \varepsilon_y < \varepsilon_s \leq \varepsilon_u \\ 0 & \varepsilon_s > \varepsilon_u \end{cases} \quad (C.1.2-2)$$

Where:

E_s —the elastic modulus of steel reinforcement;

σ_s ——the stress of steel reinforcement;

ε_s ——the strain of steel reinforcement;

$f_{y,r}$ ——the representative value of yield strength of steel reinforcement, its value may be taken as f_y , f_{yk} or f_{ym} respectively in accordance with the demand of actual structural analysis.

$f_{st,r}$ ——the representative value of ultimate strength of steel reinforcement, its value may be taken as f_{st} , f_{stk} or f_{stm} respectively in accordance with the demand of actual structural analysis;

ε_y ——the yield strain of steel reinforcement corresponding to $f_{y,r}$, which may be $f_{y,r}/E_s$;

ε_{uy} ——the strain at hardening starting point of steel reinforcement;

ε_u ——the peak strain of steel reinforcement corresponding to $f_{st,r}$;

k ——the slope of the hardening segment of steel reinforcement, $k=(f_{st,r}-f_{y,r})/(\varepsilon_u-\varepsilon_{uy})$.

C.1.3 The stress-strain constitutive relation curve of the repeated loading of steel reinforcement (Figure C.1.3) should be determined according to the following formulae or may be expressed by adopting simplified broken line form.

$$\sigma_s = E_s(\varepsilon_s - \varepsilon_a) - \left(\frac{\varepsilon_s - \varepsilon_a}{\varepsilon_b - \varepsilon_a} \right)^p [E_s(\varepsilon_b - \varepsilon_a) - \sigma_b] \quad (C.1.3-1)$$

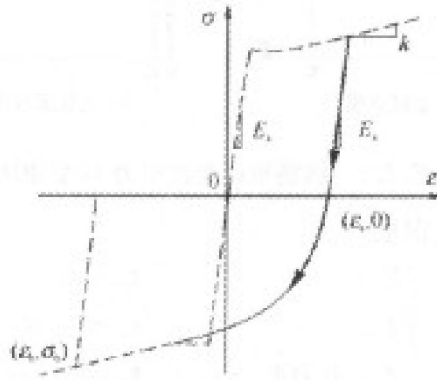


Figure C.1.3 Stress-strain Curve of the Repeated Loading of Steel Reinforcement

$$p = \frac{(E_s - k)(\varepsilon_b - \varepsilon_a)}{E_s(\varepsilon_b - \varepsilon_a) - \sigma_b} \quad (C.1.3-2)$$

Where:

ε_a ——the strain corresponding to the starting point of reloading route;

σ_b , ε_b ——respectively the stress and strain corresponding to the end point of reloading route. If the steel reinforcement in the reloading direction has not yet yielded, then σ_b and ε_b shall be taken as the stress and strain at the initial yield point of steel reinforcement. If the steel reinforcement in the reloading direction has already yielded, then ε_b shall be taken as the maximum historical strain of steel reinforcement in this direction.

C.2 Constitutive Relation of Concrete

C.2.1 the mean values f_{cm} and f_{tm} of compressive strength and tensile strength of concrete may be calculated according to the following formulae:

$$f_{cm} = f_{ck} / (1 - 1.645\delta_c) \quad (C.2.1-1)$$

$$f_{tm}=f_{tk}/(1-1.645\delta_c) \quad (C.2.1-2)$$

Where:

f_{cm} , f_{ck} —respectively the mean value and standard value of the compressive strength of concrete;

f_{tm} , f_{tk} —respectively the mean value and standard value of the tensile strength of concrete;

δ_c —the variation coefficient for concrete strength, which should be determined in accordance with the experimental statistics.

C.2.2 The concrete constitutive model specified in this section shall be applicable to the following conditions:

- 1 Concrete strength grade: C20~C80;
- 2 Mass density of concrete: 2,200kg/m³~2,400kg/m³;
- 3 Environment with normal temperature and humidity;
- 4 Normal loading rate.

C.2.3 The stress-strain curve of concrete under uniaxial tension (Figure C.2.3) may be determined according to the following formulae:

$$\sigma=(1-d_t)E_c\varepsilon \quad (C.2.3-1)$$

$$d_t = \begin{cases} 1 - \rho_t [1.2 - 0.2x^5] & x \leq 1 \\ 1 - \frac{\rho_t}{a_t(x-1)^{1.7} + x} & x > 1 \end{cases} \quad (C.2.3-2)$$

$$x = \frac{\varepsilon}{\varepsilon_{t,r}} \quad (C.2.3-3)$$

$$\rho_t = \frac{f_{t,r}}{E_c \varepsilon_{t,r}} \quad (C.2.3-4)$$

Where:

a_t —the parameter value for the descending segment of stress-strain curve of concrete under uniaxial tension, which shall be adopted in accordance with Table C.2.3;

$f_{t,r}$ —the representative value of uniaxial tensile strength of concrete, its value may be taken as f_t , f_{tk} or f_{tm} respectively in accordance with the demand of actual structural analysis;

$\varepsilon_{t,r}$ —the peak tension strain of concrete corresponding to the representative value of uniaxial tensile strength $f_{t,r}$, which shall be adopted in accordance with Table C.2.3;

d_t —the parameter of concrete damage evolution under uniaxial tension.

Table C.2.3 Values of the Stress-strain Curve Parameters of Concrete Under Uniaxial Tension

$f_{t,r}(\text{N/mm}^2)$	1.0	1.5	2.0	2.5	3.0	3.5	4.0
$\varepsilon_{t,r}(10^{-6})$	65	81	95	107	118	128	137
a_t	0.31	0.70	1.25	1.95	2.81	3.82	5.00

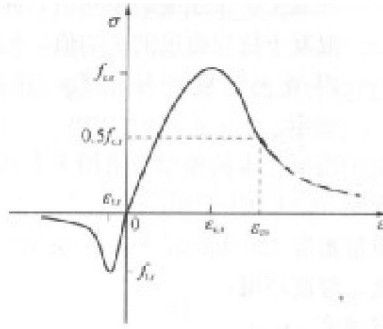


Figure C.2.3 Uniaxial Stress-strain Curve of Concrete

Note: The schematic diagrams of the stress-strain curve of concrete under tension and compression are drawn in one same coordinate system but in different scales. The signs shall be “positive for concrete under tension and negative for concrete under compression”

C.2.4 The stress-strain curve of concrete under uniaxial compression (Figure C.2.3) may be determined according to the following formulae:

$$\sigma = (1 - d_c) E_c \varepsilon \quad (\text{C.2.4-1})$$

$$d_c = \begin{cases} 1 - \frac{\rho_c n}{n - 1 + x^n} & x \leq 1 \\ 1 - \frac{\rho_c}{a_c (x - 1)^2 + x} & x > 1 \end{cases} \quad (\text{C.2.4-2})$$

$$\rho_c = \frac{f_{c,r}}{E_c \varepsilon_{c,r}} \quad (\text{C.2.4-3})$$

$$\eta = \frac{E_c \varepsilon_{c,r}}{E_c \varepsilon_{c,r} - f_{c,r}} \quad (\text{C.2.4-4})$$

$$x = \frac{\varepsilon}{\varepsilon_{c,r}} \quad (\text{C.2.4-5})$$

Where:

a_c ——the parameter value for the descending segment of stress-strain curve of concrete under uniaxial compression, which shall be adopted in accordance with Table C.2.4;

$f_{c,r}$ ——the representative value of uniaxial compressive strength of concrete, its value may be taken as f_c , f_{tk} or f_{cm} respectively in accordance with the demand of actual structural analysis;

$\varepsilon_{c,r}$ ——the peak compression strain of concrete corresponding to the uniaxial compressive strength $f_{c,r}$, which shall be adopted in accordance with Table C.2.4;

d_c ——the parameter of concrete damage evolution under uniaxial compression.

Table C.2.4 Values of the Stress-strain Curve Parameters of Concrete Under Uniaxial Compression

$f_{c,r}$ (N/mm ²)	20	25	30	35	40	45	50	55	60	65	70	75	80
$\varepsilon_{c,r}$ (10 ⁻⁶)	1470	1560	1640	1720	1790	1850	1920	1980	2030	2080	2130	2190	2240
a_c	0.74	1.06	1.36	1.65	1.94	2.21	2.48	2.74	3.00	3.25	3.50	3.75	3.99
$\varepsilon_{cu}/\varepsilon_{c,r}$	3.0	2.6	2.3	2.1	2.0	1.9	1.9	1.8	1.8	1.7	1.7	1.7	1.6

Note: ε_{cu} is the compression strain of concrete when the stress in the descending segment of stress-strain curve equals to

C.2.5 Under the action of repeated load, the unloading and reloading stress routes (Figure C.2.5) of concrete under compression may be determined according to the following formulae:

$$E_r = \frac{\sigma_{un}}{\epsilon_{un} - \epsilon_z} \quad (C.2.5-2)$$

$$\varepsilon_{ca} = \max \left(\frac{\varepsilon_c}{\varepsilon_c + \varepsilon_{un}}, \frac{0.09 \varepsilon_{un}}{\varepsilon_c} \right) \sqrt{\varepsilon_c \varepsilon_{un}} \quad (\text{C.2.5-4})$$

ε_c —the strain of concrete corresponding to the peak compression stress.

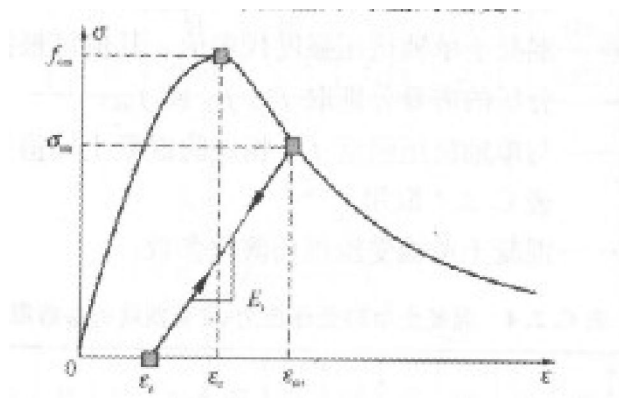


Figure C.2.5 Stress-strain Curve of Concrete under the Action of Repeated Load

C.2.6 The constitutive relation of concrete under the conditions of biaxial loading and unloading may be adopted with damage model or elastoplastic model. The elastoplastic constitutive relation may be adopted with constitutive theory of elastoplastic increment, and the damage constitutive relation may be determined according to the following formulae:

- ### 1) Loading equation

$$\begin{Bmatrix} \sigma_1 \\ \sigma_2 \end{Bmatrix} = (1-d_t) \begin{Bmatrix} \sigma_1' \\ \sigma_2' \end{Bmatrix} \quad (\text{C.2.6-1})$$

$$\varepsilon_{t,e} = -\sqrt{\frac{1}{1-\nu^2}[(\varepsilon_1)^2 + (\varepsilon_2)^2 + 2\nu\varepsilon_1\varepsilon_2]} \quad (\text{C.2.6-2})$$

$$\begin{Bmatrix} \sigma_1' \\ \sigma_2' \end{Bmatrix} = \frac{E_c}{1-\nu^2} \begin{bmatrix} 1 & \nu \\ \nu & 1 \end{bmatrix} \begin{Bmatrix} \varepsilon_1 \\ \varepsilon_2 \end{Bmatrix} \quad (\text{C.2.6-3})$$

Where:

d_t ——the parameter of tension damage evolution, which may be calculated according to

Formula (C.2.3-2), therein, $x = \frac{\varepsilon_{t,e}}{\varepsilon_t}$;

$\varepsilon_{t,e}$ ——the equivalent strain of tension energy;

σ_1', σ_2' ——the effective stress;

ν ——the Poisson's ratio of concrete, which may be taken from 0.18 to 0.22.

2) Unloading equation

$$\begin{Bmatrix} \sigma_1 - \sigma_{un,1} \\ \sigma_2 - \sigma_{un,2} \end{Bmatrix} = (1-d_t) \frac{E_c}{1-\nu^2} \begin{bmatrix} 1 & \nu \\ \nu & 1 \end{bmatrix} \begin{Bmatrix} \varepsilon_1 - \varepsilon_{un,1} \\ \varepsilon_2 - \varepsilon_{un,2} \end{Bmatrix} \quad (\text{C.2.6-4})$$

Where:

$\sigma_{un,1}, \sigma_{un,2}, \varepsilon_{un,1}, \varepsilon_{un,2}$ ——the stress and strain at the two-dimensional unloading points.

In the loading equation, the parameter of damage evolution shall be calculated by adopting with the equivalent strain of energy that is converted from the instantaneous strain; in the unloading equation, the parameter of damage evolution shall be calculated by adopting the equivalent strain of energy that is converted from the strain at unloading point and shall remain constant during the whole process of unloading and reloading.

2 Biaxial compression zone ($\sigma_1' \geq 0, \sigma_2' \geq 0$)

1) Loading equation

$$\begin{Bmatrix} \sigma_1 \\ \sigma_2 \end{Bmatrix} = (1-d_c) \begin{Bmatrix} \sigma_1' \\ \sigma_2' \end{Bmatrix} \quad (\text{C.2.6-5})$$

$$\varepsilon_{c,e} = \frac{1}{(1-\nu^2)(1-a_s)} [a_s(1+\nu)(\varepsilon_1 + \varepsilon_2) + \sqrt{(\varepsilon_1 + \nu\varepsilon_2)^2 + (\varepsilon_2 + \nu\varepsilon_1)^2 - (\varepsilon_1 + \nu\varepsilon_2)(\varepsilon_2 + \nu\varepsilon_1)}] \quad (\text{C.2.6-6})$$

$$a_s = \frac{r-1}{2r-1} \quad (\text{C.2.6-7})$$

Where:

d_c ——the parameter of compression damage evolution, which may be calculated

according to Formula (C.2.4-2), therein, $\chi = \frac{\varepsilon_{c,e}}{\varepsilon_c}$;

$\varepsilon_{c,e}$ ——the equivalent strain of compression energy;

a_s ——the shear yielding parameter;

r ——the improvement coefficient for biaxial compressive strength, which may be taken from 1.15 to 1.30 and may be determined in accordance with experimental data, and may be taken as 1.2 if the experimental data are insufficient.

2) Unloading equation

$$\begin{Bmatrix} \sigma_1 - \sigma_{un,1} \\ \sigma_2 - \sigma_{un,2} \end{Bmatrix} = (1 - \eta_d d_c) \frac{E_c}{1 - \nu^2} \begin{bmatrix} 1 & \nu \\ \nu & 1 \end{bmatrix} \begin{Bmatrix} \varepsilon_1 - \varepsilon_{un,1} \\ \varepsilon_2 - \varepsilon_{un,2} \end{Bmatrix} \quad (C.2.6-8)$$

$$\eta_d = \frac{\varepsilon_{c,e}}{\varepsilon_{c,e} + \varepsilon_{ca}} \quad (C.2.6-9)$$

Where:

η_d ——the plastic factor;

ε_{ca} ——the additional strain, which shall be calculated according to Formula (C.2.5-4).

3 Biaxial tension-compression zone ($\sigma'_1 < 0, \sigma'_2 \geq 0$) or ($\sigma'_1 \geq 0, \sigma'_2 < 0$)

1) Loading equation

$$\begin{Bmatrix} \sigma_1 \\ \sigma_2 \end{Bmatrix} = \begin{bmatrix} (1 - d_t) & 0 \\ 0 & (1 - d_c) \end{bmatrix} \begin{Bmatrix} \sigma'_1 \\ \sigma'_2 \end{Bmatrix} \quad (C.2.6-10)$$

$$\varepsilon_{t,e} = -\sqrt{\frac{1}{(1 - \nu^2)}} \varepsilon_1 (\varepsilon_1 + \nu \varepsilon_2) \quad (C.2.6-11)$$

Where:

d_t ——the parameter of tension damage evolution, which may be calculated according to

Formula (C.2.3-2), therein, $\chi = \frac{\varepsilon_{t,e}}{\varepsilon_t}$;

d_c ——the parameter of compression damage evolution, which may be calculated

according to Formula (C.2.4-2), therein, $\chi = \frac{\varepsilon_{c,e}}{\varepsilon_c}$;

$\varepsilon_{t,e}, \varepsilon_{c,e}$ ——the equivalent strain of energy, therein, $\varepsilon_{c,e}$ shall be calculated according to Formula (C.2.6-6) and $\varepsilon_{t,e}$ may be calculated according to Formula (C.2.6-11).

2) Unloading equation

$$\begin{Bmatrix} \sigma_1 - \sigma_{un,1} \\ \sigma_2 - \sigma_{un,2} \end{Bmatrix} = \frac{E_c}{1 - \nu^2} \begin{bmatrix} (1 - d_t) & (1 - d_t)\nu \\ (1 - \eta_d d_c)\nu & (1 - \eta_d d_c) \end{bmatrix} \begin{Bmatrix} \varepsilon_1 - \varepsilon_{un,1} \\ \varepsilon_2 - \varepsilon_{un,2} \end{Bmatrix} \quad (C.2.6-12)$$

Where:

η_d ——the plastic factor.

C.3 Bond-slip Constitutive Relations between Steel Reinforcement and Concrete

C.3.1 The constitutive bond stress-slip (τ — s) relation curve between concrete and hot-rolled ribbed steel bar (Figure C.3.1) may be determined according to the following requirements, and the parameter values at the characteristic points of this curve may be selected in accordance with Table C.3.1.

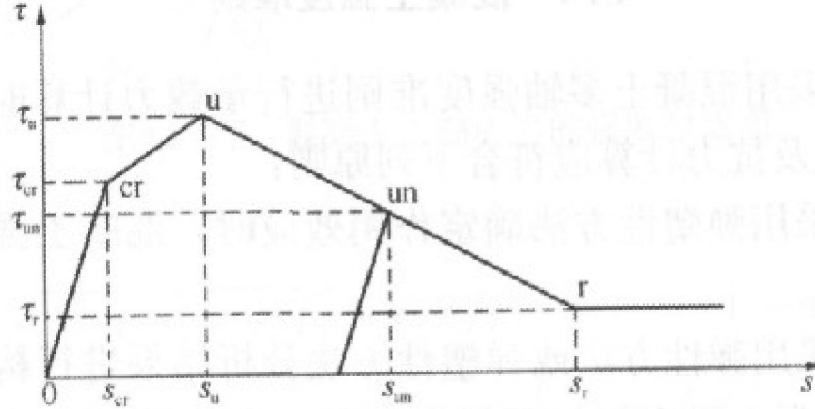


Figure C.3.1 Bond Stress-slip Curve between Concrete and Steel Reinforcement

Linear segment

$$\tau = k_1 s \quad 0 \leq s \leq s_{cr} \quad (\text{C.3.1-1})$$

Split segment

$$\tau = \tau_{cr} + k_2 (s - s_{cr}) \quad s_{cr} < s \leq s_u \quad (\text{C.3.1-2})$$

Descending segment

$$\tau = \tau_u + k_3 (s - s_u) \quad s_u < s < s_r \quad (\text{C.3.1-3})$$

Residual segment

$$\tau = \tau_r \quad s > s_r \quad (\text{C.3.1-4})$$

Unloading segment

$$\tau = \tau_{un} + k_1 (s - s_{un}) \quad (\text{C.3.1-5})$$

Where:

τ ——the bond stress between concrete and hot-rolled ribbed steel bar (N/mm^2);

s ——the relative slip between concrete and hot-rolled ribbed steel bar (mm);

k_1 ——the slope of linear segment, τ_{cr}/s_{cr} ;

k_2 ——the slope of split segment, $(\tau_u - \tau_{cr})/(s_u - s_{cr})$;

k_3 ——the slope of descending segment, $(\tau_r - \tau_u)/(s_r - s_u)$;

τ_{un} ——the bond stress at unloading point (N/mm^2);

s_{un} ——the relative slip of unloading point (mm).

Table C.3.1 Values of Parameters of Bond Stress-slip Curve between Concrete and Steel Reinforcement

Characteristic point	Split (cr)		Peak (u)		Residual (r)	
Bond stress (N/mm^2)	τ_{cr}	$2.5f_{tr}$	τ_u	$3f_{tr}$	τ_r	f_{tr}
Relative slip (mm)	s_{cr}	$0.025d$	s_u	$0.04d$	s_r	$0.55d$

Note: In this Table, d is diameter of steel reinforcement (mm); f_{tr} is the characteristic value of concrete tensile strength (N/mm^2).

C.3.2 Except for the hot-rolled ribbed steel bars, the parameter values of constitutive bond

stress-slip relation curve of other types of steel reinforcement may be determined in accordance with tests.

C.4 Concrete Strength Criterion

C.4.1 Where the calculation of load-bearing capacity is carried out by adopting with multi-axial strength criterion for concrete, the value of material strength parameter and the calculation of resistance shall meet the following principles:

1 Where the action effect is determined by adopting elastic-plastic method, the concrete strength index should take the mean value;

2 Where the calculation of load-bearing capacity of members is carried out by adopting with the analysis results by elastic method or elastic-plastic method, the concrete strength index may taken the design value (f_c or f_t) or standard value (f_{ck} or f_{tk}) of strength as required.

3 After obtaining the stress distribution and principal stress of concrete through elasticity analysis or elastic-plastic analysis, the checking of multi-axial strength of concrete shall meet the following requirements:

$$|\sigma_i| \leq |f_i| \quad (i=1, 2, 3) \quad (\text{C.4.1})$$

Where:

σ_i ——the value of the principal stress of concrete, which is negative when the concrete is under tension and is positive when the concrete is under compression, and $\sigma_1 \geq \sigma_2 \geq \sigma_3$;

f_i ——the value of the multi-axial strength of concrete, which is negative when the concrete is under tension and is positive when the concrete is under compression, and $f_1 \geq f_2 \geq f_3$.

C.4.2 Under the condition of biaxial stresses, the biaxial strength of concrete shall be determined by the closed curve (Figure C.4.2) that is formed by the following four curves or may also be adopted by interpolation method based on the values listed in Table C.4.2-1, Table C.4.2-2 and Table C.4.2-3.

The envelope curve equation of strength shall meet those specified in the following formulae:

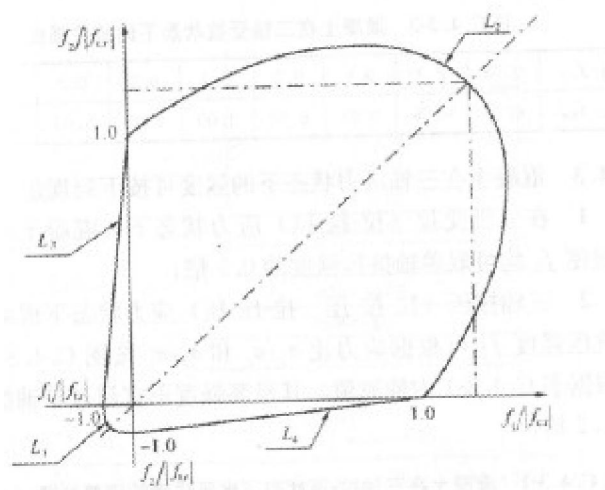


Figure C.4.2 Strength Envelope Diagram of Concrete Under Biaxial Stresses

$$\left\{ \begin{array}{l} L1: \quad f_1^2 + f_2^2 - 2\nu f_1 f_2 = (f_{t,r})^2 \\ L2: \quad \sqrt{f_1^2 + f_2^2 - f_1 f_2} - a_s (f_1 + f_2) = (1 - a_s) |f_{c,r}| \\ L3: \quad \frac{f_2}{|f_{c,r}|} - \frac{f_1}{|f_{t,r}|} = 1 \\ L4: \quad \frac{f_1}{|f_{c,r}|} - \frac{f_2}{|f_{c,r}|} = 1 \end{array} \right. \quad (C.4.2)$$

Where:

a_s ——the shear yielding parameter, which shall be determined by Formula (C.2.6-7).

Table C.4.2-1 Tensile Strength of Concrete Under Biaxial Tension-compression Stresses

$f_2/f_{t,r}$	0	-0.1	-0.2	-0.3	-0.4	-0.5	-0.6	-0.7	-0.8	-0.9	-1.0
$f_1/f_{c,r}$	1.00	0.90	0.80	0.70	0.60	0.50	0.40	0.30	0.20	0.10	0

Table C.4.2-2 Tensile Strength of Concrete Under Biaxial Compression

$f_1/f_{c,r}$	1.0	1.05	1.10	1.15	1.20	1.25	1.29	1.25	1.20	1.16
$f_2/f_{c,r}$	0	0.074	0.16	0.25	0.36	0.50	0.88	1.03	1.11	1.16

Table C.4.2-3 Tensile Strength of Concrete Under Biaxial Tension

$f_1/f_{t,r}$	-0.79	-0.7	-0.6	-0.5	-0.4	-0.3	-0.2	-0.1	0
$f_2/f_{t,r}$	-0.79	-0.86	-0.93	-0.97	-1.00	-1.02	-1.02	-1.02	-1.00

C.4.3 The strength of concrete under triaxial stresses may be determined according to the following requirements:

1 Under the state of triaxial tension (tension-tension-tension) stresses, the triaxial tensile strength f_3 of concrete may be taken as 0.9 times of the uniaxial tensile strength;

2 The triaxial compressive strength f_1 of concrete under the triaxial tension-compression (tension-tension-compression, tension-compression-compression) stresses may be determined according to Figure C.4.3-1 based on the stress ratio σ_3/σ_1 and σ_2/σ_1 , or may be adopted by interpolation method based on Table C.4.3-1, and the maximum strength should not exceed 1.2 times of the uniaxial compressive strength;

Table C.4.3-1 Adjustment Coefficient ($f_1/f_{c,r}$) for Compressive Strength of Concrete under Triaxial

Tension-compression

σ_2/σ_1	σ_3/σ_1	-0.75	-0.50	-0.25	-0.10	-0.05	0	0.25	0.35	0.36	0.50	0.70	0.75	1.00
-1.00	-1.00	0	0	0	0	0	0	0	0	0	0	0	0	0
-0.75	-0.75	0.10	0.10	0.10	0.10	0.10	0.10	0.05	0.05	0.05	0.05	0.05	0.05	0.05
-0.50	-0.50	—	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10
-0.25	-0.25	—	—	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20
-0.12	-0.12	—	—	—	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
-0.10	-0.10	—	—	—	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40
-0.08	-0.08	—	—	—	—	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
-0.05	-0.05	—	—	—	—	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60
-0.04	-0.04	—	—	—	—	—	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70
-0.02	-0.02	—	—	—	—	—	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.80
-0.01	-0.01	—	—	—	—	—	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.90
0	0	—	—	—	—	—	1.00	1.20	1.20	1.20	1.20	1.20	1.20	1.20

Note: Positive sign represents compression and negative sign represents tension.

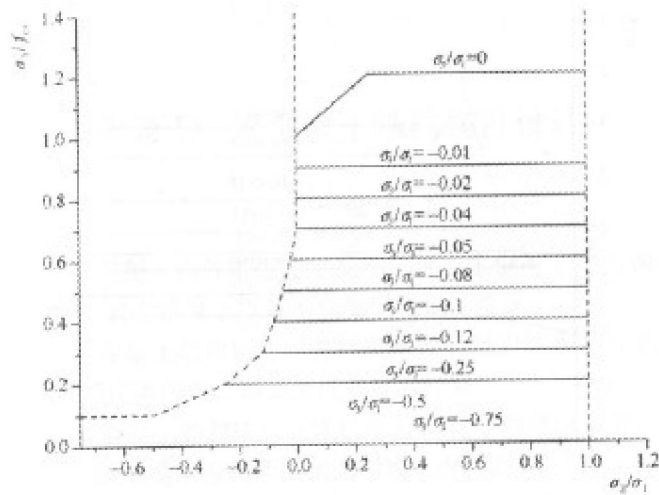


Figure C.4.3-1 Triaxial Compressive Strength of Concrete under Triaxial Tension-compression Stresses

3 The triaxial compressive strength f_1 of concrete under the triaxial compression (compression-compression-compression) stresses may be determined according to Figure C.4.3-2 based on the stress ratio σ_3/σ_1 and σ_2/σ_1 , or may be adopted by interpolation method based on Table C.4.3-2, and the maximum strength should not exceed 5 times of the uniaxial compressive strength;

Table C.4.3-2 Improvement Coefficient ($f_1/f_{c,r}$) for Compressive Strength of Concrete under Triaxial Compression

$\sigma_2/\sigma_1 \backslash \sigma_3/\sigma_1$	0	0.05	0.10	0.15	0.20	0.25	0.30	0.40	0.60	0.80	1.00
0	1.00	1.05	1.10	1.15	1.20	1.20	1.20	1.20	1.20	1.20	1.20
0.05	—	1.40	1.40	1.40	1.40	1.40	1.40	1.40	1.40	1.40	1.40
0.08	—	—	1.64	1.64	1.64	1.64	1.64	1.64	1.64	1.64	1.64
0.10	—	—	1.80	1.80	1.80	1.80	1.80	1.80	1.80	1.80	1.80
0.12	—	—	—	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
0.15	—	—	—	2.30	2.30	2.30	2.30	2.30	2.30	2.30	2.30
0.18	—	—	—	—	2.72	2.72	2.72	2.72	2.72	2.72	2.72
0.20	—	—	—	—	3.00	3.00	3.00	3.00	3.00	3.00	3.00

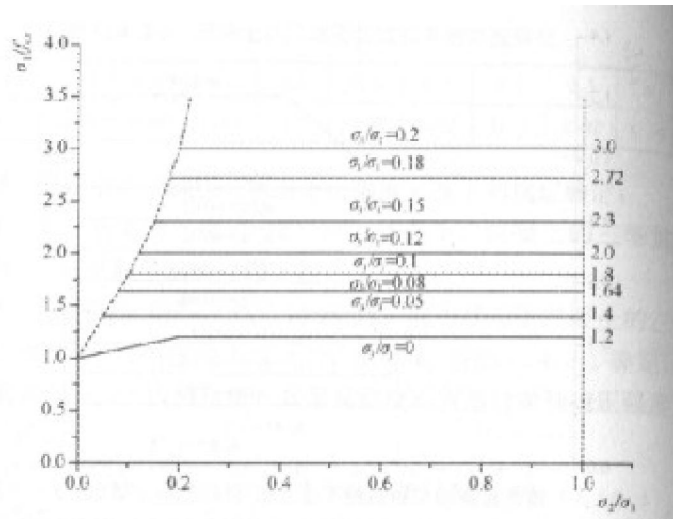


Figure C.4.3-2 Triaxial Compressive Strength of Concrete under Triaxial Compression

Appendix D Design of Plain Concrete Structural Members

D.1 General

D.1.1 The plain concrete members are mainly used as compression members. The plain concrete flexural members are only allowed to be used for the condition of horizontal located on the base and bearing no live load.

D.1.2 The plain concrete structural members shall be carried out with the calculation of normal section load-bearing capacity; and the positions bearing local loads still shall be carried out with the calculation of partial load-bearing capacity.

D.1.3 The calculated length l_0 of the plain concrete walls and columns may be adopted according to the following requirements:

1 $l_0=H$ when both ends of the wall or column are supported on rigid horizontal structures;

2 $l_0=1.25H\sim 1.50H$ when the wall or column has elastic mobile support;

3 $l_0=2H$ for the free and independent walls and columns.

Hereof, H is the height of the wall or column and is measured by the storey height.

D.1.4 The maximum spacing of the expansion joints of plain concrete structure may be selected according to those specified in Table D.1.4.

As for the entire piece of plain concrete wall structure, the expansion joints should be built into through type to disconnect the foundation.

Table D.1.4 Maximum Spacing between Expansion Joints of Plain Concrete Structure (m)

Structure type	Indoor or in soil	In the open air
Precast structure	40	30
Cast-in-situ structure (with constructional steel reinforcement)	30	20
Cast-in-situ structure (without constructional steel reinforcement)	20	10

D.2 Compression Members

D.2.1 As for the plain concrete compression members, if the calculation is done in accordance with the compression load-bearing capacity, the action of concrete in the tension zone is not considered and the normal stress shape in the compression zone is assumed to be rectangular, their stress value shall be taken as the design value of axial compressive strength of plain concrete, by this time, the action point of axial force is superposed with the point of concrete resultant forces in the compression zone.

The compression load-bearing capacity of plain concrete compression member shall meet the following requirements:

1 Section that is symmetric to the action plane of bending moment

$$N \leq \varphi_{cc} A'_c \quad (\text{D.2.1-1})$$

The depth of compression zone (x) shall be determined according to the following conditions:

$$e_c = e_0 \quad (\text{D.2.1-2})$$

Here, the distance e_0 between the action point of axial force and the centroid of section shall meet the following requirement:

$$e_0 \leq 0.9y'_0 \quad (\text{D.2.1-3})$$

2 Rectangular section (Figure D.2.1)

$$N \leq \varphi f_{cc} b (h - 2e_0) \quad (\text{D.2.1-4})$$

Where:

N ——the design value of axial pressure;

φ ——the stability coefficient of plain concrete members, which shall be adopted according to Table D.2.1;

f_{cc} ——the design value of the axial compressive strength of plain concrete, which shall be adopted according to the design value of the axial compressive strength of concrete (f_c) as specified in Table 4.1.4-1 of this code multiplied by the coefficient of 0.85;

A ——the area of the concrete compression zone;

e_c ——the distance between the point of resultant forces of concrete in compression zone and the centroid of section;

y'_0 ——the distance between the centroid of section and the edge of compression zone;

b ——the width of section;

h ——the depth of section.

If the compression load-bearing capacity is calculated according to Formula (D.2.1-1) or Formula (D.2.1-4), the concrete tension zone shall be distributed with constructional steel reinforcement in relation to the compression members with e_0 no less than $0.45y'_0$. The ratio of reinforcement shall not be less than 0.05% of the sectional area of member. However, if the conditions of Formula (D.2.2-1) or Formula (D.2.2-2) in this code are met, the member may not be applied with the constructional steel reinforcements stated in this clause.

Table D.2.1 Stability Coefficient φ of Plain Concrete Member

l_0/b	<4	4	6	8	10	12	14	16	18	20	22	24	26	28	30
l_0/i	<14	14	21	28	35	42	49	56	63	70	76	83	90	97	104
φ	1.00	0.98	0.96	0.91	0.86	0.82	0.77	0.72	0.68	0.63	0.59	0.55	0.51	0.47	0.44

Note: To calculate l_0/b , the b value: shall be taken as the depth of section of the acting plane of bending moment for the eccentric compression members; and shall be taken as the short side size of the section for the axial compression members.

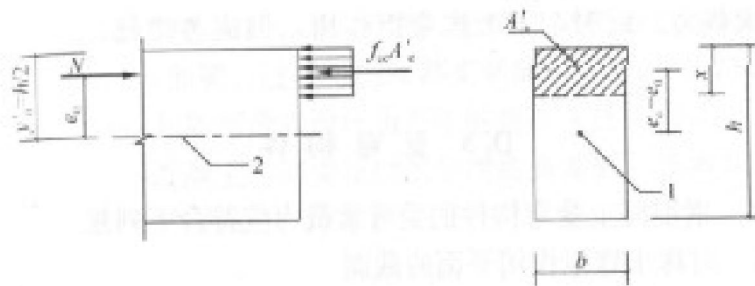


Figure D.2.1 Calculation of Compression Load-bearing Capacity of the Plain Concrete Compression Members with Rectangular Section

1—Gravity center; 2—Gravity center line

D.2.2 As for the plain concrete compression members that are disallowed cracking (for instance, the compression members and parapet walls under liquid pressure, etc), if e_0 is not less than $0.45 y_0'$, their compression load-bearing capacity shall be calculated according to the following formulae:

1 Section that is symmetric to the acting plane of bending moment

$$N \leq \varphi \frac{\gamma_{\alpha} A}{\frac{e_0 A}{W} - 1} \quad (\text{D.2.2-1})$$

2 Rectangular section

$$N \leq \varphi \frac{\gamma_{\alpha} b h}{\frac{6 e_0}{h} - 1} \quad (\text{D.2.2-2})$$

Where:

γ_{α} —the design value of axial tensile strength of plain concrete, which shall be adopted according to the design value of axial tensile strength of concrete (f_t) as specified in Table 4.1.4-2 of this code multiplied by the coefficient of 0.55;

γ —the plastic influence coefficient for the section resistance moment, which shall be adopted in accordance with Article 7.2.4 of this code;

W —the moment of elastic resistance of the tension edge of section;

A —the sectional area.

D.2.3 The plain concrete eccentric compression members not only shall be calculated with the compression load-bearing capacity of the acting plane of bending moment, but also shall be checked with the compression load-bearing capacity perpendicular to the acting plane of bending moment in accordance with axial compression members. Here, the action of bending moment shall not be considered, but the influence of stability coefficient φ shall be considered.

D.3 Flexural Members

D.3.1 The flexural load-bearing capacity of plain concrete flexural members shall meet the following requirements:

1 Section that is symmetric to the acting plane of bending moment

$$M \leq \gamma f_{\alpha} W \quad (\text{D.3.1-1})$$

2 Rectangular section

$$M \leq \frac{\gamma_{\alpha} b h^2}{6} \quad (\text{D.3.1-2})$$

Where:

M —the design value of bending moment.

D.4 Local Constructional Steel Reinforcement

D.4.1 The plain concrete structures shall be distributed with local constructional steel reinforcement at the following positions:

- 1 Position where the sectional dimension of the member changes suddenly;
- 2 Position where the wall height varies (distributing the steel reinforcement within a scope of no less than 1m);
- 3 Around the opening in concrete wall.

Note: After configuring the local constructional steel reinforcement, the spacing of expansion joints still shall be adopted in accordance with the cast-in-situ structures without constructional steel reinforcement as stated in Table D.1.4 of this code.

D.5 Local Compression

D.5.1 The partial load-bearing capacity of plain concrete members shall meet the following requirements:

- 1 Local compression face only with the action of local load

$$F_l \leq w\beta_l f_{cc} A_l \quad (D.5.1-1)$$

- 2 Local compression face still with the action of nonlocal load

$$F_l \leq w\beta_l (f_{cc} - \sigma) A_l \quad (D.5.1-2)$$

Where:

F_l ——the design value of the local load or local pressure acting on the local compression face;

A_l ——the local compression area;

w ——the influence coefficient of load distribution: when the loads on the local compression face are of uniform distribution, $w=1$; when the local loads are of nonuniform distribution (such as the end supporting surfaces of beam and lintel, etc), $w=0.75$;

σ ——the concrete compression stress produced by the design values of nonlocal loads;

β_l ——the strength improvement coefficient of concrete under local compression, which shall be calculated in accordance with Formula (6.6.1-2) of this code.

Appendix E Calculation for Load-bearing Capacity of Arbitrary Sections, Circular and Annular Normal Sections of Members

E.0.1 For the reinforced concrete and prestressed concrete members of arbitrary sections, their load-bearing capacity of normal section may be calculated according to the following methods:

1 The section is divided into multiple finite concrete units, longitudinal steel reinforcement units and prestressing tendon and/or bar units (Figure E.0.1a), the internal strain and stress of the units is regarded as being of uniform distribution approximately, and the point of their resultant forces is adopted at the gravity center of the unit;

2 The strain of each unit is determined by the following formulae in accordance with the assumption of section-strain retention plane specified in Article 6.2.1 of this code (Figure E.0.1b):

$$\varepsilon_{ci} = \varphi_u [(x_{ci} \sin \theta + y_{ci} \cos \theta) - r] \quad (\text{E.0.1-1})$$

$$\varepsilon_{sj} = -\varphi_u [(x_{sj} \sin \theta + y_{sj} \cos \theta) - r] \quad (\text{E.0.1-2})$$

$$\varepsilon_{pk} = -\varphi_u [(x_{pk} \sin \theta + y_{pk} \cos \theta) - r] + \varepsilon_{p0k} \quad (\text{E.0.1-3})$$

3 The ultimate curvature φ_u of the section when reaching the ultimate limit state shall be determined in accordance with the following two kinds of conditions:

1) Where the compression strain ε_c of concrete on the outer edge of the compression zone of section reaches the ultimate compression strain ε_{cu} of concrete and the strain ε_{sl} of the outermost row of steel reinforcements in the tension zone is less than 0.01, φ_u shall be calculated according to the following formula:

$$\varphi_u = \frac{\varepsilon_{cu}}{x_n} \quad (\text{E.0.1-4})$$

2) Where the strain ε_{sl} of the outermost row of steel reinforcement in the tension zone of section reaches 0.01 and the compression strain ε_c of concrete on the outer edge of compression zone is less than the ultimate compression strain ε_{cu} of concrete, φ_u shall be calculated according to the following formula:

$$\varphi_u = \frac{0.01}{h_{01} - x_n} \quad (\text{E.0.1-5})$$

4 The compression stress of concrete unit and the stress of steel bar unit and prestressing tendon and/or bar unit shall be determined in accordance with the fundamental assumption stated in Article 6.2.1 of this code;

5 The normal section load-bearing capacity of members shall be calculated according to the following formulae (Figure E.0.1):

$$N \leq \sum_{i=1}^l \sigma_{ci} A_{ci} - \sum_{j=1}^m \sigma_{sj} A_{sj} - \sum_{k=1}^n \sigma_{pk} A_{pk} \quad (\text{E.0.1-6})$$

$$M_x \leq \sum_{i=1}^l \sigma_{ci} A_{ci} x_{ci} - \sum_{j=1}^m \sigma_{sj} A_{sj} x_{sj} - \sum_{k=1}^n \sigma_{pk} A_{pk} x_{pk} \quad (\text{E.0.1-7})$$

$$M_y \leq \sum_{i=1}^l \sigma_{ci} A_{ci} y_{ci} - \sum_{j=1}^m \sigma_{sj} A_{sj} y_{sj} - \sum_{k=1}^n \sigma_{pk} A_{pk} y_{pk} \quad (\text{E.0.1-8})$$

Where:

N ——the design value of axial force, N adopts positive value when the force is pressure and adopts negative value when the force is tensile force;

M_x, M_y ——respectively the design value of bending moment in directions of x axis and y axis of the section of eccentrically stressed member: if the member is under eccentric compression, the additional bending moment caused by additional eccentric distance shall be considered; M_y adopts positive value when the axial pressure acts on the upside of x axis and M_x adopts negative value when the axial pressure acts on the right side of y axis; if the member is under eccentric tension, the influence of additional eccentricity shall not be considered;

$\varepsilon_{ci}, \sigma_{ci}$ ——Respectively the strain and stress of the i^{th} concrete unit: when the concrete unit is under compression, both ε_{ci} and σ_{ci} adopt positive values; when the concrete unit is under tension, the stress $\sigma_{ci}=0$; the serial number i is 1, 2, ..., l , hereof, l is the number of concrete units;

A_{ci} ——the area of the i^{th} concrete unit;

x_{ci}, y_{ci} ——respectively the distance values from the gravity center of the i^{th} concrete unit to y axis and x axis, which shall adopt positive values when x_{ci} is on the right side of y axis and y_{ci} is on the upside of x axis;

$\varepsilon_{sj}, \sigma_{sj}$ ——respectively the strain and stress of the j^{th} steel bar unit, which adopt positive values when the steel bar unit is under tension, the stress σ_{sj} shall meet the condition of Formula (6.2.1-6) in this code; and the serial number j is 1, 2, ..., m , hereof, m is the number of steel bar units;

A_{sj} ——the area of the j^{th} steel bar unit;

x_{sj}, y_{sj} ——respectively the distance values from the gravity center of the j^{th} steel bar unit to y axis and x axis, which shall adopt positive values when x_{sj} is on the right side of y axis and y_{sj} is on the upside of x axis;

$\varepsilon_{pk}, \sigma_{pk}$ ——respectively the strain and stress of the k^{th} prestressing tendon and/or bar unit, which adopt positive values when the unit is under tension, and stress σ_{pk} shall meet the condition of Formula (6.2.1-7) in this code, the serial number k is 1, 2, ..., n , hereof, n is the number of prestressing tendon and/or bar units;

ε_{p0k} ——the strain of the k^{th} prestressing tendon and/or bar unit at its gravity center when the normal stress of concrete equals to zero, its value shall be σ_{p0k} divided by the elastic modulus of prestressing tendon and/or bar shall adopt positive value when the unit is under tension; and σ_{p0k} shall be calculated according to Formula (10.1.6-3) or Formula (10.1.6-6) of this code;

A_{pk} ——the area of the k^{th} prestressing tendon and/or bar unit;

x_{pk}, y_{pk} ——respectively the distance values from the gravity center of the k^{th} prestressing tendon and/or bar unit to y axis and x axis, which shall adopt positive values when x_{pk} is on

the right side of y axis and y_{pk} is on the upside of x axis;

x, y —respectively the two coordinate axes of the rectangular coordinate system with the centroid of section as origin;

r —the distance between centroid and neutral axis of section;

h_{01} —the distance, perpendicular to the neutral axis, between the outer edge of the compression zone of section and the outermost row steel bars of the tension zone;

θ —the included angle between x axis and neutral axis, which adopts positive value in clockwise direction;

x_n —the distance between the neutral axis and the outermost edge of compression zone.

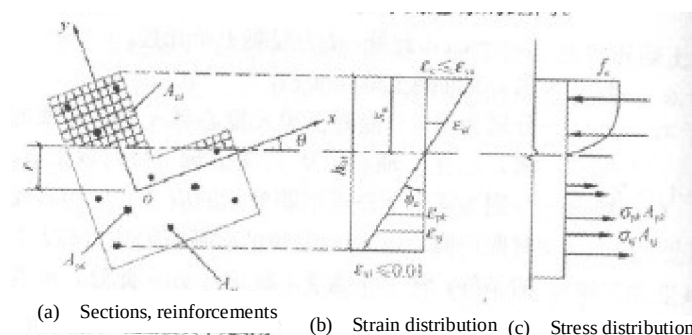


Figure E.0.1 Calculation of the Normal Section Load-bearing Capacity of Members with Arbitrary Sections

E.0.2 The normal section flexural load-bearing capacity of the flexural members with annular and circular sections shall be calculated in accordance with those specified in Article E.0.3 and Article E.0.4 of this code. However, in calculation, Formula (E.0.3-1), Formula (E.0.3-3) and Formula (E.0.4-1) shall take the equal sign, and the design value of axial force $N=0$; meanwhile, N_{ei} in Formula (E.0.3-2), Formula (E.0.3-4) and Formula (E.0.4-2) shall be replaced by the design value of bending moment M .

E.0.3 As for the eccentric compression member that is in annular section and is uniformly configured with longitudinal steel reinforcement along its periphery (Figure E.0.3), its normal section compression load-bearing capacity should meet the following requirements:

1 Reinforced concrete member

$$N \leq a a_1 f_c A + (a - a_t) f_y A_s \quad (\text{E.0.3-1})$$

$$N_{ei} \leq a_1 f_c A (r_1 + r_2) \frac{\sin \pi a}{2\pi} + f_y A_s r_s \frac{(\sin \pi a + \sin \pi a_t)}{\pi} \quad (\text{E.0.3-2})$$

2 Prestressed concrete member

$$N \leq a a_i f_c A - \sigma_{p0} A_p + a f_{py}' A_p - a_t (f_{py} - \sigma_{p0}) A_p \quad (\text{E.0.3-3})$$

$$N e_i \leq a_1 f_c A (r_1 + r_2) \frac{\sin \pi a}{2\pi} + f_{py}' A_p r_p \frac{\sin \pi a}{\pi} + (f_{py} - \sigma_{p0}) A_p r_p \frac{\sin \pi a_t}{\pi} \quad (\text{E.0.3-4})$$

The coefficient and eccentric distance in the above-mentioned formulae shall be calculated according to the following formulae:

$$a_t = 1 - 1.5a \quad (\text{E.0.3-5})$$

$$e_i = e_0 + e_a \quad (\text{E.0.3-6})$$

Where:

A ——the section of annular section;

A_s ——the sectional area of all the longitudinal steel bars;

A_p ——the sectional area of all the longitudinal prestressing tendons and/or bars;

r_1, r_2 ——the inside and outside radius of the annular section;

r_s ——the radius of the circle in which the gravity center of longitudinal steel bars located;

r_p ——the radius of the circle in which the gravity center of longitudinal prestressing tendons and/or bars located;

e_0 ——the eccentric distance of axial pressure to centroid of section;

e_a ——the additional eccentric distance, determined according to Article 6.2.5 of this code;

a ——the ratio of the concrete sectional area of compression zone to the total sectional area;

a_t ——the ratio of the sectional area of longitudinal tension steel reinforcements to the sectional area of all the longitudinal steel reinforcements, and a_t shall be taken as 0 when a is larger than 2/3.

3 When a is less than $\arccos\left(\frac{2r_1}{r_1+r_2}\right)/\pi$, the eccentric compression member with

annular section may be calculated according to the formula for normal section compression load-bearing capacity of eccentric compression member with circular section stated in Article E.0.4 of this code.

Note: This article is applicable to the condition that the quantity of longitudinal steel reinforcements in the section is not less than 6 and r_1/r_2 is not less than 0.5.

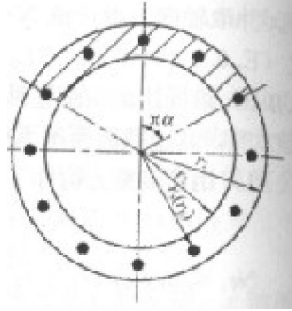


Figure E.0.3 Annular Section with Uniformly Distributed Reinforcements along Its Periphery

E.0.4 As for the reinforced concrete eccentric compression member that is with circular section and is uniformly configured with longitudinal steel bars along its periphery (Figure E.0.4), its normal section compression load-bearing capacity should meet the following requirements:

$$N \leq aa_1 f_c A \left(1 - \frac{\sin 2\pi a}{2\pi a} \right) + (a - a_t) f_y A_s \quad (\text{E.0.4-1})$$

$$Ne_i \leq \frac{2}{3} a_1 f_c A r \frac{\sin^3 \pi a}{\pi} + f_y A_s r_s \frac{\sin \pi a + \sin \pi a_1}{\pi} \quad (\text{E.0.4-2})$$

$$a_i = 1.25 - 2a \quad (\text{E.0.4-3})$$

$$e_i = e_0 + e_a \quad (\text{E.0.4-4})$$

Where:

A ——the section of circular section;

A_s ——the sectional area of all the longitudinal steel bars;

r ——the radius of circular section;

r_s ——the radius of the circle in which the gravity center of longitudinal steel bars located;

e_0 ——the eccentric distance of axial pressure to centroid of section;

e_a ——the additional eccentric distance, determined according to Article 6.2.5 of this code;

a ——the ratio of the central angle (rad) corresponding to the concrete sectional area in compression zone to 2π ;

a_t ——the ratio of the sectional area of longitudinal tension steel bars to the sectional area of all the longitudinal steel bars, and a_t shall be taken as 0 when a is larger than 0.625.

Note: This article is applicable to the condition that the quantity of longitudinal steel bars allocated in the section is not less than 6.

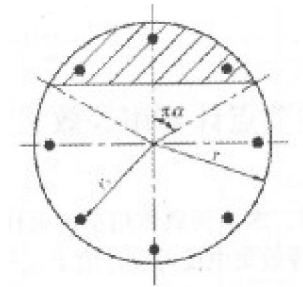


Figure E.0.4 Circular Section with Uniformly Distributed Reinforcements along Its Periphery

E.0.5 As for the eccentric tension members that are with annular or circular section and are uniformly configured with longitudinal steel reinforcements along the periphery, their normal section tension load-bearing capacity shall meet the requirements of Formula (6.2.25-1) of this code, the design value of normal section flexural load-bearing capacity M_u in this formula may be calculated according to those specified in Article E.0.2 of this code, but this formula shall take a equal sign and N_{ei} shall be replaced by M_u .

Appendix F Design Value of Equivalent Concentrated Reaction

Used for Calculation of Slab-column Joints

F.0.1 As for the slab-column joints under the action of vertical load and horizontal load, the design value of equivalent concentrated reaction ($F_{l,eq}$) used for the calculation of punching shear bearing capacity may be determined in accordance with the following conditions:

1 For the slab-column joint transferring unidirectional unbalanced bending moment:

Where the acting plane of unbalanced bending moment is supposed with one of the two axial lines of the rectangular section of column, $F_{l,eq}$ may be calculated in accordance with the following two conditions:

1) For the unidirectional unbalanced bending moment $a_0 M_{unb}$ transferred by the joint under shear force, when its acting direction points to the AB edge in Figure F.0.1, the design value of the equivalent concentrated reaction may be calculated according to the following formulae:

$$F_{l,eq} = F_l + \frac{a_0 M_{unb} a_{AB}}{I_c} u_m h_0 \quad (F.0.1-1)$$

$$M_{unb} = M_{unb,c} - F_l e_g \quad (F.0.1-2)$$

2) For the unidirectional unbalanced bending moment $a_0 M_{unb}$ transferred by the joint under shear force, when its acting direction points to the CD edge in Figure F.0.1, the design value of the equivalent concentrated reaction may be calculated according to the following formulae:

$$F_{l,eq} = F_l + \frac{a_0 M_{unb} a_{CD}}{I_c} u_m h_0 \quad (F.0.1-3)$$

$$M_{unb} = M_{unb,c} + F_l e_g \quad (F.0.1-4)$$

Where:

F_l ——the storey difference of the design value of axial compressive force endured by column minus the design value of loads endured by slab within the scope of punching failure cone at column top, under the action of vertical loads and horizontal loads;

a_0 ——the calculation coefficient, which shall be calculated in accordance with Article F.0.2 of this code;

M_{unb} ——the design value of unbalanced bending moment caused by vertical load and horizontal load to the gravity axis (axial line 2 as shown in Figure F.0.1) of the perimeter of critical section;

$M_{unb,c}$ ——the design value of unbalanced bending moment caused by vertical load and horizontal load to the gravity axis (axial line 1 as shown in Figure F.0.1) of the column section;

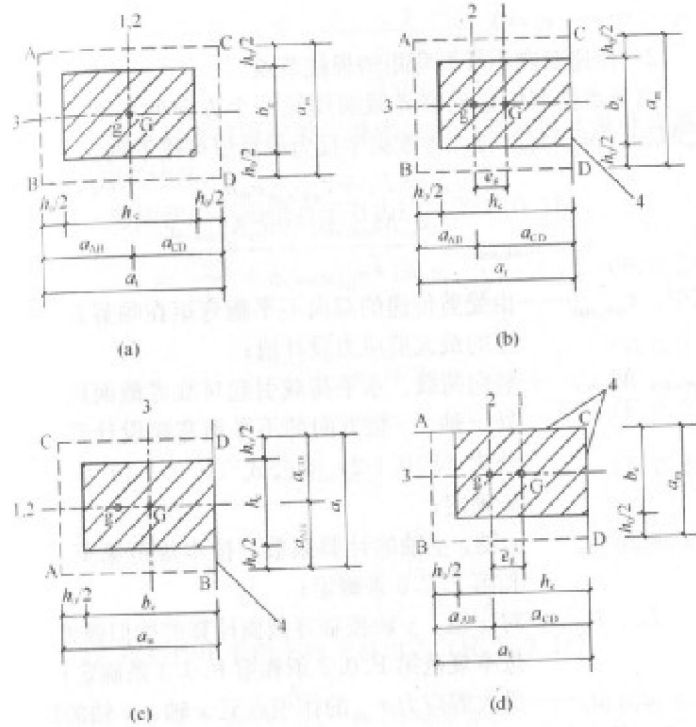


Figure F.0.1 Geometric Parameters for Rectangular Column and for Calculation of Punching Shear Bearing Capacity

- (a) Section of Central Column; (b) Section of side column (the acting plane of bending moment is perpendicular to the free edge)
(c) Section of side column (acting plane of bending moment is parallel to the free edge); (d) Section of corner column
1—Axial line of the gravity center G of column section; 2—Axial line of the perimeter gravity center g of critical section;
3—Acting plane of unbalanced bending moment; 4—Free edge

a_{AB} , a_{CD} —respectively the distance from the perimeter gravity axis of critical section to the AB and CD edges;

I_c —the similar polar moment of inertia calculated in accordance with critical section, which shall be calculated in accordance with Article F.0.2 of this code;

e_g —the distance from the gravity axis of column section in the acting plane of bending moment to the perimeter gravity axis of critical section, which shall be calculated in accordance with Article F.0.2 of this code; for section of central column section as well as the section of side column with the acting plane of bending moment parallel to the free edge, $e_g=0$.

2 For the slab-column joint transferring bidirectional unbalanced bending moment:

Where the unbalanced bending moment transferred by the joint under shear force to the two directions of the perimeter of critical section are $a_{0x}M_{unb,x}$ and $a_{0y}M_{unb,y}$, the design value of equivalent concentrated reaction may be calculated according to the following formulae:

$$F_{l,eq} = F_l + \tau_{unb,max} u_m h_0 \quad (F.0.1-5)$$

$$\tau_{unb,max} = \frac{a_{0x} M_{unb,x} a_x}{I_{cx}} + \frac{a_{0y} M_{unb,y} a_y}{I_{cy}} \quad (F.0.1-6)$$

Where:

$\tau_{\text{unb,max}}$ ——the design value of maximum shear stress produced on the critical section by the bidirectional unbalanced bending moment that is transferred due to shear force;

$M_{\text{unb,x}}, M_{\text{unb,y}}$ ——the design value of the bending moment at the perimeter gravity center of critical section respectively in directions of x axis and y axis caused by vertical load and horizontal load, which shall be determined according to the same method as Formula (F.0.1-2) or Formula (F.0.1-4);

a_{0x}, a_{0y} ——respectively the calculation coefficient for x axis and y axis, which shall be determined in accordance with Article F.0.2 and Article F.0.3 of this code;

I_{cx}, I_{cy} ——respectively the similar polar moment of inertia for x axis and y axis as calculated in accordance with critical section, which shall be calculated in accordance with Article F.0.2 and Article F.0.3 of this code;

a_x, a_y ——respectively the distance from the action point of maximum shear stress τ_{max} to the x axis and y axis.

3 Where different load combinations are considered, the larger value therein shall be taken as the design value of equivalent concentrated reaction to be used for the calculation of the punching shear bearing capacity of slab-column joint.

F.0.2 In the calculation of the punching shear bearing capacity of slab-column joint with consideration of the unidirectional unbalanced bending moment transferred due to shear force, the parameters related to the design value of equivalent concentrated reaction ($F_{l,\text{eq}}$) and the geometric dimensions as shown in Figure F.0.1 of this appendix may be calculated according to the following formulae:

1 The similar polar moment of inertia, geometric dimensions and calculation coefficient of the critical section at central column may be calculated according to the following formulae (Figure F.0.1a):

$$I_c = \frac{h_0 a_t^3}{6} + 2h_0 a_m \left(\frac{a_t}{2} \right)^2 \quad (\text{F.0.2-1})$$

$$a_{AB} = a_{CD} = \frac{a_t}{2} \quad (\text{F.0.2-2})$$

$$e_g = 0 \quad (\text{F.0.2-3})$$

$$a_0 = 1 - \frac{1}{1 + \frac{2}{3} \sqrt{\frac{h_c + h_0}{b_c + h_0}}} \quad (\text{F.0.2-4})$$

2 The similar polar moment of inertia, geometric dimensions and calculation coefficient of the critical section at side column may be calculated according to the following formulae:

1) The acting plane of bending moment is perpendicular to the free edge (Figure F.0.1b)

$$I_c = \frac{h_0 a_t^3}{6} + h_0 a_m a_{AB}^2 + 2h_0 a_t \left(\frac{a_t}{2} - a_{AB} \right)^2 \quad (\text{F.0.2-5})$$

$$a_{AB} = \frac{a_t^2}{a_m + 2a_t} \quad (\text{F.0.2-6})$$

$$a_{CD} = a_t - a_{AB} \quad (\text{F.0.2-7})$$

$$e_g = a_{CD} - \frac{h_c}{2} \quad (\text{F.0.2-8})$$

$$a_0 = 1 - \frac{1}{1 + \frac{2}{3} \sqrt{\frac{h_c + h_0/2}{b_c + h_0}}} \quad (\text{F.0.2-9})$$

2) The acting plane of bending moment is parallel to the free edge (Figure F.0.1c)

$$I_c = \frac{h_0 a_t^3}{12} + 2h_0 a_m \left(\frac{a_t}{2} \right)^2 \quad (\text{F.0.2-10})$$

$$a_{AB} = a_{CD} = \frac{a_t}{2} \quad (\text{F.0.2-11})$$

$$e_g = 0 \quad (\text{F.0.2-12})$$

$$a_0 = 1 - \frac{1}{1 + \frac{2}{3} \sqrt{\frac{h_c + h_0}{b_c + h_0/2}}} \quad (\text{F.0.2-13})$$

3 The similar polar moment of inertia, geometric dimensions and calculation coefficient of the critical section at corner column may be calculated according to the following formulae (Figure F.0.1d):

$$I_c = \frac{h_0 a_t^3}{12} + h_0 a_m a_{AB}^2 + h_0 a_t \left(\frac{a_t}{2} - a_{AB} \right)^2 \quad (\text{F.0.2-14})$$

$$a_{AB} = \frac{a_t^2}{2(a_m + a_t)} \quad (\text{F.0.2-15})$$

$$a_{CD} = a_t - a_{AB} \quad (\text{F.0.2-16})$$

$$e_g = a_{CD} - \frac{h_c}{2} \quad (\text{F.0.2-17})$$

$$a_0 = 1 - \frac{1}{1 + \frac{2}{3} \sqrt{\frac{h_c + h_0/2}{b_c + h_0/2}}} \quad (\text{F.0.2-18})$$

F.0.3 In the calculation of the punching shear bearing capacity of slab-column joint in accordance with Formula (F.0.1-5) and Formula (F.0.1-6) of this appendix with consideration of transferring the bidirectional unbalanced bending moment, if those specified in Article F.0.2 of this appendix are regarded as the similar polar moment of inertia, geometric

dimensions and calculation coefficient of x axis (or y axis), then the corresponding similar polar moment of inertia, geometric dimensions and calculation coefficient of y axis (or x axis) may be determined by replacing the relevant aforementioned parameters of x axis (or y axis).

F.0.4 Where the cantilever slabs are set at the positions of side column and corner column, the perimeter of critical section may be counted till the slab end that is perpendicular to the free edge, and the perimeter of critical section that is calculated based on this shall be compared with the perimeter of critical section that is calculated in accordance with central column, and the smaller value between them shall be taken. On this basis, the relevant parameters for the design value of equivalent concentrated reaction ($F_{l,eq}$) that is used for the calculation of the punching shear bearing capacity of slab-column joints with consideration of transferring unbalanced bending moment due to shear force shall be determined according to the principles stated in Article F.0.2 and Article F.0.3 of this code.

Appendix G Deep Flexural Members

G.0.1 The simply-supported reinforced concrete single-span deep beam may be carried out with the sectional design by adopting with the internal force that is calculated with general method; the reinforced concrete multi-span continuous deep beam may be carried out with the sectional design by adopting with the internal force that is calculated through two-dimensional elasticity analysis.

G.0.2 The normal section flexural load-bearing capacity of reinforced concrete deep flexural member shall meet the following requirements:

$$M \leq f_y A_s z \quad (\text{G.0.2-1})$$

$$z = a_d (h_0 - 0.5x) \quad (\text{G.0.2-2})$$

$$a_d = 0.80 + 0.04 \frac{l_0}{h} \quad (\text{G.0.2-3})$$

When $l_0 < h$, the internal force arm $z = 0.6l_0$.

Where:

x ——the depth of the compression zone of section, which shall be calculated in accordance with Section 6.2 of this code; when $x < 0.2h_0$, $x = 0.2h_0$;

h_0 ——the effective depth of section: $h_0 = h - a_s$, therein, h is the depth of section; when $l_0/h \leq 2$, a_s of mid-span section will be $0.1h$ and a_s of support section will be $0.2h$; when $l_0/h > 2$, a_s shall be taken in accordance with the actual distance from the centroid of the section of longitudinal steel reinforcement in tension zone to the tension edge.

G.0.3 The shear section of reinforced concrete deep flexural member shall meet the provisions the following conditions:

When $h_w/b \leq 4$

$$V \leq \frac{1}{60} (10 + l_0/h) \beta_c f_c b h_0 \quad (\text{G.0.3-1})$$

When $h_w/b \geq 6$

$$V \leq \frac{1}{60} (7 + l_0/h) \beta_c f_c b h_0 \quad (\text{G.0.3-2})$$

When $4 < h_w/b < 6$, V shall be taken according to linear interpolation method.

Where:

V ——the design value of shear force;

l_0 ——the calculated span, if $l_0 < 2h$, then $l_0 = 2h$;

b ——the width of rectangular section as well as the web thickness of T-shaped and I-shaped section;

H, h_0 ——the depth of section and the effective depth of section;

h_w ——the web depth of section: for rectangular section, h_w will be the effective depth h_0 ; for T-section, h_w will be the effective depth minus the flange height; for I-shaped section and box section, h_w will be the net depth of web;

β_c ——the influence coefficient of concrete strength, which shall be taken in accordance with those specified in Article 6.3.1 of this code.

G.0.4 For the deep flexural member of rectangular, T-shaped or I-shaped section under the action of uniformly distributed loads, if it is built with the vertically distributed steel

reinforcement and horizontally distributed steel reinforcement, the shear bearing capacity of its inclined section shall meet the following requirements:

$$V \leq 0.7 \frac{(8-l_0/h)}{3} f_t b h_0 + \frac{(l_0/h-2)}{3} f_{yv} \frac{A_{sv}}{s_h} h_0 + \frac{(5-l_0/h)}{6} f_{yh} \frac{A_{sh}}{s_v} h_0 \quad (\text{G.0.4-1})$$

As for the deep flexural members under the action of concentrated loads (including the case that diversified loads act and the shear force produced by concentrated load therein to the support section accounts for over 75% of the total shear force), the shear bearing capacity of its inclined section shall meet the following requirements:

$$V \leq \frac{1.75}{\lambda+1} f_t b h_0 + \frac{(l_0/h-2)}{3} f_{yv} \frac{A_{sv}}{s_h} h_0 + \frac{(5-l_0/h)}{6} f_{yh} \frac{A_{sh}}{s_v} h_0 \quad (\text{G.0.4-2})$$

Where:

λ ——the calculated ratio of shear span to effective depth: when l_0/h is not larger than 2.0, $\lambda=0.25$; when l_0/h is larger than 2 and is less than 5, $\lambda=a/h_0$, therein, a is the horizontal distance from the concentrated load to the support of deep flexural member; the upper limit value of λ is $(0.92l_0/h-1.58)$ and the lower limit value of λ is $(0.42l_0/h-0.58)$;

l_0/h ——the span-depth ratio, $l_0/h=2.0$ if $l_0/h<2$.

G.0.5 The reinforced concrete deep beams that are generally required not appearing inclined crack shall meet the following conditions:

$$V_k \leq 0.5 f_{tk} b h_0 \quad (\text{G.0.5})$$

Where:

V_k ——the shear force value calculated according to the standard combination of load effects.

In this case, the calculation of shear bearing capacity of inclined section may not be carried out, however, the distribution steel reinforcement shall be configured in accordance with those specified in Article G.0.10 and Article G.0.12 of this code.

G.0.6 For positions of reinforced concrete deep beam enduring the action of support counter force and concentrated load, the calculation of partial load-bearing capacity shall be carried out in accordance with those specified in Section 6.6 of this code.

G.0.7 The sectional width of deep beam shall not be less than 140mm. When l_0/h is not less than 1, h/b should not be larger than 25; when l_0/h is less than 1, l_0/b should not be larger than 25. The concrete strength grade of deep beam shall not be less than C20. Where the deep beam is supported on the reinforced concrete column, the column shall be extended to the top of deep beam. The top of deep beam shall be connected reliably with such horizontal members as floor slab.

G.0.8 The longitudinal tension steel reinforcement of the reinforced concrete deep beam should adopt smaller diameter and should be arranged according to the following requirements:

1 The lower longitudinal steel reinforcement of single-span deep beam and continuous deep beam should be uniformly arranged in the scope above the lower edge of beam by 0.2h (Figure G.0.8-1 and Figure G.0.8-2).

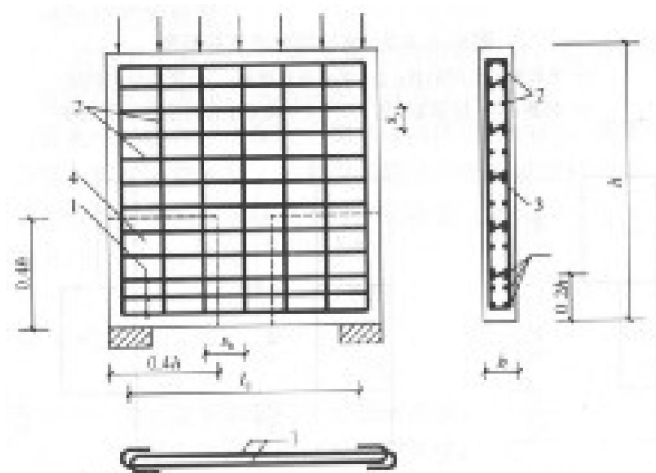


Figure G.0.8-1 Configuration of Steel Reinforcement for Single-span Deep Beam

1—Lower longitudinal tension steel reinforcement and bent anchorage; 2—Horizontally and vertically distributed steel reinforcement; 3—Tie bar; 4—Densified area of tie bars

2 The longitudinal tension steel reinforcement in the section of the intermediate support of continuous deep beam should be uniformly arranged in the corresponding altitude range according to the altitude range and ratio of reinforcement specified in Figure G.0.8-3. As for the continuous deep beam with l_0/h less than 1, the ratio of reinforcement of those longitudinal tension steel reinforcement in the altitude range above the bottom surface of intermediate support by $0.2l_0 \sim 0.6l_0$ still should not be less than 0.5%. The horizontally distributed steel reinforcement may be used for upper longitudinal tension steel reinforcement at the position of support, and the part with insufficient horizontally distributed steel reinforcement may be supplemented with additional horizontal steel reinforcement, and the extending length of additional horizontal steel reinforcement from the support to the midspan should not be less than $0.4l_0$ (Figure G.0.8-2)

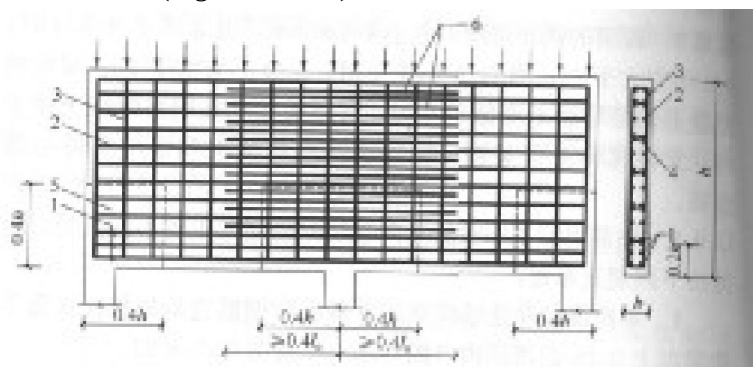


Figure G.0.8-2 Configuration of Steel Reinforcement for Continuous Deep Beam

1—Lower longitudinal tension steel reinforcement; 2—horizontally distributed steel reinforcement;
3—Vertically distributed steel reinforcement; 4—Tie bar; 5—Densified area of tie bars;
6—Additional horizontal steel reinforcement at upper part of support section

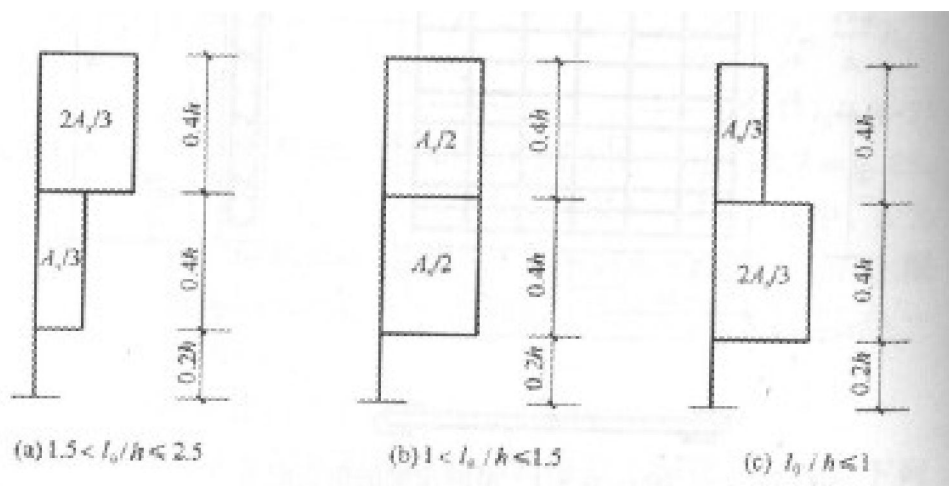


Figure G.0.8-3 Distribution Ratio of Longitudinal Tension Steel Reinforcement in Different Altitude
Ranges in the Section of Intermediate Support of Continuous Deep Beam

G.0.9 The lower longitudinal tension steel reinforcement of deep beam shall all be extended into the support and shall not be bent up or cut off at the midspan. At the support of simply-supported single-span deep beam and the simply-supported support at the bend of continuous deep beam, the longitudinal tension steel reinforcement shall be bent and anchored along the horizontal direction (Figure G.0.8-1), their anchorage length shall be adopted in accordance with the anchorage length l_a of tension steel reinforcement specified in Article 8.3.1 of this code multiplied by a coefficient 1.1; where the above-mentioned requirements on anchorage length can not be met, the effective anchoring measures, such as welding anchorage steel plate on the steel reinforcement or welding the tail end of steel reinforcement into closed type, shall be taken. The lower longitudinal tension steel reinforcement of continuous deep beam all shall be passed through the center line of intermediate support, and their anchorage length starting from the edge of support shall not be less than l_a .

G.0.10 The deep beam shall be configured with double-row steel mesh reinforcements, the diameter of both the horizontally and vertically distributed steel reinforcements shall not be less than 8mm and their spacing shall not be larger than 200mm.

Where the column is built along the vertical edge at end of deep beam, the horizontally distributed steel reinforcement shall be anchored into the column. At the upper and lower edges of deep beam, the vertically distributed steel reinforcement should be made into closed type.

Tie bars shall be arranged between the double-row steel reinforcements of deep beam, and the spacing of tie bars along both the vertical and horizontal directions should not be larger than 600mm, and the quantity of tie bars still shall be increased appropriately in the scope with height of $0.4h$ in the support zone and width of $0.4h$ extruding out of the support (the dotted portion in Figure G.0.8-1 and Figure G.0.8-2).

G.0.11 Where the deep beam is acted with uniformly distributed loads along its full span, the additional vertical suspenders shall be uniformly distributed along the full span of beam, and the spacing of suspenders should not be larger than 200mm.

If there is concentrated load acting within $3/4$ altitude range at the lower part of deep beam, this concentrated load all shall be endured by the additional suspenders, and the suspenders shall adopt the vertical suspender or inclined suspender. The horizontal

distribution length s of vertical suspenders shall be determined according to the following formulae (Figure G.0.11a):

When $h_1 \leq h_b/2$

$$s = b_b + h_b \quad (\text{G.0.11-1})$$

When $h_1 > h_b/2$

$$s = b_b + 2h_1 \quad (\text{G.0.11-2})$$

Where:

b_b ——the sectional width of the member that transfers the concentrated load;

h_b ——the sectional depth of the member that transfers the concentrated load;

h_1 ——the height from the lower edge of deep beam to the bottom edge of the member that transfers the concentrated load.

The vertical suspenders shall be distributed along both sides of the beam, they shall reach the beam top from the beam bottom and shall be made into closed type at the beam top and beam bottom.

The gross sectional area A_{sv} of additional suspenders shall be calculated in accordance with Section 9.2 of this code, however, the design strength f_{yv} of suspender shall be multiplied by the additional coefficient 0.8 that is used for the calculation of load-bearing capacity.

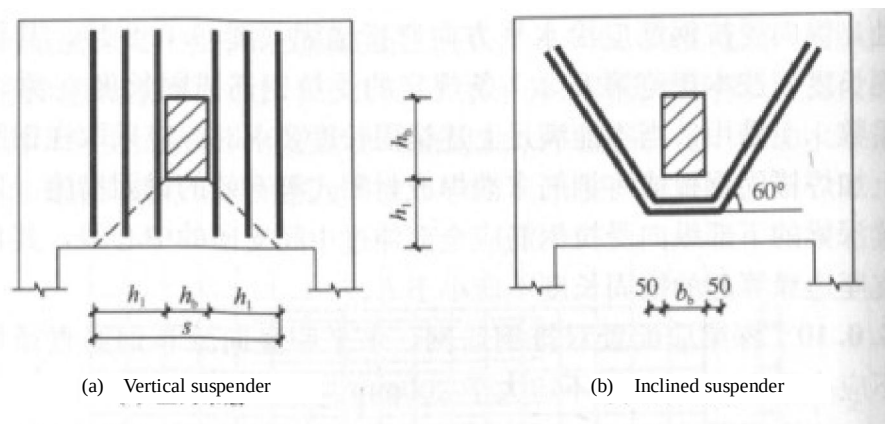


Figure G.0.11 Additional Suspenders when the Deep Beam Endures the Action of Concentrated Load

Note: Dimensional unit in the Figure is mm.

G.0.12 For the deep beam, the ratio of reinforcement $\rho \left(\rho = \frac{A_s}{bh} \right)$ of longitudinal tension

steel reinforcement, the ratio of reinforcement $\rho_{sh} \left(\rho_{sh} = \frac{A_{sh}}{bs_v} \right)$ of horizontally distributed

steel reinforcement (s_v is the spacing of horizontally distributed steel reinforcement) and the

ratio of reinforcement $\rho_{sv} \left(\rho_{sv} = \frac{A_{sv}}{bs_h} \right)$ of vertically distributed steel reinforcement (s_h is the

spacing of vertically distributed steel reinforcement) should not be less than the values specified in Table G.0.12.

Table G.0.12 Minimum Percentage of Reinforcement in Deep Beam (%)

Type of steel reinforcement	Longitudinal tension steel	Horizontally distributed steel	Vertically distributed steel
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	reinforcement	reinforcement	reinforcement
HPB300	0.25	0.25	0.20
HRB400, HRBF400, RRB400, HRB335, HRBF335	0.20	0.20	0.15
HRB500, HRBF500	0.15	0.15	0.10

Note: Where the concentrated load acts within the range of $1/4$ depth at the upper portion of continuous deep beam and l_0/h is larger than 1.5, the minimum percentage of reinforcement of the vertically distributed steel reinforcement shall be increased by 0.05.

G.0.13 As for the deep flexural members except for deep beam, the detailing requirements of the longitudinal stressed steel reinforcements, stirrups and longitudinal constructional steel reinforcements are same as those of general beams, however, the longitudinal constructional steel reinforcements distributed within a range of $1/2$ depth at the lower part of their section and those distributed within a range of $1/2$ depth at the upper portion of intermediate support should be reinforced appropriately by comparing with the general beams.

Appendix H Composite Beam and Slab Without Support

H.0.1 As for the composite flexural members (beam and slab) without support during the construction stage, the internal force shall be calculated respectively in accordance with the following two stages.

1 First stage: The stage before the post-cast concrete of composite layer reaching the design value of strength. The loads are born by the precast members, the precast members shall be calculated in accordance with simply-supported members; the loads include the deadweight of precast member, the deadweight of precast floor slab, the deadweight of composite layer and the live construction loads of this stage.

2 Second stage: The stage after the concrete of composite layer reaching the strength value specified in the design. The composite member shall be calculated in accordance with integral structure; and the load shall be taken as the larger value by considering the following two conditions:

Construction stage: The deadweight of composite member, precast floor slab, surface layer and suspended ceiling as well as the live construction loads in this stage shall be taken into consideration;

Service stage: The deadweight of composite member, precast floor slab, surface layer and suspended ceiling as well as the variable loads in service stage shall be taken into consideration;

H.0.2 The normal section flexural load-bearing capacity of the precast members and composite members shall be calculated in accordance with Section 6.2 of this code, therein, the design value of bending moment shall be taken according to the following requirements:

Precast member

$$M_1 = M_{1G} + M_{1Q} \quad (\text{H.0.2-1})$$

Positive bending moment segment of composite member

$$M = M_{1G} + M_{2G} + M_{2Q} \quad (\text{H.0.2-2})$$

Negative bending moment segment of composite member

$$M = M_{2G} + M_{2Q} \quad (\text{H.0.2-3})$$

Where:

M_{1G} ——the design value of the bending moment produced in the calculated section by the deadweight of precast member, precast floor slab and composite layer;

M_{2G} ——the design value of the bending moment produced in the calculated section by the deadweight of surface layer, suspended ceiling and others during the second stage;

M_{1Q} ——the design value of the bending moment produced in the calculated section by the live construction loads during the first stage;

M_{2Q} ——the design value of the bending moment produced in the calculated section by the variable loads during the second stage, which shall be taken as the larger value between the design values of bending moment produced in the calculated section by the live construction loads during this stage and the variable loads during the service stage.

In the calculation, the concrete strength grade of positive bending moment segment shall be taken in accordance with the composite layer; the concrete strength grade of negative bending moment segment shall be taken in accordance with the actual conditions of the

compression zone of calculated section.

H.0.3 The inclined section shear bearing capacity of precast members and composite members shall be calculated in accordance with those specified in Section 6.3 of this code. Therein, the design value of shear force shall be taken according to the following requirements:

Precast member

$$V_1 = V_{1G} + V_{1Q} \quad (\text{H.0.3-1})$$

Composite member

$$V = V_{1G} + V_{2G} + V_{2Q} \quad (\text{H.0.3-2})$$

Where:

V_{1G} ——the design value of the shear force produced in the calculated section by the deadweight of precast member, precast floor slab and composite layer;

V_{2G} ——the design value of the shear force produced in the calculated section by the deadweight of surface layer, suspended ceiling and others during the second stage;

V_{2Q} ——the design value of the shear force produced in the calculated section by the live construction loads during the first stage;

V_{2Q} ——the design value of the shear force produced by the variable loads during the second stage, which shall be taken as the larger value between the design values of shear force produced in the calculated section by the live construction loads during this stage and the variable loads during the service stage.

In the calculation, the design value of shear bearing capacity V_{cs} of the concrete and stirrups on the inclined section of composite member shall be calculated by adopting with the lower concrete strength grade among the composite layer and precast members and shall not be less than the design value of shear bearing capacity of precast members; as for the prestressed concrete composite member, $V_p=0$, without considering the favorable influence of prestress on the shear bearing capacity.

H.0.4 Where the composite beam meets the detailing requirements of beam as stated in Section 9.2 of this code, the shear bearing capacity of its composite surface shall meet the following requirements:

$$V \leq 1.2 f_t b h_0 + 0.85 f_{yv} \frac{A_{sv}}{s} h_0 \quad (\text{H.0.4-1})$$

Where, the design value of concrete tensile strength f_t shall be taken as the smaller value between the composite layer and precast member.

As for the composite slab without stirrups, if it meets the detailing requirements on the roughness of composite interface stated in this code, the shear strength of its composite surface shall meet the requirements of the following formula:

$$\frac{V}{b h_0} \leq 0.4 \text{ (N/mm}^2\text{)} \quad (\text{H.0.4-2})$$

H.0.5 As for the prestressed concrete composite flexural members, their precast members and composite members shall be carried out with the crack resistance checking of normal section. In this case, in relation to the characteristic combination of loads, the tension stress of edge concrete for the crack resistance checking shall not be larger than the standard value of concrete tensile strength f_{tk} of the precast members. The normal stress of edge concrete for

crack resistance checking shall be calculated according to the following formula:

Precast member

$$\sigma_{ck} = \frac{M_{1k}}{W_{01}} \quad (\text{H.0.5-1})$$

Composite member

$$\sigma_{ck} = \frac{M_{1Gk}}{W_{01}} + \frac{M_{2k}}{W_0} \quad (\text{H.0.5-2})$$

Where:

M_{1Gk} ——the bending moment value produced in the calculated section by the deadweight of precast member, the deadweight of precast floor slab and the standard value of deadweight of composite layer;

M_{1k} ——the bending moment value produced in the calculated section under the characteristic combination of loads during the first stage, $M_{1k}=M_{1Gk}+M_{1Qk}$, and herein, M_{1Qk} is the bending moment produced in the calculated section by the standard value of live construction loads during the first stage;

M_{2k} ——the bending moment value produced in the calculated section by characteristic combination of loads during the second stage, $M_{2k}=M_{2Gk}+M_{2Qk}$, and herein, M_{2Gk} is the bending moment value produced in the calculated section by the standard values of the deadweight of surface layer, suspended ceiling and others; M_{2Qk} is the bending moment value produced in the calculated section by the standard value of the variable loads during the service stage;

W_{01} ——the moment of elastic resistance at the tension edge of the transformed section of precast member;

W_0 ——the moment of elastic resistance at the tension edge of the transformed section of composite member, here, the sectional area of concrete at the composite layer shall be converted to the sectional area of concrete of precast member in accordance with elastic modulus ratio.

H.0.6 The prestressed concrete composite members shall be carried out with crack resistance checking of inclined section in accordance with those specified in Article 7.1.5 of this code; the principal tensile stress and principal compressive stress of concrete shall be calculated in accordance with those specified in Article 7.1.6 of this code by considering with the loading-bearing characteristics of composite members.

H.0.7 Under the quasi-permanent combination of loads, the stress σ_{sq} of longitudinal tension steel reinforcement for the reinforced concrete composite flexural member shall meet the following requirements:

$$\sigma_{sq} \leq 0.9f_y \quad (\text{H.0.7-1})$$

$$\sigma_{sq} = \sigma_{s1k} + \sigma_{s2q} \quad (\text{H.0.7-2})$$

Under the action of bending moment M_{1Gk} , the stress σ_{slk} of longitudinal tension steel reinforcement for precast member may be calculated according to the following formula:

$$\sigma_{slk} = \frac{M_{1Gk}}{0.87A_s h_{01}} \quad (\text{H.0.7-3})$$

Where:

h_{01} ——the effective depth of section of precast member.

Under the action of bending moment M_{2q} corresponding to the quasi-permanent combination of loads, the increment σ_{s2q} of the stress of longitudinal tension steel reinforcement for composite member may be calculated according to the following formula:

$$\sigma_{s2q} = \frac{0.5 \left(1 + \frac{h_1}{h} \right) M_{2q}}{0.87 A_s h_0} \quad (\text{H.0.7-4})$$

When $M_{1Gk} < 0.35 M_{1u}$, the value of $0.5 \left(1 + \frac{h_1}{h} \right)$ in Formula (H.0.7-4) shall be equal to

1.0; here, M_{1u} is the design value of the normal section flexural load-bearing capacity of precast member and shall be calculated in accordance with Section 6.2 of this code, however, the formula shall take the equal sign and M shall be replaced by M_{1u} .

H.0.8 For the concrete composite members, the checking of crack width shall be carried out, and the maximum crack width $w_{\max}$ calculated in accordance with the quasi-permanent combination or characteristic combination of loads and considering the influence of long-term action shall not exceed the limit for maximum crack width as specified in Section 3.4 of this code. The maximum crack width $w_{\max}$, in accordance with the quasi-permanent combination or characteristic combination of loads and considering the influence of long-term action, may be calculated according to the following formulae:

Reinforced concrete member

$$w_{\max} = 2 \frac{\varphi(\sigma_{slk} + \sigma_{s2q})}{E_s} \left(1.9c + 0.08 \frac{d_{eq}}{\rho_{tel}} \right) \quad (\text{H.0.8-1})$$

$$\varphi = 1.1 - \frac{0.65 f_{tkl}}{\rho_{tel} \sigma_{slk} + \rho_{te} \sigma_{s2q}} \quad (\text{H.0.8-2})$$

Prestressed concrete member

$$w_{\max} = 1.6 \frac{\varphi(\sigma_{slk} + \sigma_{s2k})}{E_s} \left(1.9c + 0.08 \frac{d_{eq}}{\rho_{tel}} \right) \quad (\text{H.0.8-3})$$

$$\varphi = 1.1 - \frac{0.65 f_{tkl}}{\rho_{tel} \sigma_{slk} + \rho_{te} \sigma_{s2k}} \quad (\text{H.0.8-4})$$

Where:

d_{eq} ——the equivalent diameter of longitudinal steel reinforcement in tension zone, which shall be calculated in accordance with those specified in Article 7.1.2 of this code;

ρ_{tel} , ρ_{te} ——the ratio of reinforcement of longitudinal tension steel reinforcements calculated in accordance with the area of effective tension concrete section of precast member and composite member, which shall be calculated in accordance with Article 7.1.2 of this code;

f_{tkl} ——the standard value of concrete tensile strength of precast member.

H.0.9 The composite members shall be carried out with the deflection checking at serviceability limit state in accordance with those specified in Article 7.2.1 of this code. Therein, the stiffness composite flexural members, in accordance with the quasi-permanent combination or characteristic combination of loads and considering the influence of long-term action, may be calculated according to the following formulae:

Reinforced concrete member

$$B = \frac{M_q}{\left(\frac{B_{s2}}{B_{s1}} - 1 \right) M_{1Gk} + \theta M_q} B_{s2} \quad (\text{H.0.9-1})$$

Prestressed concrete member

$$B = \frac{M_k}{\left(\frac{B_{s2}}{B_{s1}} - 1 \right) M_{1Gk} + (\theta - 1) M_q + M_k} B_{s2} \quad (\text{H.0.9-2})$$

$$M_k = M_{1Gk} + M_{2k} \quad (\text{H.0.9-3})$$

$$M_q = M_{1Gk} + M_{2Gk} + \varphi_q M_{2Qk} \quad (\text{H.0.9-4})$$

Where:

θ ——the influence coefficient to deflection increment by considering long-term action of loads, which shall be taken according to Article 7.2.5 of this code;

M_k ——the bending moment value of composite member that is calculated in accordance with the characteristic combination of loads;

M_q ——the bending moment value of composite member that is calculated in accordance with the quasi-permanent combination of loads;

B_{s1} ——the short-term stiffness of precast member, which shall be taken in accordance with Article H.0.10 of this code;

B_{s2} ——the short-term stiffness of composite member during the second stage, which shall be taken in accordance with Article H.0.10 of this code;

φ_q ——the coefficient of quasi-permanent value of variable loads during the second stage.

H.0.10 Under the action of the quasi-permanent combination or characteristic combination of loads, the short-term stiffness of composite flexural member in its positive bending moment segment may be calculated according to the following requirements.

1 Reinforced concrete composite member

- 1) The short-term stiffness B_{s1} of precast member may be calculated in accordance with Formula (7.2.3-1) in this code.
- 2) The short-term stiffness of composite member during the second stage may be calculated according to the following formula:

$$B_{s2} = \frac{E_s A_s h_0^2}{0.7 + 0.6 \frac{h_1}{h} + \frac{45 a_E \rho}{1 + 3.5 \gamma_f'}} \quad (\text{H.0.10-1})$$

Where:

a_E ——the ratio of the elastic modulus of steel reinforcement to the elastic modulus of

concrete of the composite layer: $a_E = E_s/E_{c2}$.

2 Prestressed concrete composite member

- 1) The short-term stiffness B_{s1} of precast member may be calculated in accordance with Formula (7.2.3-2) in this code.
- 2) The short-term stiffness of composite member during the second stage may be calculated according to the following formula:

$$B_{s2} = 0.7E_{c1}I_0 \quad (\text{H.0.10-2})$$

Where:

E_{c1} ——the elastic modulus of concrete of precast members;

I_0 ——the inertia moment of the transformed section of composite member, here, the sectional area of concrete at the composite layer shall be converted to the sectional area of concrete of precast member in accordance with elastic modulus ratio.

H.0.11 Under the action of quasi-permanent combination or characteristic combination of loads, the short-term stiffness B_{s2} during the second stage in the negative bending moment segment of composite flexural member may be calculated according to Formula (7.2.3-1) in this code, therein, the ratio of elastic modulus shall be $a_E = E_s/E_{c1}$.

H.0.12 The inverted camber value of the prestress prestressed concrete composite member during its service stage may be calculated according to the stiffness of precast member with the method of structural mechanics. In the calculation, the stress of prestressed steel reinforcement shall be deducted by all the prestress loss; with consideration of the long term effect of prestress, the calculated inverted camber value of prestress shall be multiplied by a amplified coefficient 1.75.

Appendix J Prestress Loss of Curved Post-tensioned Prestressing

Tendons and/or Bars Due to Anchorage Deformation and Tendon

Shrinkage

J.0.1 In post-tensioned members, the prestress loss of curved prestressing tendon and/or bar due to anchorage deformation and tendon shrinkage shall be calculated.

1 The influence length l_f (mm) (Figure J.0.1) of reverse friction may be calculated according to the following formulae:

$$l_f = \sqrt{\frac{a \cdot E_p}{\Delta\sigma_d}} \quad (\text{J.0.1-1})$$

$$\Delta\sigma_d = \frac{\sigma_0 - \sigma_l}{l} \quad (\text{J.0.1-2})$$

Where:

a ——the value of the deformation of tensioning end anchorage and the inner shrinkage of prestressing tendon and/or bar (mm), which shall be adopted in accordance with Table 10.2.2 of this code;

$\Delta\sigma_d$ ——the prestress loss in unit length caused by the friction of pipe (MPa/mm);

σ_0 ——the control stress under the tensioning end anchorage, which shall be adopted according to those specified in Article 10.1.3 of this code;

σ_l ——the stress at anchoring end of prestressing tendon and/or bar after excluding the friction loss along the line;

l ——the distance from the tensioning end to anchoring end (mm).

2 When $l_f \leq l$, the prestress loss σ_{l1} of prestressing tendon and/or bar, after considering the reverse friction, at the position where is x away from the tensioning end may be calculated according to the following formulae:

$$\sigma_{l1} = \Delta\sigma \frac{l_f - x}{l_f} \quad (\text{J.0.1-3})$$

$$\Delta\sigma = 2\sigma_d l_f \quad (\text{J.0.1-4})$$

Where:

$\Delta\sigma$ ——the prestress loss value of prestressing tendon and/or bar under the tensioning end anchorage after considering the reverse friction.

3 When $l_f > l$, the prestress loss σ'_{l1} of prestressing tendon and/or bar, after considering the reverse friction, at the position where is x' away from the tensioning end may be calculated according to the following formula:

$$\sigma'_{l1} = \Delta\sigma' - 2x'\Delta\sigma_d \quad (\text{J.0.1-5})$$

Where:

$\Delta\sigma'$ —the prestress loss value of prestressing tendon and/or bar under the tensioning end anchorage after considering the reverse friction, which may be obtained according to the following method: In Figure J.0.1, it is supposed that the area of "ca'bd" isosceles trapezoid $A=a \cdot E_p$, and the cd value may be calculated, then $\Delta\sigma'=cd$.

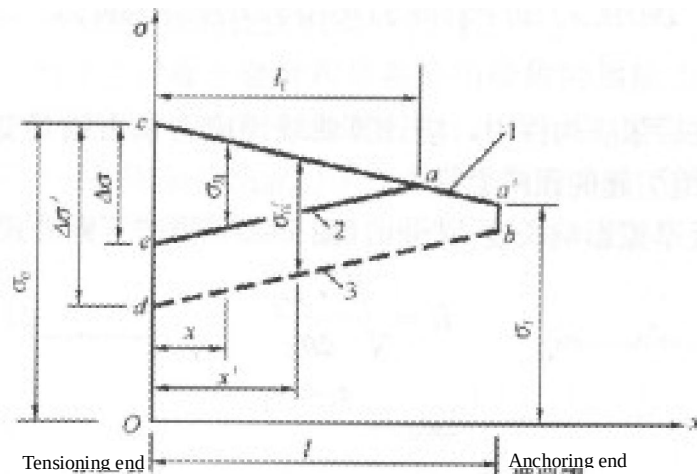


Figure J.0.1 Calculation of Prestress Loss after Considering the Reverse Friction

Notes:

- 1 caa' means the stress distribution line of prestressing tendon and/or bar after deducting the positive friction loss of pipe;
- 2 $ea'a'$ means the stress distribution line of prestressing tendon and/or bar after deducting the loss due to positive friction and inner shrinkage (considering the reverse friction) of pipe when $l_f \leq l$;
- 3 db means the stress distribution line of prestressing tendon and/or bar after deducting the loss due to positive friction and inner shrinkage (considering the reverse friction) of pipe when $l_f > l$.

J.0.2 Where the loss influence length of tensioning at both ends (tensioning in several times or simultaneously) is overlapped with the that of reverse friction, then the stress of prestressing tendon and/or bar in one same section within the overlapped scope after deducting the loss of positive friction and rebounding reverse friction may be taken as: the larger value, which is obtained from the control stress under the tensioning end anchorage minus the above-mentioned stress calculation results, by respectively tensioning and anchoring both ends and calculating the loss of positive friction and rebound reverse friction.

J.0.3 As for the curved post-tensioned prestressing tendon and/or bar or broken-line prestressing tendon and/or bar of the common tendon shapes, the value of their prestress loss σ_{l1} within the scope of reverse friction influence length l_f due to the anchorage deformation and the inner shrinkage of prestressing tendon and/or bar may be calculated according to the following formula:

1 The prestressing tendon and/or bar of parabolic shape may be considered approximately according to the curved prestressing tendon and/or bar of circular arc shape (Figure J.0.3-1). Where the corresponding central angle $\theta \leq 45^\circ$ ($\theta \leq 90^\circ$ for the unbonded prestressing tendon and/or bar), the value of prestress loss σ_{l1} may be calculated according to the following formula:

$$\sigma_{l1} = 2\sigma_{con} l_f \left(\frac{\mu}{r_c} + k \right) \left(1 - \frac{x}{l_f} \right) \quad (J.0.3-1)$$

The influence length l_f (m) of reverse friction may be calculated according to the following formula:

$$l_f = \sqrt{\frac{aE_s}{1000\sigma_{con}(\mu/r_c + k)}} \quad (\text{J.0.3-2})$$

Where:

r_c —the curvature radius of the curved prestressing tendon and/or bar of circular arc shape (m);

μ —the friction coefficient between the prestressing tendon and/or bar and pore path walls, which shall be adopted in accordance with Table 10.2.4 of this code;

k —the friction coefficient in consideration of partial deviation in each meter length of the pore path, which shall be adopted in accordance with Table 10.2.4 of this code;

x —the distance from the tensioning end to the calculated section (m);

a —the value of the deformation of tensioning end anchorage and the inner shrinkage of prestressing tendon and/or bar (mm), which shall be adopted in accordance with Table 10.2.2 of this code;

E_s —the elastic modulus of prestressing tendon and/or bar.

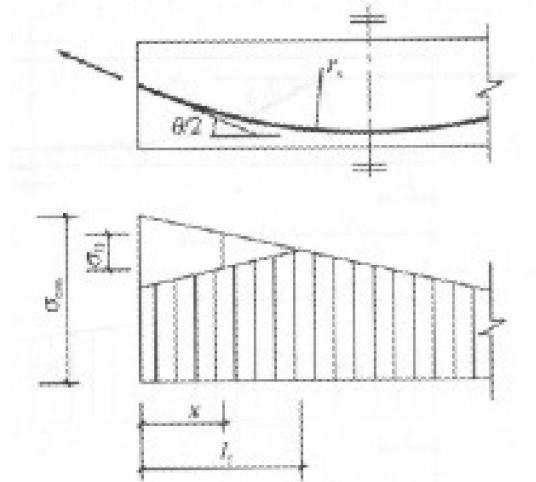


Figure J.0.3-1 Prestress Loss σ_{11} of the Curved Prestressing Tendon and/or Bar of Circular Arc Shape

2 As for the prestressing tendon and/or bar (Figure J.0.3-2) that is straight lines at ends (the length of straight line is l_0) and is composed of two circular curves subsequently (corresponding to the circular arc, the central angle $\theta \leq 45^\circ$, and $\theta \leq 90^\circ$ for the unbonded prestressing tendon and/or bar), the value of its prestress loss σ_{11} may be calculated according to the following formulae:

When $x \leq l_0$

$$\sigma_{11} = 2i_1(l_1 - l_0) + 2i_2(l_f - l_1) \quad (\text{J.0.3-3})$$

When $l_0 < x \leq l_1$

$$\sigma_{11} = 2i_1(l_1 - x) + 2i_2(l_f - l_1) \quad (\text{J.0.3-4})$$

When $l_1 < x \leq l_f$

$$\sigma_{11} = 2i_2(l_f - x) \quad (\text{J.0.3-5})$$

The influence length l_f (m) of reverse friction may be calculated according to the following formulae:

$$l_f = \sqrt{\frac{aE_s}{1000i_2} - \frac{i_1(l_1^2 - l_0^2)}{i_2}} + l_1^2 \quad (\text{J.0.3-6})$$

$$i_1 = \sigma_a(k + \mu/r_{c1}) \quad (\text{J.0.3-7})$$

$$i_2 = \sigma_b(k + \mu/r_{c2}) \quad (\text{J.0.3-8})$$

Where:

l_1 ——the horizontal projection length from the starting point of the tensioning end of prestressing tendon and/or bar to the inflection point;

i_1, i_2 ——respectively the slope of the approximate variation straight line of stress in the first and second segments of circular curved prestressing tendon and/or bar;

r_{c1}, r_{c2} ——respectively the curvature radius of the first and second segments of circular curved prestressing tendon and/or bar;

σ_a, σ_b ——respectively the stress of prestressing tendon and/or bar at Point a and b.

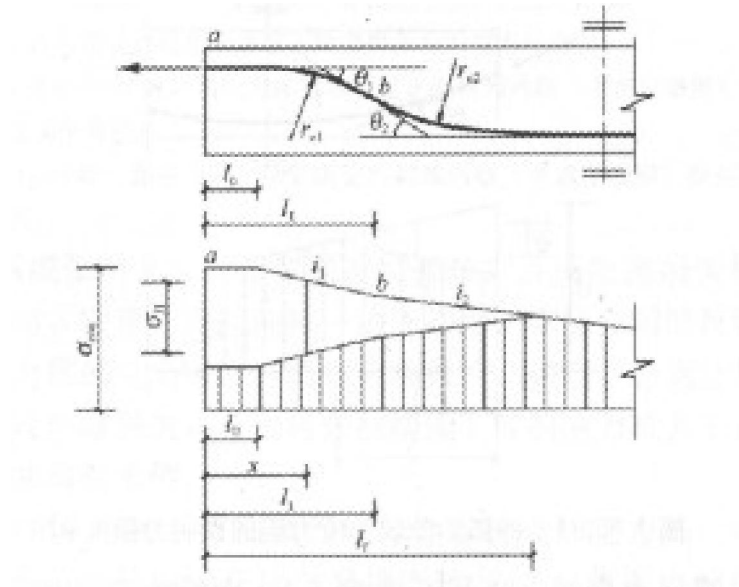


Figure J.0.3-2 Prestress Loss σ_n of the Prestressing Tendon and/or Bar Composed of Two Circular Curves

3 Where the anchorage loss of broken-line prestressing tendon and/or bar disappears outside the break point c (Figure J.0.3-3), the value of prestress loss σ_{l1} may be calculated according to the following formulae:

When $x \leq l_0$

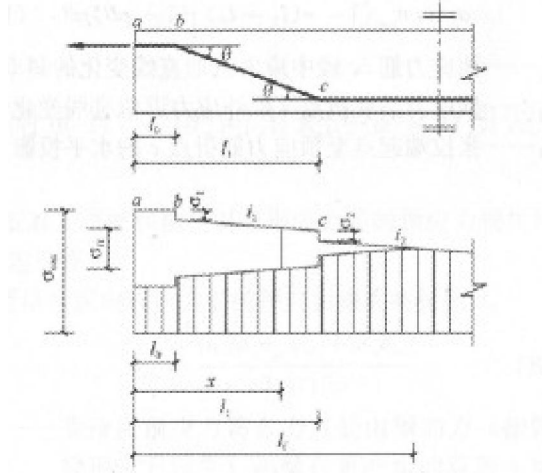


Figure J.0.3-3 Prestress Loss σ_{n1} of the Broken-line Prestressing Tendon and/or Bar

$$\sigma_{n1} = 2\sigma_1 + 2i_1(l_1 - l_0) + 2\sigma_2 + 2i_2(l_f - l_1) \quad (\text{J.0.3-9})$$

When $l_0 < x \leq l_1$

$$\sigma_{n1} = 2i_1(l_1 - x) + 2\sigma_2 + 2i_2(l_f - l_1) \quad (\text{J.0.3-10})$$

When $l_1 < x \leq l_f$

$$\sigma_{n1} = 2i_2(l_f - x) \quad (\text{J.0.3-11})$$

The influence length l_f (m) of reverse friction may be calculated according to the following formulae:

$$l_f = \sqrt{\frac{aE_s}{1000i_2} - \frac{i_1(l_1 - l_0)^2 + 2i_1l_0(l_1 - l_0) + 2\sigma_1l_0 + 2\sigma_2l_1}{i_2}} + l_1^2 \quad (\text{J.0.3-12})$$

$$i_1 = \sigma_{\text{con}}(1 - \mu\theta)k \quad (\text{J.0.3-13})$$

$$i_2 = \sigma_{\text{con}}[1 - k(l_1 - l_0)](1 - \mu\theta)^2k \quad (\text{J.0.3-14})$$

$$\sigma_1 = \sigma_{\text{con}}\mu\theta \quad (\text{J.0.3-15})$$

$$\sigma_2 = \sigma_{\text{con}}[1 - k(l_1 - l_0)](1 - \mu\theta)\mu\theta \quad (\text{J.0.3-16})$$

Where:

i_1 ——the slope of the approximate variation straight line of stress in bc segment of prestressing tendon and/or bar;

i_2 ——the slope of the approximate variation straight line of stress outside the break point c of prestressing tendon and/or bar;

l_1 ——the horizontal projection length from the starting point of tensioning end to the break point c of prestressing tendon and/or bar.

Appendix K Time-dependent Loss of Prestress

K.0.1 The ultimate value of the prestress loss of prestressing tendon and/or bar caused by concrete shrinkage and creep may be calculated according to requirements:

1 The ultimate value σ_{l5} of prestress loss of the longitudinal prestressing tendon and/or bar in tension zone

$$\sigma_{l5} = \frac{0.9a_p \sigma_{pc} \varphi_{\infty} + E_s \varepsilon_{\infty}}{1 + 15\rho} \quad (\text{K.0.1-1})$$

Where:

σ_{pc} ——the normal compression stress of concrete produced at the point of resultant force of prestressing tendon and/or bar in tension zone due to the pre-applied force (excluding the prestress loss of the corresponding stage) and the deadweight of beam, its value shall not be larger than $0.5 f'_{cu}$; for the simple-supported beam, σ_{pc} may take the mean value of the mid-span section and the section at 1/4 span; as for continuous beam and frame, σ_{pc} may take the mean value of several representative sections;

φ_{∞} ——the ultimate value of the creep coefficient of concrete;

ε_{∞} ——the ultimate value of the shrinkage strain of concrete;

E_s ——the elastic modulus of prestressing tendon and/or bar;

a_p ——the ratio of the elastic modulus of prestressing tendon and/or bar to the elastic modulus of concrete;

ρ ——the ratio of reinforcement of prestressing tendons and/or bars and steel bars in the tension zone: as for the pre-tensioned members, $\rho = (A_p + A_s)/A_0$; as for post-tensioned members, $\rho = (A_p + A_s)/A_n$; as for the members symmetrically distributed with prestressing tendons and/or bars and steel bars, the ratio of reinforcement ρ shall be one half of the gross sectional area of steel reinforcement.

If there is no reliable information, the values of φ_{∞} and ε_{∞} may be adopted in accordance with Table K.0.1-1 and Table K.0.1-2. If the structure is in such environment where the annual average relative humidity is less than 40%, the values listed in these two tables shall be increased by 30%.

Table K.0.1-1 Ultimate Value of the Shrinkage Strain of Concrete $\varepsilon_{\infty} (\times 10^{-4})$

Annual average relative humidity RH		40% ≤ RH < 70%				70% ≤ RH ≤ 99%			
Theoretical thickness 2A/u (mm)		100	200	300	≥600	100	200	300	≥600
Concrete age under prestress t_0 (d)	3	4.83	4.09	3.57	3.09	3.47	2.95	2.60	2.26
	7	4.35	3.89	3.44	3.01	3.12	2.80	2.49	2.18
	10	4.06	3.77	3.37	2.96	2.91	2.70	2.42	2.14
	14	2.73	3.62	3.27	2.91	2.67	2.59	2.35	2.10
	28	2.90	3.20	3.01	2.77	2.07	2.28	2.15	1.98
	60	1.92	2.54	2.58	2.54	1.37	1.80	1.82	1.80
	≥90	1.45	2.12	2.27	2.38	1.03	1.50	1.60	1.68

Table K.0.1-2 Ultimate Value of the Creep Coefficient of Concrete φ_{∞}

Annual average relative humidity RH		$40\% \leq RH < 70\%$				$70\% \leq RH \leq 99\%$			
Theoretical thickness $2A/u(\text{mm})$		100	200	300	≥ 600	100	200	300	≥ 600
Concrete age under prestress $t_0(\text{d})$	3	3.51	3.14	2.94	2.63	2.78	2.55	2.43	2.23
	7	3.00	2.68	2.51	2.25	2.37	2.18	2.08	1.91
	10	2.80	2.51	2.35	2.10	2.22	2.04	1.94	1.78
	14	2.63	2.35	2.21	1.97	2.08	1.91	1.82	1.67
	28	2.31	2.06	1.93	1.73	1.82	1.68	1.60	1.47
	60	1.99	1.78	1.67	1.49	1.58	1.45	1.38	1.27
	≥ 90	1.85	1.65	1.55	1.38	1.46	1.34	1.28	1.17

Notes:

- 1 The concrete age under prestress may be taken between 3d and 7d for pre-tensioned members and between 7d and 28d for post-tensioned members;
- 2 A is the sectional area of member, u is the peripheral length of this section contacting with the atmosphere; if the member is of variable section, both A and u may be taken as their mean values;
- 3 This table is applicable to the concrete prepared with common portland cement or rapid hardening cement; the values in the above table are calculated according to the concrete with strength grade of C40, and they shall be multiplied by

$$\sqrt{\frac{32.4}{f_{ck}}}$$

in relation to the concrete of or above C50, therein, f_{ck} is the standard value of axial compressive strength of concrete (MPa);

- 4 This table is applicable to the cases that the seasonal mean temperature change is between -20°C and $+40^\circ\text{C}$;
- 5 Where the theoretical thickness of actual member and the concrete age under prestress are the medium values of those values listed in the table, they may be determined according to linear interpolation method.

2 The ultimate value σ'_{l5} of prestress loss of the longitudinal prestressing tendon and/or bar in compression zone

$$\sigma'_{l5} = \frac{0.9a_p \sigma'_{pc} \varphi_\infty + E_s \varepsilon_\infty}{1 + 15\rho'} \quad (\text{K.0.1-2})$$

Where:

σ'_{pc} ——the normal compression stress of concrete produced at the point of resultant force of prestressing tendons and/or bars in compression zone due to the pre-applied force (excluding the prestress loss of corresponding stage) and the deadweight of beam, and its value shall not be larger than $0.5 f'_{cu}$, and $\sigma'_{pc} = 0$ when σ'_{pc} is tension stress;

ρ' ——the ratio of reinforcement of prestressing tendons and/or bars and steel bars in the compression zone: as for pre-tensioned members, $\rho' = (A'_p + A'_s) / A_0$; as for post-tensioned members, $\rho' = (A'_p + A'_s) / A_n$.

Note: As for the member, configured with prestressing tendon and/or bar A'_p and steel bar A'_s in the compression

zone, σ_{pc} and σ'_{pc} in Formula (K.0.1-1) and Formula (K.0.1-2) shall be calculated according to the gross pre-applied force of the section.

K.0.2 The value of prestressing loss due to concrete shrinkage and creep with consideration of the time influence may be determined by the ultimate value σ_{l5} and σ'_{l5} of prestress loss calculated in Article K.0.1 multiplied by the corresponding coefficient in Table K.0.2.

The value of prestressing loss due to stress relaxation of prestressing tendon and/or bar with consideration of the time influence may be determined by the prestress loss value σ_{l4} calculated in Article 10.2.1 of this code multiplied by the corresponding coefficient in Table K.0.2.

Table K.0.2 Coefficient of Time-dependent Loss of Prestress

Time (d)	Coefficient of loss due to relaxation	Coefficient of loss due to shrinkage and creep
2	0.50	—
10	0.77	0.33
20	0.88	0.37
30	0.95	0.40
40	1.00	0.43
60		0.50
90		0.60
180		0.75
365		0.85
1095		1.00

Notes:

- 1 For the pre-tensioned prestressed concrete members, the time of relaxation loss shall be calculated starting from the completion of tensioning and the time of shrinkage and creep loss shall be calculated starting from the completion of extension;
- 2 For the post-tensioned prestressed concrete members, the time of both the relaxation loss and the shrinkage and creep loss shall be calculated starting from the completion of tensioning.

Explanation of Wording in This Code

1 Words used for different degrees of strictness are explained as follows in order to mark the differences in executing the requirements in this code:

- 1) Words denoting a very strict or mandatory requirement:
"Must" is used for affirmation, "must not" for negation;
- 2) Words denoting a strict requirement under normal conditions:
"Shall" is used for affirmation, "shall not" for negation.
- 3) Words denoting a permission of a slight choice or an indication of the most suitable choice when conditions permit:
"Should" is used for affirmation, "should not" for negation;
- 4) "May" is used to express the option available, sometimes with the conditional permit.

2 "shall comply with..." or "shall meet the requirements of..." is used in this code to indicate that it is necessary to comply with the requirements stipulated in other relative standards and codes.

List of Quoted Standards

- 1 "Unified Standard for Reliability Design of Engineering Structures" (GB 50153)
- 2 "Unified Standard Reliability Design of Building Structures" (GB 50068)
- 3 "Load Code for the Design of Building Structures" (GB 50009)
- 4 "Code for Seismic Design of Buildings" (GB 50011)
- 5 "Steel for the Reinforcement of Concrete" (GB 1499)
- 6 "Steel Wires for the Prestressing of Concrete" (GB/T 5223)
- 7 "Steel Strand for Prestressed Concrete" (GB/T 5224)
- 8 "Standard for Evaluation of Concrete Compressive Strength" (GB/T 50107)
- 9 "Code for Construction of Concrete Structures" (GB 50666)
- 10 "Code for Acceptance of Constructional Quality of Concrete Structures" (GB 50204)